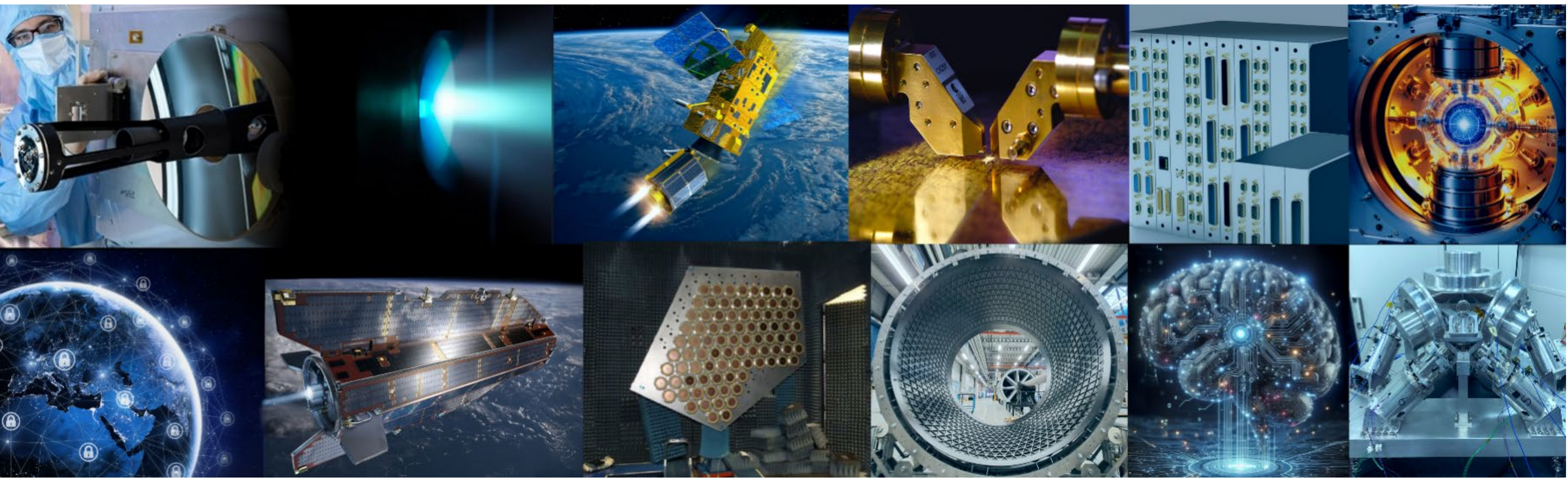
GSTP Element 1 Develop Compendia 2026 - 2028

Generic Technologies
Artificial Intelligence
Quantum Technologies
Innovative Propulsion
Sustainable Space
(Cyber)security
Serialisation (Standardised Sub-Systems)
Technologies for VLEO (Very Low Earth Orbit)



GSTP Management
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1. Introduction

The GSTP E1 Develop Compendia 2026 - 2028 include a list of candidate activities for the GSTP E1 Develop Work Plan, in line with ESA Strategy 2040 and the ESA Technology Vision 2040. Activities presented in this document have been selected following a streamlined R&D definition process, including programmatic and application domain screening and ensuring consistency with THAG Roadmaps.

The aim of the GSTP E1 Develop Compendia is to provide to industry and Delegations a consolidated overview of the priorities for ESA in the development of technologies within the GSTP. These compendia foster competition and cooperation among economic operators of the GSTP Participating States and provides inspiration for tailored industry driven activities.

This document follows the previous GSTP Element 1 Compendia of Potential Activities. In particular, the following documents:

- GSTP Element 1 “Develop” Compendium 2022: Generic Technologies – (ref. ESA-GSTP-TECT-PL-2022-002816), issued in October 2022.
- GSTP Element 1 “Develop” Compendium 2022: Artificial Intelligence – (ref. ESA-GSTP-TECT-PL-2022-002817), issued in October 2022.
- GSTP Element 1 “Develop” Compendium 2022: Digitalisation – (ref. ESA-GSTP-TECT-PL-2022-002818), issued in October 2022.
- GSTP Element 1 “Develop” Compendium 2022: Quantum – (ref. ESA-GSTP-TECT-PL-2022-002819), issued in October 2022.
- GSTP Element 1 “Develop” Compendium 2022: Cybersecurity – (ref. ESA-GSTP-TECT-PL-2022-003078), issued in November 2022.

This document provides a list and descriptions of 202 (~280M€) candidate activities for the Work Plan of the GSTP Element 1 Develop targeted for implementation in the 2026 - 2028 period. The GSTP Element 1 Develop Compendia 2026-2028 edition is structured in 3 areas:

- **Generic Technology Areas** - Generic activity developments that cover all space applications.
- **Technology Focus Areas** – Activity developments targeting technology themes that need special attention, as highlighted in the context of the ESA Strategy 2040:
    - Artificial Intelligence
    - Quantum Technologies
    - Innovative propulsion
    - Sustainable Space
- **Specific Areas** – Activity developments targeting specific areas, as recently introduced in GSTP Element 1, providing the first mechanism to launch and prioritise these topics.
    - (Cyber)security
    - Serialisation / Standardised Sub-Systems
    - Technologies for VLEO (Very Low Earth Orbit)

These compendia are issued to Delegations of the GSTP Participating States and their companies and academia for expression of interest. This expression of interest will be considered in the following updates of the work plan for this GSTP Element 1 Develop. The objective is to have a good indication of the activities the GSTP Participating States intend to support to present updates of the GSTP E1 Develop work plan with consolidated set of activities to the Industrial Policy Committee (IPC) for approval.

Industry and academia can indicate their expression of interest per activity in the OSIP campaign: RFI - GSTP Element 1 "Develop" Compendia 2026 – 2028.

The following requirements have driven the definition and selection of activities for this GSTP Element 1 Develop Compendia 2026 - 2028:

- Technology developments corresponding to the GSTP objectives, notably in terms of Technology Readiness Levels (TRLs) and application domains.
- The ESA Strategy 2040 issued in January 2025.

- The ESA Technology Vision 2040 issued in November 2024.

In addition, elements of critical importance were also considered: fostering state of art technology adoption, innovation, technology push, spin–in of commercial technologies, supporting industrial sustainability, capacity building, European competitiveness and risk obsolescence management of critical parts.

ACCELERATE Initiative

With the **ACCELERATE** initiative ESA plans to speed-up the procurement of the activities (improving “Time-to-Contract”) and to increase the predictability of the procurement (Invitations to tenders) publication.

ACCELERATE proposes several mechanisms to achieve these goals. One of this is the **bundling of procurements**, which is planned within this Compendia.

In this context a set of technology development activities will be proposed to be bundled into synchronised procurements. It is proposed to cover 2 to 5 activities per procurement batch “Bundle”, addressing a specific sub-set of activities for a given theme or addressing a common system or product development.
Such a programming of procurements in a concentrated manner (i.e., Bundles) will ensure as well the parallel implementation of the development activities ensuring achieving results in a coordinated manner.

In addition, the intention is to indicate the targeted Invitation to Tender publication dates of the Bundles proposed in this Compendia once they are presented and approved as part of the GSTP E1 WP, this will enhance the timeliness and the schedule predictability of the Invitations to Tender.

The activities proposed to be implemented in batches are presented in **Annex I – Proposed bundled activities**.

Mission Classification

The activities in these compendia show an indicative mission classification, according to the table below.

Mission Characteristics Criteria & Related Weighting Factors	Class Level			
	Alpha	**Beta**	**Gamma**	**Delta**
Acceptable Risk Risk of not fulfilling some or all of the mission objectives	**LOW**			**HIGH**
Criticality to Agency Objectives, Strategy and Image Flagship mission, international co-operation, impact on strategic ESA goals and image	**Extremely Critical**	**Highly Critical**	**Medium Criticality**	**Low Criticality**
Cost Cost at completion inc. Phase E1	**> 400 M€**	**200 - 400 M€**	**25 – 200 M€**	**< 25 M€**
Mission Lifetime Nominal mission life duration	**> 7 years**	**5 -7 years**	**2-5 years**	**< 2 years**
Mission complexity Design interfaces, unique payloads, new technology development.	**Extremely Complex**	**Highly Complex**	**Medium Complexity**	**Low Complexity**

Technology Domain

The technology domain has been identified for the activities in this compendia, following the ESA Technology Tree v.4.1 (https://esamultimedia.esa.int/multimedia/publications/STM-277/STM-277.pdf).


Generic
Technologies

2.1. Introduction and selection criteria

The different generic technology areas and their selection criteria for the activities in this GSTP compendium are presented hereafter.

GNC, AOCS and Pointing

The GNC, AOCS and Pointing systems are evolving rapidly to meet the increasing demands of future space missions, characterized by higher autonomy, agility, robustness, and system complexity. Well-established techniques for Estimation, Control, Navigation, and Guidance are being augmented by a new generation of methods that integrate fault tolerance, AI-driven autonomy, and digital engineering practices. These advances are essential for addressing challenges such as multi-actuated vehicle configurations, constellation-scale operations, and new mission scenarios requiring real-time decision-making.

Recent and upcoming activities focus on developing scalable, integrated control architectures capable of managing actuator and sensor faults without relying on simplified decoupling assumptions. At the same time, modular and enhanced systems are being introduced to optimize agility, trajectory control, and on-board decision-making under uncertainty. These include learning-based controllers, model predictive control, and embedded solvers tailored for constrained, real-time environments.

A major emphasis is placed on enabling mission-vehicle management for autonomous rendezvous and proximity operations, integrating collision avoidance, resource management, and fault detection into onboard systems. Similarly, digital twins and advanced uncertainty propagation tools are being developed to improve the validation, and in-flight performance of GNC, AOCS and Pointing subsystems—especially when flight data is scarce.

Hardware and software co-design is becoming increasingly important. Modular test facilities and space-qualified platforms are being established to prototype and validate novel algorithms. GNC, AOCS and Pointing equipment hardware evolution encompass high-performance CMOS detectors, radiation-tolerant magnetometers, and embedded control logic on representative processing units. These developments are key to support missions for Earth observation, planetary exploration, and in-orbit assembly missions.

Across the board, verification and validation methods are being adapted to support the growing complexity of AOCS and GNC systems. This includes enhanced simulation fidelity, digital engineering workflows, and agile hardware-in-the-loop testing, enabling faster development cycles and improved assurance for new mission classes. These activities collectively strengthen European autonomy and technological leadership in the domain of advanced spacecraft control systems.

Avionics and Electronics, Electrical, and Electromechanical (EEE)

The Avionics and EEE division focus on advancing avionics technologies, innovative EEE components, software frameworks, and standardized hardware modules essential for future space missions. Key areas include AI for avionics, reliable and radiation-hardened electronic components, Commercial Off-The-Shelf (COTS) adoption, data handling and processing, and sustainable space solutions.

A core part of the division’s work supports the Advanced Data Handling Architecture (ADHA) through the creation of standardized Ethernet input/output and router modules. These enable scalable, high-speed data transfer and ensure interoperability via open standards such as Time-Sensitive Networking (TSN). Complementary Engineering Ground Support Equipment (EGSE) tools provide realistic testing environments to validate ADHA units and modules.

Central to the roadmap is the development of AI-enabled flight software observability frameworks designed to improve onboard autonomy. These frameworks offer predictive maintenance, real-time diagnostics, and reduce operator workload, meeting the increasing demands of complex space missions.

Additionally, the division advances autonomous collision avoidance by developing event-based data fusion methods and Spiking Neural Networks (SNNs). This includes generating realistic datasets and ground-based testbeds to enable real-time detection and tracking of space debris in Low Earth Orbit (LEO).

The division also develops Model-Based Design (MBD) tools for heterogeneous System-on-Chip Field-Programmable Gate Arrays (SoC-FPGAs), enabling efficient computational load balancing across Central Processing Units (CPUs), programmable logic, Graphics Processing Units (GPUs), and AI cores, thereby supporting reliable autonomous operation. Advancements in wireless communication include the development of space-grade Impulse-Radio Ultra-Wideband (IR-UWB) radio modules, which support robust, high-precision inter-satellite links essential for future satellite constellations.

Radiation resilience remains a priority through efforts to design low-resource radiation monitors for internal spacecraft integration, as well as flexible test setups for mitigating Single Event Effects (SEE) in FPGAs and SoCs. These developments enable the dependable use of COTS components, especially for AI-driven applications in space.

In the photonics domain, the division focuses on creating space-qualified Vertical Cavity Surface Emitting Lasers (VCSELs) tailored for quantum devices. These meet stringent space requirements and pave the way for compact atomic sensors and quantum clocks.

Sustainability is addressed by optimizing supercapacitor cells that provide safer, higher-power energy storage compared to traditional batteries.

The selected activities align with strategic roadmaps from the Technology Harmonisation Advisory Group (THAG), Component Technology Board (CTB), and the EC-ESA-EDA Joint Task Force (JTF) on Critical Space Technologies for European Strategic Non-Dependence.

Through these efforts, the Avionics and EEE division strengthens Europe’s autonomy and leadership in resilient, interoperable, and sustainable avionics and electronics technologies for future space exploration.

RF Payloads / Technology

The Radio Frequency Payloads and Technology Area is responsible for RF payloads and technologies for space applications and associated laboratory facilities. More specifically, the responsibilities encompass at subsystem and instrument level:

- Payloads with RF interface exploiting different technologies (e.g., analogue, digital, optical) including design and performance analysis tools and testing.
- RF active and passive instruments, including design and performance analysis, engineering and testing up to sub-millimetre waves.
- Telemetry, tracking and control (TT&C) subsystems, payload data transmission (PDT) subsystems including deep space and near-Earth transponders, proximity and intersatellite (ISL) link antennas and equipment, high speed downlink modulators, digital communication equipment.
- Wave-propagation and interaction, including signal impairments and regulatory aspects.
- Performance assessment of RF and Optical remote sensing data products (including spurious effects mitigation and calibration techniques) at Level-1 and Level-2, and associated data processing techniques.
- Antenna systems, architecture, technologies, and techniques for all space applications, including space vehicle TT&C and user segment terminals, submillimetre wave instruments and associated technologies, as well as antenna engineering, and RF testing of antenna and materials.
- RF technologies and RF equipment, also including vacuum electronics and high-power RF phenomena (multipactor, corona and passive intermodulation).
- RF Payload digital equipment, and on-board Payload Signal and Data Processing algorithms and techniques for RF payloads and instruments.
- Time and frequency references, modelling, design tools, measurements, performance characterisation and calibration techniques.

Power systems, EMC and Space Environment

The activities were selected based on their adherence to reviewed strategy of the Power Systems Architects, Solar Generators, Energy Storage & Nuclear, EMC & Harnes and Space Environments and Effects domains, and, when applicable, to the Harmonisation dossiers for the specific domains. Eight activities were selected among the ones in high or medium urgency and criticality of the Harmonisation dossiers, as well as considering the very recent evolution and needs of European Space Industry and missions, also emphasizing technology cost reduction.

In the field of power management, efforts are centered on increasing the integration level of power electronics circuits to enable more efficient and compact conversion stages. Achieving higher performance requires increasingly sophisticated electronics, which, when implemented discretely, occupy excessive board space. The solution lies in developing integrated circuits that consolidate these functions into minimal silicon area.

A pivotal component in this evolution is the integrated high-side GaN driver, which facilitates the creation of smaller, more efficient DC/DC converters using GaN switches. Additionally, the Integrated Standard TM/TC Interface based on CAN technology streamlines the interconnection of power boards within the backplane, offering fast and highly reliable data communication.

The Adaptive Synchronous Rectifier (SR) controller is designed to implement synchronous rectification in converter output stages. While its discrete implementation is complex and prone to timing delays, its integrated form offers a compact and high-performance solution—particularly beneficial for low-voltage outputs required by microcontrollers and FPGAs.

Together, these advanced integrated circuits form the foundation for next-generation commercial DC/DC converter building blocks. The overarching goal is to enable European industry to produce competitive, high-performance converters, aligning with strategic objectives for technological sovereignty and innovation in Europe.

The proposed activities in the solar generator domain target flexible, high-power solar arrays for multi-launch missions, cost-effective solar cells optimised for LEO constellations, and standardised PVA manufacturing for scalable production. Together, they address critical cost, performance, and industrialisation challenges, enabling Europe to meet growing market demands and reduce reliance on non-European suppliers.

The innovative activity proposed in the energy storage systems domain focus on designing containment solutions to prevent space debris propagation in case of batteries overheat or collision with other space debris.

Optics, Robotics, Life Science

The Optics, Robotics and Life Sciences area encompasses a broad spectrum of key technological domains and strategic priorities that lie at the heart of TEC and ESA's mission. From lidar systems and photonics to advanced optical instrumentation and detection technologies; from environmental control and life support systems to foundational robotics components; from optical communications to in-orbit assembly and modular space architectures—this domain holds significant promise for enhancing future mission capabilities and boosting commercial competitiveness.

In this new edition of the Generic Technologies Compendium, the selected activities for this technological area highlight critical building blocks and processes that will enable next-generation missions and payloads across diverse application areas. The focus includes:

- Optical technologies that support low-cost, rapid manufacturing.
- Innovations in lidar enabling compact, high-performance missions and novel applications.
- Advancements in detectors and optical coatings for next-generation infrared payloads.
- Integrated photonics for high-power optical communication systems.
- Key developments in in-situ resource utilisation (ISRU) and production technologies.
- Essential building blocks for adaptable and flexible robotic systems.

Structures, Mechanisms, Materials

This technology domain covers the following technical areas:

- Structures: Structural design and verification methods and tools; High stability and high precision spacecraft structures; Inflatable and deployable structures; Hot structures; Active/adaptive structures; Damage tolerance and health monitoring; Launchers, re-entry vehicles, planetary vehicles; Crew Habitation, Safe Haven and EVA suits; Meteoroid and Debris shield design and analysis; Advanced structural concepts and materials.
- Mechanisms: Mechanism core technologies; Nonexplosive release technologies; Exploration tool technologies; Control electronics technologies; MEMS technologies; Tribology technologies; Mechanism engineering; Pyrotechnics technologies.
- Materials and Processes: Novel Materials and Materials Technology; Materials Processes; Cleanliness and Sterilisation; Space Environmental Effects on Materials and Processes; Modelling of Materials Behaviour and Properties; Non-Destructive Investigation; Materials and Processes Obsolescence; Materials for Electronic Assembly.

The activities proposed in the current Compendium are targeting several highly innovative and crucial developments in space technology. These include advancements in materials to reduce atmospheric drag on satellites, hybrid manufacturing techniques, additive manufacturing for ceramics, and the assessment of European-sourced tow-pregs. Additionally, the proposals explore smart sensing technologies for space textiles, passive dust-repellent coatings for lunar surfaces, generic relubrication devices, ultra-fine positioning actuators, and nanostepping rotary piezoelectric actuators. Collectively, these projects aim to enhance European autonomy, resilience, and competitiveness in space technologies.

Thermal

The thermal activities presented in the current Compendium span a diverse range of breakthrough technologies. They include advancements in cryogenic systems for in-situ resource utilisation, enabling long-term storage of propellants for exploration missions; the development of continuous sub-Kelvin cooling capabilities to meet the demanding requirements of next-generation scientific instruments; and innovative co-simulation approaches for thermal analysis, streamlining collaboration across the supply chain and enhancing model accuracy. Together, these developments push the boundaries of thermal engineering and open new pathways for sustainable and resilient space missions.

E2E Systems

As space systems evolve toward greater complexity, autonomy, and service diversity, the traditional satellite-centric paradigm is giving way to dynamic, interconnected architectures. These systems must support a growing array of applications while operating in increasingly contested and congested environments, underscoring the need for resilience, interoperability, and early validation of innovative concepts.

In this context, End-to-End (E2E) simulators and demonstrators are indispensable. They enable early-stage validation of novel architectures, technologies, and operational concepts, significantly reducing development risks and lifecycle costs. The following activities exemplify this approach:

- End-to-End Testbed for Resilient System Architecture: modern space systems must be intelligent, adaptive, and service-oriented, integrating heterogeneous assets and advanced payloads. This testbed enables the seamless orchestration of diverse components, validating their interoperability and resilience in real time. It supports the transition from monolithic systems to flexible, distributed architectures, ensuring uninterrupted service delivery even under degraded conditions.
- E2E Space System Performance Monitoring Building Blocks: as systems grow in complexity, real-time performance monitoring and predictive analytics become critical. This activity aims to establish a generalised and scalable architecture for integrating internal telemetry with external observations. It enables anomaly detection, classification, and predictive maintenance, ensuring mission reliability and supporting autonomous decision-making.

- E2E RF Constellation Simulator for Resilience Sensing Systems: this simulator directly supports ESA Strategic Action #7, enabling RF-level assessment of the end-to-end behaviour of sensing technologies—such as radar and SAR—under realistic channel impairments and adverse conditions like interference, jamming, and spoofing. It provides a controlled environment to test resilience strategies, ensuring that future European space systems can maintain autonomy and operational continuity during crises—vital for both civilian and security-oriented missions.
- E2E Demonstrator for Cooperative and Non-Cooperative Multistatic Radar for Space Object Detection: with the proliferation of LEO satellites and debris, space situational awareness (SSA) and space traffic management (STM) are more critical than ever. This demonstrator validates the use of multistatic radar configurations to detect objects—both cooperative and non-cooperative—by opportunistically leveraging powerful signals emerging from a variety of applications. It enhances spatial coverage and detection reliability, supporting the development of scalable SSA solutions for the NewSpace era.
- End-to-End Joint Satellite Communications and RF Sensing System Demonstrator: with spectrum scarcity becoming a pressing issue, this demonstrator explores the opportunistic 5G concept of RF sensing using satellite communication signals. It enables dual use of infrastructure for both communications and sensing, supporting emerging applications such as maritime surveillance and UAV traffic control. This approach maximizes resource efficiency and opens new avenues for integrated services.
- End-to-End Simulation and Validation of Quantum-Enhanced Localisation Systems: conventional navigation systems are vulnerable to signal degradation and interference. This activity explores the integration of quantum-enhanced sensors to improve redundancy, fault detection, and integrity monitoring. It is particularly relevant for safety-critical applications such as autonomous vehicles, planetary exploration, and secure navigation in GNSS-denied environments.
- E2E Interoperable Signal-In-Space Interfaces Simulator Tools for SSI Navigation Systems: as ESA looks beyond Earth orbit, interoperability across planetary navigation systems becomes essential. This simulator supports the ESA Technology Vision 2040 by enabling compatibility testing between Signal-in-Space (SSI) PNT systems and deep-space user terminals. It lays the groundwork for a cohesive Solar System navigation infrastructure, supporting future lunar and Martian missions.

Together, these activities form a strategic portfolio that addresses the technical, operational, and strategic challenges of next-generation space systems. By leveraging E2E simulation and demonstration, ESA can de-risk innovation, accelerate development, and ensure mission success in an increasingly complex and contested space environment.

At the same time, these platforms offer new space players a valuable opportunity to demonstrate technical capabilities and foster early-stage partnerships, strengthening Europe’s leadership in space innovation and resilience.

Future Engineering

The objective of the Future Engineering axis is to prepare ESA and industry to apply advanced engineering and manufacturing approaches for future systems. The backbone of that engineering is based on the concept of a Digital Twin, representing the real system in the different phases of the life-cycle and linking it to the data produced.

The Digital Twin concept relies strongly on achievements already reached in the areas of MBSE, digitalization and simulation. However, the underlying data management and interoperability across organizational boundaries still present challenges. Also, the user-friendliness and impact of upcoming standards needs to be addressed to ensure smooth adoption in the industrial context.

The proposed activities therefore focus in the areas of digital continuity, which needs to be reached on one side by addressing the interoperability of the underlying data spaces, leveraging on emerging standards and tools, and supporting data hubs. The operational use of the data spaces in model-based approaches is culminating in pilot activities, addressing the challenges of the industrial context for space missions, such as complex collaboration setups and supply chain management. The third element aims to improve the user interaction with the data and system design, in the different phases from concept to operations.

Ground Segment

Spacecraft and their operational demands require ground segments to be smarter and more sophisticated than ever before. As identified also in the ESA Technology Vision 2040, these ground segments shall allow fully autonomous and secure operations and exploitation of assets in space, facilitating unprecedented data downlink rates and almost real-time data exploitation and dissemination. At the same time, space system architectures are becoming more diverse and complex requiring constant innovation and flexibility in ground segment technologies to support them.

The proposed activities drive innovation and industrial competitiveness in the following key ground segment areas:

- Ground Software and Data Processing Systems: Ground software and data processing systems are fundamental for monitoring and control, real-time spacecraft simulation, mission planning, as well as data processing and dissemination of housekeeping and payload telemetry. The proposed activities focus on innovation leading to systems that are more autonomous, faster, and more flexible in terms of mission architecture. The activities address the needs of both industrial ground segment system providers as well as industrial and institutional mission operators.
- Ground-based Space Communication Systems: This encompasses all elements of the ground segment intended to enable communication with the near-Earth and deep space segment, including radiofrequency as well as optical communication systems. The proposed activities focus on digitalisation of previously hardware-based system components, developing further communication as-a-service, and mature optical communication systems. The proposed activities address the needs of both industrial and institutional ground/space communication systems and de-risk key component development.
- Astrodynamics Systems: Astrodynamics systems are essential to determine, propagate and control the trajectories and attitude of assets in space. The proposed activities will lead to more autonomous and powerful astrodynamics systems, exploiting novel computing technologies, benefitting both industrial and institutional operators.

2.2. List of activities

2.2.1. GNC & AOCS & Pointing

Programme Reference	Activity Title	Budget (k€)
GT17-800EC	Efficient embedded solver for optimal guidance and control on spacecraft processors	700
GT17-801EC	Efficient uncertainty propagation for autonomous spacecraft robust guidance and control	700
GT17-802EC	Maturation of onboard fault-tolerant control techniques	600
GT17-803EC	Layered mission-vehicle management for autonomy enhancement of rendezvous and proximity operations	900
GT17-804EC	Third generation high performance CMOS detector for AOCS	5,000
GT17-805EC	High volume radiation-tolerant magnetometer	1,500
GT17-806EC	CMG based AOCS performance improvement for agility	500
GT17-807EC	Digital twin of attitude and orbit sensors	600
GT17-808EC	Modular facility for hardware-in-the-loop testing of nanosat AOCS	600
GT17-809EC*	Guidance, navigation and control system development for in-orbit assembly of very large antennas	800
	Total GNC, AOCS & Pointing	**11,900**

* **Proposed Bundle 1 – Enabling in orbit assembly of large antennas:**
- GT17-809EC - Guidance, navigation and control system development for in-orbit assembly of very large antennas
- GT17-821EF - In-orbit assembly and performance demonstration of a large antenna
- GT17-848MS - Deployable support structure with feed horn and metrology functionality

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2.2.2. Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT17-810ED	Power management integrated circuit for space FPGAs and microprocessors	1,200
GT17-811ED	Deterministic Ethernet space time-sensitive networking soft IP core	500
GT17-812ED	Parallel processing for high-throughput and low power space applications	800
GT17-813ED	Re-usable and flexible SRAM based FPGA radiation test set-up	850
GT17-814ED	Radiation hardened digital signal processor IP core	1,200
GT17-815ED	Flight software observability framework for autonomy	600
GT17-816ED	Computational load partitioning tool for model-based design payloads	500
	Total Avionics & EEE	**5,650**

2.2.3. RF Payloads / Technology

Programme Reference	Activity Title	Budget (k€)
GT17-817EF	Deployable reflector for cube- and smallsats	1,500
GT17-818EF	Integrated submillimetre wave antenna-coupled dual-polarization focal plane array and detection chain	7,000
GT17-819EF	Next generation phased array antennas at K-band and Frequency Range 2	1,500
GT17-820EF	Low-cost and compact packaged high-power amplifier with integrated antenna for active antenna arrays	1,000
GT17-821EF*	In-orbit assembly and performance demonstration of a large antenna	1,900
GT17-822EF	High performance digital RF back-end	1,500
GT17-823EF	Digital validation standard antenna	800
GT17-824EF	Next generation distributed and modular digital payload	1,800
GT17-825EF	G-band receiver monolithic microwave integrated circuit	750
GT17-826EF	Advanced multi-input receiver for next-generation space transponders	1,000
GT17-827EF	Compact submillimetre wave radiometer	1,000
GT17-828EF	Critical technologies for satellite formation flying at very close range	600
	Total RF Payloads & Technology	**20,350**

*** Proposed Bundle 1 – Enabling in orbit assembly of large antennas:**
- GT17-809EC – Guidance, navigation and control system development for in-orbit assembly of very large antennas
- GT17-821EF – In-orbit assembly and performance demonstration of a large antenna
- GT17-848MS – Deployable support structure with feed horn and metrology functionality

2.2.4. Power Systems, EMC and Space Environment

Programme Reference	Activity Title	Budget (k€)
GT17-829EP	Development of standardised low cost PVA manufacturing processes	2,000
GT17-830EP	Low volume high power flexible solar array	2,000
GT17-831EP	Adaptive synchronous rectifier controller	750
GT17-832EP	Cost-competitive solar cell development	2,000
GT17-833EP	Containment techniques for batteries stored energy at the end of life	850
GT17-834EP	Integrated standard telemetry and telecommand interface based on CAN	2,000
GT17-835EP	Isolated high side GaN driver	2,000
GT17-836EP	Advanced DC/DC converters building blocks	2,000
Total Power Systems, EMC & Space Environment		**13,600**

2.2.5. Optics, Optoelectronics Robotics, Life Sciences

Programme Reference	Activity Title	Budget (k€)
GT17-837MM	Fibre optic distribution for phased array antenna frontends	1,500
GT17-838MM*	Large duolithic mirrors for snap-together telescope assembly	1,000
GT17-839MM*	Low energy / high pulse repetition frequency based space lidars	1,000
GT17-840MM***	Alexandrite based Lidar for the development of vegetation monitoring techniques	1,000
GT17-841MM**	Infrared detector for next generation sounder instruments	1,500
GT17-842MM	Prismatic joint for planetary robotic manipulators	1,000
GT17-843MM**	Coating development for lens-like samples in the infrared	500
GT17-844MM***	Detector array for EO lidar applications	1,000
GT17-845MM	Inert anode development for molten salt electrolysis	500
GT17-846MM*	Integrated chip based high power C-band optical amplifier	1,500
	Total Optics, Robotics & Life Sciences	**10,500**

*** Proposed Bundle 2 - Enabling New space - miniaturisation and serialisation:**
- GT17-838MM – Large duolithic mirrors for snap-together telescope assembly
- GT17-839MM – Low energy / high pulse repetition frequency based space lidars
- GT17-846MM – Integrated chip based high power C-band optical amplifier

**** Proposed Bundle 3 - Enabling high Infrared performances:**
- GT17-841MM – Infrared detector for next generation sounder instruments
- GT17-843MM – Coating development for lens-like samples in the infrared

***** Proposed Bundle 4 - LIDAR developments for enhanced vegetation monitoring:**
- GT17-840MM – Alexandrite based Lidar for the development of vegetation monitoring techniques
- GT17-844MM – Detector array for EO Lidar applications

2.2.6. Structures, Mechanisms, Materials

Programme Reference	Activity Title	Budget (k€)
GT17-847MS	Develop an ultra-fine positioning actuator for adaptive optics	550
GT17-848MS*	Deployable support structure with feed horn and metrology functionality	900
GT17-849MS	Material accelerator platform for space systems	1,000
GT17-850MS	Hybrid manufacturing for space application	1,000
GT17-851MS	Development of a passive dust-repellent coating for lunar surface application	500
GT17-852MS	Smart sensing technologies for space textiles inspired by terrestrial sectors	500
GT17-853MS	Development of sourced tow-pregs for space applications	900
GT17-854MS	Generic relubrication device for long lifetime mechanisms	550
GT17-855MS	Development of a nanostepping rotary piezo electric actuator based on walking principle	550
GT17-856MS	Innovative ceramic manufacturing for space applications	1,200
	Total Structures, Mechanisms & Materials	**7,650**

*** Proposed Bundle 1 - Enabling in orbit assembly of large antennas**
- GT17-809EC – Guidance, navigation and control system development for in-orbit assembly of very large antennas
- GT17-821EF – In-orbit assembly and performance demonstration of a large antenna
- GT17-848MS – Deployable support structure with feed horn and metrology functionality

2.2.7. Thermal

Programme Reference	Activity Title	Budget (k€)
GT17-857MT	Cryogenic system for in situ resource utilization liquefaction	500
GT17-858MT	Continuous sub-Kelvin cooling	750
GT17-859MT	Thermal analysis demonstrator using cosimulation	500
	Total Thermal	**1,750**

2.2.8. E2E Systems

Programme Reference	Activity Title	Budget (k€)
GT17-860SE	End-to-end demonstrator radar for space object detection	800
GT17-861SE	End-to-end simulation and validation of quantum-enhanced localisation systems for safety critical functions	800
GT17-862SE	End-to-end joint satellite communications and radio frequency sensing system demonstrator	2,000
GT17-863SE	End-to-end space system performance monitoring building blocks	1,000
GT17-864SE	Interoperable signal-in-space interfaces simulator tools for the solar system internet navigation systems	2,000
GT17-865SE	RF constellation simulator for sensing systems	2,000
GT17-866SE	Technology building blocks for an end-to-end space system security verification	2,500
GT17-867SE	End-to-end testbed for robust system architecture	1,500
	Total End-to-End Systems	**12,600**

2.2.9. Future Engineering

Programme Reference	Activity Title	Budget (k€)
GT17-868SF	Maturation of extended reality solutions for space environments	900
GT17-869SF	Facility-less setup for enhanced virtual presence	900
GT17-870SF	SysML v2 for space systems engineering projects	1,500
GT17-871SF	Open end-to-end engineering portal	800
GT17-872SF	System model segregation, integration and exchange	750
GT17-873SF	Towards modular re-usable system design architectures	1,000
GT17-874SF	Space system ontology development for engineering domains	1,200
GT17-875SF	Adaptive human-machine interfaces for enhanced data monitoring and operations	1,200
GT17-876SF	Digital continuity for digital spacecraft and digital twin spacecrafts	1,000
	Total Future Engineering	**9,250**

2.2.10. Ground Segment

Flight Dynamics

Programme Reference	Activity Title	Budget (k€)
GT17-877GF	Rapid radiofrequency assisted optical communication acquisition system	1,000
GT17-878GF	Streaming and automation for modern flight dynamics operations	1,000
	Total Flight Dynamics	**2,000**

Ground Station Engineering

Programme Reference	Activity Title	Budget (k€)
GT17-879GS	Open monitoring and control protocols and user interfaces for ground stations	1,100
GT17-880GS	Ground station scheduling as a service	3,000
GT17-881GS	Deep-space optical communication ground laser transceiver	5,000
GT17-882GS	All digital photon starved optical modem	1,500
GT17-883GS	All-digital ground station	5,000
	Total Ground Station Engineering	**15,600**

System and Applications Engineering

Programme Reference	Activity Title	Budget (k€)
GT17-884GA	Mission planning capabilities	1,200
GT17-885GA	On-board software execution in simulators for future missions with faster on-board computers	1,000
GT17-886GA	Enhancing operational platforms with modern compute resources	1,200
GT17-887GA	Hierarchical data models for ground segments	2,800
GT17-888GA	Data foundations for ground segment operations	3,000
GT17-889GA	Digital ground segment	2,000
	Total System and Applications Engineering	**11,200**

Total Ground Segment	**28,800**

Total Generic Technologies	**122,050**

2.3. Description of activities

2.3.1. GNC & AOCS & Pointing

Generic Technologies

GNC, AOCS & Pointing

GT17-800EC – Efficient embedded solver for optimal guidance and control on spacecraft processors

Budget: 700 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To leverage the latest developments in numerical optimization to develop a modular software package for solving constrained nonlinear optimal control problems that can be deployed on spacecraft processors.
Description:
Advances in numerical methods and embedded computing capabilities have opened new opportunities for deploying optimisation-based guidance and control algorithms directly onboard spacecraft. Nonlinear model predictive control (NMPC) and real-time trajectory optimisation techniques hold significant potential for space missions but have not yet been widely applied due to the high computational cost and strict requirements of onboard execution. These methods are particularly valuable in mission contexts such as autonomous rendezvous, precision planetary landing, and constrained attitude manoeuvres. However, existing solvers are often not designed with the limited hardware resources of spacecraft in mind, leading to challenges in meeting real-time performance and robustness.

There is a strong need for a high-performance, embedded-capable optimisation solver that can reliably solve a wide variety of nonlinear, constrained optimal control problems (OCPs) on space processors. Such a tool would significantly improve the autonomy, performance, and robustness of onboard guidance, navigation, and control (GNC) systems. The proposed activity directly supports this need by enabling advanced guidance and control strategies that are currently impractical for onboard execution due to computational constraints. The activity will develop a modular software package that enables rapid and reliable execution of onboard optimal control problems (OCPs), tailored for embedded deployment on space-qualified processors. The tool shall be flexible enough to allow users to quickly specify the constraints and objectives for their particular use case. The tool shall leverage modern optimization toolchains that support symbolic modelling, automatic differentiation, and efficient code generation.

This activity encompasses the following tasks:
- Design and implement a numerical optimal control software that allows interfacing to a variety of existing low-level convex optimisation solvers capable of running on space processors.
- Integrate symbolic modelling and automatic differentiation tools for efficient optimal control problem formulation and code generation.
- Develop a modular and lightweight functional interface suitable for real-time GNC integration and automatic code generation pipelines.
- Validate solver performance on representative mission scenarios involving complex constrained control tasks (e.g. proximity operations and precision landing).
- Use the high-level framework to perform in-depth testing and profiling of different existing low-level solvers on the target flight processors.
- Develop a tailored, low-level convex optimization solver for the use case and interface with it from the high-level tool.

Deliverables: Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity D09 “Numerically robust & computationally efficient embedded solver for nonlinear optimal control on spacecraft processors”.

Generic Technologies

GNC, AOCS & Pointing

GT17-801EC – Efficient uncertainty propagation for autonomous spacecraft robust guidance and control

Budget: 700 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To develop computationally efficient and scalable methods for uncertainty quantification and propagation applicable to a wide range of space missions.
Description:
The demand for autonomous space missions necessitates advanced on-board decision-making and control capabilities in complex and uncertain environments. These uncertainties stem from various sources, including model inaccuracies, navigation errors, and external disturbances. Traditional trajectory optimization methods often overlook these uncertainties, relying on simplified nominal models and conservative design margins to construct trajectories, leading to sub-optimal performance and increased resource requirements.

To address this, this activity aims to develop scalable and efficient methods for uncertainty quantification and propagation to support trajectory optimization and robust guidance, navigation, and control (GNC) across various space missions. These methods will compute the reachable set of system states in real-time within a closed-loop configuration over a finite planning horizon, dynamically adapting guidance and feedback control strategies. Key sources of uncertainty, including model inaccuracies, navigation and state estimation errors, and predicted disturbances. Implementation will target representative space-grade processors, with performance enhancements explored through hardware acceleration technologies, such as GPUs or other suitable alternatives. These hardware accelerators would be used to efficiently compute bounds on the reachable sets by simulating in parallel many possible future scenarios under different uncertainty realisations. The methods will be compatible with modern GNC design practices relying on automatic code generation, providing a functional interface for computing reachable sets in the form of ellipsoidal bounds across a predefined horizon, facilitating seamless integration into existing GNC development pipelines. The framework will incorporate linear covariance analysis and intelligent sampling-based approaches, allowing users to select the most appropriate algorithm based on computational resources. Automatic differentiation frameworks will enable efficient computation of necessary derivatives.

This activity encompasses the following tasks:
- Develop a real-time ellipsoidal uncertainty propagation framework combining linear covariance and sampling-based approaches.
- Implement the methods within a functional interface compatible with automatic code generation workflows.
- Validate the methods in two mission scenarios involving complex, constrained trajectory planning and control under uncertainty (formation flying, in-orbit servicing, guided atmospheric re-entry, and precision planetary landing).
- Benchmark the execution performance of the uncertainty propagation algorithm on representative space processors and hardware accelerators (GPUs).
- Produce reference implementations and integration guidelines for trajectory optimisation and advanced GNC applications.

Deliverables: Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity D14 “Accelerated methods for closed-loop uncertainty propagation in spacecraft control systems for robust motion planning and optimization”.

Generic Technologies

GNC, AOCS & Pointing

GT17-802EC – Maturation of onboard fault-tolerant control techniques

Budget: 600 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To consolidate and demonstrate the effectiveness of control techniques that enable to cope with actuator and sensor failures, particularly in systems with a multi-actuated configuration.

Description:
The inherent redundancy present in spacecraft and launchers equipped with a multi-actuated configuration (such as multiple engines, multiple thrusters and aerodynamic surfaces) yields substantial advantages concerning the robustness of the system against operational faults and component failures. This built-in redundancy allows the vehicle to maintain functionality and stability in the presence of malfunctions. However, effectively controlling a vehicle that uses multiple actuators in an integrated manner is a challenging task. This challenge is increased when the objective extends to developing fault-tolerant control techniques that can seamlessly employ actuators of potentially very different types and characteristics. This necessitates a control architecture capable of managing diverse actuation sources under both nominal and fault conditions.

Conventional fault-tolerant control methodologies often rely on a simplifying decoupling assumption. This assumption postulates a separation between the individual actuators and the specific control modes, objectives, or axes of motion they are responsible for. This activity aims to depart from this decoupling assumption by blending state-of-the-art multivariable and fault-tolerant control design techniques, and to demonstrate the potential gains that can be realised through a more integrated approach to the control of multi-actuated systems in the presence of faults and failures.

To depart from the abovementioned decoupling simplification, this activity will target the application and blending of state-of-the-art multivariable and fault-tolerant control design and analysis techniques. These include but are not limited to model predictive control, robust control techniques, feedback linearization, data-driven methods and digital twin models. The demonstration of the achievable gains will be performed through thorough comparative simulations between conventional and integrated fault-tolerant approaches on representative real-world mission scenarios. Applicable scenarios include those where precise trajectory and orbit control is key, such as launcher stages, re-entry vehicles, planetary landers and interplanetary missions.

This activity encompasses the following tasks:
- Requirements definition under nominal and failure conditions.
- Failure impact analysis and scenario definition.
- Design of nominal multivariable control algorithms.
- Design of health monitoring and fault detection algorithms.
- Design of fault-tolerant multivariable control algorithms.
- Implementation of the algorithms in a software prototype and verification at simulation level.
- Implementation of selected algorithms on a space-graded processor and processor-in-the-loop testing.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity D13 “Maturation of Fault-tolerant Techniques for Launchers”.

Generic Technologies

GNC, AOCS & Pointing

GT17-803EC – Layered mission-vehicle management for autonomy enhancement of rendezvous and proximity operations

Budget: 900 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To develop innovative mission vehicle management (MVM) systems relying upon sensor data fusion, layered autonomous guidance, control and decision-making techniques.

Description:
Autonomous vehicle management is believed to be a key functionality in future rendezvous and proximity operations to ensure fast reactivity in the event of non-nominal conditions and to relieve ground from the intensive monitoring of possibly a large number of spacecrafts in operation.

In order to achieve this goal, it is essential to achieve a higher integration of spacecraft functions, and especially of the various decision layers (i.e. lower level control tasks such as attitude control or thermal control, mid-level planning tasks, such as path planning or trajectory guidance, and higher level management of on-board resources including the system FDIR) that execute the mission plan. The goal is to combine the two domains of mission and path planning with collision risk assessment and anomaly detection to achieve a versatile mission and ensure a seamless continuation in the event of failures.

For the specific case of rendezvous and proximity operations (RPO), the mission plan is particularly sensitive to the need to limit the collision risk and contrary to standard spacecraft operations, where this risk is handled through ground-based services, such operation needs to be performed on board. The guidelines and best practices for RPO often suggest the use of passively safe trajectories to limit the collision risk, however, these are not always suitable and, especially in case of capture, are not applicable to the final phases of the mission, since the capture is a controlled collision between the target and the chaser. For these reasons, innovative MVM systems need to be developed to process information from multiple sensors and multiple spacecraft systems in order to ensure that operations are executed that achieve mission optimisation while keeping the collision risk under control.

To achieve this, on-board collision detection algorithms running on the chaser spacecraft shall be developed, with the following benefits:
- Increased mission reliability: the chaser can detect unsafe conditions and take remediation action autonomously.
- Increased mission flexibility: relaxing the requirement for a passively safe trajectory allows to use more advanced trajectories, including target motion synchronization, expanding the range of potential missions.

This activity encompasses the following tasks:
- Identify the most suitable technique, or set of techniques, for mission and path planning, including collision detection, that fit the purpose of a RPO mission, taking into account the limitations of an on-board processor.
- Consolidate the mission scenario and requirements that are driving the need for extended on-board autonomy.
- Prototype the on-board algorithms that enable autonomous MVM functionalities and evaluate them in a representative numerical simulator.
- Deploy the algorithms into a representative processor and test its performances in processor-in-the-loop and hardware-in-the-loop test setups.

Deliverables: Report, Software

Application/Need Date: All missions involving autonomous rendezvous and close proximity operations. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity D12 “Mission-vehicle-management for autonomy enhancement in rendezvous and docking operations”.

Generic Technologies

GNC, AOCS & Pointing

GT17-804EC – Third generation high performance CMOS detector for AOCS

Budget: 5000 k€ - **Duration**: 36 months – **Current / Targeted TRL:** 3 / 7 – **TD** 5

Objective:
To design, manufacture and evaluate a Complementary Metal-Oxide-Semiconductor (CMOS) visible image sensor to be embedded in the next generation of star trackers, as well as supporting high-end navigation cameras.
Description:
Significant advancements have been made in terrestrial imaging technologies. From the first-generation CMOS detectors developed for AOCS applications, i.e. the HAS2 detector, which, although now considered obsolete, is still in use, through the second generation represented by the Faint Star 2, the improvements in performance have been substantial. However, to meet the new performance requirements, the development of a third-generation CMOS detector is necessary. This new generation will incorporate a significantly higher pixel count, reduce angular pixel size, and introduce global shutter readout modes.

An additional key requirement for the third generation is the establishment of a fully European supply chain, encompassing design, manufacturing (foundry), assembly (encapsulation), and testing (lot acceptance and screening). Building upon the insights gained from a preceding TDE activity, T717-702SA – High Resolution Detector Prototype Characterisation for Star Trackers, which involved the characterization of several high-resolution detectors, a further development effort will be undertaken to solidify Europe's leadership in third-generation CMOS image sensors for AOCS and star tracker applications.

The third generation CMOS would be designed with several targeted features. It would include a pixel array that exceeds 2k x 2k, with an overall size of approximately 14 mm x 14 mm. It would have a global shutter with read noise below 4 e-. The DCNU and PSNL would be minimized at end-of-life. The pixel full well capacity would be above 40 ke-. It would have an on-board sequencer supporting at least 25 windows and pre-processing tailored for AOCS. The full-frame readout would be at 10 Hz, with windowed readout above 30 Hz. The power consumption in windowed mode would be below 200 mW.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 3,000 k€, duration: 20 months):
- Finalization of specification and trade-offs for the key on-chip blocks.
- Preliminary design of detector architecture (pixel type and layout, readout scheme, ADC architecture, ESD protection structure, interfaces and packaging), preliminary performance and compliance assessment and identification of key risk points.
- Critical design of the detector, finalizing the detailed design, analysis and simulations of the chip, package, testing equipment, component assembly procedures and PID.
- Prototype manufacturing and procuring wafers, packages and cover glass, and manufacturing, as well as tape out, ensuring the completeness and correctness of the final layout, timing, simulation and other design verification activities.
- Preliminary functional and performance verification of the prototype.

Phase 2 (budget: 2000 k€, duration: 16 months):
- Comprehensive characterization of the prototype, including emphasis on radiation performance.
- Design re-spin in view of qualification devices manufacturing for evaluation.
- Qualification of the final detector.
Deliverables: Engineering/Qualification Model, Report

Application/Need Date: TRL6 by 2028 for the development of the next generation of star trackers. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Generic Technologies

GNC, AOCS & Pointing

GT17-805EC – High volume radiation-tolerant magnetometer

Budget: 1500 k€ - **Duration**: 24 months – **Current / Targeted TRL:** 4 / 7 – **TD** 5

Objective:
To design, manufacture and qualify a magnetometer, designed to be produced in large quantities at an affordable recurring price while maintaining sufficient reliability for LEO constellations.

Description:
Magnetometers play a critical role in attitude and orbit control systems (AOCS), enabling spacecraft to perform essential functions such as magnetically actuated detumbling or maintaining safe mode operations in Low Earth Orbit (LEO). By utilizing these devices, the reliance on gyroscopes is reduced, particularly in scenarios where telecommunication constellations are involved. These constellations frequently depend on magnetometers in their AOCS. A capability gap persists in Europe regarding the availability of low-cost, radiation-tolerant magnetometers suitable for high-volume constellation deployment.

The objective of this activity is to develop a European-manufactured magnetometer that is:
- Radiation-tolerant for LEO environments.
- Cost-effective for large-scale production (e.g. up to 4 per week).
- Small (below 300g), with low power consumption (below 1W) and digital interface.
- Fully compliant with AOCS requirements for constellation platforms.
- Demonstrating the industrial capability to manufacture and test 4 to 8 units at a time.

This activity encompasses the following tasks:
- Specification, including industrialisation requirements (manufacturing and testing capability).
- Design and testing of an engineering model for design freeze.
- Manufacturing and test of two qualification models.
- Validation of EGSEs (Electrical Ground Support Equipment) focusing on testing capabilities of several units in parallel.

Deliverables: Engineering model, Two qualification models, Report

Application/Need Date: TRL7 by 2027 for LEO missions. **Mission Classification:** beta, delta, gamma

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Generic Technologies

GNC, AOCS & Pointing

GT17-806EC – CMG based AOCS performance improvement for agility

Budget: 500 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To design, develop, breadboard and test optimized attitude guidance and control for enhanced agility and mission images throughput with increased autonomy and reduced operations.

Description:
Jointly optimized guidance and control enable not to oversize the control momentum gyros (CMGs) and to reduce the mission cost while benefiting of an enhanced agility and mission images throughput, with a shorter re-visit time and a more flexible adaptation to in-orbit observation variation. Such an optimized design can also enable increased autonomy and reduced operations. In previous agile missions, guidance and control functions from attitude and orbit control system (AOCS) systems were mostly designed and optimised separately, missing the performance growth opportunity of a higher degree of joined optimisation. However, the joined G&C (guidance and control) non-convex optimisation needs further development, being currently limited by the challenges of real time use of convexification techniques.

Taking benefits of the advances in Artificial Intelligence (AI) using deep neural networks (DNN) taught with reinforcement learning (RL), guided policy search (GPS) and imitation learning (IL) to mimic the optimal outputs, the proposed development aims at developing an on-board optimization functionality, together with its training and tuning process for reducing the mission operations, and most of all enabling an overall optimization of the attitude guidance control and allocation (kernel guidance), for an enhanced agility to increase mission throughput. This solution should also provide robustness with ability to learn and adapt to differences between the actual system and its training. Implementation tasks will make use of rapid prototyping on real time processing targets to demonstrate on board feasibility and consolidate the real-time usability of such G&C techniques on board.

This activity encompasses the following tasks:
- Mission definition and requirements of the relevant AI techniques and possible complementary techniques.
- Flow-down to AOCS (guidance and control) requirements, architecture and training and tuning pipeline trade-offs including CMG selection and accommodation.
- Preliminary design of the AI-based guidance, control and CMG allocation together with its training objectives definition and setting, possibly with nonlinear programming algorithm for optimal solution definition combined with its initialization and the grid sampling of the initial state with its optimal initial state search.
- Detailed design with system modelling (spacecraft and CMG dynamics, disturbances), attitude guidance and control and CMG allocation detailed design and its training and tuning.
- Detailed simulations and analysis of the guidance and control system performance, with sensitivity analysis to the disturbances and control errors.
- Implementation on real time target, representative of future processing target for AI-based algorithms.

Deliverables: Breadboard, Prototype, Report, Software

Application/Need Date: Earth observation missions with possibly increased throughput. TRL 5 by 2027. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity F19 “CMG based AOCS performance improvement via AI techniques”.

Generic Technologies

GNC, AOCS & Pointing

GT17-807EC – Digital twin of attitude and orbit sensors

Budget: 600 k€ - **Duration**: 15 months – **Current / Targeted TRL:** 3 / 5 – **TD** 5

Objective:
To develop and test a model for a digital twin of attitude and orbit control system (AOCS) sensors as well as to assess compatibility of this multi-physics model-based digital twin with on-board self-calibration employing data-driven techniques and with the simplification of ground verification processes of new satellite platforms.
Description:
Advances in model based engineering and digital twin (DT) development has led to growth in the interest of multi-physic modelling of space systems, including, for example, AOCS sensors. The AOCS sensor DTs include functional and performance behaviour, complemented by physical behaviour (thermal, power, electronic, etc.), as well as failure and degradation modes. This detailed modelling is based on analytic and background supplier models. The emergence of artificial intelligence (AI) / machine learning (ML) combined techniques, potentially augmented with analytical a priori knowledge, offer the potential to improve the overall accuracy and validity of this modelling, together with a specific interest for statistical interpretation in the case of mass production (sensors for constellations, for example, gyros, star trackers, sun sensors). These new engineering techniques aimed at better fidelity in the AOCS sensor modelling offer the prospect of extending the benefits to on-board performance, for example, by adapting these techniques to the ultimate AOCS sensor calibration: self-calibration on board the spacecraft (e.g. constellation or low-cost platform). The following techniques and use cases will be studied:
- Digital Twin for AOCS sensors, preferably from sensors used on platform series: architecture; interoperability needs, multi-physic modelling trade-off and key components; use of data-driven techniques (AI/ML) to populate the DT models; statistical interpretation and validation of AOCS sensor performance (mass manufacturing, ground self-calibration based on sample measurements and statistic extrapolation).
- Use of ground models and knowledge for on-board self-calibration; data fusion techniques for on-board use to consolidate ground self-calibration with flight measurements, autonomously; autonomous verification of on-board self-calibration (fault detection).

This activity will bring tools, methods, and processes for increasing the end-to-end autonomous handling of AOCS sensors in the platform development and verification chain. It includes predictive maintenance monitoring to inform on the overall health status and to prevent anomalies/failures. This activity aims at providing digital data and models to improve knowledge of the units in flight, allowing correction and improvement by the supplier and better integration in model-based system engineering (MBSE) or digital production.

This activity encompasses the following tasks:
- Define the system into which the deliverable items will be integrated, establish the end user requirements and quantify the advantages of utilising the new items with respect to state-of-the-art alternatives.
- Define a complete, self-standing and traceable set of technical requirements for the deliverable items.
- Select the technical baseline that offers the best potential to achieve the activity objectives.
- Detailed design and demonstrate, by appropriate analyses, simulations, tests or other means, that it can satisfy the technical requirements.
- Establish all detailed plans necessary to successfully implement and test the deliverable items and verify their compliance to the technical specifications.
- Implement the deliverable items, quantify their performances by means of a suitable test campaign and demonstrate compliance to the technical requirements.

Deliverables: Report, Software

Application/Need Date: All missions. TRL5 by 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity G01 “Building blocks for digital twin of attitude and orbit control system equipment (P0)”.

Generic Technologies

GNC, AOCS & Pointing

GT17-808EC – Modular facility for hardware-in-the-loop testing of nanosat AOCS

Budget: 600 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 6 – **TD** 5

Objective:
To design, develop, prototype and test a cost-efficient modular platform enabling hardware-in-the-loop testing of nanosat AOCS.

Description:
The AOCS verification and validation (V&V) represents an important part of the cost of development of an AOCS subsystem. For low-cost mission, cost saving on V&V phase are needed, however it increased risk of issue in LEOP (launch and early operations phase) and sometimes loss of mission in case of AOCS issue in contingency modes. Hardware (HW)-in-the-loop (HIL) testing, in particular closed loop testing, is often lacking due to AOCS units ground support equipment (GSE) and dynamics emulator partly or not available.

This activity aims at developing a novel and flexible AOCS verification facility for prototyping flight software (SW) and testing it with AOCS hardware-in-the-loop (HIL), with the use of modular simulation and flight software libraries, together with automated instantiation on processing board hardware, enabling a wide range of missions to benefit from HIL testing with shorter development time. In addition, the portable targets for processing and possibly sensor and actuator hardware and ground support equipment (GSE) should provide a powerful verification platform for nanosat missions, keeping all along the definition and development of the test facility the economic target price objective.

This activity encompasses the following tasks:
- Definition of the flexible AOCS V&V facility requirements, encompassing the platform hardware, software and interfaces.
- Architecture trade-off and definition covering the modularity, configurability and possible evolution needs and the identification of the sensor and actuator hardware and GSEs.
- Design and development of the modular facility, including the modular simulation and flight software libraries and their configuration and automated instantiation, and including the hardware integration with the processing board targets and the sensor and actuator hardware and GSEs.
- Identification of AOCS flight software or its prototype to be tested.
- AOCS equipment integration tests, integration of the AOCS flight software, and AOCS functional tests with HIL closed loop testing.

Deliverables: Prototype, Report, Software

Application/Need Date: SmallSat missions. TRL 6 by 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity C07 “Modular Platform for Hardware-in-the-Loop Testing of nanosat AOCS”.

Generic Technologies

GNC, AOCS & Pointing

GT17-809EC – Guidance, navigation and control system development for in-orbit assembly of very large antennas

Budget: 800 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 4 / 6 – **TD** 5

Objective:
To develop and validate a GNC system tailored for the in-orbit assembly, deployment, and stabilization of large reflector antennas, in the order of 10 m diameter.
Description:
In-orbit assembly (IOA) of large antennas introduces unique challenges for GNC systems. Traditional approaches, optimized for compact, monolithic spacecraft, are not suitable for managing the dynamics and control of large, lightweight, and flexible structures assembled in space. The assembly, deployment, and operational phases of a ~10 m diameter reflector antenna demand a novel GNC strategy that ensures precise structural configuration, pointing accuracy, and dynamic stability under microgravity conditions. This activity addresses the maturation of a dedicated GNC subsystem specifically designed to support the end-to-end lifecycle of large reflector antenna structures assembled in-orbit. It will consider the mechanical characteristics of modular, electrically connected panels, and the structural flexibility of the assembled system. The GNC design will consider the coupling effects between structural dynamics and control forces/torques, especially during deployment and pointing operations. The work will focus on developing robust algorithms and sensor-actuator architectures capable of supporting autonomous or semi-autonomous in-orbit assembly operations, monitoring and controlling dynamic stability across multiple structural configurations and ensuring high pointing accuracy and maintaining alignment of the RF feed and reflector under various orbital and thermal conditions. In parallel, advanced modelling and simulation tools will be developed to capture the coupled dynamics of the structure and validate control strategies. This includes rigid-flexible multibody dynamics, metrology-based feedback control, and fault-tolerant autonomy for long-duration operations. Breadboarding and hardware-in-the-loop testing will be carried out to demonstrate GNC functionality in a representative environment.

This activity encompasses the following tasks:
- Identification of mission phases (assembly, deployment, operation), performance needs, and environmental constraints.
- Evaluation of control architectures, sensor configurations, and actuation strategies. Development of simulation models including structural dynamics.
- Implementation of control laws for stabilization, pointing, and disturbance rejection, with emphasis on flexibility compensation and robustness.
- Development of a representative GNC testbed including sensors (e.g., IMU, metrology), actuators, and embedded control system.
- Hardware-in-the-loop simulation and dynamic testing to validate the performance of the GNC solution under relevant conditions.
- Assessment of readiness level and recommendations for future maturation or integration with full antenna assembly demonstrators.

This activity is proposed to be implemented in Bundle 1 - Enabling in orbit assembly of large antennas, together with the activities GT17-821EF – In-orbit assembly and performance demonstration of a large antenna and GT17-848MS – Deployable support structure with feed horn and metrology functionality.

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions in need of in-orbit assembly, deployment, and stabilization of large reflector antennas **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity B01 "Guidance, navigation and control of in-orbit assembly of large antennas".

2.3.2. Avionics & EEE

Generic Technologies

Avionics & EEE

GT17-810ED – Power management integrated circuit for space FPGAs and microprocessors

Budget: 1200 k€ - **Duration**: 24 months – **Current / Targeted TRL:** 3 / 4 – **TD** 1

Objective:
To develop a Power Management Integrated Circuit (PMIC) that provides regulated supplies, protections, and power sequencing for Field-Programmable Gate Arrays (FPGAs) and microprocessors used in space applications.

Description:
Power Management and Distributions (PMD) systems are widely employed to generate regulated, protected, controlled and filtered supplies which can provide stable voltages to subsystems, and specifically to FPGAs and microcontrollers. However, such systems are normally built with discrete components which require high volume and have limited power efficiency over the full operational range. Therefore, applications would benefit from high power efficiency with improved integration level and performance at reduced size and cost.

In this activity, the use of GaN (gallium nitride) and advanced silicon power control technologies within a System in Package architectures are expected to improve the current state of the art power solutions. Such a multi-output PMIC development will target the design of integrated DC-DC converters and Points of Load (PoL), while implementing latching Current Limiters and different protections (under/over, voltage/currents), as well as a power up and down logic. The output signals produced by the PMIC will provide stable, optimum and protected supplies to FPGAs and microprocessors. Furthermore, such a PMIC protects against transients, voltage variations and peaks that could lead to multi-supply FPGA and microcontroller failures.

This activity encompasses the following tasks:
- Requirements definition and architectural design and evaluation of the space grade PMIC to the target specifications, including protections against over/under current/voltage conditions and power up and down logic.
- Manufacturing of the PMIC Breadboard, following either a System in Package (SiP) or a monolithic approach.
- Electrical and radiation test and characterization of the PMIC.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Power Management Distribution (2025) – Consistent with Strategic Objective 1 “Improve the integration level of power units”.

Generic Technologies

Avionics & EEE

GT17-811ED – Deterministic Ethernet space time-sensitive networking soft IP core

Budget: 500 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 4 – **TD** 1

Objective:
To develop a technology independent soft IP (Intellectual Property) core that complies with the aerospace profile of TSN (Time Sensitive Networking).

Description:
TSN (Time-Sensitive Networking) and TTE (Time Triggered Ethernet) are the two technologies that allow to deliver determinism and quality of service on top of standard Ethernet networks. Since TTE can be considered relatively closed technology (i.e. proprietary solution, patents pending on the technology), from a few years TSN is emerging as a potential successor of that technology. TSN is IEEE supported set of standards that implements wide range of quality of services (i.e. reliability, determinism, mixed criticality traffic, redundancy, time synchronization) on top of the standard Ethernet protocols. TSN benefits from Ethernet Commercial Off-The-Shelf (COTS) tools/ecosystem, it is free of patents and is being introduced step-by step in industrial applications (automotive/industrial) and aerospace domain (IEEE P802.1DP – TSN for Aerospace Onboard Ethernet Communications). Although TSN is very successful in the automotive/industrial sector, in space applications TSN is still being considered as low TRL technology. Many current and future Exploration and Space Transportation missions are considering Ethernet-based networks for the data-handling communication bus. Having this technology available for future ESA missions is seen as a key driver for future missions.

The aim of this activity Is to develop an IP core that contains the core functionalities of Time-Sensitive Networking needed for space applications. Although the set of TSN standards is very wide, there are several on-going activities aiming at tailoring of TSN for space applications. Having an TSN IP tailored to space applications could be considered as a key enabler of this technology in future ESA and national missions.

This activity encompasses the following tasks:
- TSN IP core requirements identification and development of the high-level architecture of the TSN soft IP core.
- Prototype development and integration into a representative Field-Programmable Gate Array (FPGA) based network platform.
- Verification and validation targeting high speed up to 10G.
- TSN soft IP core release and related documentation.

Deliverables: Prototype, Report, Software

Application/Need Date: Exploration and Space Transportation missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap On Board Computers, Data Handling Systems and Microelectronics.

Generic Technologies

Avionics & EEE

GT17-812ED – Parallel processing for high-throughput and low power space applications

Budget: 800 k€ - **Duration**: 24 months – **Current / Targeted TRL:** 3 / 4 – **TD** 1

Objective:
To develop vector extensions and parallel accelerators in processor architectures to achieve high performance with low power consumption and implement solutions for RISC-V.

Description:
A parallel processing extension (accelerator) can significantly enhance the performance of space processors. Current off-the-shelf solutions, such as Vector Processing Units (VPUs), Graphics Processing Units (GPUs), and AI Engines, are somewhat radiation-tolerant but often too costly in terms of power consumption.

This activity aims to develop an architecture for vector extensions and an accelerator co-processor that provides high-performance processing while maintaining low power consumption. The benefits of this solution will be demonstrated across various applications, including communications, Synthetic Aperture Radar (SAR) processing, image compression, and Artificial Intelligence (AI). The activity will also address the optimal integration of the proposed accelerator with existing space microprocessors, System-on-Chip (SoC) solutions, and software toolchains.

The activity encompasses the following tasks:
- Requirements identification among existing application needs: communications, SAR processing, image compression, AI.
- Propose and compare different architectural solutions and different interconnect solutions (tightly and loosely coupled; network on chip) of the accelerator.
- Design the architectural solution based on hardware modelling tools.
- Implementation, verification and prototyping of the architectural accelerator solution for RISC-V.
- Comparison and benchmarking of the accelerator solution, including power estimates.

Deliverables: Prototype, Report

Application/Need Date: All missions and space applications requiring high-throughput and low power.

Mission Classification: alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap On Board Computers, Data Handling Systems and Microelectronics.

Generic Technologies

Avionics & EEE

GT17-813ED – Re-usable and flexible SRAM based FPGA radiation test set-up

Budget: 850 k€ - **Duration**: 24 months – **Current / Targeted TRL:** 4 / 7 – **TD** 1

Objective:
To develop a re-usable and flexible test setup for Single Event Effect of advanced Field-Programmable Gate Array (FPGA) and System on Chip (SoC), allowing the verification of Single Event Effects (SEE) mitigation and error injection techniques.

Description:
FPGAs/APSoC (All Programmable SoC) components are highly complex and very sensitive to SEE, especially the configuration memory. Different techniques can be used to mitigate SEE in configuration memory: internal scrubbing, readback and external scrubbing, blind reconfiguration. The heavy ion or proton testing of these techniques is very difficult because of specific characteristics of the different particle accelerators (accelerated flux, pulsed beam, scanned beam, etc.) and the complexity of the devices / systems under test.

In addition, very often the assessment of SEE sensitivity is based on results available in literature or based on tests with a specific test setup (i.e. vendor demonstration board). Test conditions are often unknown or not directly comparable to the intended application.

A flexible test setup will allow to mimic the SEE mitigation scheme intended in an application. Then, an accurate assessment of the mitigation implemented is possible. This test setup will allow independent testing reproducing exactly SEE mitigation scheme intended on a given application and will be particularly useful for prime contractors and equipment manufacturers by homogenizing radiation testing of state of the art Commercial Off-The-Shelf (COTS) SoC and processors / microcontrollers, often used for Artificial Intelligence applications. It will also allow the performance of fault injection with realistic error signatures (as observed during irradiation tests) allowing for verification of mitigation strategies before the actual beam testing.

This activity encompasses the following tasks:
- Development of a re-usable and flexible FPGA based test setup that will be designed and built to test last generation FPGAs.
- Test setup validation:
    - Proton and heavy ion validation testing in several configuration memories FPGA based.
    - Fault injection testing.

Deliverables: Test set-up and radiation reports

Application/Need Date: All mission. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Radiation Environment and Effects (2022) – Consistent with activity J05 “SRAM Based FPGA radiation test set-up”.

Generic Technologies

Avionics & EEE

GT17-814ED – Radiation hardened digital signal processor IP core

Budget: 1200 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 1

Objective:
To develop a radiation hardened DSP (Digital Signal Processor) IP core for space front-end Field-Programmable Gate Array (FPGA) signal processing for Artificial Intelligence, Communication, Image and Radar applications.

Description:
The front-end payload signal processing for artificial intelligence, communication, Image and radar increase continuously in complexity as well as throughput. Currently this functionality is designed with radiation mitigation ad-hoc onto the front-end FPGA and even common functionality like filtering, correlations, convolutions, Fourier transforms and telemetry packet encapsulation among others require extensive debugging on the hardware before full operational functionality is achieved and validated. A parallel scalable DSP IP core with radiation mitigation and extensive debugging functionality together with a signal processing library would increase data processing throughput, reduce latency and allow validation of its functionality before implementation onto the FPGA hardware. Furthermore, the parallel scalable nature of the DSP IP core would maximise the reuse of the functionality as the complexity and throughput increase for each new generation of front-end.

The activity encompasses the following tasks:
- Requirements identification of space front-end FPGAs signal processing for Artificial Intelligence, Communication, Image and Radar applications.
- Selection of suitable DSP IP cores for the above applications fields.
- Development of Radiation Hardened DSP IP core.
- Integration of the DSP IP core into the targeted front-end FPGAs application.
- Radiation testing of the DSP IP core.

The DSP IP core and signal processing library are envisaged to be made available to the space industry for enhanced front-end signal processing performance and dependability.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: On-Board Computers, Data Handling Systems and Microelectronics (2021) – Consistent with AIM J “Improve and expand the European IP Cores catalogue and respective Design Methods and Tools”.

Generic Technologies

Avionics & EEE

GT17-815ED – Flight software observability framework for autonomy

Budget: 600 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 2

Objective:
To develop a flight software observability framework to improve the monitoring capabilities on board future satellites.
Description:
New space missions demand advanced autonomy due to increased onboard complexity, faster response times, extensive onboard data processing, and high availability requirements. Advanced autonomy necessitates smart observability, suitable for automatic, potentially AI-driven ground processing and analysis, which triggers operator action only when necessary.

The development of a standardized observability framework for flight software is essential to enhance mission reliability and efficiency. By implementing smart observability, missions can achieve better data processing, timely responses, and improved system performance. This framework will provide concrete benefits such as reduced operator workload, enhanced data accuracy, and predictive maintenance capabilities.

The technology to be developed involves creating a comprehensive observability service for flight software from early design of the SW until in-flight observation. This service will go beyond basic event logging and will be specified at the system level, addressing various aspects of software execution. Key features will include the collection of execution times, production of standardized boot and death reports, and monitoring of network buffers to trace issues back to their source. It will possibly also address the support for visual observability through cameras monitoring sensitive mechanisms on board.

This activity encompasses the following tasks:
- Specification of a standardized set of observability requirements applicable across platforms.
- Development of observability capabilities for single-core and multi-core systems, including support for time and space partitioned environments.
- Integration of advanced network monitoring features, including metrics such as throughput, latency, and jitter.
- Management and monitoring of all memory and buffer usage for trend analysis and early anomaly detection.
- Improvement of tools and methods to support software tuning, trending analysis, and early-stage failure prediction.

Deliverables: Report, Software: Validated Observability Framework

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Functional verification and Mission Operations (2025) – Consistent with activity R&D Line 2.2 “Autonomous Monitoring, Testing and Operations”.

Generic Technologies

Avionics & EEE

GT17-816ED – Computational load partitioning tool for model-based design payloads

Budget: 500 k€ - **Duration**: 18 months – **Current / Targeted TRL:** 3 / 5 – **TD** 1

Objective:
To develop a tool that helps in partitioning the computational load of a subsystem in heterogeneous processing devices, thus complementing current model-based design (MBD) approaches.
Description:
MBD approaches for digital systems have been in the market already for several years. First, they were used only to ease designing, then they stepped into Software (SW) auto-coding. The next step has been Hardware Description Language (HDL) auto-coding of Hardware (HW) programmable devices and code for embedded hard Graphics Processing Units (GPUs), when available. Lastly, with the irruption of Artificial Intelligence (AI), some can also cope with parametrization of some of the AI cores that some devices include.

The main idea behind this activity would be to develop a parametrizable profiling application that can help in the task of partitioning and balancing the computational load of an algorithm/solution among the different processing hard cores that can be available in a System-on-Chip (SoC) Field-Programmable Gate Array (FPGA) device: multi-core Central Processing Units (CPUs), Programmable Logic, GPUs and/or AI cores.

It should be technology agnostic and parametrizable in the sense that it shouldn't be linked to any device, nor development flow/toolset. It should help answering the question of what to accelerate and where, giving trustworthy estimations that ensure robust autonomous operation in space.

The validation of such tool with a use case based in a complex on-board algorithm and targeting an EU based technology is deemed mandatory. Use cases might come from Robotics, GNC or Telecom, where complex algorithms with high performance needs are often deployed.

This activity encompasses the following tasks:
- Target technologies assessment including the heterogeneous processing devices that it will be able to cope with and their parametrization. At least 3 devices shall be considered.
- Generic algorithm complexity parametrization and requirements capture.
- Tool architecture and development.
- Tool validation using a hardware breadboard mounting a heterogeneous device that runs an algorithm in at least 3 processing architectures among (multi-core CPUs, Programmable Logic, GPU, AI) and whose splitting has been envisaged using the developed tool.
- Software tool delivery along with User Manual and explained example.

Deliverables: Breadboard, Report, Software

Application/Need Date: All type of missions. Applications where computational load of the solution can be shared among different processing elements. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Related to the Roadmap On-Board Computers, Data Handling Systems and Microelectronics.

2.3.3. RF Payloads / Technology

Generic Technologies

RF Payloads & Technology

GT17-817EF – Deployable reflector for cube- and smallsats

Budget: 1500 k€ - **Duration**: 24 months – **Current / Targeted TRL:** 3 / 6 – **TD** 7

Objective:
To design, manufacture and test a 1m deployable offset fed reflector antenna for small/cube satellites suitable for missions up to Ka-band.

Description:
A strong need for deployable Ka-band high gain antennas for cubesat and small satellites for various applications (TT&C, payload data downlink, black hole observation VLBI (very long baseline interferometry), rain/precipitation observation and more) has been identified, hence, the importance to develop this technology to further maturity, which is currently not available in Europe.

The antenna to be developed is 1m (scalable up to 3m) diameter deployable reflector antenna in an offset configuration (reduction of aperture blockage) suitable for missions up to Ka-band. An engineering model including the reflector, the boom and the feed will be manufactured and tested demonstrating the critical functions of the element in a relevant environment. The testing will include as a minimum surface accuracy measurement after deployment, deployment test in representative environment (TVAC), vibration tests, thermal cycling, RF performance tests, passive intermodulation (PIM) tests at sample level.

This activity encompasses the following tasks:
- Preliminary design (including de-risking activities).
- Detailed design (including de-risking activities).
- Antenna manufacture and integration.
- Test campaign and development roadmap.

Deliverables: Critical breadboards, Engineering Model, Reports

Application/Need Date: Cubesat/small satellite missions operating up to Ka-band. **Mission Classification:** gamma, delta

THAG Roadmap: Reflector Antennas (2023) – Consistent with activity D13 “Deployable Reflector Antenna for CubeSat missions”.

Generic Technologies

RF Payloads & Technology

GT17-818EF – Integrated submillimetre wave antenna-coupled dual-polarization focal plane array and detection chain

Budget: 7000 k€ - **Duration**: 36 months – **Current / Targeted TRL:** 3 / 5 – **TD** 7

Objective:
To develop an integrated submillimeter wave antenna-coupled dual-polarization focal plane array and detection chain for space and ground applications.

Description:
Submillimeter wave focal plane array (FPA) based on very sensitive detectors like transition edge sensor (TES) technology have been proposed in several cosmic microwave background missions like SPICA and LiteBIRD as well as ground experiments. The European technology for the realization of such FPA and receiver chains has been demonstrated as individual building blocks. However, further developments are required to optimize the design and further implement the full receiver chain. To achieve scientific objectives, large-format millimeter and submillimeter arrays (~100-500GHz) with moderate sensitivity (~10-18 WHz-1/2) are required in two linear polarizations and high polarimetric purity over large fractional bandwidths. The most promising solution consists of using microstrip-coupled TESs and waveguide horns to maintain clean beams with a high degree of frequency independence and straylight rejection. Miniature on-chip superconducting RF components, such as directional couplers and bandpass filters, will be used to provide additional functionality.

This activity is divided into phases, it is expected to be run by a prime being responsible for the full system, with contributions of different institutes for the following sub-systems and it encompasses the following tasks per phase:
Phase 1 (budget: 4000 k€, duration: 24 months):
- Design and development of an antenna-coupled TES bolometers, including the ortho-mode transducer (OMT) and on-chip bandpass filter and TES bolometers.
- Development and packaging of lightweight low frequency feedhorns, considering additive manufacturing or advanced manufacturing techniques (including modelling, development of manufacturing processes and the manufacturing of test demonstration devices).
- Development and packaging of corrugated silicon platelet feed horn arrays, considering drilled profiled horn arrays, optical and thermal CTE optimization and the refinement of manufacturing processes.
- Development of frequency division multiplexing (FDM) cold readout, including a two stage SQUID (superconducting quantum interference device) system.

Phase 2 (budget: 3000 k€, duration: 12 months):
- Implementation of a testbed detection chain, considering the TES noise and the noise equivalent power (NEP) demonstration in single-pixel and multiplexing and the development of algorithms for 1/f noise correlation among pixels in same readout chain.
- Demonstration of the integration and the cryogenic performance of the FPA, including a fully representative FPA breadboard model (50%-pixel population), the FPA packaging and integration, feedhorn integration and full module testing (cryogenic optical testing).

Deliverables: Two breadboards of the integrated FPA, Reports

Application/Need Date: Future missions including scientific applications like LiteBIRD. **Mission Classification:** alpha, beta.

THAG Roadmap: Relevant to the Roadmap Technologies for Passive Millimetre & Submillimetre Wave Instruments.

Generic Technologies

RF Payloads & Technology

GT17-819EF – Next generation phased array antennas at K-band and Frequency Range 2

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop an elegant breadboard of a compact integrated spaceborne K-band and frequency range FR2 phased array elegant breadboard with significant integration to optimize size, weight, power and cost.
Description:
Recent developments in active phased array technology have proven the potential for high savings in mass and size as well as power efficiency using the integration of RF, DC and digital functionalities. These new designs bring additional cost and time saving benefits through mass-production processes compatible with space-qualification and automation of testing. Furthermore, the modularity and inherent scalability of this approach offer a high degree of flexibility in mission design and concepts. Extension of X-band designs to frequency ranges like K-band and FR2 pose significant challenges that need to be addressed and investigated, including careful consideration of materials, shielding, and impedance matching to maintain signal integrity. In addition, as power handling capabilities of components degrades at higher frequencies, thermal management with higher density becomes more challenging. Based on this, the main elements to be addressed within this activity are:

- System and subsystem level:
    - Critical review of system requirements to reduce the challenges on hardware level.
    - Review of classical AIT approaches, shifting e.g. building block performance verification to higher levels by exploiting the capabilities of digital controlled built-in calibration methods.
    - Modular array antenna designs configurable by element building blocks.
    - Review of relevant commercial technologies and elements and mapping potential space-borne usage.
    - Trade-off analog vs digital vs hybrid beamforming.
- Hardware and technology utilized for the phased array:
    - Improved radiator integration with transmit/receive (TR) electronic modules.
    - Single chip integration of TR electronic modules.
    - Metal only vs dielectric-based trade-off for antenna components.
    - Reducing the mass of energy storage elements by using advanced power conditioner.
    - Data transmission within the array by means of optical fibres.
    - Lightweight mechanical structures and deployment technology.

The activity is divided into phases and encompasses the following tasks:
Phase 1 (budget: 375 k€, duration: 10 months):
- Consolidation of requirements for the phased array and identification of the appropriate technologies to set the goal for mass and cost optimization through integration.
- Investigation of the application of advanced technologies and AIT procedures.
- Selection and design of preliminary array architecture.

Phase 2 (budget: 1,125 k€, duration: 14 months):
- Detailed design and implementation of an elegant breadboard to demonstrate signification mass and cost reduction with respect to the current generation of phased arrays.
- Testing of the EBB based on the pre-defined verification and test plan.
- Analysis of test results and extrapolation of expected performance at phased array system level.

Deliverables: Elegant Breadboard of the K/FR2 phased array, Reports

Application/Need Date: Future missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

RF Payloads & Technology

GT17-820EF – Low-cost and compact packaged high-power amplifier with integrated antenna for active antenna arrays

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop new concepts of integration of packaged high-power amplifier (HPA) with radiating element or waveguide launcher for active arrays.

Description:
Nowadays, there is a clear need for higher levels of integration as the technologies evolve to more complex structures and higher frequencies. Especially, highly integrated RF functionality to meet the periodicity/lattice spacing requirements with associated low-cost is becoming a critical element for future LEO missions such as high-density millimetre-wave active antenna arrays in Q-/V-/W-bands.

The proposed activity aims to develop new and novel manufacturing, packaging and advanced RF integration techniques which provide improved, highly integrated, and repeatable performance. Novel techniques will be necessary to allow a simplified and efficient connection between the low-cost packaged high-power amplifier and the array element, maintaining RF low-losses transition whilst managing the thermal heat dissipation. The package could integrate a radiating element or a waveguide launcher and could be manufactured by using new manufacturing and integration techniques such as antenna-in-package, precision additive manufacturing, micromachining, etc. This low-cost solution is compact and should offer better signal integrity (reduced crosstalk, better impedance matching) than wire bonding-based connections.

The objective of this activity is to design, manufacture and test a packaged high power amplifier breadboard with a radiating element or a waveguide launcher for active antenna array application.

This activity encompasses the following tasks:
- Definition of an operational scenario and end-user requirements.
- Definition of the active antenna array architecture in which the breadboard will be integrated.
- State-of-the-art review of advances and trends in packaging technologies and a survey of relevant implementation techniques.
- Identification of potential solutions and preliminary trade-off analysis.
- Detailed electrical design and thermal analysis of the breadboard in the down-selected technologies, considering critical areas requiring specific attention during the manufacturing.
- Definition of a manufacturing and test plan.
- Procurement, manufacturing, and assembly of the breadboard, tests in relevant environment, collection and evaluation of the results.
- Critical evaluation of the outcome of the activity, proposing updates or potential improvements.

Deliverables: Breadboard of the compact packaged HPA with integrated antenna, Reports

Application/Need Date: Phased array antennas in future missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Active Critical RF Technologies (2025) – Consistent with activity R&D Line 4.2 “Develop packaging technologies”.

Generic Technologies

RF Payloads & Technology

GT17-821EF – In-orbit assembly and performance demonstration of a large antenna

Budget: 1900 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 7

Objective:
To design, manufacture, and test a large reflector antenna and feed, in the order of 10 m, made out electrically connected shaped panels for in-orbit assembly applications.
Description:
In Europe, there are currently no antennas with large diameters. A deployable antenna with a diameter close to 10 m is being developed under the CIMR project, but it still requires space qualification. Moreover, deployable solutions beyond 15 - 25 m are not expected in Europe. While some initial in-orbit-assembly (IOA) and in-orbit-manufacturing (IOM) activities have been conducted, focusing on mechanical and robotic aspects, no one has concentrated on the actual antenna from an electromagnetic point of view.

The proposed activity aims to address all critical technologies necessary to realize a large antenna in space, in the order of 10 m diameter, including the feed as an integral part of the antenna and fully RF representative. It will cover RF, mechanical, thermal, materials, structures, and Guidance, Navigation and Control (GNC) aspects. The activity will incorporate advanced metrology techniques to ensure structural and functional accuracy, such as electrical connectivity, feed design and deployment, and the corresponding radiation pattern. Advanced modelling will also be required to address this challenge. Additionally, novel approaches to GNC for such a large structure will need to be considered.

This activity will cover the design, manufacturing and testing of a large reflector antenna made of electrically connected shaped panels. The design phase will cover a trade-off between operational frequency and size of the panels, aiming at an antenna in the order of 10 m diameter. Following a trade-off and selection of materials, manufacturing and joining processes for the panels, the full design of the antenna, including the illuminating feed will be performed. Technologies for electrical connectivity will be selected and demonstrated with measurements. After the full assembly, the RF performance of the antenna will be demonstrated by test.

This activity will encompass the following tasks:
- Requirements consolidation, architecture and manufacturing trade-off.
- Preliminary design and critical breadboarding.
- Detailed design and analysis.
- Manufacturing and integration.
- Testing.

This activity is proposed to be implemented in Bundle 1 - Enabling in orbit assembly of large antennas, together with the activities GT17-848MS - Deployable support structure with feed horn and metrology functionality and GT17-809EC - Guidance, navigation and control system development for in-orbit assembly of very large antennas.

Deliverables: Elegant breadboard, Reports

Application/Need Date: Large antenna reflectors for future missions. **Mission Classification:** alpha, beta

THAG Roadmap: Relevant to the Roadmap Reflector antennas.

Generic Technologies

RF Payloads & Technology

GT17-822EF – High performance digital RF back-end

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To design, develop and test the breadboard of a hardware module capable of high-speed data conversion and high-performance reconfigurable processing, for next-generation RF payloads.

Description:
Advances in data conversion and digital processing technology are paving the way for highly integrated digital RF hardware modules. This is achieved by embedding in the same form factor the capabilities for multi-channel waveform generation and digitization through direct sampling, together with powerful data processing for functions ranging from digital beamforming to cognitive systems.

Such integration and flexibility also come with new challenges beyond the exploitation of the EEE components themselves, especially from a system design perspective: sophisticated high-speed PCB layouts; wide bandwidth/high load-step power supplies; and robust thermal architecture, among others.

This activity aims to take a systemic approach to the design of a next-generation RF payload back-end module, exploiting state-of-the-art EEE components, capable of operating in different scenarios.

This activity encompasses the following tasks:
- Investigation of the EEE components to be used as building blocks of the digital processing unit for RF payloads (data acquisition and conversion, digital data processing, clock generation and distribution, high-speed links), including supporting toolchains and potential European technologies.
- Definition of at least two RF payload and/or microwave instrument use-cases and derive requirements for breadboarding and processing.
- Development of an elegant Breadboard (EBB) of the representative digital RF module using next-generation EEE technologies and evaluating such technologies.
- Development of the relevant software and firmware for implementing use-cases, evaluating toolchains' quality and maturity to deploy the use-cases in next-generation Field-Programmable Gate Arrays (FPGAs) and/or System on Chips (SoCs) where appropriate.
- Identification of technology gaps and limitations taking also into account existing and future European EEE technology, and preparation of the "Lessons learnt" for future roadmaps on European technology EEE and support ecosystem developments.

Deliverables: High Performance Digital RF Module Elegant Breadboard, Reports.

Application/Need Date: future RF payloads including RF microwave instruments for Earth Observation and advanced signal generation units for Navigation. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Active Critical RF Technologies (2025) – Consistent with the activity R&D Line 5.1 "Develop RF functions for Telecommunications".

Generic Technologies

RF Payloads & Technology

GT17-823EF – Digital validation standard antenna

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 7

Objective:
To develop and validate a phased array VAST antenna that can offer European test facilities benchmarks to implement the facility updates and upgrades in the most efficient way.

Description:
Validation antennas are special reference antennas that provide the means to accurately characterize and validate antenna test facilities.

Space antenna technologies are making wide steps to active phased arrays and digital antennas that combine analogue elements with complex digital backends, capable of generating thousands of beams. However, today's European facilities are not prepared to test these types of antennas as it requires digital interfaces instead of an RF connector and adaptation of acquisition systems with fast multibeam software-based switching and triggering. Moreover, the measurements are heavily dependent on the phase array calibration which can be performed inside the facility and bring some errors in main parameters from the start if the facility is not adapted for this.

This activity will develop, manufacture and validate a VAST antenna with the main purpose to offer to European test facilities benchmarks in order to implement the facility updates and upgrades in the most efficient way, making them ready for the antennas of the future. The VAST antenna to be developed in the activity will be an active phased array with 500+ beams and have the possibility to be tested without RF connectors at two different frequencies. The phase array coefficients will be known and how well the VAST can be calibrated in the European's facilities will be a key point in the assessment of their performance. The VAST antenna due to its robustness and stability, aims to become the reference antenna for facilities measuring phased array antennas.

This activity encompasses the following tasks:
- Development of a VAST antenna.
- Manufacturing the VAST antenna.
- Validation of the VAST antenna.

Deliverables: VAST antenna breadboard, Report

Application/Need Date: All missions with phased array antennas. TRL 5/6 by 2028. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Array Antennas and Periodic Structures.

Generic Technologies

RF Payloads & Technology

GT17-824EF – Next generation distributed and modular digital payload

Budget: 1800 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To design and develop an elegant breadboard (EBB) for a distributed modular digital payload, embedding data conversion and digital functions in the front-end.

Description:
Current generation of phased array antenna-based payloads are successfully implementing digital beamforming capabilities to meet the ever more challenging requirement. However, the present classical technology of centralized digitization, processing and control, used to enable high accuracy, fast time adaptable, multi-beam power distribution over a given access range is reaching its limits in terms of scalability as well as efficiency, when it comes to processing power, SWaP and on-board data rates. The idea of distribution of the centralized data conversion and digital functionalities over the array aperture offers several options to further enable scalability of the general principles of beamforming arrays that are necessary to ensure the evolution of performance of the next-generation RF payloads. A distributed and modular digital active antenna offers the potential for adaptable high bandwidth beamforming, while simultaneously offering the potential to overcome the limiting factors of the current generation microwave instruments like complex calibration, thermal hotspots and high mass due to the need of an analogue signal distribution network, opening degrees of freedom for the design, manufacture, integration testing and even operational concepts.

This activity will assess the potential of a microwave payload architecture based on an active phased array antenna in which the up-/down-conversion, digitisation and first steps of digital beamforming and processing are distributed over the front-end, very close to the aperture itself, supported by a compact fully digital unit that is able to manage the distributed elements on the next scale. This activity aims to develop, design, manufacture and test an EBB of a phased array based on the proposed approach of the distribution of digital functionalities. An important aspect to be evaluated and demonstrated during this activity will be the distribution and maintenance of clock synchronization between the digital components on antenna level, as well as the actual implementation of the beamforming capabilities, including the potential of non- Field-Programmable Gate Array (FPGA) beamforming implementation on next generation data conversion devices or specialised digital beamformer chips.

This activity encompasses the following tasks:
- Selection and definition of an application scenario to be investigated and appropriate system level requirements.
- Assessment of the suitability of the identified technologies and methods of the distributed digital antenna approach for other application scenarios.
- Payload design based on the distributed digital processing approach.
- Trade and selection of different technologies applicable for selected designs.
- Trade-off analysis and concept selection to be evaluated by means of the EBB preliminary design.
- Definition of a preliminary architecture for the active antenna and digital module.
- Definition of the critical functionalities (e.g. performance, clock synchronisation and control, calibration and beamforming) of the payload that will be demonstrated by the EBB.
- Establishment of a test and verification plan that will ensure the demonstration of the selected functionalities.
- EBB manufacturing, test and verification of the performance and the critical functionalities.

Deliverables: Breadboard of the distributed and modular digital payload demonstrator, Report

Application/Need Date: Next Generation Copernicus, Future Earth Explorers, IRIS2 & Galileo Evolutions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

RF Payloads & Technology

GT17-825EF – G-band receiver monolithic microwave integrated circuit

Budget: 750 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop ultra-compact receiver front-end payloads by integrating the major functions of a heterodyne / direct detection receiver onto a single MMIC (monolithic microwave integrated circuit).

Description:
Mm-wave heterodyne and/or direct detection receivers such as the one developed for MetOp-SG and AWS include modular-based architectures with many interfaces between the fundamental oscillator, active multipliers, mixers and Low Noise Amplifiers. This results in complex RF chains which are difficult to scale to extended arrays. It is therefore critical to develop highly integrated receiver MMICs for future multi-pixel heterodyne and interferometric instruments. Recent advances in Silicium Germanium (SiGe) technologies allow to generate local oscillator (LO) signals up to W-band including Phased Locked-Loop (PLL) electronics, but also low noise amplification and mixing up to G-band.

This activity will combine all functions required in the RF frontend down-conversion into a single MMIC. These functions include fundamental oscillator, amplifier, mixer, filters and possibly some back-end elements. Technologies such as Silicon germanium (SiGe), BiCMOS or Gallium Arsenide (GaAs) mHEMT are particularly suited for this approach. Within the G-band, several bands are of interest for Earth Observation, such as the ones centered at 165.5 GHz for altimeter correction, 183.3 GHz for water vapor remote sensing and 235 GHz also for atmospheric sounding. An improvement of one order of magnitude in SWaP (Size, Weight and Power) will be targeted compared to current flight solutions. This range is of particular interest for future atmospheric remote sensing missions and GEO-sounders.

This activity will encompass the following tasks:
- Identification of the key building blocks to be integrated into a single MMIC, and most suitable technologies.
- Design, fabrication and chip-level/module testing of a MMIC receiver front-end breadboard using the most appropriate technologies
- Module level reliability assessment including radiation and life testing on an elegant breadboard.

Deliverables: Breadboard and Elegant Breadboard of an integrated 183 GHz receiver module using packaged MMIC; Reports

Application/Need Date: Future Earth Observation missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Technologies for Passive Millimetre & Submillimetre Wave Instruments (2024) – Consistent with activity A03 “183 GHz MMIC Front-end Design”.

Generic Technologies

RF Payloads & Technology

GT17-826EF – Advanced multi-input receiver for next-generation space transponders

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 1

Objective:
To develop and test an Advanced Multi Input Receiver (AMIR) elegant breadboard for next generation of Space Transponders (STs), demonstrating multi-source RF input processing, external LNA support, and unified data/inter-unit interface, validating key architectural concepts.

Description:
The proposed activity will focus on the development and validation of an Advanced Multi Input Receiver (AMIR) Elegant Bread Board (EBB). The AMIR EBB will serve as a critical foundational step towards realizing advanced, European-sovereign STs, addressing limitations in current architectures related to performance, flexibility, and integration.

Developing this AMIR technology is paramount for reducing reliance on non-European sources for critical space communications, thereby strengthening European independence and security in orbit. While demanding future planetary exploration missions represent a key driver, advanced capabilities of AMIR such as enhanced sensitivity via external Low Noise Amplifier (LNA) support, improved link reliability through novel inter-transponder processing, and greater operational flexibility offer substantial benefits to a broader spectrum of European space applications. The activity will focus on evaluating the feasibility of incorporating key European technologies further underpins this strategic autonomy.

The AMIR Elegant Bread Board (EBB) will demonstrate an innovative architecture featuring:
- Multiple Radio Frequency (RF) inputs for simultaneous signal reception from various antenna sources.
- Support for external Low Noise Amplifiers (LNAs), allowing placement near antennas for optimized sensitivity and link margin.
- A unified, high-speed data interface (e.g., SpaceWire, SpaceFibre, or space-qualified Ethernet) for streamlined communication with the On-Board Computer (OBC), Payload Data Handling Unit (PDHU), and for inter-transponder communication. A key innovation is the inter-transponder link, enabling shared signal, data exchange, and functional coordination between redundant units. This significantly improves overall link reliability and operational flexibility, particularly in challenging deep space environments.

This activity will encompass the following main tasks:
- Definition of AMIR EBB detailed architectural design.
- Hardware and firmware/software detailed design for the modular EBB.
- Procurement of components, manufacturing of the Printed Circuit Boards (PCBs), and assembly of the EBB.
- Functional and performance testing of the AMIR EBB, validating:
    - Multi-RF-input signal processing capabilities.
    - Support to external LNA interface for performance and sensitivity benefits.
    - Inter-transponder link functionality and shared processing concepts.
    - Unified data interface performance with representative OBC/PDHU emulators.
- Execution of a feasibility study on the integration of European key enabling technologies for future flight units.
- Development of a technology roadmap for next-generation AMIR-based STs.

Deliverables: Elegant Breadboard of the AMIR; Reports

Application/Need Date: Building block for next generation transponders for all missions / TRL5/6 by 2028.
Mission Classification: alpha, beta, gamma

THAG Roadmap: TT&C Transponders and Payload Data Transmitter (2025) – Consistent with activity R&D Line 1.8 “New Generation Agile Multi-Input Transponder with Enhanced Security and Performance”.

Generic Technologies

RF Payloads & Technology

GT17-827EF – Compact submillimetre wave radiometer

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - TD 7

Objective:
To design, manufacture and test an ultra-compact submillimeter radiometer breadboard that will be compatible with the mass and power consumption budgets of a typical CubeSat or SmallSat.
Description:
In recent years, CubeSats have been gaining considerable popularity for Science and EO applications. Mostly, European radiometer instruments have been deployed on larger platforms like Metop-SG (MWS and MWI) and JUICE (SWI). Artic Weather Satellite mission is already a push towards smaller platforms and more compact radiometers. Even reducing the submillimeter instruments further, would enable the implementation of a European constellation of radiometers embarked on CubeSats.

This activity will focus on the demonstration of the feasibility of an ultra-compact multi-frequency channel radiometer for future smallsat platforms or as a secondary payload on next generation Science or EO missions. To achieve the objective, previous subsystem technology developments will form the basis for the instrument design and so it is seen as a next step in achieving a state-of-the-art compact submillimeter radiometer.

After the instrument requirements are consolidated, a design of a submillimeter radiometer will be performed. The exact frequencies will have to be defined but 183 GHz, 325 GHz or even higher frequencies are envisaged. The design will be based on the technology developments that have been made over the last years and will focus on mass and power consumption budgets compliant with typical CubeSats/SmallSats. The final submillimeter radiometer breadboard, consisting of the optics, RF front-end and back-end hardware, will be designed, manufactured and tested for both RF and thermal-mechanical aspects.

This activity will encompass the following tasks:
- Requirements of sub-millimetre wave radiometer instrument.
- Preliminary design and critical breadboarding of the submillimeter radiometer.
- Detailed design of the submillimeter radiometer demonstrator.
- Manufacturing and testing of submillimeter radiometer for both RF and thermo-mechanical performance.

Deliverables: Breadboard, Reports

Application/Need Date: All missions. Future cubesats missions (including constellations) with submillimeter wave instruments. The technology push will enable to fly a fully European radiometer instruments in space for various EO (now-casting, tracking sever weather events) or Science mission (future L4 mission might include a submmw instrument). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Technologies for Passive Millimetre and Sub-Millimetre Wave Instruments.

Generic Technologies

RF Payloads & Technology

GT17-828EF – Critical technologies for satellite formation flying at very close range

Budget: 600 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop an elegant breadboard comprising of three stackable uni-directional laser inter-satellite link units capable of supporting 1 Gbps data rate each, enabling 100 Gbps by stacking 100 of those units, and an optical sensor providing relative distance and angle with 1 mm and 0.03 degree precision respectively, both breadboard elements to work over 1-10 m spacecraft separation.

Description:
There is a strong interest in remote sensing, radio-astronomy, navigation and communications, to deploy payloads distributed on board a set of satellites, with the objective of creating a large coherent synchronized system that can achieve a far superior performance to that of each of the instruments aboard each satellite when operated individually.

Critical to such a concept of distributed payloads are the inter-satellite links and the relative positioning and orientation between spacecraft. The inter-satellite links (ISL) are needed to exchange the local oscillator reference signal for syntonization and synchronization, payload scientific raw data, spacecraft telemetry for relative navigation and calibration signals. An optical sensor (OS) can provide accurate relative position and orientation between spacecraft to enable the control and maintenance of the right geometry of the set of satellites.

The activity will comprise of the following tasks:
- Analysis and design of ISL and the OS including, for the ISL, the selection of the laser wavelength, laser type, laser modulator, optical fiber types for transmitting and receiving, transmitting and receiving collimators and detector type, collimators stacking mechanical approach, and for the OS, the selection of the LEDs, camera detectors and target arrangement.
- Manufacturing of an elegant breadboard of both the ISL and the OS, as well as a mock-up of the lateral spacecraft panels which would host the units; testing of the data rate of each of the three ISL data rate (1 Gbps) and its robustness against lateral displacements of 12 mm and angular mis-alignments of 0.03 degrees between the transmitting and receiving collimators; testing of the OS for distance and angular precision against the 1 mm and 0.03 degrees requirements respectively.
- Assembly and end-to-end testing of the ISL and the OS; integration of the three stacked ISLs and the OS elements onto the panel; testing of the data rate of the ISLs and the relative distance and angle precision of the OS in final configuration; verification of robustness of both the ISL and the OS to lateral displacements and angular misalignments in final configuration.
- Environmental testing: the ISL and the OS in their final configuration shall be tested inside a vacuum thermal chamber for a distance of 1 m between spacecraft panels.

Deliverables: Elegant breadboard of the ISLs and the OS, Reports

Application/Need Date: All missions requiring the coherent and synchronized operation of several satellites to achieve, for example, very high-resolution remote sensing observations, space radio-astronomy free of atmospheric effects, coherent operation of navigation satellites when extending the range of the ISLs, and, in general, highly accurate beamforming. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: TT&C Transponders and Payload Data Transmitter (2025) – Consistent with activity R&D Line 3.6 “Improve the capabilities of ISL solution”.

2.3.4. Power Systems, EMC and Space Environment

Generic Technologies

Power Systems, EMC & Space Environment

GT17-829EP – Development of standardised low cost PVA manufacturing processes

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop a standardised photovoltaic assembly (PVA) manufacturing processes with the aim to achieve a 30% cost reduction in the manufacturing of PVA.
Description:
The deployment of satellite constellations demands a paradigm shift in the manufacturing approach for satellite subsystems. In this scenario, a significant reduction in the cost of the photovoltaic assembly (PVA) is essential to ensure economic viability. To achieve this goal, traditional manufacturing methods, designed primarily for low-volume production, must be replaced with highly standardized processes. Current state-of-the-art approaches often rely on manual manufacturing steps and on a huge variety of customized components and design configurations. While this offers flexibility for small-scale production, it significantly limits the potential for automation and scalability. In contrast, this activity aims to pursue the opposite approach developing standardized PVA manufacturing processes by minimizing design variability. These new processes shall be specifically engineered for compatibility with semi-automated or fully automated systems, adopting industrial-scale production models like those used in automotive or consumer electronics manufacturing.

In the frame of this activity, standardization not only enables high-throughput manufacturing but also simplifies the subsequent assembly, integration and testing of the final solar array. Moreover, standardized and automated processes generate consistent data streams that can be leveraged (e.g. through machine learning techniques) to continuously optimize production efficiency, detect anomalies early and improve the reliability of the product. The manufacturing processes could be compatible with a modular approach (e.g. plug-and-play modules designed as building blocks for mass production of solar arrays) and be suitable for both flexible and rigid solar array designs, ensuring versatility across various satellite architectures.

This activity encompasses the following tasks:
- Definition of specifications for low cost PVA manufacturing processes emphasizing standardization and compatibility for semi or fully automated manufacturing.
- Development of technology building blocks (solar cell design, interconnector design, welding technology, coverglassing, stringing, …) derived from the previously defined specifications.
- Description of manufacturing processes including automation strategy and quality control methods.
- Manufacturing of an engineering model based on the developed modular PVA concept followed by preliminary testing both at component and PVA level.

Deliverables: Engineering Model, Report

Application/Need Date: LEO missions. TRL 6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Solar Generators and Solar Cells (2025) – Consistent with activity R&D Line 3.3 “Develop testing, characterisation and qualification techniques for low-cost solar arrays and subassembly components”.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-830EP – Low volume high power flexible solar array

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To design a flexible solar array with a low stowed volume and high performance, which modular and standardized design is compatible with a multi-launch satellite approach, reducing the overall system cost by up to 50%.
Description:
The concept of a solar array with low stowed volume and high power can be an asset for different mission types. In the case of GEO and interplanetary missions, they are becoming more power hungry, and for this a solar array concept with a high power per stowed volume (kW/m3) is of great interest. In addition, satellite constellations, require solar arrays concepts which are compatible with a multiple satellite launch scheme and, again here, a solar array with a high kW/m3 figure of merit would be necessary.

This low stowed volume high power development can be achieved by using flexible solar array technologies in combination with fully or semi-flexible photovoltaic assemblies (PVAs). In addition, and to keep the cost low, the technologies used in manufacturing shall be suitable for a high level of modularity, automation and use of standardized processes.

This activity aims to build upon previous existing solar array concepts in which additional elements require further development, like the mechanism, supporting blanket, photovoltaic assembly or harness.

The activity encompasses the following tasks:
- Identification of flexible solar array concepts allowing for reaching power per volume figures higher than 30 kW/m3 and respective building blocks.
- Mechanical analysis of identified technology, including strength and stiffness analysis of the array in both stowed and deployed configurations.
- Development and breadboarding of technology building blocks (mechanisms, supporting blankest, photovoltaic assembly, harness, frames, structures, etc.) required to meet the identified solutions in the previous task.
- Manufacturing of representative mock-ups, breadboards and engineering models (e.g. DVT coupon) allowing for critical function testing validating the mechanical analysis.
- Engineering tests (deployment test, sine vibration, thermal cycling).

Deliverables: Engineering Model, Report

Application/Need Date: LEO missions. TRL 6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Solar Generators and Solar Cells (2025) – Consistent with activity R&D Line 2.2: “Develop and qualify next generation compact solar array and/or photovoltaic assembly technology”.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-831EP – Adaptive synchronous rectifier controller

Budget: 750 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop an adaptive synchronous rectifier (SR) controller to enhancing functional and power density of payloads and/or power supplies for satellite standard platforms.
Description:
Synchronous rectification significantly increases efficiency and reduces the size of DC/DC converters, particularly in low-voltage, high-current applications. This is achieved by replacing traditional diodes with highly efficient silicon (Si) or gallium nitride (GaN) switches. To simplify the control of these switches and further enhance their performance, Adaptive Synchronous Rectifier Controller integrated circuits (ICs) are widely used in industrial and commercial sectors. However, in the space industry, the lack of a European-developed Synchronous Rectifier Controller presents a gap in the supply chain and technology sovereignty. The development of a European Adaptive Synchronous Controller for space applications would bring substantial benefits: component manufacturers would gain from producing a high-value part, electronic equipment suppliers could deliver higher-performance and more efficient systems, and prime contractors would ultimately benefit from the resulting improvements in efficiency, power density, and reliability.

By sensing the voltage and/or the current through the switching element (SiC, GaN, Si) the IC detects when the Metal-Oxid-Semiconductor Field-Effect Transistor (MOSFET) must be switched on and off. The IC drives the Field-Effect Transistor (FET) with a gate driver. This emulates an ideal diode character, lowest conduction losses and lowest switching losses. This enables high frequency switching with easy achievable soft switching characteristics. This soft switching behavior is only achievable with an ideal diode characteristic. The soft switching will lead to good EMC performance.

This activity encompasses the following tasks:
- Final consolidation of the preliminary requirements.
- Identify potential solutions (find suitable topologies for design blocks like LDO (low dropout regulator), gate driver, parameter drift) and perform dedicated trade-offs to select baseline.
- Simulate and optimize design for needed performance in spec, also optimize TID/SEE hardness.
- Perform the routing and layout of the IC. Simulate the implementation and optimize for electrical and radiation performance.
- Prepare all the inputs for the prototype manufacturing of the adaptive synchronous rectifier controller integrated circuits and manufactured prototypes.
- Prepare and execute all test activities (test set-up, test procedures), including Total Ionizing Dose (TID) / Single Event Effect (SEE).
- Analyze test results, identify and check deviations, implement corrections on the prototype if any to cover deviation.

Deliverables: Prototype, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Power Management and Distribution (2025) – Consistent with R&D Line 1.1: "Development of mixed-signal ASICS for specific power needs”.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-832EP – Cost-competitive solar cell development

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop a cost-competitive solar cell either III-V-based or silicon-based for LEO satellite constellations.
Description:
The satellite industry is undergoing a major shift from traditional geostationary (GEO) platforms to large-scale low Earth orbit (LEO) constellations. In this new landscape, solar cell requirements are increasingly driven by the need for scalability, cost competitiveness, and radiation hardness tailored to the LEO environment. III-V multi-junction solar cells remain the benchmark for efficiency and space reliability, but their high cost and manufacturing complexity limit their suitability for mass production. In contrast, silicon-based solar cells, widely used in terrestrial solar arrays, offer a more scalable and cost-efficient option, though they are not optimized for the space environment and typically have lower radiation resistance. Future development of III-V cells should focus on reducing complexity and cost. For silicon technologies, the priority is improving tolerance to radiation-induced degradation, which is generally more severe than in III-V cells.

Within this activity, it is foreseen to develop **one** solar cell technology that can support the LEO constellations in view of cost competitiveness and production capability. Technology options for this activity are either III-V-based or silicon-based, with the following specific objectives:

- **Option 1 - III-V-based technology development**, to reduce production costs by up to 50% and double manufacturing throughput to meet large-scale deployment needs: there is substantial potential to reduce costs and increase manufacturing throughput by optimizing existing processes or implementing new ones. Potential strategies include increasing epitaxial growth rates, integrating terrestrial PV techniques (e.g., screen-printing for contact metallization), and replacing photolithography steps with more cost-effective alternatives.
  For this option, the activity encompasses the following tasks:
  - Develop an optimized manufacturing route for III-V solar cells, incorporating both improvements to current processes and introduction of new techniques.
  - Fabricate solar cell samples using the newly developed manufacturing process.
  - Conduct an engineering test campaign to evaluate and validate the electrical and mechanical performance of the produced cells.

- **Option 2 - Silicon-based technology development,** with the aim to achieve 21% efficiency at beginning-of-life (BOL) and 18% at end-of-life (EOL) and to improve reliability in the LEO space environment. The development activity will focus on both design innovation and process optimization, aiming to enhance radiation tolerance and thermal cycling robustness while achieving the required beginning-of-life (BOL) and end-of-life (EOL) efficiency targets. For this option, the activity encompasses the following tasks:
  - Implement cell design optimizations already identified for existing silicon solar cell concepts to improve radiation resistance to meet the 21% BOL and 18% EOL efficiency targets.
  - Investigate alternative metallization methods, anti-reflective coatings (ARCs), and interconnection techniques to improve thermal cycling robustness.
  - Explore compatibility of silicon structures with potential self-annealing properties to mitigate radiation-induced degradation with industrial manufacturing framework.
  - Validate the final solar cell design through an engineering test campaign.

Deliverables: Engineering Model, Report

Application/Need Date: LEO missions. TRL 6 by 2028. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Solar Generators and Solar Cells (2025) – Consistent with activity R&D Line 1.2: “Develop and qualify low cost high efficient solar cells”.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-833EP – Containment techniques for batteries stored energy at the end of life

Budget: 850 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop containment techniques for batteries stored energy at the end of life, considering solutions that are light and have an easily demisable and minimal re-entry footprint on the spacecraft.

Description:
Over the course of a spacecraft's mission, batteries provide vital power during periods without sunlight. After the mission concludes, however, these batteries can create additional considerations for safe disposal and management. Space debris mitigation rules require the complete deactivation of the electrical power sources and the discharge of the batteries, to prevent future recharge, overcharge and explosion that can create dangerous space debris. However, this soft passivation doesn't enable fully deplete the energy inside the batteries.

Consequently, there is still a risk of generating more debris in space. To mitigate this risk, the development of containment techniques is crucial to avoid the spreading of additional space debris, in case of overheating of batteries or collision with existing space debris.

During a previous TDE activity, the preliminary design of containment techniques for batteries stored energy at the end of life has shown interesting results in terms of materials choices (several thermoplastics have been studied) and in terms of design of the containment for cubesat batteries. This activity aims to develop containments techniques for the overall range of spacecrafts, with a particular focus on small and large spacecraft.

This activity encompasses the following tasks:
- Assess battery cells used in small and large spacecrafts explosion and corresponding behavior with respect to worst-case levels of charge at end of life and temperature.
- Analyse the energy remaining in various battery cells used in small and large spacecrafts.
- Analyse the energy released by the explosion of the batteries used in small and large spacecraft, considering various electrochemistries and cell format.
- Define the requirements of a containment system for small and large spacecraft.
- Design a containment system for small and large spacecraft.
- Manufacture and test a mock-up for small spacecraft (TVAC, mechanical environment).
- Assess the scalability of the design for batteries used in large spacecraft.

Deliverables: Engineering Model, Report

Application/Need Date: All spacecraft in LEO. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-834EP – Integrated standard telemetry and telecommand interface based on CAN

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop a radiation hardened application-specific integrated circuit (ASIC) for a telemetry and telecommand (TM/TC) interface based on CAN.
Description:
Internal unit communications, between different modules, are a proprietary feature of each manufacturer. Only external communications with on-board computer or with other units is standardized to a certain physical layer and protocol (MIL-STD-1553, Space-wire, CAN). Standardization of internal unit communications using CAN has many advantages:
- Scalability: all modules are connected using only two signal lines independently of the total number of modules.
- Noise immunity: all analog telemetries are digitalized locally and sent to the controller module using the digital serial bus.
- Simplification of backplane design due to the reduction of the number of lines.
- Potential reuse of module test equipment.

One of the main drawbacks of having a CAN-based internal TM/TC interface is the impact on the cost of some modules, where a more cost-effective and less performant solution could be used. A standard TM/TC interface ASIC adopted by all unit manufacturers would have a high volume and therefore a lower cost. This feature will make this solution very attractive to use in any kind of module or unit. The main functions to be included in the ASIC are single supply, plastic package, internal oscillator with only external crystal, two CAN-FD transceivers and controllers, analog interface: voltage reference, ADC and DAC, digital I/O, non-volatile memory and internal temperature sensing.

This activity is divided in phases and encompasses the following tasks:
Phase 1 - ASIC development (budget: 700k€, duration: 10 months):
- Requirement consolidation.
- Trade-off analysis of potential solutions and assessment of the total PCB area and cost including external components.
- Assessment of ASIC technologies suitable for space including European supply chain sovereignty analysis.
- Design and optimization of the ASIC design considering the external components.
- Simulate and optimize design for needed performance including radiation TID/SEE hardness.
- ASIC manufacturing and validation including TID/SEE testing

Phase 2 - ASIC improvement and qualification (budget: 1000k€, duration: 10 months):
- Requirement consolidation from phase 1.
- Improved design of the ASIC to mitigate shortcomings from phase 1.
- ASIC manufacturing and validation.
- ASIC qualification tests (radiation, thermal and EMC).

Phase 3 - Development of a technology demonstrator with several ASIC and a master module emulating the final configuration of the TM/TC interface (budget: 300k€, duration: 4 months):
- Development of the controller IP to be implemented in the master module.
- Definition of a technology demonstrator with several ASICs and a master module.
- Prototype manufacturing activities.
- Test activities (setup, procedures…).
- Test result analysis and design update.

Deliverables: Breadboard, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Avionics Embedded Systems.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-835EP – Isolated high side GaN driver

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop a rad-hard isolated high side Gallium nitride (GaN) driver to enable a fully European GaN technology for space power conversion.
Description:
The use of GaN devices for power processing in space has become essential for the industry to remain competitive, enabling high efficiency, high power density, and lower-cost solutions. However, the high switching frequencies enabled by GaN devices make it particularly challenging to drive high-side GaN FETs, due to the large and rapid voltage transients involved. The use of galvanically isolated GaN drivers with advanced integrated functionality simplifies device control, enhances performance, significantly reduces design time, and lowers development risk. Currently, there is no space-qualified, radiation-hardened GaN gate driver developed or manufactured in Europe. This reliance on non-European sources represents a critical vulnerability in the technological sovereignty, limiting the development to fully independent space systems and comply with export control regulations.

The development of a European radiation-hardened, isolated high-side GaN driver would offer significant advantages across the space industry. Component manufacturers would enter a new market for high-value, space-grade GaN solutions, while electronic equipment suppliers could enhance their offerings with higher performance and more efficient systems. Prime contractors, in turn, would benefit from improved system capabilities, helping to strengthen European leadership in the space sector.

The activity is divided in phases and encompasses the following tasks:
Phase 1 - ASIC Development (budget: 700k€, duration: 12 months):
- Requirement consolidation.
- Trade-off analysis of potential solutions. Assessment of total area and cost including external components, delays, current capability and transient immunity to dv/dt.
- Assessment of ASIC technologies suitable for space including European supply chain sovereignty analysis.
- Design and optimization of the ASIC design considering the external components. Simulate and optimize design for needed performance including radiation TID/SEE hardness.
- ASIC manufacturing and validation including TID/SEE testing
- Assessment report on future improvements

Phase 2 - ASIC qualification (budget: 1300k€, duration: 12 months):
- Improved design of the ASIC to mitigate shortcomings from phase 1 and to apply lessons learned.
- ASIC manufacturing and validation.
- ASIC qualification tests.
- Development of a technology demonstrator with the ASIC and all the required components to drive a high side GaN FET.

Deliverables: Breadboard, Prototype, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Power Management and Distribution.

Generic Technologies

Power Systems, EMC & Space Environment

GT17-836EP – Advanced DC/DC converters building blocks

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To define the essential building blocks of a comprehensive line of DC/DC converters, identify the integrated circuits required to minimize both the size and the cost of these converters, and ultimately design and construct a functional breadboard incorporating all the developed components.

Description:
Standard DC/DC converters are off-the-shelf converters that are available in the market. They have frozen specifications and designers can select them if their characteristics match their needs. However, non-European parts are available. Updating the derating rules of the converters has become a fundamental problem for European competitiveness in all electrical units. Since EEE parts cannot be utilised to their maximum capabilities, the European units are typically bigger and have worse performance. Updating the overall knowledge about EEE components, failure modes and derating rules is of relevance for the European industry to become more competitive in this field.

This activity also encompasses the integration of all basic functions of DC/DC converters in one or several ASICs: start-up circuit, control, protections, commanding and telemetry. The use of GaN switches could enable smaller and more performing and cost-effective designs.

The objective of the activity is to define the building blocks of a comprehensive line of DC/DC converters to cover a large percentage of applications. This will be based on a number of specifications mainly on input and output voltage, power level and isolation requirements. The idea is to come up with a comprehensive line of converters that could cover most of the requirements. In addition, this activity aims to define the integrated circuits (ICs) needed to reduce size and cost of the converters, integrating in one or several ICs all basic functions needed, to design and build a breadboard of all building blocks, including a breadboard of the ICs and finally, to test the breadboards in all needed configurations and identify gaps, possible improvements and to define the future development up to qualification.

This activity encompasses the following tasks:
- Trade off building blocks to cover the specifications.
- Define the functions that can be integrated in ASICs.
- Design all building blocks of the DC/DC converters.
- Design the ASICs.
- Build breadboards of the building blocks.
- Perform functional and performance tests of the breadboards.

Deliverables: Breadboard, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Power Management and Distribution.

2.3.5. Optics, Optoelectronics Robotics, Life Sciences

Generic Technologies

Optics, Robotics & Life Sciences

GT17-837MM – Fibre optic distribution for phased array antenna frontends

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 17

Objective:
To develop and test in temperature a novel phased array frontend using fiber optics to implement critical functions for high-precision clocks, digital and RF signals and distributed fiber sensing

Description:
Today's antenna frontends for spaceborne radars use copper-based signal distribution implementations for various types of signal transfer like command and control, timing and RF. Such cables are bulky, and the resulting bunches are difficult to route e.g. via hinge lines. Further, special care needs to be taken in the electrical design to achieve Electromagnetic Interference (EMI) / Electromagnetic Compatibility (EMC) immunity and stable RF performance versus temperature. Modern future phased array antennas will be based on a hybrid beamforming architecture. Such architectures with multiple digital channels offer enormous advantages on Synthetic Aperture Radar (SAR) system level and total performance. It allows for parallel signal acquisition in receive by multiple subarrays and by dedicated processing the radar assets, like e.g. swath width and/or resolution, may be multiplied and offer by factors better performance than conventional phased arrays. However, the number of signals to be routed and transferred will increase, drastically.

Photonics / optics offer enormous advantages compared to pure copper-based solutions. Besides huge bandwidth capabilities (>> 1 GHz), a fibre can carry multiple signals and EMI / EMC immunity as well as electrical isolation is guaranteed by nature. Further, a reduction in volume and weight is expected leading to a smooth harness routing across the phased array antenna. Reference signals for the ADCs and DACs with a rms jitter in the order of a few femtoseconds only can be generated. Hence, signal generation and distribution via optical fibres is a prerequisite for modern phased array radar frontends used in future spaceborne SAR mission enabling new features on instrument level like hybrid SAR mode support and by factors improved performance like sensitivity, swath width and azimuth resolution.

The activity aims to develop a representative phased array front end making optimized use of fiber optic technology. The activity will include the implementation of low jitter optical clock generation and distribution, and optical down conversion together with RF over fiber and digital traffic over fiber including a demonstration of distributed fiber optic sensing of strain and temperature. The activity will test a relevant thermal environment to assess the thermal sensitivity of the solution and identify suitable compensation strategies. This activity aims to develop the hardware demonstrator, perform a full system characterization and temperature compensation.

This activity encompasses the following tasks:
- Clock and synchronization and RF over fiber approaches, incl. skew / time-delay compensation.
- Comparison of approaches and architectures.
- Design of higher level architecture and requirements consolidation.
  - Clock & synchronization signal distribution.
  - RF over fibre distribution.
  - Command and control and low-speed timing signal distribution.
- Define / derive detailed design requirements.
- Concept development for skew / time-delay compensation vs. temperature.
- Detailed design, including schematics and layouts.
- Breadboard, manufacturing and testing tests in relevant environment.

Deliverables: Elegant breadboard tested in a representative thermal environment, Reports

Application/Need Date: Future SAR missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Photonics.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-838MM – Large duolithic mirrors for snap-together telescope assembly

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 16

Objective:
To develop technological processes for manufacturing large freeform duolithic mirrors and methodologies for snap-together telescope assembly and alignment verification.

Description:
The performance of image quality delivered by a reflective telescope is directly linked to the surface shape and co-alignment accuracies of the individual mirrors constituting the telescope. Duolithic mirrors are produced by manufacturing two individual mirrors on to a common substrate, in a single high-precision machining process. Each individual mirror is precisely referenced to markers engraved on to the common substrate. A telescope featuring duolithic mirrors allows to reduce the number of mirrors to be co-aligned. Several duolithic mirrors can be assembled with high-precision relying on the referenced markers enabling a faster and high-accurate telescope assembly, a so-called 'snap-together' technique. Snapped-together duolithic mirrors therefore enables an enhanced system alignment by reducing the time needed for the Assembly, Integration and Test process.

In addition, the mass and the dimensions of optical instruments for space applications can be reduced using freeform mirrors, which provides more flexibility for the optical designer. Finally, the use of a common substrate material for duolithic mirrors in a snap-together enables for an athermal telescope assembly (i.e. optical mirror and mechanical structure), which is a great advantage for optical instruments working at cold temperature for instance.

Machining processes shall be developed in order to push the current limitations of size to enable the production of large freeform duolithic mirrors. Metrology processes shall be developed accordingly in order to verify the surface shape of each mirror within a duolithic mirror as well as to verify the co-alignment of the duolithic mirrors assembled in a 'snap-together' fashion.

The activity encompasses the following tasks:
- Development of accurate and reliable manufacturing and metrology processes for the production and verification of large freeform duolithic mirrors.
- Manufacturing of large freeform duolithic mirrors and characterization of the optical performance (mirrors surface shape accuracy).
- Assembly of two duolithic mirrors in a 'snap-together' fashion and characterization of the residual co-alignment errors.
- Assessment of the production time and way forward.

This activity is proposed to be implemented in Bundle 2 - Enabling New space - miniaturisation and serialisation, together with the activities GT17-839MM - Low energy / high pulse repetition frequency based space lidars and GT17-846MM - Integrated chip based high power c-band optical amplifier.

Deliverables: Breadboard, Engineering Model, Report

Application/Need Date: Earth Observation and Science missions. TRL 5/6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-839MM – Low energy / high pulse repetition frequency based space lidars

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 17

Objective:
To develop high repetition rate lidars to unlock the possibility of small size Lidar missions.

Description:
Low energy lidar systems based on high pulse repetition frequency (PRF) (> 1kHz) pulses at 1550 nm use accumulation or averaging of many return pulses to compensate the lower signal of the single pulse, bringing the accuracy back to good levels. In fact, the Signal-to-Noise Ratio (SNR) can be considered at first approximation proportional to the square root of the overall laser optical power.

On the one hand, low energy Lidars are much less critical in terms of all problems related to coating and material damage risk, such as Laser Induced Damage (LID), Laser Induced Contamination (LIC), etc. They benefit from the technological maturity achieved for ground optical telecommunications and their recent space engineering for space laser communications. They also ensure a higher eye safety. On the other hand, the vertical resolution could be degraded due to the shorter time between pulses, unless some coding is employed. For applications at wavelengths different from 1550 nm, like UV scattering detection or chemical species detection, they suffer a lower conversion efficiency. They also require a high rejection of Earth radiometric background.

They would particularly fit on small platform (<20 cm telescope) and could be a secondary payload on several kinds of payloads. In this way the global coverage could be increased and ensure a much higher spatial resolution. The potential improvement in the prediction of intense atmospheric phenomena is of particular interest today.

The activity shall start from a trade-off analysis for a typical wind lidar application, define an architecture that can be useful for the proposed applications, provide a baseline design of the possible implementation, produce a breadboard of the critical parts and test it.

The activity shall encompass the following main tasks:
- Requirements review and trade-off.
- Design of baseline solution.
- Breadboarding of critical parts.
- Testing.
- Final reporting.

This activity is proposed to be implemented in Bundle 2 - Enabling New space - miniaturisation and serialisation, together with the activities GT17-838MM - Large duolithic mirrors for snap-together telescope assembly and GT17-846MM - Integrated chip based high power C-band optical amplifier.

Deliverables: Engineering Model, Report, Software

Application/Need Date: Earth Observation Lidar missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Lidar Critical Subsystems.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-840MM – Alexandrite based Lidar for the development of vegetation monitoring techniques

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 17

Objective:
To develop an engineering model of an Alexandrite Lidar to be used for vegetation monitoring and carbon storage analysis.
Description:
Alexandrite lasers are a class of tuneable solid-state lasers capable of emitting light across a wavelength range of approximately 700 to 850 nm. This spectral region is particularly valuable for remote sensing applications, as it encompasses wavelengths that are sensitive to vegetation characteristics—especially the chlorophyll absorption features that are key indicators of plant health. Utilizing this capability within a Light Detection and Ranging (Lidar) system allows for the active probing of vegetation by analysing the light that is backscattered from plant surface.

By quantifying chlorophyll content and analysing the spatial and temporal variations in vegetation cover, researchers can estimate parameters such as carbon storage within biomass and overall plant productivity. These metrics are crucial for understanding the role of terrestrial ecosystems in the global carbon. Insights gained from such measurements can help refine climate models and inform policy decisions regarding climate change mitigation and adaptation strategies.

The proposed activity will focus on the design, development, and validation of an Alexandrite-based Lidar Engineering Model (EM). The EM shall enable precise measurements of backscattered laser signals targeting different types of vegetation and structures. The collected data will help to evaluate the instrument's sensitivity to variations in vegetation parameters and its potential for future spaceborne applications.

The main tasks shall include:
- Conduct a review of scientific and technical requirements for vegetation monitoring using Lidar instruments.
- Design and manufacture of the Engineering Model (EM) of the Alexandrite Lidar system, taking into account performance specifications, optical design, laser source integration, and signal detection components.
- Testing with various vegetation structures, using a range of vegetation targets with differing densities, heights, and species compositions.
- Develop a roadmap outlining the steps necessary for adapting the system for spaceborne operation.

This activity is proposed to be implemented in Bundle 4 - LIDAR developments for enhanced vegetation monitoring techniques, together with the activity GT17-844MM - Detector array for EO Lidar applications.

Deliverables: Engineering Model, Report

Application/Need Date: Future Earth Observation vegetation monitoring missions. TRL7 by 2028. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Lidar Critical Subsystems.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-841MM – Infrared detector for next generation sounder instruments

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 17

Objective:
To hybridise and develop mid wave infrared and long wave infrared (MWIR/LWIR) detectors with high frame rate and high dynamic range.

Description:
Future Earth Observation mission studies have identified the need for high-frame rate, low noise, high dynamic range, 2D array detectors with high efficiency in the MWIR to LWIR waveband to support Fourier transform spectrometer based instrumentation.

In previous developments, specialised readout integrated circuits (ROICs) have been developed targeting these challenging requirements. This activity aims to continue development of the ROICs by hybridising them with suitable photo-sensitive detection layers in the Mid Wave Infrared and Long Wave Infrared wavebands.

The detectors shall be characterised at representative operational temperatures providing a detailed assessment of their electro-optic performance. A radiation test campaign shall also be performed on the detectors, with the aim of de-risking the radiation tolerance of the detector assembly.

This activity encompasses the following tasks:
- Requirements review and design of photosensitive detection layers.
- Manufacturing of detection layers.
- Hybridisation of ROICs into detectors.
- Electro-optic characterisation of detectors at representative operational temperatures.
- Radiation testing of detector assembly.

This activity is proposed to be implemented in Bundle 3 - Enabling high Infrared performances, together with the activity GT17-843MM - Coating development for lens-like samples in the infrared.

Deliverables: Engineering Model, Report

Application/Need Date: Earth Observation infrared sounders. TRL5/6 by 2028. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Optical Detectors.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-842MM – Prismatic joint for planetary robotic manipulators

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 13

Objective:
To develop a prismatic (linear) joint for robotic manipulators compatible with the demanding environments of planetary robotics, capable to support future Moon/Mars missions.

Description:
Development prismatic joints is key for planetary robotic manipulators because they can enable the design of flexible robotic systems that can be easily reconfigured and redeployed based on the specific mission need, using proven and tested building blocks. This will allow shorter development schedules for future flight robotic manipulators developments.

Prismatic joints provide higher precision and stiffness, enabling high repeatability and high-speed operations. This development can aid complex robotic systems in the execution of a wide range of relevant activities for future missions, e.g. insertions/retrieval operations, tool handling (e.g. in a tool rack, to present the right tool to another system or an astronaut), cargo sorting and storage, payload deployments (on the surface or onto other surface assets). High TRL has been achieved for rotary joints, but a similar level for prismatic ones, that need to operate on Moon or Mars, is currently missing in Europe.

The activity will cover the following tasks:
- Definition of the requirements for a prismatic joint that can be integrated in robotic manipulators, including: high force-to-mass ratio, high linear positioning accuracy, high lateral/bending load bearing capabilities, flex harness spooling capability, compatibility with Mars and Moon environmental conditions (thermal and dust), compatibility with mechanical loads generated by launch/landing events.
- Design and manufacturing of an engineering model of the prismatic joint.
- Testing of the prismatic joint engineering model covering at least the following aspects:
  - Thermal vacuum and/or Mars atmosphere compatibility.
  - Motorization losses characterization as a function of temperature.
  - Thermal control system behaviour characterization.
  - Performance characterization in presence of dust and at end of life.
  - Static load and random vibration testing.

Deliverables: Engineering Model, Report

Application/Need Date: Mars and Moon missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-843MM – Coating development for lens-like samples in the infrared

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 16

Objective:
To develop high-performance and durable infrared lenses (coating and substrate) in the midwave, longwave and thermal infrared.

Description:
It is commonly accepted that coating developments are always characterized on standard flat-disk samples, even if the final application of the process is on lens-like curved elements. However, it is known that the curvature of a sample can significantly affect the performance and durability of a coating previously validated on a flat sample.

Building partly on the results of a precursor activity (Advanced coating processes and materials for high performance infrared optics), this activity aims at developing and characterizing IR coatings on lens-like elements, i.e. samples with a curvature (e.g. plano-convex or plano-concave).

The deformation of the lens surface form due to the stress induced by the coating would be modelled and characterized. Aspects of coating delamination, imperfections and degradation would be assessed carefully, accounting for relevant environmental conditions (e.g. vacuum, thermal, radiations).

The activity encompasses the following tasks:
- Design of high-performance coating for lens-like samples in the infrared, modelling the lens deformation due to stress induced by the coating.
- Review and implementation of improved deposition processes tailored to curved samples.
- Manufacturing of the selected coatings on both flat-disks and curved-lenses, for cross-comparison of the performances.
- Characterization of the spectral and optical performances (e.g. wavefront error) of the lens-like samples.
- Characterization of the coating cosmetics and durability in environment for curved samples.
- Assessment of the production time and way forward.

This activity is proposed to be implemented in Bundle 3 - Enabling high Infrared performances, together with the activity GT17-841MM - Infrared detector for next generation sounder instruments.

Deliverables: Breadboard, Report

Application/Need Date: TRL6 by 2030 for meteo and sounding missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-844MM – Detector array for EO lidar applications

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 17

Objective:
To develop and characterise a detector array for earth observation vegetation Lidar applications.
Description:
Earth observation lidar systems have had limited resolution in the across-track direction with a conventional single pixel detector providing ranging information in only one spatial sample. Multi-element arrays offer a solution with each element receiving the light from a separate sampling point in the observed swath. Space-qualified solutions for the 1064nm wavelength targeted by future vegetation lidar payload studies are not yet mature and require specific developments. This activity targets to make available a mature solution suitable for adoption into the designs of future vegetation lidar payload designs.

A custom detector array operating will enable simultaneous detection across tens of across-track points, whilst maintaining the time resolution and low photon count sensitivity required by the application. This activity will assess the focal plane requirements of one or more instrument concept designs to derive the critical specifications to target for the detector array. Focal plane concepts to meet the requirements shall be subject to a trade-off and a detector array shall be designed. Following manufacture and assembly, the performance of the detector array shall be characterised, and its tolerance to the space radiation environment evaluated.

This activity encompasses the following tasks:
- Requirements review involving input from one or more instrument concept designs.
- Detector preliminary and detailed design.
- Detector manufacturing.
- Detector test and characterisation.
- Radiation environment testing.

This activity is proposed to be implemented in Bundle 4 - LIDAR developments for enhanced vegetation monitoring, together with the activity GT17-840MM - Alexandrite based Lidar for the development of vegetation monitoring techniques.

Deliverables: Engineering Model, Report

Application/Need Date: TRL6 by 2029 for Earth Observation Lidar applications **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Optical Detectors.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-845MM – Inert anode development for molten salt electrolysis

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 14

Objective:
To develop an inert anode system which can be used for the production of refractory and high performance alloys using molten salt electrolysis (MSE).

Description:
Classically carbon anodes are used to extract metals using MSE, which form CO2 and CO. Furthermore, carbon anodes lower the FFC process efficiency and metal product quality, as CO2 can dissolve in the CaCl2/CaO electrolyte leading to carbonate cycling. Inert anodes are essential in producing a higher quality product (for instance Titanium ally grades) with high efficiencies and at the same time reducing the carbon footprint of this process.

So far, the application of a wide variety of (noble) metals and costume ceramics failed to produce an anode with the following desired properties: a) CaCl2 salt compatibility at 950°C b) Electrical conductivity c) Oxygen and chlorine side product resistance d) Long service life e) Stability towards thermal cycling f) Stability towards feedstock (and its impurities). This activity aims at identifying and assessing new inert anode materials and manufacturing processes, producing and testing anode systems (engineering models) in the process conditions of metal extraction.

This activity encompasses the following tasks:
- Investigation of new inert anode materials and novel approaches (e.g. coatings).
- Production of anodes shall be subjected towards the conditions of the MSE FFC Cambridge process for the extraction of metals.
- Performance evaluation regarding the above-mentioned desired properties, enabling a more efficient and sustainable MSE process potentially to be used for titanium and refractory materials production.
- Assessment of potential transfer to other MSE processes (e.g. Hall-Heroult).

Deliverables: Breadboard, Engineering Model, Report

Application/Need Date: Science missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Optics, Robotics & Life Sciences

GT17-846MM – Integrated chip based high power C-band optical amplifier

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 17

Objective:
To design, manufacture and test a packaged C-band high power integrated chip based laser diode with more than 2W optical output suitable for free space optical communication applications.

Description:
Optical terminals are finding applications in generic space applications spanning low cost Earth Observation (EO) missions to Science missions, where the possibility to dump large amounts of data to an optical ground station or orbiter is an interesting prospect to open up new EO applications or enable challenging Science missions.

One of the challenges of this technology is to improve the reliability and power efficiency of the optical source. Today the optical source is typically a combination of several components, a Distributed Feedback (DFB) laser, modulator and amplifier, which together provide the high-power laser transmitter.

Some non-European companies have demonstrated that a highly efficient semiconductor optical amplifier can be developed that can emit several Watts of power and be directly modulated at several Gbps using a PIC (photonic integrated circuits) design that combines several elements in the one device, namely a laser section, a pulse shaper and a tapered amplifier section launched into a single mode optical fiber. This activity seeks to develop an European solution and mature an European source for such a device, which could be widely used across missions that make use of optical free space optical communications to provide a source that could emit several Watts of power with a wall plug efficiency exceeding 25%.

This activity encompasses the following tasks:
- Perform a state-of-the-art review of main integrated photonic building blocks (seed laser, modulator, high power amplifier).
- Preliminary design of the concept.
- Design maturation through building block manufacture and testing.
- Detailed design of the final device together with detailed test plan.
- Manufacture of laser diode prototype and test in a relevant environment.

This activity is proposed to be implemented in Bundle 2 - Enabling New space - miniaturisation and serialisation, together with the activities GT17-838MM - Large duolithic mirrors for snap-together telescope assembly and GT17-839MM - Low energy / high pulse repetition frequency based space lidars.

Deliverables: Prototype, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Optical Communications for Space.

2.3.6. Structures, Mechanisms, Materials

Generic Technologies

Structures, Mechanisms & Materials

GT17-847MS – Develop an ultra-fine positioning actuator for adaptive optics

Budget: 550 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 15

Objective:
To develop an ultra-fine positioning actuator for adaptive optics with resolution around 1-10nm and centimetric stoke.
Description:
Precision optical systems operating in space environments increasingly rely on adaptive and deployable optics to maintain optimal performance throughout their mission lifetime. These systems require actuators capable of extremely fine positional control to correct for misalignments, thermal distortions, and dynamic changes in optical geometry. The challenge lies in achieving nanometer-level resolution over millimeter-scale strokes, while ensuring mechanical robustness and reliability under both ground-based handling and in-orbit conditions.

This activity addresses the need for a high-performance actuator that can support ultra-fine positioning functions within optical payloads. The actuator must be compatible with typical deployment and adjustment scenarios, such as segmented mirror alignment, deformable surface control, and precision optical element positioning.

This activity aims to develop, manufacture and test an actuator elegant breadboard model (EBB), suitable for ultra fine positioning functions to cover deployable / adaptive optics in orbit, with the following requirements:
- Stroke: 1 - 10 mm.
- Resolution: ~ 1 – 10 nm.
- Load capability for typical ground operations.

This activity encompasses the following tasks:
- State of the art of existing technology.
- Selection of the most promising actuator technology based on a trade-off addressing key requirement (resolution, stroke, stability, thermal dissipation, magnetic cleanliness, low exported microvibration, life, space environment compatibility, radiation).
- Design, manufacture and test an EBB.
- Design, manufacture and commission GSE for the actuator EBB test campaign.

Deliverables: elegant breadboard model (EBB), reports.

Application/Need Date: Missions requiring adaptive optics in orbit. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Actuators Building Blocks for Mechanisms (2022) – Consistent with activity D12 “Ultra-fine positioning actuator”.

Generic Technologies

Structures, Mechanisms & Materials

GT17-848MS – Deployable support structure with feed horn and metrology functionality

Budget: 900 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - **TD** 20

Objective:
To develop a deployable support structure to place a feed horn and metrology device at 10 m standoff from a large reflector dish, with a diameter of 10 m and to demonstrate stable attachment to the reflector.

Description:
In Europe, there are currently no antennas with large diameters. A deployable antenna with a diameter around 10 m is being developed under the Copernicus Imaging Microwave Radiometer (CIMR) mission. Moreover, deployable solutions beyond 15 - 25 m are not expected in Europe. While some initial in-orbit-assembly and manufacturing (IOA/M) activities focused on mechanical and robotic aspects were done, none have concentrated on the actual antenna from an electromagnetic point of view. This proposal aims to address all critical technologies necessary to realize a large antenna in space, in the order of 10 m diameter, including the feed as an integral part of the antenna and fully RF representative. It will cover RF, mechanical, thermal, materials, structures, and guidance, navigation and control (GNC) aspects. The antenna assembly will serve as a focused use case, incorporating advanced metrology techniques to ensure structural and functional accuracy, such as electrical connectivity, feed design and deployment, and the corresponding radiation pattern. Advanced modelling will also be required to tackle this challenge. Additionally, novel approaches to GNC for such a large structure need to be considered.

This requires the development of a large reflector parabolic dish that can be assembled from parts, in-orbit, by automated systems, and then the assembly to this reflector of a deployable support structure that will lift the antenna feed horn package and electrical connections to 10 metres from the reflector aperture. The design of the reflector will be done out of the scope of this activity. The electronic package will also contain a LIDAR based metrology system that will perform in-orbit metrology of the reflector surface and provide near-real-time feedback of the deployed surface shape and pointing of the antenna.

This activity encompasses the following tasks:
- Design a deployable support structure to provide dimensionally stable and strong attachment whilst carrying the electronic package (feed horn, metrology, harnesses).
- Manufacturing the support structure and integration of electronic modules.
- Mechanical testing in lab conditions followed by deployment of the support structure on its own.
- Demonstration of the attachment strategy to a representative reflector.
- Mechanical testing and metrology of the support structure attached to the reflector.

This activity is proposed to be implemented in Bundle 1 - Enabling in orbit assembly of large antennas, together with the activities GT17-809EC - Guidance, navigation and control system development for in-orbit assembly of very large antennas and GT17-821EF - In-orbit assembly and performance demonstration of a large antenna.

Deliverables: Breadboard, Report

Application/Need Date: Missions requiring in orbit assembly of large antennas. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Structures, Mechanisms & Materials

GT17-849MS – Material accelerator platform for space systems

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 24

Objective:
To develop a multi-scale simulation framework to predict mechanical, high strain-rate, and chemical degradation behaviours of space-relevant materials and integrate with robotic high-throughput coupon fabrication, testing, and inspection for closed-loop model validation and optimization.
Description:
Material Acceleration Platforms (MAPs) are an emerging paradigm in materials science that integrate artificial intelligence (AI), high-throughput experimentation, automation, and advanced simulation to dramatically speed up the discovery, development, and deployment of new materials. The aerospace sector is undergoing a transformative shift in materials innovation through the adoption of Materials Acceleration Platforms (MAPs). These platforms integrate AI, high-throughput experimentation, automation, and advanced simulation to significantly reduce the time required for materials discovery and deployment—from decades to just a few years.

The proposed activity aims to develop a physics-based simulation framework for mesoscale dynamics, integrating multiple numerical techniques (e.g. Kinetic Monte Carlo, Molecular Dynamics, Computational Fluid Dynamics, Finite Element Method, Phase Field, among others) to support AI/ML-accelerated materials discovery. The framework will include:
- A data-exchange and interfacing layer compatible with diverse meshing/ALE formats.
- Workflow examples for key aerospace alloys, including compositional tailoring and property optimization.
- A use case focused on critical alloy classes such as High-Entropy Alloys or CuCrZr-IN718 bonded structures.
- Integration of open-source and proprietary codes to ensure broad accessibility and avoid vendor lock-in, leveraging existing EU-funded software developments.

This activity encompasses the following tasks:
- Develop a core physics-based simulation framework.
- Develop data-exchange and Interfacing layer.
- AI/ML accelerated design of experiment for material discovery and property engineering.
- Creation of materials database for space relevant materials.
- Synthesis of material coupons.

Deliverables: Software, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Structures, Mechanisms & Materials

GT17-850MS – Hybrid manufacturing for space application

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - TD 24

Objective:
To advance hybrid manufacturing strategies that integrate conventional and advanced techniques to enhance the performance, functionality and competitiveness of space hardware.

Description:
Hybrid manufacturing combines subtractive methods (e.g. milling, turning, grinding) with additive processes (e.g. powder bed processes, direct energy deposition). It may also include fabrication/joining steps like welding, bonding or brazing. This approach aims to overcome limitations of individual processes, enabling the production of high-performance, competitive components. Such methodologies are of growing relevance particularly in propulsion, thermal management and structural systems. The key drivers include:
- Geometric complexity: enabling the manufacture of intricate, topology-optimised structures difficult to realise by conventional means alone.
- Material efficiency: by adding material only where needed, hybrid manufacturing reduces waste and optimizes material usage.
- Functional performance: enabling graded material properties, local reinforcements or tailored functionality.
- Cost and lead time: consolidating steps, minimising waste and enabling complex geometries, though it may require specialised planning, equipment and qualification.
- Design flexibility: greater customization and flexibility in design, making it ideal for producing bespoke or low-volume parts.

This activity encompasses the following tasks:
- Survey of relevant techniques, materials and equipment capabilities relevant to a hybrid manufacturing approach.
- Selection of two representative case studies with associated requirements (e.g. one propulsion component, one thermal management part).
- Design and definition of the hybrid manufacturing route for each case study.
- Manufacturing and functional testing of demonstrators. The utilisation of new digital methods for supporting hybrid manufacturing decisions is welcomed. This may include artificial intelligence techniques, such as deep/federated learning techniques, as well as a focus on use-cases of generatively designed parts (which inherently benefit from the hybrid manufacturing techniques), alongside the artificial intelligence and machine learning data-driven learning.
- Economic analysis, including cost comparison with conventional approaches and life-cycle considerations.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Structures, Mechanisms & Materials

GT17-851MS – Development of a passive dust-repellent coating for lunar surface application

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 24

Objective:
To design, manufacture and test a coating to be applied on space-grade materials in order to maintain surfaces clean of lunar dust for exploration missions.

Description:
In lunar environment, dust is one of the main challenges. It adheres to every exposed surface, impairs operations, reduces lifetime and performances. It is important to create surface's compatible coatings to prevent passive dust adhesion. Application could target solar array, radiators, optics and / or fabrics.

This activity shall develop dust-repellent coating on representative surfaces for space applications and demonstrate its functionality for a selected mission scenario.

This activity encompasses the following tasks:
- Propose dust adhesion mitigation strategies achievable by passive coatings.
- Filter down and categorised propositions for each-sub systems.
- Select a mission scenario and use case relevant for lunar exploration.
- Select at least two relevant substrate materials,
- Manufacture the coatings on procured coupons.
- Characterize gain of performance in term of lunar dust adhesion mitigation in relevant environments (in terms of vacuum, temperature, irradiation), considering surface coverage with dust in comparison to a selected reference substrate and possible surface degradation.

Deliverables: Breadboard, Report

Application/Need Date: TRL 5, by 2028. Exploration missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Coatings (2023) – Consistent with Aim H “Protect spacecraft surfaces and materials from degradation in the space environment”.

Generic Technologies

Structures, Mechanisms & Materials

GT17-852MS – Smart sensing technologies for space textiles inspired by terrestrial sectors

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 4 - TD 24

Objective:
To develop e-textile technologies for selected use cases in aerospace applications, inspired by terrestrial sectors, ensuring compatibility with space environment.

Description:
Flexibility and versatility of textile materials have led to numerous applications in automotive industry, incorporating additional functionalities, such as sensing, lighting or heating capabilities. Such e-textiles have led e.g. to increase of comfort and reliability in automotive vehicles. Previous activities focusing on future EVA suit materials have identified several such technologies that could be spun into aerospace applications (e.g. sensory wrapping yarns or textile-embedded tactile sensors). However, the application cases could be much wider, including unmanned spaceflight cases. This justifies the need to explore the field of e-textiles further and to verify such materials for aerospace applications, enabling new designs, mass and cost savings. The e-textiles functionalities could include sensing (e.g. motion, light, oxygen sensing) or other textile-embedded functionalities (e.g. flexible heaters). Development of these e-textiles shall consider both fabric and the integrated sensing/functional part, ensuring compatibility with space environment.

This activity encompasses the following tasks:
- Survey of state of the art to identify e-textiles used in terrestrial sectors, e.g. automotive industry, with the corresponding sensing technologies.
- Selection of mission scenario that would be enhanced via use of e-textiles.
- Definition of the corresponding requirements on material level, considering the foreseen space environment.
- Definition of breadboard demonstrator showing the selected functionality under representative conditions, for instance a sensing textile sheet (both textile material and sensor) for thermal blanket.
- Development of a manufacture and test plan where the identified sensing technologies are applied in space-qualified textiles and the defined requirements are verified by testing.
- Manufacture and testing of breadboard demonstrators.

Deliverables: Samples of e-textiles, Breadboard, Report

Application/Need Date: TRL 4, by 2028. All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Structures, Mechanisms, Materials, Thermal

GT17-853MS – Development of sourced tow-pregs for space applications

Budget: 900 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3/5 - **TD** 24

Objective:
To develop tow-pregs suppliers for use in critical space applications and qualify these materials in a relevant space application.

Description:
Tow-pregs—pre-impregnated fiber tows—are essential for producing lightweight and high-strength structural components used in satellites, launchers, and other spacecraft systems, particularly when employing automated manufacturing techniques such as filament winding (FW) and automated tape placement (ATP). However, the majority of high-performance tow-pregs currently in use are sourced from outside Europe, leading to strategic dependencies.

This activity proposes a thorough evaluation of tow-pregs produced by European suppliers to determine their technical suitability and readiness for space applications. The activity will focus on mechanical characterization, thermal performance, outgassing behaviour, and conformity with ESA standards, aiming to support the development of a robust and independent European supply chain. The selected material system will then be employed to realise a space structure breadboard that will be demonstrated in a space relevant mechanical environment.

The activity encompasses the following tasks:
- Identify benchmark space structures of which a design is already available and suitable for tow-preg application, associated manufacturing methods (e.g., filament winding, automated tape placement) and initial material characterization test plan.
- Conduct a market survey of the European tow-preg supply chain, including material producers and system integrators.
- Select a minimum of two independent suppliers for which the complete supply chain is entirely European and assess their material systems and processing compatibility with the target applications.
- Perform material characterization and testing.
- Adapt the benchmark structure design to the new selected European material system.
- Benchmark results against ESA-qualified, non-European material systems to identify performance gaps and qualification needs.
- Develop a tow-preg materials database and provide recommendations for standardization, further development, and potential qualification paths strength driven.
- Build and test the benchmark structure using the most promising European tow-preg system.

Deliverables: Space benchmark structure and development coupons, Reports

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Composite Materials (2025) – Consistent with activity F01 “Assessment of European sourced towpregs for space application”.

Generic Technologies

Structures, Mechanisms & Materials

GT17-854MS – Generic relubrication device for long lifetime mechanisms

Budget: 550 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 15

Objective:
To design, develop, manufacture and test an engineering model (EM) of a generic relubrication device (simple/reliable/one or multi-shot) for space mechanism applications to periodically re-lubricate with fresh oil the fluid lubricated tribological contact in ball bearings/gear-train/Harmonic Drive, compatible with different oils.
Description:
Most of the space mechanisms with long lifetime requirements in terms of number of operational cycles (inter-satellite link, antenna pointing mechanism, thruster pointing mechanism, reaction wheel) or duration (actuation at very end of life, deep-space exploration) rely on liquid lubrication, usually based on oils or greases, the latter being a mixture of base oil and solid additives. The lifetime and performance of space mechanisms are primarily related to the presence of efficient lubricant and not to the fatigue or life limitation of the surfaces in contact. During operation and over life of the mechanisms, the base oil is subjected to not well predictable phenomenon (like ageing/degradation, evaporation.

The introduction of a generic relubrication device (simple/reliable/one or multi-shot product), able to periodically re-lubricate the tribological contact with fresh oil shall overcome the above limitations and significantly improve the life of tribology contact in space mechanisms. The proposed activity aims to develop a relubrication device concept, easy to integrate into mechanisms architecture, expected to be compatible with most common space oils (PFPE, MAC-based lubricants or minerals oils). The overall approach will allow an in-orbit regular fresh oil refill. This will make more robust the mechanism performances, and drastically increase life capability, at least +200%. Considering that the concept will be designed for being used in ball bearings, gear train or harmonic drive of different sizes, it is expected that large production will reduce the overall cost of the relubrication device.

This activity encompasses the following tasks:
- Requirement review, concept trade-off and concept choice review.
- Design and development of an engineering model (EM) and breadboard design review.
- Testing of the Engineering Model, including compatibility with different space oil and functionally tested in representative ground conditions (ambient pressure), and in-flight representative condition (vacuum, mechanical environment, thermal range).

Deliverables: Engineering Model, Report

Application/Need Date: All missions. TRL 5/6 by 2028/2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Structures, Mechanisms & Materials

GT17-855MS – Development of a nanostepping rotary piezo electric actuator based on walking principle

Budget: 550 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 15

Objective:
To develop and test in relevant environment a nano-stepping rotary piezoelectric actuator based on a walking principle, including the necessary long-life shaft-actuator friction interface.

Description:
The piezo walking technology is interesting for positioning of optical elements (e.g. refocusing mechanisms) in high performance optical instruments. The advantages of the piezo walking technology are small volume, high resolution and unpowered position holding. A competitive technology is inertial piezo actuator. Walking piezo actuator is expected to have larger life, holding and powered torque and produce much lower exported microvibrations. The walking piezoelectric actuators exist in the ground industry but have very limited use in space because vacuum tribology is complex, the resistance to vibrations and shock may require specific design solutions and redundancy is not implemented (failed walking piezo motor would block the shaft).

The aim of this activity is to address the technology gaps and develop a breadboard of a rotary walking piezo actuator and demonstrate its performance in a relevant environment (mechanical vibrations, thermal vacuum Chamber (TVAC)). The control electronics and a position sensor are not in the scope as existing COTS products exist and can be used.

This activity encompasses the following tasks:
- Develop the friction interface for the walking piezo actuator in case of an adaptation of industrial ground product.
- Adaptation of materials and processes to space standards.
- Implement redundancy into the actuator.
- Design a walking piezo breadboard actuator with a COTS feedback sensor and control electronics. Low magnetic moment should be targeted as this is inherently possible for this technology.
- Evaluate scalability of the product in terms of size and delivered torque
- Breadboard manufacturing.
- Breadboard testing, including mechanical vibration and shock, TVAC cycling, exported micro vibrations and life test in vacuum.

Deliverables: Breadboard, Report

Application/Need Date: All mission. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Actuators Building Blocks for Mechanisms.

Generic Technologies

Structures, Mechanisms & Materials

GT17-856MS – Innovative ceramic manufacturing for space applications

Budget: 1200 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 24

Objective:
To develop and optimise additive manufacturing techniques for ceramic components, as well as developing post-processing, improving crack detection and establishing a robust European supply chain.

Description:
Ceramics play a crucial role in spacecraft for their unique properties, namely high thermal stability/resistance, low density and high strength. Ceramics like silicon carbide (SiC) and silicon nitride (Si3N4) are employed in structural components, delivering high stiffness with relatively low mass. They are vital for optical applications for a stable and accurate placement of sensors and mirrors. They are also used for thermal protection systems, microwave and radio-frequency equipment. Recently, additive manufacturing (AM) has further expanded the applications of ceramics in spacecraft, enabling the production of unprecedented complex shapes. Overall, ceramics are becoming indispensable for modern spacecraft.

Nevertheless, ESA recognises the need to enhance several key aspects of the ceramic manufacturing chain to support industrialisation and establish a robust supply. Despite the advancements in AM, specific challenges persist, particularly when attempting to achieve thin walls, improving surface finish or enabling more intricate lattice geometries with high reliability and precision. The feasibility of assembling larger structures from smaller ceramic parts is a way to overcome AM equipment size and process limitations. In parallel, there is a need to improve data for crack analysis and damage tolerance, coupled with the development of viable Non-Destructive Testing (NDT) solutions for crack detection. Downstream operations are critical, including precision machining for mounting interfaces, coatings and polishing. Additionally, exploration of alternatives to SiC and Si3N4 may yield new opportunities. The use of ceramics poses challenges for spacecraft demisability, which can limit their suitability for satellites in large constellations adhering to the zero-debris charter. This constraint must be considered early in the design phase and accounted for in mission architecture.

The aim of this activity is to further develop AM processes for ceramics. It can be expected that this will involve the use of novel materials, achieving higher geometric complexity, improved surface finish, larger part size and/or thinner walls.

This activity encompasses the following tasks:
- Trade-off potential use cases (e.g. parts, AM processes and materials) which would allow further developing ceramic manufacturing technologies for space applications.
- Select the most promising application and develop the corresponding AM processes, progressing towards a representative part prototype. This will likely involve complementary steps such as surface treatments, joining, coating, polishing and NDT.
- Develop numerical simulation tools to support design for manufacturing, for example by predicting deformation and crack formation during fabrication or sintering.
- Manufacture and test the prototype under relevant conditions.
- Define end-of-life strategies.

Deliverables: Breadboard, Prototype part, Report

Application/Need Date: All missions. TRL6 by 2028. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Additive Manufacturing.

2.3.7. Thermal

Generic Technologies

Thermal

GT17-857MT – Cryogenic system for in situ resource utilization liquefaction

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 21

Objective:
To develop sustainable energy storage system based on densified cryogen for In Situ Resource Utilization (ISRU) for propellant production or regenerative fuel cell.

Description:
To enable resource utilization on the Moon, Mars, or other planetary bodies, a substantial amount of stored energy is required. Since transporting energy from Earth is not feasible, it must be generated locally. Once produced, gases must be densified through liquefaction, and boil-off minimized to allow for long-term storage.

A promising solution is the development of a cryogenic system for ISRU liquefaction, an advanced technology designed to support the production of propellants and regenerative fuel cells by leveraging in-situ resources. Operating at extremely low temperatures, this system enables the liquefaction of critical gases such as hydrogen, oxygen, and methane, which are essential for sustainable space missions.

This activity will build upon re-used existing cryogenic technologies, including for example cryocoolers, low-temperature heat exchangers, and cryogen storage, by adapting and developing them to meet the demanding requirements of lunar and Martian ISRU scenarios.

This activity encompasses the following tasks:
- Requirements consolidation.
- Technology trade-off analysis and baseline selection of the cryogen system.
- Design and manufacturing of a breadboard demonstrator.
- Functional testing of the breadboard demonstrator.
- Definition of the way forward for industrialization.

Deliverables: Engineering Model, Report

Application/Need Date: Lunar and Martian ISRU missions. **Mission Classification:** alpha

THAG Roadmap: Cryogenics and Focal Plane Cooling (2024) – Consistent with activity G14 "In situ propellant liquefaction transfer and storage for Moon and Mars".

Generic Technologies

Thermal

GT17-858MT – Continuous sub-Kelvin cooling

Budget: 750 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 21

Objective:
To develop a sub-Kelvin cooler reaching 0.1 K continuously, using a sink temperature of 4 K.

Description:
Future science missions will need continuous sub-Kelvin cooling. The need for continuous sub-Kelvin cooling is a hurdle to the currently available state-of-the-art. The Adiabatic Demagnetisation Refrigerator (ADR) is a type of cryo-refrigerator that can be used to achieve sub-Kelvin temperatures and a functioning breadboard has been developed in previous activities, including the development of the critical cryogenic heat switch that enables the continuous operation.

The aim of this activity is to develop a qualification model of a fully functional continuous ADR, together with associated heat switch.

This activity encompasses the following tasks:
- Review of state of the art, and assessment of needed developments including technology trade-offs.
- High level trade-offs and baseline architecture consolidation.
- Low level trade-offs and technical baseline configuration.
- Interface and system requirements consolidation.
- Detailed design and manufacturing plans.
- Manufacture, assembly and integration of the ADR qualification model.
- Performance and environmental testing of the qualification model.

Deliverables: Qualification Model, Report

Application/Need Date: Future missions including scientific applications. **Mission Classification:** alpha

THAG Roadmap: Cryogenics and Focal Plane Cooling (2024) – Consistent with Aim C “Developing Sub-Kelvin Cryo Coolers”.

Generic Technologies

Thermal

GT17-859MT – Thermal analysis demonstrator using cosimulation

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 21

Objective:
To demonstrate co-simulation of thermal analysis models coming from heterogeneous sources and apply to realistic analysis scenarios, drawing conclusions on engineering effort and accuracy.

Description:
Exchange of thermal analysis models continues to be a major effort in the supply chain. The TDE activity T721-602MT “Efficient Integration of Space Thermal Analysis Models” looked at new approaches to ease model exchange and led to development of a demonstrator using co-simulation. This technology can significantly reduce the effort to exchange models as well as enable use of new techniques (e.g. hybrid surrogate and physics-based models).

In particular, the effort required for model conversion between different tools can be drastically reduced by wrapping the native solver inside a co-simulation container. This can be used for standard thermal model integration at system level, but also other applications such as launcher coupled analysis. The activity builds upon this and focuses on a real work demonstrator, combining multiple subassembly models and running representative load cases. Tentatively the activity would focus on the new Functional Mock-Up Interface (FMI) version 3 for the underlying exchange and seek to combine models coming from several sources (i.e. different tools and companies).

This activity encompasses the following tasks:
- Requirements consolidation and selection of demonstration models.
- Design of co-simulation architecture.
- Development of the demonstrator.
- Testing of the demonstrator and assessment of results.

Deliverables: Co-simulation demonstration models, Software, Reports

Application/Need Date: TRL 5 by 2028 for all missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

2.3.8. E2E Systems

Generic Technologies

End-to-End Systems

GT17-860SE – End-to-end demonstrator radar for space object detection

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop and validate an end-to-end demonstrator of a multistatic radar system for space object detection and tracking using cooperative and non-cooperative signals from diverse satellites. The activity evaluates the integration within NewSpace LEO constellations to support advanced surveillance, monitoring, and situational awareness services.

Description:
The activity focuses on the development and validation of a complete end-to-end testbed demonstrator for multistatic radar object detection. The system shall operate by receiving and processing reflected signals from multiple satellites, regardless of whether they are designed for cooperative use (e.g. hosted beacons or known signal providers) or non-cooperative (e.g. opportunistic use of signals transmitted for other purposes). The aim is to build a prototype capable of operating in representative environments, addressing the practical implementation of hardware and software elements, as well as system integration within NewSpace LEO constellations.

The system designed in this activity focuses on detecting, localizing, and tracking space objects through the reception and processing of reflected signals. The activity targets to validate detection performance in controlled settings, assess spatial coverage and scalability, and propose strategies for real-world deployment. A key objective is to enable seamless integration into existing or planned LEO constellations via flexible architectures, such as hosted payloads or onboard processing modules. The use of a multilayer sensing approach will also be evaluated. The results of this activity aims to contribute to space capabilities in Space Situational Awareness (SSA) and Space Traffic Management (STM), Resilience and Security of Space Assets.

The system developed in this activity shall be designed to detect and correlate both direct and reflected signals, supported by the development of advanced algorithms for object detection, tracking, and clutter suppression.

The activity encompasses the following tasks:
- Provide a comprehensive state-of-the-art and feasibility baseline to drive the design of the radar demonstrator architecture.
- Provide a complete end-to-end system architecture specification and a technical reference model for demonstrator implementation.
- Develop a modular, end-to-end software simulator that models the transmission processing chain, signal-object interaction, and receiver-side processing.
- Develop the functional breadboard implementations of both the transmitter and receiver chains.
- Execute a controlled environment validation of the breadboards and provide experimental validation results.

Deliverables: Breadboard, Report

Application/Need Date: Space object detection in LEO constellations **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-861SE – End-to-end simulation and validation of quantum-enhanced localisation systems for safety critical functions

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 8

Objective:
To design, simulate, and validate an end-to-end localization system architecture that integrates quantum-enhanced sensor for improved integrity and reliability used for safety-critical functions. The activity targets to develop system-level tools and methodologies for performance assessment and validation of the entire navigation chain, from signal reception to final localisation output, enabling the evaluation of quantum sensor integration benefits, and ensuring compliance with integrity and safety requirements.

Description:
This activity assesses the feasibility, performance, and system-level impact of integrating quantum-enhanced sensors into the localization systems, with a specific focus on meeting integrity assurance requirements of safety-critical functions. The activity focuses on operational contexts where conventional navigation sensors, experience performance degradation, signal loss, or interference, which are common challenges in terrestrial environments. In response to these limitations, the activity defines integrity requirements at subsystem-level, identify critical failure modes, and model relevant threat scenarios. Emphasis shall be placed on the integration of quantum-enhanced sensors (e.g., ultra-stable accelerometers and gyroscopes), analyzing how their superior stability and drift-free characteristics can enhance redundancy, ensure continuity of service, and strengthen fault detection and integrity monitoring mechanisms. This approach aims to support the development of resilient hybrid architectures capable of delivering reliable navigation performance even under degraded or denied sensor conditions.

The final output of the activity is a system-level validation and simulation testbed, supporting both offline and real-time performance assessment. This activity shall evaluate hybrid localization systems in a modular way, assess the added value of quantum sensors under nominal and degraded operational conditions, and de-risk future deployment in regulated transport environments.

The activity encompasses the following tasks:
- Development of a functional architecture combining quantum sensor data with conventional localization technologies within a multi-sensor fusion framework.
- Modelling and simulation of error propagation and integrity threats across the full navigation chain.
- Design and implementation of integrity monitoring and fault detection mechanisms.
- Development of an end-to-end system testbed tool capable of simulating realistic scenarios, injecting faults, and validating system performance against integrity and reliability requirements.

Deliverables: Breadboard, Report

Application/Need Date: Safety-critical localization functions; enabler for drift-free inertial navigation. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-862SE – End-to-end joint satellite communications and radio frequency sensing system demonstrator

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop a radio frequency system demonstrator for joint satellite communication and sensing purposes in representative system scenarios and for a commonly selected frequency range. The aim is to explore opportunistic sensing with satellite communication signals and enhance the communication sub-systems benefiting both the communication and the sensing applications.

Description:
The forecast deficit of available spectrum on one side, and possible reusability of radio frequency infrastructure on the other side, have made the coexistence of satellite communications and radio frequency sensing systems an appealing value proposition. Several emerging application scenarios such as space situational awareness (SSA), maritime surveillance, autonomous shipping, air traffic control of UAVs as well as in-land geo-localizations and machine-to-machine communications in remote areas are examples of possible joint satellite communications and sensing applications. Current research results indicate that advanced signal processing techniques in the air interface design would allow coexistence of radio detection and ranging to share spectrum with satellite communications systems.

As the starting point, the state-of-the art solutions for satellite communications and space-borne RADAR remote sensing along with the frequency allocation with potential use for joint satellite communications and sensing will be short listed. Regulatory aspects on the spectrum will also be examined. Synergies between existing space-borne remote sensing and satellite communications system scenarios will be identified to consolidate system requirements for joint radio frequency air interface development.

The following tasks are foreseen in this activity:
- Consolidate relevant use cases for joint space-based communications and sensing and derive corresponding system and payload requirements.
- Establish key sources of error parameter estimation and corresponding error budgets along with detail trade-offs.
- Join communication and sensing system architecture, definition and trade-offs according to selected minimum number of two use cases.
- Define concepts of operation for joint space-based communication and sensing payloads.
- Design of corresponding satellite payloads and assess the complexity, according to selected use cases and scenarios, with respect to state-of-the art solutions.
- Carry out implementation and verification of the joint communication and sensing solution in an end-to-end testbed.
- Assess the compliance with the key functional and performance requirements of the design by means of a suitable test campaign and demonstrate compliance to the technical requirements.

Deliverables: Breadboard, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-863SE – End-to-end space system performance monitoring building blocks

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop building blocks for an end-to-end monitoring function to predict performance, detect faults, and ensure data integrity of a generic Space System. The monitoring should be based on internal system data and external user observations.

Description:
Performance monitoring is a critical function for the reliability of space applications. As systems become more complex, there is a growing need for real-time performance monitoring and prediction. Traditional performance models rely on predefined models and post anomaly analysis, which could be limiting. There is limited capability today to integrate both internal system data (e.g. telemetry, onboard logs) and external user observations for continuous system performance assessment. This activity aims to develop a modular, autonomous monitoring that enables real-time performance monitoring and prediction. By fusing external user observation with internal system data (if available), the platform will generate realistic, data-driven models. These models will allow real-time performance prediction, anomaly detection and classification. The inclusion of continuous data and anomalies will allow to learn and adapt autonomously. Based on this, the main modules include:
- Real time performance prediction module - Generation of realistic data driven models (e.g. based on telemetry and environmental data); Use of data driven models to predict performance; Forecast performance degradation.
- Fault detection and autonomous anomaly identification - including autonomous ability to flag unexpected behaviour, categorization of detected anomalies and mitigation mechanism based on fault detection.
- End-to-end data integrity and verification - ensuring all collected data, including telemetry are verified and crosschecked multiple data sources.
- Adaptive digital twin of the end-to-end system - based on real telemetry and environmental data from operational space assets. Includes ability to update user model based on digital twin and anomaly injection and resilience testing (e.g autonomously generate thousands of test scenarios covering failure modes such as GNSS spoofing, telecom interference, propagation effects, and etc.).
- Monitoring station - allowing on site performance monitoring and enabling interactive dashboards to investigate anomalies.

This activity encompasses the following tasks:
- Definition of preliminary design of the architecture and the detailed requirements for the full system considering e.g. Navigation, Earth Observation among other use cases, and full design tailored to specific chosen application.
- Development of real-time performance prediction module development with implementation of algorithms for real-time performance estimation and degradation forecasting.
- Development of data integrity and verification module including crosschecking algorithms and alerting system.
- Development of fault detection and anomaly identification module including autonomous fault detection mechanisms (threshold, statistical, ML-based).
- Development of monitoring station including interactive tools for anomaly investigation and performance visualization.
- Development of adaptive digital twin module with built initial digital twin models reflecting system behaviour.
-System integration of all modules into a monitoring platform, after individual modules validation.
- End-to-end testing using real and simulated data and evaluation of system performance against KPI and tunning.

Deliverables: Breadboard, Report, Software
Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Big Data from Space (2023) – Consistent with activity AIM C04 “Cloud-based Data Platform for GNSS Anomalies Detection and Mitigation based on Machine learning algorithms”.

Generic Technologies

End-to-End Systems

GT17-864SE – Interoperable signal-in-space interfaces simulator tools for the solar system internet navigation systems

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop an end-to-end (E2E) RF simulator tools for the RF signal-in-space interfaces between the building blocks of the solar system Internet navigation systems.

Description:
The solar system internet (SSI) navigation and communications systems by 2040 are expected to incrementally expand the current and planned planetary navigation systems and communication networks into the solar system in a secure and interoperable way. This system-of-systems will build on, and further complement, the existing ground and space infrastructure, and the assets to be developed under Moonlight, MARCONI, ASSIGN, and other institutional and commercial infrastructure.

The objective of this activity is to design and develop an E2E RF simulator tools for all the RF signal-in-space interfaces between the different building blocks of the SSI positioning, navigation and Timing (PNT) systems, supporting the development of the future interoperable SSI PNT system-of-systems, composed by both existing/planned and new ground and space infrastructure. This will enable to test the compatibility and interoperability between the SSI PNT systems providing navigation services and the deep space PNT user terminals navigating across the Solar System exploiting the different systems (Moonlight, MARCONI, new SSI PNT nodes, etc.) and derive the user performance achievable in different system and user scenarios. A flexible software define radio (SDR) implementation is targeted to enable it is enhanced with additional features / concepts as the multiple technologies and system concepts evolve. The activity is divided in two phases, phase 1 will focus on the design of the overall E2E RF simulator tools, including the simulation of the RF signal-in-space from existing/planned and new ground and space infrastructure, as well as a test deep space user receiver platform. A first preliminary breadboard of the E2E RF simulator tools will be implemented, including full capabilities for interfaces of existing/planed infrastructure, and limited capabilities of the new SSI PNT nodes and the deep space user receiver platform. The phase 2 of the activity will focus on the full breadboarding of the E2E RF simulators tools, and the complete verification of compatibility, interoperability and user performance in relevant environments across the Solar System for different user and system scenarios.

This activity encompasses the following tasks:
Phase 1 (800 k€, 12 months):
- Define and design the RF simulator tools considering the SSI PNT system building blocks defined above.
- Implement a preliminary breadboard, including full capabilities for interfaces of existing/planed infrastructure, and limited capabilities of the new SSI PNT nodes and the deep space user receiver platform.
- Perform preliminary compatibility, interoperability and performance tests in a limited number of relevant conditions.

Phase 2 (1200 k€, 12 months):
- Implement the final breadboard, including full capabilities for the new SSI PNT nodes and the deep space user receiver platform.
- Perform the complete verification of compatibility, interoperability and user performance in relevant environments across the solar system for different user and system scenarios.

Deliverables: Breadboard, Report, Software

Application/Need Date: Solar System Internet navigation systems **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-865SE – RF constellation simulator for sensing systems

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To design and develop an end-to-end RF constellation simulator (RFCS) for future sensing systems, covering synthetic aperture radar (SAR) and ground moving target indication (GMTI) satellite configurations and technologies, with focus on the RF signal-in-space (SIS) interfaces both in transmission and reception (sensing) for different operational modes and representative scenarios (including interference, jamming and spoofing).
Description:
Satellite sensing systems based on synthetic aperture radar (SAR) are unique in their imaging capabilities, since they provide high-resolution images independently from daylight, cloud coverage and weather conditions, and can be exploited to detect, monitor and geolocate ground moving targets. The availability of timely geospatial information with a global coverage is becoming very important to react quickly to crisis and threats anywhere on Earth. This will require to go from the current typical small constellations of two or few satellites (with relatively large revisit times), to large constellation of satellites with very low revisit times (down to few hours or even minutes), enabling from quick disaster monitoring to reconnaissance and security related tasks. The end-to-end RFCS is key to support the consolidation of future resilience systems operating in different constellation configurations and bands. Also, a flexible software define radio (SDR) implementation is targeted to enable the emulation and assessment of additional features / concepts as the multiple technologies and system concepts evolve. The system developed in this activity covers features such as configurable constellations, multi-layer and multi-band configurations, single and formation-flying satellite configurations enabling advanced applications (e.g., moving target monitoring and geolocation), support standard operation modes: stripmap, scanSAR, spotlight, support advanced modes and architectures relying on digital beamforming, multiple baselines, MIMO-SAR, etc.

The activity is divided into two phases. In phase 1, the activity addresses the design of the overall E2E RFCS, and the implementation and test of a preliminary breadboard partially covering the overall system features, including standard SAR system configurations and operational modes, and a limited number of constellation configurations. Upon successful completion of the Phase 1, the activity will address full breadboarding of the E2E RFCS, expanding with the complete SAR and GMTI system configurations and operational modes, including formation-flying configurations and advanced architectures/modes, and the full verification and performance assessment in relevant environments for different constellation configurations.

The activity encompasses the following tasks:
Phase 1 (800 k€, 12 months):
- Define and design the end-to-end RFCS covering the features described above.
- Develop a preliminary breadboard, including standard SAR system configurations and technologies, and a limited number of constellation configurations.
- Perform preliminary verification and performance tests in a limited number of relevant conditions.
Phase 2 (1200 k€, 12 months):
- Develop the final breadboard, including full capabilities for the resilient sensing system including advanced SAR and GMTI features.
- Perform the complete verification and user performance in relevant environments.

Deliverables: Breadboard (preliminary breadboard for Phase 1, and final breadboard for Phase 2), Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-866SE – Technology building blocks for an end-to-end space system security verification

Budget: 2500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 8

Objective:
To identify and develop the technology building blocks needed to implement a capability suitable for performing an end-to-end security verification of space systems.

Description:
To improve the security level of future space missions, on top of other capabilities, is essential for the industry to have the means to verify the security of their system(s) before delivering them for acceptance or operation. While for some parts of a space system a security verification can rely on already existing available tools, for other, new building blocks need to be developed. These building blocks must be tailored to fit the needs of typical subsystems and components used in space systems (and especially on space segment). These building blocks can be used to develop their own capability to perform security verification of space systems, offering it as a service to space systems designers/implementers.

This activity aims at filling this gap, by developing the building blocks needed to implement their own security verification capability for space systems. Such building blocks can be tools, software and hardware, needed to assess in a lab environment the security of space platforms (on-board computers, avionics busses, on board software, etc.), communication links (from RF layer and above, following CCSDS protocol stack implementation), and ground segment aspects that potentially cannot be addressed by commercial-off-the-shelf or open source software available tools.

This activity is split into two phases and encompasses the following tasks:

Phase 1 (500 k€, 8 months):
- Following a security risk analysis and based on a security reference architecture, identification of the space subsystems and components that need to undergo security verification, and assessment of the currently available tools that can be used (including a trade-off analysis).
- Design an end-to-end security verification capability and definition of its requirements, identifying the building blocks that needs to be developed.
- Design a testbed suitable to be used for the testing of the security verification building blocks.
- Proof-of-concept development of selected building blocks, and their demonstration using suitable simulators, emulators or other implementations.

Phase 2 (2000 k€, 16 months):
- Develop the selected security verification building blocks.
- Develop the testbed that will be used for testing the building blocks to be developed.
- Test, verify and demonstrate the implemented building blocks at the testbed.

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

End-to-End Systems

GT17-867SE – End-to-end testbed for robust system architecture

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 26

Objective:
To develop an end-to-end testbed to demonstrate resilient, secure, flexible system architecture, capable of supporting various types of missions.

Description:
In recent years, the space sector has been undergoing a profound transformation marked by two converging trends, firstly, modern space systems are no longer isolated, monolithic satellites. Instead, we are witnessing proliferation of large constellations; integration of heterogeneous assets (e.g., space/ ground/ airborne) into unified architectures; and advanced payloads with onboard processing, including Cybersecurity and resilience. Secondly, the stakeholders from institutional users to commercial customers are demanding seamless, uninterrupted service delivery; high levels of autonomy, with self-diagnosis/healing; rapid deployment and scalability; and user-centric operations, with hidden complexity behind intuitive interfaces. These expectations are pushing boundaries, necessitating a shift towards intelligent, adaptive, and service-oriented architectures, beyond traditional satellite-centric models, based on modularity, abstraction, and composability.

The activity proposes to develop a testbed demonstrating architecture, critical functions and building blocks with selected use cases. The approach applied in this activity is based on properties such as logical abstraction and decoupling with separation of concerns across layers (i.e. mission logic, physical, application layers); composable and interoperable modules (i.e. satellite, ground node, data processor) that are designed as a self-contained, self-defendant, interoperable components; autonomy and intelligence by design is embedded at multiple levels (i.e. from onboard satellites to global orchestration of constellations); service-centric operations with the system being designed to deliver end-user services, not just data or telemetry; flexible and reconfigurable platforms with federation and interoperability.

This activity encompasses the following tasks:
- Requirements consolidation and use case definition.
- System/Ground/Space architecture and detailed design, including functional and security architecture.
- Design of data management and transport layers and networking and routing protocols and concepts.
- Prototype architecture through core building blocks emulation and implementation (based on core functions):
  - Emulation of onboard avionics for multiplicity of instances including constellation emulation.
  - Deploy software functions including resource management (on-ground vs. in-space) and associated flexibility, including task management and load balancing between resources; software stacks and application programming interface (API).
  - Operating system and hardware.
  - Define protocols and interconnections including security protocols to ensure resilience and robustness and define algorithms for on-board and on-ground processing.
- End-to-end test and demonstration of the proposed architecture.

Deliverables: Breadboard, Report

Application/Need Date: Large constellations management; Secure-by-design; **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Functional Verification and Mission Operations Systems.

2.3.9. Future Engineering

Generic Technologies

Future Engineering

GT17-868SF – Maturation of extended reality solutions for space environments

Budget: 900 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 7 - **TD** 8

Objective:
To develop solutions to overcome the limitations of commercial off-the-shelf Extended Reality headsets for their usage in Space environments.

Description:
In recent years, Extended Reality (XR) technology has shown great potential for various applications on-orbit, including astronaut training, science, maintenance and operations.

Some Commercial Off-The-Shelf (COTS) XR devices have recently introduced different experimental “Travel Modes” that were tested during a Parabolic Flight Campaign in 2025. Results showed robust localization and tracking when relying on visual cues only (no inertial sensors were used). As a result, a Meta Quest 3 headset is planned to be delivered to the ISS in 2026 for further testing. However, some other challenges still remain, like the lack of a non-invasive solution integrated in the visor of spacesuits, environmental challenges (e.g. radiation hardening, thermal management, power and battery constraints, dust and particle contamination, device durability, etc.), limited bandwidths, weight and balance distribution affected by lower-zero-G conditions, psychological effects of using XR in Space, among others.

The present activity must address the remaining challenges by proposing XR solutions with higher maturity for (1) Intra-Extravehicular Activities (IVA/EVA) in microgravity (e.g. Launch & Entry, International Space Station) and (2) Moon / Mars environments. These XR solutions shall propose non-invasive projection systems, adapted in the visor of spacesuits to allow Mixed Reality interactions. This technology could support astronaut operations in Space by providing navigation assistance, scientific site identification, procedure guidance, equipment troubleshooting, teleoperation of robotic devices, immersive training for countermeasure, virtual mapping, data visualization or enhanced collaboration between astronauts and ground personnel, among others.

The following tasks are foreseen in this activity:
- Define specifications and guidelines for XR hardware used in space and extraterrestrial environments and develop strategies to ensure their technological maturity and readiness.
- Create a non-invasive Mixed Reality solution integrated in the visor of spacesuits.
- Create a roadmap to address the upcoming challenges, environments and use cases of using XR in Space environments.

Deliverables: Flight Model, Report, Software

Application/Need Date: Intra-Extravehicular Activities (IVA/EVA), Moon / Mars environments. TRL 7 by 2028
Mission Classification: alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Future Engineering

GT17-869SF – Facility-less setup for enhanced virtual presence

Budget: 900 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 8

Objective:
To develop a self-contained system using Extended Reality and interactive devices to allow space personnel to train and operate remotely and safely through immersive digital environments.
Description:
Virtual Presence refers to the sense of realism or immersion when interacting with a digital environment without being physically present. Human-machine interfaces and interaction technologies have evolved to evoke this feeling in users by leveraging familiar and intuitive cues. Virtual Presence is particularly advantageous in applications where interactions or complex visualizations are required from remote and safe locations. Virtual Presence can be used generically across a wide range of space use cases because it enables immersive interaction with complex systems or environments, making it valuable not only for astronaut training but also for robotic operations, telemaintenance, mission planning, payload integration, and expert support—any scenario where safe, intuitive, and effective human involvement is needed without physical presence.

The utilization of Extended Reality (XR) technologies in such space applications has attracted significant attention for their potential to enhance user interaction, improving usability, and enabling more natural interactions. However, as space exploration and missions become increasingly complex, the need for higher levels of immersion and efficient human-computer interaction has grown.

Virtual and Augmented Reality headsets and glasses are maturing with commercial off-the-shelf solutions, usually offering visual, aural and vestibular feedback. However, such stimuli are often insufficient for achieving higher levels of Virtual Presence in areas like training, where kinesthetics or force feedback are also crucial to feel object shapes, inertias or outer-Earth gravities. Therefore, the combination of XR technologies with other interaction devices like motion platforms, exoskeletons, haptic, tactile or neural devices, could potentially offer a more immersive, multimodal, scalable and embodied training experience.

The following tasks are foreseen in this activity:
- Define specifications and guidelines for XR interaction devices in space applications and outline the development of a multimodal framework to ensure their technological maturity and readiness.
- Design a self-contained, immersive hardware setup for training and teleoperation, tailored to specific space use cases, integrating XR technologies and interaction devices.
- Integrate the hardware solutions with a multimodal software framework capable of hosting multiple users while monitoring and orchestrating the XR and interaction devices.
- Collect and analyse learning analytics to develop custom training or teleoperation programs and provide personalized feedback to users.

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions. TRL6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

Future Engineering

GT17-870SF – SysML v2 for space systems engineering projects

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 8

Objective:
To address current limitations in model-based systems engineering practices by applying SysML v2 in real projects, leveraging its advanced capabilities to overcome existing inefficiencies and complexity, and paving the way for broader adoption through practical, value-driven use cases.
Description:
Designing and operating space systems—like satellites, spacecraft, and ground control infrastructure—is a highly complex process. It involves multiple teams, tools, and phases, from early concept design to mission operations and disposal. One of the biggest challenges is ensuring that all these elements work together smoothly across the entire system lifecycle. Traditionally, engineers use different tools and formats to model systems, which can lead to interoperability issues, duplication of effort, and communication gaps between teams. These problems slow down development, increase costs, and make it harder to adapt systems to new mission needs. The space industry needs a more streamlined, integrated, and flexible approach to systems engineering—one that supports collaboration across disciplines and phases, reduces manual work, and improves consistency.

The newly released SysML v2 standard offers promising features to address these needs, such as:
- A textual notation that makes models easier to read, write, and version control.
- A standard API that allows tools to interact with models programmatically.
- Support for Product Line Engineering (PLE), enabling reuse and customization of system components across different missions.

However, to truly benefit from these features, they must be applied and tested in real-world space projects. This activity will focus on applying SysML v2 in actual space projects—either one comprehensive project or several smaller ones—to solve practical engineering challenges.

This activity encompasses the following tasks:
- Identify and formalize mission objectives, constraints, and stakeholder needs within SysML v2, ensuring seamless traceability across design phases.
- Define system components, interfaces, and interactions using SysML v2 constructs to create a clear, scalable architecture model.
- Represent dynamic spacecraft behaviour, ensuring consistency in operational scenarios.
- Utilize SysML v2 enhanced analytical capabilities to validate mission-critical parameters such as thermal control, power distribution, and communication protocols.
- Link SysML v2 models with simulation environments (e.g., MATLAB, Simulink, or CAD tools) to verify system performance in real-world conditions.
- Define test cases, verification procedures, and compliance checks within SysML v2, ensuring alignment with aerospace standards (such as ECSS).
- Establish a shared model repository for interdisciplinary collaboration, enabling continuous model updates and version control through SysML v2 APIs.

Deliverables: Report, Software

Application/Need Date: All missions. TRL 6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM H01 “Proof of concept in project of SysML v2 use”.

Generic Technologies

Future Engineering

GT17-871SF – Open end-to-end engineering portal

Budget: 800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 7 - **TD** 8

Objective:
To develop a digital engineering capability, accessible to SMEs with low total cost of ownership, that can be applied along the complete space system engineering lifecycle.
Description:
In space missions, many different engineering teams work together—each using their own specialized tools and languages to design, analyse, and operate complex systems. This often leads to fragmented information, communication barriers, and inefficiencies when trying to share or reuse data across teams and project phases. To address this, the space industry has been exploring the idea of a central digital engineering portal—a kind of “hub” where models and data from different tools and domains can come together in a unified, accessible way. There is a growing need for a flexible, user-friendly platform that can:
- Connect different engineering tools and allow them to work together.
- Translate between different modelling languages and data formats, so teams don’t have to manually rework information.
- Make complex models accessible to non-experts, such as operations teams or small companies (SMEs), who may not have deep Model-Based Systems Engineering (MBSE) expertise.
- Lower the barrier to entry for digital engineering, especially for smaller stakeholders who can’t afford to build custom solutions. Without such a platform, digital engineering remains fragmented and difficult to scale across the full space system lifecycle.

This activity will focus on developing a configurable, end-to-end MBSE service, not just a standalone tool. The solution will build on the open-source Model-Based Engineering Hub and related initiatives, and will include:
- Interoperability with domain-specific tools, allowing them to plug into the portal.
- Meta-model mappings using the Space System Ontology, which helps translate and align data between different tools and engineering domains.
- Lightweight editing capabilities that allow users to view and modify models without needing to understand the underlying technical language (e.g., SysML v2).
- Multi-language support, so users can interact with models in the format or language most relevant to their role or expertise.
- A low-barrier, shared platform that supports collaboration across different organizations, making digital engineering more accessible and scalable.

The goal is to enable seamless collaboration and data exchange across the entire space system lifecycle, while making digital engineering tools easier to adopt and use for all stakeholders.

This activity encompasses the following tasks:
- Identification of main user requirements and analysis of the state-of-the-art for critical technologies.
- Design of the end-to-end capability with identification of the components to be developed and reused, and interfaces.
- Development of specific software components, and integration into the full system.
- Validation of the end-to-end system with end-users.

Deliverables: Report, Software

Application/Need Date: All missions. TRL 7 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM G10 “Open End-to-End Engineering Portal (E3P)”

Generic Technologies

Future Engineering

GT17-872SF – System model segregation, integration and exchange

Budget: 750 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 8

Objective:
To support industrialization by establishing mechanisms to maintain segregation of model information across the entire supply chain and differing security levels.
Description:
In space missions, many different organizations work together to design and build complex systems. These organizations often need to share engineering models and data, but not all information can or should be shared freely. Some data may be classified, export-controlled, or only accessible to people with a specific “Need to Know”. At the same time, engineering teams use a variety of tools (like SysML v2, Capella, Cameo, and others) to build and manage these models. Ensuring that sensitive information is properly protected—while still allowing collaboration—is a growing challenge.

There is a clear need for a secure and flexible way to manage access to engineering models across the entire supply chain. This includes:
- Controlling who can see or edit specific parts of a model, based on security levels or contractual boundaries.
- Merging model updates from different teams or tools, while respecting access restrictions.
- Supporting both online and offline collaboration, depending on the security requirements.
- Ensuring compatibility with current and future tools, by using open standards like SysML v2.

Without these capabilities, organizations risk either exposing sensitive data or creating bottlenecks that slow down collaboration and innovation.

This activity will focus on developing a framework and supporting tools to manage secure collaboration in model-based engineering environments. Specifically, it will:
- Define access control mechanisms that allow different users or organizations to work on shared models while keeping sensitive parts protected.
- Develop strategies for merging models from different sources, both online (in real-time) and offline (when working in isolated environments), depending on the security context.
- Create plugins and configuration guidelines for widely used engineering tools (e.g., SysML v2, Capella, Cameo, Enterprise Architect, COMET, Genesys, etc.) to support these mechanisms.
- Use open standards like the SysML v2 API wherever possible, to ensure long-term compatibility and ease of integration with future tools.

The result will be a secure, standards-based approach that enables ESA and its partners to collaborate effectively across the supply chain—without compromising on security or efficiency.

This activity encompasses the following tasks:
- Understand user needs: the needs from different stakeholder shall be considered as in a real project, where different supplier levels are involved. Constraints such as Intellectual Property Rights, security levels (export control) shall be considered.
- Develop a solution, by re-using existing components and developing others.
- Validate the solution, by involving different stakeholders and their data, validate that their needs are met and the data is shared according to the project constraints.

Deliverables: Report, Software

Application/Need Date: All missions. TRL 8 by 2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM I04 “Secure system model segregation, integration and exchange”.

Generic Technologies

Future Engineering

GT17-873SF – Towards modular re-usable system design architectures

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 8

Objective:
To develop modular and reusable systems engineering models, prepare for industrialization and shorten schedule and cost of engineering activities.
Description:
Designing and building space missions—like satellites, spacecraft, and ground control systems—is a complex and time-consuming process. However, many of these systems share common features. For example, different spacecraft might use similar power systems, communication modules, or software components. This repetition opens the door to a smarter way of working: instead of starting from scratch each time, we can reuse proven designs and models.

This is where Model-Based Systems Engineering (MBSE) comes in. MBSE uses digital models to design and manage complex systems. If we can create modular, reusable building blocks within MBSE, we can speed up the development of future missions and reduce errors. Currently, even though many space systems are similar, engineers often re-do the same work for each new mission. This leads to inefficiencies, higher costs, and longer timelines. There's a clear need for reusable models and architectures that can be adapted for different missions, standardized processes for how these models are created, managed, and reused, and tools and workflows that support this reuse in a structured and reliable way.

The objective of this activity is to leverage existing technologies (see list below) to define modular architectures that can be reused for space missions with commonalities and define the governance model for this framework. This activity will investigate technologies like:
- Product Line Management: Managing a family of related systems with shared components.
- Variant Management: Handling different versions or configurations of a system.
- SysMLv2 Reference Libraries: Using the latest modelling language standards to store and reuse system elements.
- Specialization Patterns: Techniques to adapt general models to specific mission needs.

This activity encompasses the following tasks:
- Analyse commonalities across different space missions to identify reusable components.
- Define modular architectures that can be reused and adapted for various missions.
- Develop workflows for how these modules can be instantiated (i.e., turned into mission-specific models).
- Investigate governance models—rules and processes for managing and updating reusable modules.
- Demonstrate feasibility by applying the approach to example missions.
- Evaluate the benefits in terms of time saved, consistency, and quality improvements.

Deliverables: Report, Software

Application/Need Date: All missions. TRL 8 by 2030 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM G13 “Instantiable modular system models”.

Generic Technologies

Future Engineering

GT17-874SF – Space system ontology development for engineering domains

Budget: 1200 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 8

Objective:
To develop, map and integrate the Space System Ontology (SSO) model as identified in the OSMoSE (Overall Semantic Modelling for Space Systems Engineering) initiative for different engineering domains with the objective of enabling semantic interoperability and digitalizing the engineering exchanges for space missions throughout the project lifecycle.
Description:
Since 2020 the OSMoSE community is targeting semantic interoperability between stakeholders and tools in support of digital co-engineering. These efforts have resulted in the space system ontology, a formal model defining the semantics of the concepts used in space engineering throughout the project lifecycle. This model defines both the vocabulary, the relationships and the constraints between any space engineering concept of interest. Currently the SSO contains all the concepts as used in the Model Based System Engineering (MBSE) domain. However also other domains have already been modelled, like AIV, OPS, thermal, QA/PA and even digital procurement. These so-called universe of discourses are currently standalone models, not yet fully integrated into the formal SSO.

This activity combines a set of efforts for the incremental development of the SSO as identified in the OSMoSE initiative and highlighted and supported by THAG dossiers on MBSE, System Modelling & Simulation Tools and Functional Verification & Mission Operations Systems. Additional engineering domains' specific ontology elements are added to the existing skeleton. Amongst others: power management, thermal control, LEOP, FDIR, security, RF link, payload, system management, AOCS, GNC, mass memory management, etc.). In addition, this activity foresees in the integration of the already produced universe of discourse ontology models on operation, on-board communication, digital procurement, quality assurance, AIV into the existing space system ontology.

This activity encompasses the following tasks:
- Semantic modelling of additional engineering domains using the object role modelling language and considering the semantic understanding of stakeholders and standards and coverage by respective digital tools. This requires domain expert knowledge.
- Mapping and integration of newly created engineering domain models into the existing space system ontology. This requires involvement of the design authority.
- Mapping and integration of the currently existing universe of discourse models into the space system ontology. This requires involvement of the design authority.
- Feedback to ECSS-NG body on the formal definition of concepts and triggering of change requests in case of mismatches.
- Demonstration of the use of the space system ontology in the creation of information system and the creation of Interface Control Documents and its transformation to existing tools.

Deliverables: Other, Report, Software

Application/Need Date: All missions. TRL5 by 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM C10 “OSMoSE Space System Ontology Development: engineering domains”.

Generic Technologies

Future Engineering

GT17-875SF – Adaptive human-machine interfaces for enhanced data monitoring and operations

Budget: 1200 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 8

Objective:
To develop an adaptive and immersive interface that enhances real-time monitoring and decision-making by integrating advanced visualization tools, extended realities (XR), digital twin data and Model Based Systems Engineering (MBSE) to support co-engineering in design, training, Assembly, Integration, and Test (AIT) and operations.

Description:
Modern aerospace operations demand continuous monitoring and interpretation of vast telemetry and design datasets to ensure mission success. Traditional monitoring methods are increasingly challenged by data overload, leading to high cognitive workload, slower response times, and reduced anomaly detection effectiveness. These challenges are especially critical during spacecraft operations, astronaut activities, testing, and mission planning.

This activity aims to develop an intelligent, adaptive, and immersive user interface that enhances operator efficiency, reduces cognitive load, and improves situational awareness and decision-making across design, integration, and operational contexts. By integrating XR, artificial intelligence, MBSE and digital twin technologies, the solution will provide advanced visualization tools capable of:
- Presenting telemetry, test, and design data in a structured, semantically rich, and interactive 3D environment.
- Dynamically highlighting critical insights using real-time data analytics and predictive models.
- Supporting collaborative engineering and reviews, root-cause analysis, and predictive maintenance.
- Enhancing spatial and temporal understanding through immersive AR/VR environments tailored to operator needs.

The outcome will be a reference approach and software architecture for next-generation human-machine interfaces in space missions. The prototype will demonstrate improved operator performance, more efficient spacecraft control, and faster, more informed decision-making.

The activity includes the following tasks:
- Analyse use cases, state-of-the-art solutions, user requirements, and trade-offs for enhanced monitoring across design, AIT, and operations.
- Develop a prototype adaptive interface leveraging XR, AI, and digital twin integration with MBSE and ontology-based semantics.
- Evaluate the solution’s performance across multiple domains and define a roadmap for technology maturation.

Deliverables: Prototype, Report, Software

Application/Need Date: All missions. TRL6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM H12 “Model Based System Engineering Explorer (MB-SEE)”.

Generic Technologies

Future Engineering

GT17-876SF – Digital continuity for digital spacecraft and digital twin spacecrafts

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 8

Objective:
To develop a unified and traceable data management system that enables the consistent integration of data, simulation models, and tools related to digital twin spacecraft across multiple disciplines, mission phases, and mission types. The solution shall be based on the Space System Ontology (SSO) for its semantics yet shall allow for flexibility in the mapping towards different sources and authoring tools.
Description:
The digital twin spacecrafts represent virtual representations of the spacecraft throughout the project lifecycle with synchronised interaction means. Examples are the digital twin spacecraft "as designed", "as built" but also "as operated". The data and information into (and out of) these digital twins come from the underlying digital spacecraft context, a federated set of databases and system information systems that capture all the spacecraft data and knowledge available. Within this eco-system, comprising and supporting various stakeholders with different roles, digital continuity refers to the consistent and traceable integration of data, models, and simulation tools across different disciplines, mission phases, and even across different missions. This means that all the data, simulation models and discipline-specific tools are reusable and interoperable throughout the entire spacecraft lifecycle—from early design through to operations. More specifically it includes the mapping of the lifecycle usage of models and data and defining how simulation assets contribute to various phases and ensuring their continuous integration. The simulation artefacts include the mission scenarios as input to the test scenarios and time-series data as output of the simulation and AIV test campaigns. Within the RatioSim community already a system-level simulator reference architecture exists that serves as a basis. Within ongoing Digital Twin Spacecraft activities an overview is created of the spacecraft data from requirements to mission/system design, up to PLM and CAD data. The SSO defines the semantics of the data and the domain specific concepts. By taking the Overall Semantic Modelling for Space Systems Engineering (OSMoSE) methodology, the generation of the information system in line with the data and semantics is ensured and to large extent automated.

This activity encompasses the following tasks:
- Analysis of the lessons learned, including:
  - Formal semantic approach of space system ontology and OSMoSE for overall semantic interoperability,
  - RatioSim system level simulator reference architecture, specifying the ecosystem, specification, development and usage of any system level simulator as basis for digital twins.
  - Use-cases and data specifics coming from pilot digital twin activity. This includes the different roles of different stakeholders
  - Digital twin spacecraft and digital spacecraft data management aspects.
- High level architecture definition of data management system supporting the digital twin spacecrafts.
- Definition of methodology to define the semantics of the data into space system ontology model (top down and bottom-up approach, allowing for configuration and flexibility).
- Development of the data management system, following the OSMoSE guidelines for generation of the information system based on the created SSO.
- Testing and Verification of the developed system using real-life data/simulators for a number of digital twin use-cases.

Deliverables: Prototype, Report, Software

Application/Need Date: All missions. TRL6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based for System Engineering (2024) – Consistent with activity AIM I05 "Digital spacecraft data management".

2.3.10. Ground Segment

Generic Technologies

Flight Dynamics

GT17-877GF – Rapid radiofrequency assisted optical communication acquisition system

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - **TD** 6

Objective:
To develop and operationally demonstrate a miniaturized optical communication acquisition assistance system based on low-rate radio frequency communication. This optical ground station sub-system will enable ad-hoc optical communication link acquisition without prior accurate orbital ephemeris knowledge.
Description:
Successful signal acquisition by an optical ground station (OGS), requires an accurate prediction of the spacecraft's position and visibility time. In commercial contexts, especially for small satellites in low Earth orbit (LEO), two-line element sets (TLEs) commonly sourced from platforms, are typically used for this purpose. However, TLEs have limited accuracy (position errors on the order of ca. 1 km), which is insufficient for reliable optical link acquisition, particularly given the narrow field of view of optical systems. The current operational workaround, as demonstrated for missions like Alisio-1, involves scheduling an RF (e.g., UHF or S-band) pass approximately 6 hours before the optical pass. During this pass, onboard GNSS navigation data is downlinked and rapidly processed to generate a high-accuracy ephemeris. This enables updated predictions of orbital position with errors of ca. 120 meters, sufficient to ensure first acquisition by the optical ground station. While effective, this approach is operationally complex, incurs cost, and requires dedicated orbit determination (OD) expertise, which is often beyond the capabilities of commercial ground station operators.

To reduce reliance on specialized flight dynamics operations and streamline optical link acquisition, this activity aims to develop and operationally demonstrate a self-contained system to support first optical acquisition. The expected benefits of this activity are:
- enabling autonomous, low-cost optical link acquisition without needing an RF pre-pass from other ground stations.
- reducing reliance on specialist flight dynamics teams for short-term optical planning.
- increasing operational efficiency and accessibility of optical communications for commercial satellite operators.
- facilitating broader use of OGS networks in missions with tighter resource constraints.

This activity encompasses the following tasks:
- Install a small UHF antenna at the optical ground station to receive low-bit-rate telemetry (TM), specifically GNSS navigation messages or state vectors. This enables acquisition of position data starting from low elevation angles (e.g. 5° to 10°).
- Implement a lightweight orbit determination toolchain to rapidly process the downlinked GNSS data and refine the spacecraft's predicted trajectory. This process generates a high-accuracy CCSDS orbit ephemeris message (OEM) to replace or improve the baseline TLE-based prediction.
- Integrate the improved OEM into the OGS's pointing control system. This updated trajectory information enables refined pointing in time for the upcoming optical contact, typically around 15° elevation.
- Closed-loop support. Ensure the system continues refining pointing predictions during the approach until optical lock is achieved, enabling robust acquisition even with initially degraded TLE accuracy.
Deliverables: Breadboard, Report

Application/Need Date: All missions, especially for small satellites in LEO, significant simplification and therefore cost of operations required for reliable station acquisition. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Optical Communications for Space.

Generic Technologies

Flight Dynamics

GT17-878GF – Streaming and automation for modern flight dynamics operations

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 5 / 8 - **TD** 10

Objective:
To develop a data streaming and automation framework for flight dynamics (FD) computations that enables integration of the expertise of flight dynamics teams directly into operations workflows within the next-generation European ground control systems.
Description:
Flight dynamics (FD) specialists provide essential expertise and make critical contributions to spacecraft operations. FD activities are traditionally carried separate from the rest of the flight control team. Data is analysed, results validated, responses formulated and finally the rest of the control team informed via text file or voice loop communication. This activity aims to empower FD teams to deliver their expertise more directly and efficiently to the end users of their services. By embedding FD processes within Europe's next-generation ground systems e.g. MCS-CC (Mission Control System Common Core) and GSMC-CC (Ground System Monitoring and Control Common Core), based on the European Ground Segment Common Core and its service-oriented, message-driven architecture, we can reduce latency, enhance situational awareness, and enable a faster, more integrated operational response. New challenges for Europe need to be addressed, large constellations, as well as more cost-conscious operations and more demanding mission complexity. Continuous modernization of the ground segment needs to be reflected in flight dynamics activities to allow them to evolve efficiently together. The proposed system puts the essential FD skills at the centre of a more modern, collaborative, and customer-focused environment. FD experts will retain full control over the design, validation, and maintenance of their data processing pipelines. At the same time, their outputs can be delivered automatically and in near real time to service users across the operations team, improving handover and accelerating decision making.

The expected benefits of this activity are:
- Having a framework for flight dynamics services, enabling increased commercialization and standardization for existing commercial applications.
- Integrating flight dynamics service provision into core flight control teams.
- More efficient, near-real-time provision of results of flight dynamics data analysis, to facilitate faster, robust decision making.
- Allowing increased commercialization of development and maintenance of FD service provision.

This activity encompasses the following tasks:
- Identification of suitable missions for initial implementation targets, both ESA and commercial.
- Identification and prioritization of use cases for system, leveraging domain and end-user expertise.
- Definition of framework technologies: database, data streaming, internal API interfaces.
- Definition of robust computational elements for flight dynamics activities, to achieve use cases, maximizing modularity and potential re-use.
- Development of core framework and initial MVP computational elements.
- Develop robust computational elements for flight dynamics, based on prioritized use cases.
- Implement user APIs (Application Programming Interface), e.g. RESTful API definitions and implementations, end user visualization and data analytics.
Deliverables: Report, Software

Application/Need Date: Next generation ground control systems on the cusp of operational use, commercial use of fully compatible FD systems for missions requiring higher degree of autonomy. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic

→ THE EUROPEAN SPACE AGENCY

Generic Technologies

Ground Station Engineering

GT17-879GS – Open monitoring and control protocols and user interfaces for ground stations

Budget: 1100 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 12

Objective:
To define and develop open monitoring and control (M&C) protocols and user interfaces for ground stations.
Description:
While for space missions the protocols for communication on the space link and over terrestrial networks have been standardized to a good extent by CCSDS and others, this is not true for the monitoring and control of equipment, for ground stations. This lack of a common approach to M&C drives integration costs for ground station equipment and prevents easy reuse of equipment and ultimately efficient industrial productization. This activity shall design and demonstrate an approach to effectively address the problem described above. The analysis shall take into account the constraint, that a certain level of heterogeneity of M&C protocols has to be accepted among different organizations, i.e. a fully harmonized M&C protocol architecture can most likely not be achieved among different organizations in the medium or even long term.

This activity aims to analyse currently used and potentially available M&C protocols for equipment like those based on HTTP (REST), IoT protocols like MQTT and standards like NETCONF, RESTCONF and CORECONF. One of the design goals is an easy adoption and exchange of M&C protocols, shielding the equipment from details of M&C protocols. This activity will assess the potential of an abstract M&C model which can be bound to specific M&C protocols and equipment specific M&C programming interfaces. In addition, the activity will cover an M&C model driven approach to automatically create extensible web-based user interfaces to enable remote M&C for human operators. The approach shall allow an easy replacement of potentially changing web technologies and aim at preventing technology lock-in.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (600K€, 14 months):
- Analysis and development of an open M&C protocol middleware and M&C model.
- Implementation of an early adopter version for M&C protocol middleware and M&C model.

Phase 2 (500K€, 10 months):
- Design of automatic web user interface generation.
- Implementation of generated web user interfaces.

Deliverables: Open M&C middleware software, Report

Application/Need Date: Evolving space mission technologies require corresponding updates of ground stations and involved station equipment. The latter updates must be as simple and cost effective as possible. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic

Generic Technologies

Ground Station Engineering

GT17-880GS – Ground station scheduling as a service

Budget: 3000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 4 / 6 - **TD** 9

Objective:
To develop a service-oriented approach for all types of ground stations allocation planning: legacy space missions, spacecraft constellations and opportunistic planning for missions using optical direct to earth communication.
Description:
For of the ESA Tracking Network (ESTRACK), a central planning and scheduling has been developed and operated for more than 15 years. In the context of ESTRACK, ground station planning and scheduling is done for ESA and non-ESA missions over ESA and non-ESA ground stations in a one-stop-fashion. Therefore, it is proposed to extend and evolve the successful underlying principles towards latest developments and demands, and to generalise the approach beyond the scope of ESTRACK.

This approach implies transitioning to a standardized service-based approach, where arbitrary space missions can integrate ground station planning and scheduling as true vendor independent services into their potentially automated planning process, including a sandbox approach to exercise what-if scenarios. While mega constellations tend to have dedicated ground segments, for many other missions sharing ground station resources is a requirement to optimize availability, efficiency and ultimately costs. Moreover, optical direct-to-earth (DTE) missions will require opportunistic ground station planning and scheduling to deal with uncertainties imposed by changing weather conditions. At the same time, during the last years new planning and scheduling approaches have been developed. A massive rise of cloud computing technologies has been observed. This activity aims to analyse relevant technology evolutions with their potential of adoption to minimize the amount of bespoke software and hardware. Finally, the activity will address original technology obsolescence, consider the benefit of latest AI technics, and aim at easing the setup and configuration.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 1500 k€, duration: 18 months):
- Analyse current technology, systems and limitations and define concepts and requirements.
- Design and implement the following functionalities:
  - Service interfaces, allowing consumption of provided station scheduling services for missions and scheduling stations 'on-behalf' of missions for (external) stations and networks.
  - Core planning and scheduling.
  - Ease configurability by simplifying as much as possible the configuration of missions and ground stations.
- Update latest technology standards:
  - Storage of plans (e.g. NoSQL databases).
  - Files (e.g. Ceph).

Phase 2 (budget 1500 k€, duration: 18 months):
- Design and implement the following functionalities:
  - Support constellation planning
  - Opportunistic planning for optical missions (direct to Earth communications).
  - Support missions to exercise in 'what-if scenarios' to validate new scheduling requirements.
- Upgrade the planning AI kernel to latest planning and scheduling technologies in the field.
- Update latest technology standards:
  - Design for cloud ready deployments.
  - Web based UI for scheduling office and missions.

Deliverables: Software, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic

Generic Technologies

Ground Station Engineering

GT17-881GS – Deep-space optical communication ground laser transceiver

Budget: 5000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 12

Objective:
To develop a deep-space communication ground laser transceiver (GLTX), a dual-function beacon and receiver for deep-space optical communications.
Description:
The quantum communication ground laser transceiver (GLTX) is a next-generation dual-function beacon and receiver system designed to enable high-rate optical communications and quantum information transfer for deep-space missions. Building upon the ground segment technology previously tested, this activity will enhance and optimize the transceiver for operational use with Lightship-1/Marconi and future interplanetary missions.

The GLTX will provide precise optical beaconing for spacecraft acquisition and tracking, alongside a high-sensitivity receiver for optical data downlinks and quantum communication signals. The GLTX will serve as both a:
- uplink optical beacon – providing high-precision tracking and acquisition support for deep-space spacecraft, ensuring robust link establishment.
- downlink optical receiver – enabling high-sensitivity photon detection for data reception and quantum-secured communication.

This activity will advance the technology to a fully operational deep-space optical ground segment:
- Supports high-data-rate optical communication for Mars and interplanetary exploration.
- Strengthens global cooperation in deep-space optical networking, leveraging European and international space agency partnerships.
- Lays the groundwork for a scalable ground network for future lunar, Mars, and deep-space exploration.

This activity is divided in phases and encompasses the following major tasks:

Phase 1 (budget: 3000 k€, duration: 14 months):
- Design update for daylight operation and with lessons learned from Deep Space Optical Communications (DSOC) 2025 campaign (incl. updated laser safety system).
- Design, implementation and testing of central measurement and command (M&C) and data delivery system.
- Breadboarding and testing of subsystems with design changes as required.

Phase 2 (budget: 2000 k€, duration: 10 months):
- Combination of GLR and GLT (and full integration of all subsystems) towards a GLTX system.
- Assembly and manufacturing of the full scale GLTX.
- Field deployment and testing with a satellite or emulated satellite.

Deliverables: Engineering Model, Software, Report

Application/Need Date: Future exploration missions (e.g. LightShip-1, Marconi) **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Ground Station Technology.

Generic Technologies

Ground Station Engineering

GT17-882GS – All digital photon starved optical modem

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 12

Objective:
To develop a high photon-efficiency optical modem prototype that enables robust, scalable, and multi-mission optical communication, leveraging all-digital signal processing techniques for photon-starved environments.
Description:
Optical communication is a critical enabler for high-data-rate, long-distance space missions, but current systems are limited by photon losses and require high-power laser sources. This activity aims to develop a high photon-efficiency, all-digital optical modem prototype optimized for photon-starved communication scenarios.

This prototype will leverage advanced digital signal processing techniques to enhance detection sensitivity, improve link robustness, and maximize data throughput under low-photon conditions. The modem will feature a modular and scalable architecture, supporting multiple wavelengths, adaptive modulation schemes, and forward error correction to ensure interoperability across deep-space, lunar, and interplanetary networks. Designed for multi-mission compatibility, it will integrate seamlessly with optical ground, supporting future lunar, planetary, and communication infrastructures.

This activity encompasses the following tasks:
- Design and prototype an all-digital optical modem capable of operating in extreme low-photon regimes, optimizing detection sensitivity and energy efficiency.
- Ensure adaptability across diverse mission architectures, supporting deep-space, lunar, and interplanetary communication scenarios.
- Develop a flexible architecture that supports various link budgets, wavelengths, and modulation schemes for different mission requirements.
- Implement cutting-edge digital techniques, such as forward error correction (FEC) and adaptive filtering, to maximize data throughput under photon-starved conditions.
- Ensure compatibility with existing and future optical communication infrastructures.

Deliverables: Hardware, Software, Report

Application/Need Date: Future exploration missions (e.g. LightShip-1, Marconi). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Ground Station Technology.

Generic Technologies

Ground Station Engineering

GT17-883GS – All-digital ground station

Budget: 5000 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 4 / 7 - **TD** 12

Objective:
To build a prototype of an all-digital ground station which will replace the current ground station modem, radio frequency converters and analog intermediate frequency distribution network.
Description:
The signal processing approach of satellite ground station back-ends is undergoing a major conceptual change worldwide: instead of processing the signal in the back-end in the classical way with a dedicated up-/ downlink chain, there is the more flexible approach of sampling the signal directly before transmission / after reception. This step also includes the up- and down-conversion. Thereafter the data is transferred via the network using the Digital Intermediate Frequency Interoperability (DIFI) protocol and processed on a Software Defined Radio (SDR) platform (instead of a classical modem). Commercialisation of this concept is on-going on both sides: the classical modem/back-end providers and the providers of commercial satellite ground station services. For the latter one, the high flexibility, the easy scalability and the interoperability are only some of the key advantages. For all sides advantageous is the high degree of modularity, that can be achieved with an all-digital ground station. This allows, to program dedicated modules for dedicated satellite missions, which can then be plugged into any ground station, without changing hardware components, just simply by a software update. The use of the open DIFI standard instead of other proprietary protocols also helps avoid vendor lock-in by allowing systems from different vendors to communicate with each other. The open DIFI standard can also be a door-opener for new and existing vendors to offer COTS components with the option to include custom developments according to customer requests.

Furthermore, the signal digitisation right at the beginning allows for much more flexibility in terms of processing. For example, the processing could even be split between a COTS SDR and a custom developed SDR part (e.g. for a dedicated mission need(s)). This allows to take profit of low-cost COTS developments, while still maintaining all the dedicated custom developed functionalities in a separate SDR system. Moreover, the given flexibility of an all-digital ground station is much larger than only this and this activity will assess additional possibilities which an all-digital ground station could provide. Additionally, it will also show in which areas there are still limitations to overcome. Currently, a TDE activity, T712-703GS - Ground Station Backend Architecture with High-Rate Sampling and Software Defined Radio, for a very limited all digital ground station is running (due to budget constraints, it will be limited in terms of scope, functionality and performance).

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 2000 k€, duration: 12 months):
- Full SDR breadboard for a satellite ground station based on DIFI standard.
  - digital modem implementing full functionality.
  - A/D and D/A converters operating at ESTRACK intermediate frequencies.

Phase 2 (budget: 3000 k€, duration: 18 months):
- All digital ground station prototype, including:
  - digital modem implementing full functionality and meeting all performance requirements.
  - digital distribution network.
  - high-rate A/D converter and digital down-converter.
  - M&C system.
  - test prototype with high-rate sampler using DIFI output.

Deliverables: Breadboard, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Ground Station Technology.

Generic Technologies

System and Applications Engineering

GT17-884GA – Mission planning capabilities

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 5 / 7 - **TD** 9

Objective:
To develop advance mission planning framework with new capabilities to be demonstrated on future missions showcasing improvements in planning performance and flexibility.
Description:
Future missions have more complex planning problems leading to the need to support an increasing amount of data. Such support for very high amount of data cannot be supported with the current state of the systems, hence the technology needs to be updated. At the same time, the gap between what is demonstrated feasible via previous R&D activities and what is used by standard missions are growing. Building on the GSTP activity, GT1O-303GD - Interactive distributed end-to-end mission planning for multi users, that provides a platform for interactive planning systems based on CCSDS standards, the next step is to ensure advanced concepts each a maturity to be adopted, hence providing real benefits for real missions. The tasks will continue the evolution based on open micro services-based architectures for mission planning systems utilizing CCSDS standards for increased interoperability and flexibility, enabling continues planning capabilities. The need for innovation in mission planning solutions also comes from the emerging use of AI. Such use within the ground segment implies a greater need to synchronize and plan ground systems in coordination with the space assets. This increased needs for multi system orchestration, planning and scheduling calls for more open interconnectable capabilities. The orchestration capabilities can build on resent activities making advanced algorithms, artificial intelligence, and machine learning tools available to improve the accuracy, efficiency, and adaptability of the overall combined space and ground planning processes. By leveraging these technologies, improvement in planning streamlines the coordination of complex systems, reduce operational risks, and maximize the scientific return of space endeavours. Capitalizing on such innovative approaches for data management allows real-time decision-making, ensuring forefront solutions are available for space exploration in Europe. These techniques are now not only R&D, but ready to be deployed once a suitable framework for their execution is adopted.

A key challenge to reach this goal is to ensure applications are sufficient flexile to avoid organization specific solutions, while remaining attractive and cost effective without the need for over complex adaptations. Furthermore, there is an increasing opportunity for spin-in of technology towards space with several emerging tools both commercial and free-to-use. Integration with such tools is key for the future.

This activity encompasses the following tasks:
- Complete the development of an open, interoperable platform for MPS (mission planning system) enabling advanced planning tools.
- Migrate existing interfaces from legacy file-based interfaces to direct services based on CCSDS enabling continuous replanning capabilities.
- Develop a wide catalogue of import/export and transformation capabilities to ensure that the solution is not organization specific, including demonstration of performance improvements with respect to current solutions required to handle more data and more complex scenarios.
- Develop complex data manipulation capabilities based on state-of-the-art mission planning open-source libraries and AI tools.
- Provide end user support, training and user manual to ensure easy adaptation.
- Support first pilot user and update system based on feedback from real users.

Deliverables: Software, Reports

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Ground Station Technology.

Generic Technologies

System and Applications Engineering

GT17-885GA – On-board software execution in simulators for future missions with faster on-board computers

Budget: 1000 k€ - **Duration**: 15 months - **Current / Targeted TRL:** 3 / 7 - **TD** 1

Objective:
To develop multiple technical solutions allowing for performant execution of the required functions of the on-board software for system-level simulators such as in software validation facilities and operational simulators for future missions with faster on-board computers.

Description:
Two core requirements of all the operational simulators developed in the last twenty years are the capability to run at least in real time to support the execution of simulation campaigns and the capability to run the central software binary image in an emulated on-board computer to have a high-fidelity representation of the spacecraft behaviour on ground. These two requirements are easily satisfied as far as the clock speed of the on-board computers is a small fraction of the clock speed of the processors available on ground. Unfortunately, the coming generations of on-board computers beyond LEON-4 are approaching the performance of commodity computers available on the market, causing a conflict between the core system-level simulation facilities (e.g. Software Validation Facility (SVF) and operational simulators) requirements, mentioned above. Also, the new on-board computers are moving toward multi core architectures, allowing partitioning of the central software and its execution of tasks previously allocated to external units (e.g. GNSS (Global Navigation Satellite System) signal processing, star tracker (STR) sky image processing). The emulation of this functionality requires complex modelling of the external units that so far was not needed (e.g., modelling of the GNSS constellations signals rather than the position, velocity and time (PVT) only, modelling of the sky image rather than the attitude quaternion only).

This activity will analyse, trade off and prototype multiple solutions allowing the execution of the required functions of the on-board software. Optionally, a mission or CubeSat shall be identified where the new concepts can be tested.

This activity encompasses the following tasks:
- Identify multiple solutions allowing the execution of the on-board functionality. These solutions can include emulation of selected partitions, re-compilation on non-native architectures of selected functionality, replacement with functional modelling of selected functionality (following ECSS-E-ST-40-07 (SMP)), etc.
- Analyse the impact that each proposed solution has on the typical use cases covered by the current generation of operational simulators (e.g. loss of possibility to patch CSW (central software) image and rerun it) and to identify new offered opportunities.
- Develop a selected subset of solutions/technologies that address simulation models, emulators, central software and simulation architecture.
- Test the software under realistic conditions, looking both at performance and functionality.

Deliverables: Prototype, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

System and Applications Engineering

GT17-886GA – Enhancing operational platforms with modern compute resources

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - **TD** 9

Objective:
To assess and implement methodologies for offloading and managing compute-intensive tasks asynchronously, enabling enhanced scalability, efficiency, and performance optimization of on-premises legacy solutions, without compromising existing infrastructure integrity.
Description:
Many modern technologies require to access large amounts of compute resources, for instance AI training processes, Monte-Carlo simulations, digital twin simulations, etc. As user and functional demands on operational systems evolve and require such technologies, organisations are faced with a need to enhance existing legacy, on-premise operational solutions to support more complex processing. However, operational or financial reasons may prohibit a complete migration to a more capable IT resource platform. Therefore, integration of modern high-powered compute clusters, including cloud-based resources and high-performance computing (HPC) into legacy solutions is required. Integration of non-native remote facilities implies unguaranteed response times for compute. Nevertheless, operations teams require verifiable systems, and non-guaranteed performance must be managed in accordance with operational needs. Different strategies can be deployed for various situations; some tasks may be satisfied by waiting for hours or days before results are ready; others may require initial results to be available with regular progress updates increasing accuracy; some may choose calls to multiple remote compute providers to mitigate localised bottlenecks.

A suitable software will then be identified and created and applied to at least these following use-cases:
- Recent developments in Planetary Defence highlight a growing need for the massive parallelization of computation-heavy tasks, such as image processing and orbit determination. This activity aims to identify systems currently in use or under development and apply massive parallelization techniques—such as CUDA (Compute Unified Device Architecture) and OpenMP (Open Multi-Processing) —to assess their applicability on hardware platforms like ESA's HPC infrastructure and general-purpose GPU compute servers.
- AI models can require many hours of training time on large compute resources; typically, AI developers are forced to compromise during model development; in managing model complexity and during operational use; in re-training of models with recent operational data, in order to allow local deployment into limited on-premise compute.
- A non-ESA case. Assessment of the performance improvements and stability and suitability of the approach for operational use will then be performed. Key issues will be non-guaranteed performance of some remote Performance metrics such as execution time, resource utilization, and system throughput will be measured across various scenarios.

The activity encompasses the following tasks:
- Assessment of requirements for use cases involving an asynchronous framework allowing to offload and manage compute-heavy tasks in support of operational solutions.
- Design approaches for use of cloud- or HPC-based resources to augment legacy platforms.
- Development a suitable framework(s) for management of asynchronous computing tasks in operational space systems.
- Demonstrate the ability to use HPC or cloud for several legacy applications.
- Verify the impact of external compute integration on system performance, reliability and operability.
Deliverables: Prototype, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

System and Applications Engineering

GT17-887GA – Hierarchical data models for ground segments

Budget: 2800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 7 - **TD** 9

Objective:
To develop key components required for transitioning to a hierarchical data model in the ground segment, including identification and capability demonstration of key expected benefits.

Description:
Current ground segment applications rely on long-standing formats and tools that often structure spacecraft databases and procedures as flat information, limiting modern model-based methods. New hierarchical data models are emerging to enhance mission automation and intuitive spacecraft modelling. As missions adopt these new formats, it is essential for the ground segment to implement them fully. The introduction of hierarchical model-based tailoring data is anticipated to create significant new opportunities across the entire ground segment. The data model will serve as a single central source of truth for pre- and post-launch phases, spanning various areas of the ground segment, including mission preparation, mission planning, ground stations, flight dynamics and mission operations.

Since the data model will be used across different application domains within the ground segment, the activity must demonstrate the end-to-end approach where the same data model is used in the mission control system, mission planning system, operational simulator and flight dynamics systems. For this activity, a minimum set of ground segment functions shall be able to validate the approach end to end, such as an operational simulator, a mission control system, a mission planning system and a mission preparation system (data model and procedure management). For the simulator, the implementation will focus on the adoption of a hierarchical data model in SIMULUS, an open source simulation infrastructure, available to ESA member states industry through ESA Community License. Applications in other parts of the ground segment will be selected in the initial phase of the study. For the other ground segment functions, a tool selection shall be made during the first phase.

The activity is divided by phases and encompasses the following tasks:

Phase 1 - Design and prototyping (budget: 1700 k€, duration: 15 months)
- Design and develop a ground segment based on a hierarchical data structure and identify gaps with respect to current approaches.
- Analyze each identified gap, and priorities the various opportunities according to expected benefit vs. complexity of implementation.
- Implement and integrate changes required for basic adoption of a hierarchical data model on selected ground segment applications
- Demonstrate basic interoperability between key systems with hierarchical data models.

Phase 2 - Software prototype validation (budget: 1100 k€, duration: 9 months)
- Improve software systems from phase 1 based on initial user feedback and optimize the tailoring process
- Validate the prototyped changes within a pilot mission with mission realistic data.

Deliverables: Prototype, Software, Report,

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Generic Technologies

System and Applications Engineering

GT17-888GA – Data foundations for ground segment operations

Budget: 3000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 7 - **TD** 9

Objective:
To develop and demonstrate a working data-centric architecture for ground segment systems, enabling exploitation of space system ground segment data for automation, artificial intelligence and autonomic based solutions.
Description:
During development of several recent automation and artificial intelligence-based solutions, the effort to access existing system data has often slowed or blocked progress. The need for a coherent approach to access to the diverse ground segment system data has been repeatedly reported. This activity will implement this required functionality, by demonstrating the capability and benefit of a data-centric approach to any space system ground segment in easing the development and operations of ground segment solutions involving automation, artificial intelligence or autonomy. A common IT approach to similar problems in other industries is the data-centric architecture. This transformative approach to data management and utilization supports and increases emphasis the interoperability, security, and governance of operational data. Within the ground segment, such an approach can provide access to siloed systems data, addressing the issue already found to be key shortcoming in current ground segment system design and implementation. This must be addressed if space systems are to increase the use of artificial intelligence solutions that will lead to increased space system autonomy. Additionally, a data-centric approach would also present opportunities and offer alternatives to current paradigms for the data and file exchange between traditional components, for instance:
- Ease organisation of information flows in any system deployments supporting multiple missions that require more intelligent and standardized approaches to manage data and file flows.
- Ease transfer of information to external entities.
- Introduce new approaches for the continued support for files delivery, which is likely to remain a fundamental aspect for correct functioning of mission data systems.

Other user level benefits of a data-centric architecture include exchange of data by various parties and increased access to all data to all users and AI tools. Additionally, it improves visualisation of data sources and pipelines, analysis for operational teams and data specialists and increases knowledge of data lineage and applicability.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 1500 k€, duration: 12 months):
- Design a suitable data-centric architecture, following a review and consolidation of the requirements.
- Review of potential available implementing platforms.
- Development and deployment of one or more platforms and components as prototypes for selected use cases, to confirm key aspects such as performance, security, resilience and usability.
- Definition of optimal platform and identification of use cases for next phase.

Phase 2 (budget: 1500 k€, duration: 12 months):
- Development and deployment of selected platforms and components.
- Migration of several ground segment systems toward to use the selected platform(s).
- Demonstration of one or more existing automation, artificial intelligence or autonomy solutions to confirm the benefits of the platform for such solutions.
- Demonstration of new analysis, automation, artificial intelligence or autonomy functions, enabled by the platform's capabilities.

Deliverables: Software, Reports

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with R&D Line 3.3 "AI techniques for data dissemination and citizen engagement".

Generic Technologies

System and Applications Engineering

GT17-889GA –Digital ground segment

Budget: 2000 k€ - **Duration**: 15 months - **Current / Targeted TRL:** 4 / 7 - **TD** 9

Objective:
To develop and verify digital twin (DT) ground segment concepts, processes, software and supporting tools, through the application of digital engineering techniques, to facilitate high fidelity emulation and digitalisation of full-lifecycle design, deployment and operation processes.
Description:
The digital ground segment builds on the concept of the digital spacecraft introduced in ESA, focusing on the ground segment development, deployment and operation. It combines the concepts of ground segment as a Service (GSaaS) with digital spacecraft, digital engineering, artificial intelligence, Model based System Engineering (MBSE), security best practise, simulator, cyber range and cloud technologies as key enablers to achieve a digital transformation concept covering all aspects to constitute a full-lifecycle spacecraft and ground segment. This is expected to radically improve the efficiency to develop, deploy and securely operate the ground segment for a mission, specifically when scaling solutions that include multiple related missions or large constellations, and by delivering additional capabilities to ground, flight and security teams in the support of situational awareness both on ground and in space in detecting and understanding anomalies that may be caused by degradation, malfunction or malicious actions. GSaaS, DevSecOps (Development, Security, and Operations) and Infrastructure-As-Code (IaC) shall enable automated deployments and testing of ground segment applications as defined by digital models of the system. For example, the tailoring of the spacecraft PUS (Packet Utilization Standard) services as defined by the digital twin of the spacecraft can be used to configure an automated deployment of the mission control system (MCS) with the required PUS tailoring. This is an enabler to easily clone a deployment of a live system, which can be used to detect and investigate live issues or test potential corrective changes and what-if scenarios via operations. This digitalisation allows the capability of a freshly deployed system configuration and state to be accurately captured. The ability to accurately clone the entire space system (space and ground segments) would allow ground, flight and security operations teams to explore responses to novel situations. For instance, such a high-fidelity model across the entire space system presents an opportunity for conducting equally high-fidelity cyber exercises including fully representative security vulnerability and resilience assessments and what-if scenarios which go beyond what is possible today with replica environments and deployed mission data systems.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 1200 k€, duration: 9 months)
- Perform a critical analysis of predecessor, ongoing and planned activities.
- Leveraging existing DT, MBSE and cloud-native and AI developments, specify, design and prototype the target DT capability and use cases including DT cybersecurity scenarios and exercises in a cyber range environment, identifying necessary extensions and wider system integration opportunities
- Identification of suitable use cases and demonstration scenarios for phase 2.

Phase 2 (budget: 800 k€, duration: 6 months):
- Develop the DT solution and verify target capability use cases.
- Perform a critical assessment of results achieved, lessons learned, and any necessary future development roadmap.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Functional Verification and Mission Operations Systems.

Technology Focus

Artificial Intelligence
Quantum Technologies
Innovative propulsion
Sustainable Space

Artificial Intelligence

3.1.1. Introduction and selection criteria

Artificial Intelligence (AI) is transforming every discipline involved in the development of space systems. Recognizing its strategic importance, ESA has established an AI Strategy Plan that outlines key objectives across four main application domains. These domains serve as the foundation for the “AISSA” Technology Harmonisation Document (THD), which translates them into targeted strategic goals and R&D lines.

This Compendium presents a curated selection of activities aligned with these domains. Each activity focuses on the development and application of AI techniques identified as having the greatest potential to enhance space system capabilities or to enable entirely new missions and applications.

Mission Engineering and Design

AI can support the digitalisation of mission development, in early stages with Model-Based System Engineering assistants, during the development and simulation, and during the verification and validation of all space subsystems like instruments, payloads, software and control. In AOCS (Attitude and Orbit Control Systems) and GNC (Guidance, Navigation and Control), AI complements model-based methods with data-driven approaches to improve robustness and performance.

Mission Operations

The ESA Directorate of Operations developed the Artificial Intelligence for Automation (A2I) Roadmap with industry to guide the development of future AI applications. Activities proposed in this compendium will ensure the continuity of the implementation of the A2I Roadmap. This implementation will entail the investigation and development of AI-based solutions for automating a broad range of mission operations tasks and activities. Among others, these include predictive infrastructure maintenance, mission planning, operational simulation, mission operations systems testing and validation, spacecraft health monitoring, problem diagnosis, and the recommendation of actions to spacecraft operators.

Insights from mission data

AI streamlines data processing, enhancing pattern recognition, and extracting meaningful insights from complex datasets. Recent advances have proven AI’s value in extracting and sharing information. Applying AI to space-generated data may lead to new discoveries and boost public engagement

AI streamlines data processing, improves pattern recognition, and uncovers insights from complex datasets. Recent advances have proven AI’s value in extracting and sharing information. Applying AI to space-generated data may lead to new discoveries and boost public engagement.

3.1.2. List of activities

Artificial Intelligence

Ground Segment - System and Applications Engineering

Programme Reference	Activity Title	Budget (k€)
GT1I-800GA	AI application suite for ground segment operations	3,000
GT1I-801GA	AI enhanced modelling for synthetic data generation	1,000
GT1I-802GA	Multi-domain multi-modal AI co-worker using an agentic architecture	2,500
GT1I-803GA	Operational qualification and explainability of AI solutions	2,000
GT1I-804GA	Structured data foundations for machine learning in space mission applications	1,500
GT1I-805GA	Security framework for AI applications	1,500
Total Ground Segment - System and Applications Engineering		**11,500**

Ground Segment - Space Debris

Programme Reference	Activity Title	Budget (k€)
GT1I-806SD	Adaptative operator-driven AI methods for collision avoidance decisions	800
Total Ground Segment - Space Debris		**800**

Ground Segment - Space Weather

Programme Reference	Activity Title	Budget (k€)
GT1I-807SW	High performance modelling with AI and data assimilation	2,100
Total Ground Segment - Space Weather		**2,100**

GNC, AOCS & Pointing

Programme Reference	Activity Title	Budget (k€)
GT1I-808EC	Robustification of feature extraction and tracking-based image processing through machine learning	700
GT1I-809EC	Data learning model predictive control	600
GT1I-810EC	Integrated large language models tools and simulation environments for faster AOCS design and prototyping	700
GT1I-811EC*	Verification and validation of AOCS/GNC using AI on-board	1,000
GT1I-812EC	Enhanced on-board autonomy and agility using AI-based guidance/control optimization	1,000
	Total GNC, AOCS & Pointing	**4,000**

*** Proposed Bundle 5 – AI for verification and validation**
- GT1I-811EC - Verification and validation of AOCS/GNC using AI on-board
- GT1I-813ED - Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform

Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT1I-813ED*	Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform	700
GT1I-814ED	AI/ML framework for on-board inference	1,000
	Total Avionics & EEE	**1,700**

*** Proposed Bundle 5 – AI for verification and validation**
- GT1I-811EC - Verification and validation of AOCS/GNC using AI on-board
- GT1I-813ED - Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform

RF Payloads & Technology

Programme Reference	Activity Title	Budget (k€)
GT1I-815EF	AI reconstructed radiation patterns from amplitude measurements	500
GT1I-816EF	Testbed demonstration of payload processor applications using AI capable RF SoC Technology	800
GT1I-817EF	Sensor with AI based processing and quality improvement	1,000
	Total RF Payloads & Technology	**2,300**

Structures, Mechanisms & Materials

Programme Reference	Activity Title	Budget (k€)
GT1I-818MS	Stable diffusion algorithms and geometrical deep learning for space systems	500
	Total Structures, Mechanisms & Materials	**500**

Future Engineering

Programme Reference	Activity Title	Budget (k€)
GT1I-819SF	AI accelerated ontology generation	600
GT1I-820SF	Spatiotemporal knowledge graph for insights from mission data	800
GT1I-821SF	Generative design for early design phases	900
GT1I-822SF	On-the-fly data access for AI foundation models	500
GT1I-823SF	AI library for holistic verification and validation of space systems	1,200
GT1I-824SF	AI based application for on-board planning and scheduling	1,200
GT1I-825SF	Federate simulator for AI system verification and validation	800
GT1I-826SF	Modular framework for enhanced perception and decision-making	600
GT1I-827SF	AI enhanced digital twin simulators for operations and AIT	1,500
GT1I-828SF	Intelligent assistant for MBSE and simulation model generation, setting and evolution for mission preparation	1,800
GT1I-829SF	AI based sensor fusion	900
GT1I-830SF	AI-powered assistant for SysML v2: enhancing MBSE	600
	Total Future Engineering	**11,400**

Total Artificial Intelligence	**34,300**

3.1.3. Description of activities

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-800GA – AI application suite for ground segment operations

Budget: 3000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 7 - **TD** 9

Objective:
To develop, test, and use the latest AI methods for spacecraft and ground station operations, including digital twins and advanced models.

Description:
The development of a suite of AI applications for accurate and reliable prognostics and health management of spacecraft represents a significant advancement and opportunity in space technology. AI systems, leveraging machine learning algorithms and data from various sensors, enhance the capability to monitor the health and performance of spacecraft in real-time, predicting potential failures before they occur and ensuring optimal operation. However, current AI tools are often separate and specialised, leading to inefficiency. There is a need for a seamless, unified suite with shared features like explainability, data handling, and consistent interfaces. This will make AI easier to use and adopt across teams. Many new technologies and approaches are in active development. Especially, use of federated learning and continual learning lead to reduced computation demands and enable more powerful AI techniques, such as prediction remaining useful lifetime of components. Additionally, whilst traditional AI/ML approaches have been proven valuable solutions to detect anomalies during routine operations, they degrade fail in accuracy, generalization and interpretability when novel conditions are encountered or manoeuvres are performed. AI-enhanced modelling and simulation tools such as Digital Twins, which integrate Operational simulators, which emulate OBC and run OBSW with AI models to enhance the fidelity of the simulations and are provide credible representativeness of the behaviour of spacecraft under various conditions. This allows for more personalized/tailored modelling which will enhance the accuracy of prognostics and health management capabilities. They can model novel conditions and consider the effect of manoeuvres and procedures in the spacecraft behaviour. In this light, this activity aims also at leveraging AI-enhanced modelling and simulation tools to detect and assess normal and abnormal conditions, by comparing the expected behaviour of high-fidelity and representative simulations with acquired monitoring data.

The activity is divided in phases and encompasses the following tasks:
Phase 1 – Assessment and Minimum Viable Products (MVPs) development (budget: 1300 k€, duration: 9 months):
- Evaluate current AI solutions in the industry and identify pain points.
- Review state-of-the-art AI, especially Spacecraft Digital Twins.
- Create MVPs addressing user issues.
- Define a flexible, modular AI Package, covering data handling, security, explainability, and validation.

Phase 2 – Implementation and Deployment (budget: 1700 k€, duration: 9 months):
- Develop and prototype AI solutions for selected MVPs.
- Implement and integrate the AI Package with new or existing AI solutions.
- Deploy the framework and prototypes to end-users.
- Assess the performance and impact of AI solutions.

Deliverables: Report, Software

Application/Need Date: Mission demand for solutions that enhance capabilities across ground segment operations, by increasing efficiency, awareness and responsiveness of teams. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity “2.1 DAI-Enhanced Health Monitoring and Control Capabilities”.

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-801GA – AI enhanced modelling for synthetic data generation

Budget: 1000 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 6 - **TD** 9

Objective:
To generate reliable synthetic data from digital replicas and models for spacecraft behaviour simulations and space weather forecasting.
Description:
AI/ML models for spacecraft autonomy, decision-making, and health management require large, high-quality datasets for training and validation. Unfortunately, in-orbit data is scarce and expensive, limiting its ability to cover all operational scenarios. Similarly, space weather forecasting relies on resource-intensive models that hinder responsiveness. Recent advancements in high-fidelity simulation models, including digital twins, provide detailed and dynamic representations of spacecraft and their environment. These digital replicas integrate physical models, simulators, and telemetry to understand spacecraft status and behaviour comprehensively. Accurate environmental models enhance these simulations, capturing complex interactions between spacecraft and their surroundings. AI techniques, such as machine learning and physics-informed AI, can create surrogate models that approximate complex simulations with less computational cost. These models, trained on detailed simulations or mission data, offer rapid predictions for real-time applications. In space weather, where high-fidelity models are slow, surrogate AI models provide faster and more responsive forecasting.

Combining AI with digital twins and high-fidelity simulations offers significant opportunity for the space sector. AI can enhance the predictive capability, speed, and adaptability of these models. These technologies promise to generate synthetic datasets that are representative, credible, and mission specific.Developing AI/ML models for spacecraft autonomy and space weather forecasting is a strategic priority to enhance mission safety, resilience, and efficiency. However, the lack of large, high-quality datasets representing diverse conditions is a major barrier. In-orbit data is limited and costly, while space weather models are complex and computationally demanding.

AI-enhanced modelling tools, including digital twins, surrogate models, and physics-informed AI, offer a solution by generating credible and representative synthetic datasets. These tools reproduce spacecraft behaviour and environmental interactions with high fidelity, supporting the development and testing of AI/ML models and decision-making algorithms. They also accelerate space weather modelling through surrogate AI models, supporting faster and more responsive forecasting.

The activity is divided in phases and encompasses the following tasks:
Phase 1 - Assessment and MVP Development (budget: 500 k€, duration: 6 months)
- Survey state-of-the-art.
- Define environmental models, operational simulators, and AI modules.
- Develop the first MVP for synthetic datasets generation for selected scenarios.

Phase 2 - Operationalization and Deployment (budget: 500 k€, duration: 6 months)
- Implement a modular application for scalable synthetic data generation.
- Integrate AI solutions like dynamic eigenvalue decompositions and surrogate models.
- Deploy the AI-enhanced synthetic data generator in a representative operational environment.
- Demonstrate the capability to produce synthetic datasets supporting the training and validation of AI/ML models and decision-making tools.

Deliverables: Report, Software

Application/Need Date: Supporting training of Machine Learning based solutions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 2.3 “AI-Enhanced Modelling and Simulations (M&S) Capabilities”.

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-802GA – Multi-domain multi-modal AI co-worker using an agentic architecture

Budget: 2500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 7 - **TD** 9

Objective:
To develop a multi-domain multi-modal AI-powered assistant based on a state-of-the-art AI agentic architecture to unify knowledge access, streamline tasks, and reduce bespoke tools.
Description:
Agentic AI architecture refers to the design and implementation of AI systems that exhibit autonomy, make decisions, and adapt their behaviour with minimal human oversight. Ground systems engineers and operators today face increasing complexity, fragmentation, and cognitive load across all the domains of ground segment engineering, mission operations and space safety. Users must navigate siloed tools, disconnected data sources, and overlapping processes, leading to inefficiencies and duplicated effort. This activity addresses these challenges by developing an AI-powered" co-worker" - an intelligent, context-aware assistant designed to support users across all phases of the ground segment, as the 'entry-point' and orchestrator for user interaction with ground segment systems. Existing solutions based on Large Language Models (LLM)-are already supporting users by ingesting and summarising information from the various systems. An agentic approach significantly enhances such solutions by adding dedicated and specialised integrations with external tools and data source as well multi-step task breakdown and orchestration. This approach provides the user with a proactive, goal-driven and autonomous support. Building on existing operational solutions already tested by various missions, this co-worker will unify access to knowledge, streamline task execution, and reduce the need for bespoke tools. It will enable users to interact with all systems using human language and receive meaningful, actionable responses.

The co-worker will be designed with intuitive interfaces and robust explainability to support decision-making and user confidence. It will integrate securely with MLOps platforms, supporting verification, validation (V&V), monitoring, and risk mitigation by design. The system will be based on a state-of-the-art AI agentic architecture, leveraging modular, explainable, goal-oriented agents capable of reasoning across heterogeneous data sources, orchestrating insights, and coordinating multi-step tasks. Employing multi-agent frameworks and model context protocol (MCP), the system will support task decomposition and specialisation, making it scalable and adaptive to varied user needs.

This activity is divided in phases and encompasses the following tasks:
Phase 1 - Minimum Viable Product Development (budget: 1250 k€, duration: 9 months):
- Selection of target use cases and appropriate architecture.
- Interface development to external tooling and data sources.
- Definition of the minimum HW and SW requirements.

Phase 2 - Validation (budget: 1250 k€, duration: 9 months):
- AI solution development for selected MVPs, creating prototype AI solution.
- Framework implementation, and adoption for MVPs and potentially existing AI solutions.
- Deployment of framework and prototype AI solutions into AI package for end-users.
- Assessment of performance; assessment of framework features as applied to each AI solution, impact of pain points.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 2.4 “AI-Automated content generation and AI-enhanced user interaction with data”.

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-803GA – Operational qualification and explainability of AI solutions

Budget: 2000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 7 - **TD** 9

Objective:
To develop a structured approach for the operational qualification of AI solutions and to validate this method with real-world AI applications in space missions, ensuring their reliability, traceability, and explainability.
Description:
As artificial intelligence technologies become increasingly integrated into ground segment operations — enabling advanced functions such as intuitive human-machine interaction, anomaly detection, mission planning, and autonomous system management — it is imperative to establish robust frameworks that ensure their deployment remains reliable and traceable. While developments and ECSS guidelines provide valuable guidance, practical methodologies for assessing AI readiness in mission contexts are lacking. This activity aims to define a structured approach for operational qualification of AI solutions and develop a qualification tool focused on validation and verification, uncertainty quantification, performance and robustness, traceability and lifecycle management. The solution will address core areas essential for trusted AI adoption, including uncertainty quantification to assess model confidence and data variability, tools for monitoring and managing AI system drift, and mechanisms for versioning and traceability across AI model updates. Verification and validation will be adapted to account for the characteristics of machine learning systems, such as non-determinism and data dependency.

Explainability methods will ensure AI outputs can be interpreted by operators and engineers, supporting trust and decision-making. Security aspects, including the resilience of AI models to adversarial inputs, will be considered as part of a comprehensive assurance approach. The structure will be tested with operational AI solutions, ensuring results align with the ESA Strategy 2040's goals of greater autonomy, resilience, and digital transformation.

This activity is divided in phases and encompasses the following tasks:
Phase 1 - Minimum Viable Product Development (budget: 750 k€, duration: 9 months):
- Identification of suitable approaches for qualification and explainability of AI components and solutions.
- Implementation of Proof of Concepts for selected approaches.
- Identification of appropriate architecture.

Phase 2 - Productisation Development and Validation (budget: 1250 k€, duration: 9 months):
- Selection of AI components and systems requiring qualification.
- Selection of AI use cases to be enhanced with explainability features.
- Creation of coherent qualification framework.
- Application of framework to selected cases.
- Assessment of performance; assessment of structure features as applied to each AI solution.

Deliverables: Report, Software

Application/Need Date: The use of artificial intelligence in systems mandates an approach to ensure the quality of the solutions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 5.2 “Verification and Validation techniques for AI systems”.

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-804GA – Structured data foundations for machine learning in space mission applications

Budget: 1500 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 6 - TD 9

Objective:
To create, curate, and disseminate standardized datasets for machine learning development and evaluation, including data collection, annotation, preparation, and validation for mission operations.
Description:
This activity aims to address the challenge of creating, curating, and disseminating standardized, representative datasets specifically designed to support the development and evaluation of Machine Learning (ML) algorithms and models in a repeatable manner. Developing AI solutions necessitates easy access to large quantities of high-quality data. Over the past few years, ESA has frequently received requests from both industry and academia for access to operational data from space systems operated by ESA. These stakeholders have shown a keen interest in leveraging this data to explore, develop, and train a wide range of AI solutions to augment or replace existing systems and processes. However, ESA has generally been unable to support these requests due to concerns regarding the security, commercial confidentiality, and personal data content of the ESA-operated equipment. This situation contrasts sharply with the open data policy applied to payload datasets, where data is typically made freely available. To partially mitigate this challenge in the context of spacecraft anomaly detection, ESA recently produced and publicly released the ESA Anomaly Dataset. This dataset has been viewed 5.4k times and downloaded 2.5k times between April and July 2025. It has also been the focus of several hackathons and competitions. This initiative is regarded as a significant success, demonstrating the high demand from industry and academia for real-world space system operational datasets. However, this dataset was produced through a one-off activity over 12 months, and the 13TB of data released is limited to only a few years of a small subset of curated and anonymized spacecraft telemetry data from just three ESA missions. It is evident that European industry would benefit from easy access to a curated, high-quality datasets spanning a wide range of space systems that is regularly updated. To achieve this goal cost-effectively, comprehensive data governance and processes must be defined and implemented through a suitable framework. These processes should facilitate the easy creation and dissemination of data while ensuring that security, confidentiality, and anonymization requirements are met. The provision of standardized datasets and evaluation benchmarks will enable the development, comparison, and validation of machine learning algorithms and models, thereby contributing to enhanced data-driven solutions for space mission operations.

The activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 600 k€, duration: 6 months):
- Identify and gather relevant data from various space systems, including historical and simulated mission data.
- Create rules for managing data securely, protecting confidentiality, and ensuring compliance with regulations.
- Implement a framework for data storage, processing, and sharing. Choose suitable technology for these tasks.
- Create framework prototype ensuring sub-set of selected data sources meet governance and format standards.
- Make data available to a limited environment.

Phase 2 (budget: 900 k€, duration: 6 months):
- Complete prototype of full data foundation framework.
- Ensure data sources meet governance standards and make data available to partners.
- Annotate and format data for machine learning, including cleaning and transforming it as needed.
- Use mission operations and existing algorithms to test and validate datasets, ensuring quality for AI development. Update datasets periodically based on new advancements and requirements.

Deliverables: Report, Software
Application/Need Date: Mature AI and ML solutions require reliable training data sets. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Enabling Artificial Intelligence for Space System Applications.

Artificial Intelligence

Ground Segment - System and Applications Engineering

GT1I-805GA – Security framework for AI applications

Budget: 1500 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 4 / 7 - **TD** 8

Objective:
To develop a framework consisting of tools, processes and datasets to mitigate the novel and specific security risks and vulnerabilities associated with AI applications for the space system domain.
Description:
Increasingly, advances in AI applications, models, and big data capabilities are making their way into productive applications. Roadmaps exist to introduce AI technologies into critical functional and operational domains. With new technology spin-ins like AI, cybersecurity must be accounted for, considering new attack surfaces and vulnerabilities. Currently, no specialized approach exists for systems with AI components, necessitating an ad-hoc approach for each system. To avoid redundant efforts, a coherent security approach for AI applications in the space domain should be defined. Common mitigation measures for identified threats and risks related to data and models need to be integrated into the AI lifecycle, ensuring a secure and AI-enabled infrastructure. Systems today demand increased security and even security certification. While security approaches for non-AI systems are well understood, AI components introduce new threats. This activity will guide developers in creating secure AI systems and assist project teams with certification bases for AI solutions. Building on earlier efforts and considering emerging policies and regulations, this activity will develop an 'MLSecOps' framework, i.e. a comprehensive set of practices and tools, processes, and datasets to address new vulnerabilities. These will be validated in a space system context. The targeted space system applications may range from engineering aids to security monitoring solutions.

The solution will be based on public standards and knowledge bases such as the ENISA report “Cybersecurity of AI and Standardisation,” MITRE ATLAS, and OWASP Gen AI Security Project. These offer a rich list of vulnerabilities and controls without considering the required adaptations for space mission specifics. This activity will establish a set of practices and create common tools for AI systems to provide standardized security features and functions, easing the certification process.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 600 k€, duration: 6 months):
- Assess previous developments and the make vs buy trade-off for the solution, considering emerging policies and standards.
- Design the AI Security Practices, including threat and risk assessments, threat and control catalogues, and assurance requirements for certified AI applications in space.
- Proof of concept activity for high-risk aspects of proposed practices in laboratory environments.

Phase 2 (budget: 900 k€, duration: 6 months):
- Develop and deploy the set of practices in representative space environments.
- Validate the solution with representative AI application use cases in the space domain.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 5.5 “End-to-end security”.

Artificial Intelligence

Ground Segment - Space Debris

GT1I-806SD – Adaptative operator-driven AI methods for collision avoidance decisions

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 11

Objective:
To enhance the reliability of AI models on the automation of collision avoidance manoeuvre decisions and design by direct and dynamic insertion of operator inputs into the models.
Description:
The lessons learned from collision avoidance operations showed the performance limitation of the AI/ML algorithms to support operators on automated collision avoidance manoeuvre decisions and design. This is mainly due to the quality of the underlying dataset (CDMs) that are used to train the ML models. If the AI algorithms can also learn directly from inputs of the operator provided in real-time/dynamically, these models can adapt their training with more reliable information from the operator. This prototype is meant to be exclusively used by the operator who drives the performance of the AI models according to his needs, allowing more transparency between the user and the software, hence higher reliability on the computed predictions. This dataset can be built up on over 10 years of operations in a mixed data-driven and human-in-the-loop environment. User-friendly guidelines on how to train the AI models need to be implemented, with a proper validation and testing of the feasibility.

The activity will develop an operator UI to decrease the human load implied due do collision avoidance operations based on AI/ML applied to the last decade of operations in a congested environment.

This activity encompasses the following tasks:
- Build an interface prototype that can automatically amalgamate different inputs (e.g. from difference surveillance data or collision avoidance services or other operators) towards an operator.
- Manage the data streams between the inputs and the processing tool that uses AI algorithms to assess conjunction risk and suggest collision avoidance manoeuvre.
- Assess and develop the capability of AI algorithms to dynamically take these inputs as for training when they are available and selection of the AI models with the best performance.
- Execute a performance test and validation of the selected algorithms with known scenarios, different input sources and real data. The test operator requires to have an available processing tool for this purpose.
- Implement user-friendly guidelines for the operator to know how to use the interface.

Deliverables: Report, Software

Application/Need Date: TRL9 in 2030 for all LEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 2.6 “AI for Space Safety”.

Artificial Intelligence

Ground Segment - Space Weather

GT1I-807SW – High performance modelling with AI and data assimilation

Budget: 2100 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 9

Objective:
The goal is to develop advanced space environment models using AI and data assimilation to improve prediction accuracy, real-time calibration, uncertainty measurement, and support for operational decision-making.
Description:
Space weather significantly affects spacecraft operations, navigation, and communications, making it crucial for space situational awareness and mission safety. Advanced numerical models simulate the Sun-Earth system and heliosphere dynamics, but their computational expense limits operational responsiveness, especially for continuous calibration with new observations. Models for spacecraft simulations also face similar challenges and often lack integration. Data assimilation, essential for enhancing model accuracy and forecast skill, is computationally demanding, especially for integrating multi-source data in real time. AI and ML techniques offer transformative potential, aiding in model deficiency learning, accelerating calibration, providing surrogate models, and delivering uncertainty quantification. Physics-informed AI and hybrid models combine physical models with AI, enhancing performance and scientific credibility. AI can also integrate environmental models into operational simulators, aiding flight dynamics and space debris risk assessments. The urgent need for advanced modelling techniques to deliver timely, accurate predictions highlights the limitations of current high-fidelity models. AI and ML can accelerate data assimilation, complement physics-based models, build surrogate models, provide uncertainty quantification, and support multi-source data fusion. Integrating AI-enhanced models into operational frameworks improves space weather forecasts' accuracy, responsiveness, and reliability, supporting real-time flight dynamics and debris risk analysis. The activity will focus on the development and demonstration of high-performance space weather and space environment modelling capabilities that combine high-fidelity physics-based models with AI-enhanced data assimilation, model calibration, and surrogate modelling.

The activity is divided in phases and encompasses the following tasks:
Phase 1 – Assessment, Architecture, and MVP Development (budget: 1000k€, duration: 12 months):
- Assess current capabilities and review the latest in space weather, space debris, and flight dynamics models.
- Design a solution to integrate AI-enhanced models with existing tools and systems.
- Prototype an AI-enhanced framework that can quickly process and integrate multi-source data, and modules for error learning, uncertainty quantification, and confidence intervals.
- Develop a basic version demonstrating AI-accelerated data assimilation, model calibration, and surrogate model generation for specific space weather scenarios.

Phase 2 – Operationalization and Deployment (budget: 1100 k€, duration: 12 months):
- Implement the AI-enhanced framework to process multi-source data in near real-time.
- Integrate AI modules for error learning, uncertainty quantification, and confidence intervals.
- Deploy AI models to speed up complex computations.
- Integrate these enhanced models into existing operational simulators for space weather prediction, flight dynamics, and space debris risk assessment.
- Demonstrate system capability in real-world scenarios.

Deliverables: Report, Software

Application/Need Date: Virtual Space Weather Modelling Centre (VSWMC), space debris and Flight dynamics
Mission Classification: alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 2.6 “AI for Space Safety”.

Artificial Intelligence

GNC, AOCS & Pointing

GT1I-808EC – Robustification of feature extraction and tracking-based image processing through machine learning

Budget: 700 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 5

Objective: To improve the performance of the pose estimation, overcoming the lack of robustness in feature extraction and tracking functions for Vision-Based Navigation (VBN) systems, using machine learning technique.
Description: Missions that are heavily relying on Vision Based Navigation (VBN), such as precision landers and rendezvous missions for applications like debris removal and in-orbit servicing, assembly and manufacturing, rely on relative pose knowledge to perform precision landing or proximity operations. The robustness of the pose estimation techniques is essential for operational safety but is often difficult to reach with VBN. As opposed to landing scenarios on mapped celestial bodies like the Moon, Mars, etc, or rendezvous with known spacecrafts where a digital model exists, landing on unknown bodies like asteroids or performing a rendezvous with a non-cooperative target (no markers or beacons) introduces difficulties in the VBN system in terms of performance and robustness of the solution, which can be solved by the careful identification and tracking of features that can be followed during a significant amount of time for getting a stable and continuous navigation solution. The interpretation of the visual information available from the images goes beyond conventional feature extraction and tracking. Deep Learning techniques could implement essential concepts from traditional VBN in a generic manner. When trained on large and diverse enough datasets, Deep Neural Networks (DNNs) could transfer well across tasks, e.g., from lunar landing to Mars landing or to spacecraft docking. To address the domain gap—primarily resulting from the reliance on synthetic training datasets due to the scarcity of real images from space operations—and to manage the wide range of operational situations in practice (such as varying lighting conditions, sensor noise characteristics, surface texture differences affecting optical properties, and optical artifacts), it is essential to develop and train DNNs for domain adaptation and randomization. This process should focus on key features while filtering out less relevant elements. Automated semantic segmentation can aid in handling specular surfaces, moving spacecraft components (such as solar panels and antennas), and partial occlusions during relative motion. DNNs should be integrated with classical systems in hybrid approaches, combining deep learning-based feature detection and tracking as measurements with traditional methods like Kalman filters or bundle adjustment for pose refinement. To enhance robustness, redundancy, and interpretability, classical techniques can be used alongside the flexibility of DNNs during both training and operational scenarios. Specific challenging areas could be the onboard update (at least partial) learning/training of DNNs or adaptive fine-tuning during different mission phases.

The activity encompasses the following tasks:
- Define the reference scenarios for the activity, together with subsystem requirements. The following scenarios shall be included: fly-around and motion synchronization with a target satellite in front of adversarial background and precision landing on the Moon, Mars or other celestial body.
- Trade-off and select a suitable algorithm pipeline and suitable HW for the subsystem. The HW shall comprise a camera and a companion processing unit, which shall both be representative of space HW.
- Develop the relative navigation subsystem including image processing, neural network(s).
- Validate the relative navigation subsystem on the reference scenarios in a numerical environment relying on representative synthetic images.
- Port the subsystem SW to the selected HW. Demonstrate replicability of the previously obtained results and compatibility with real-time processing
- Validate the relative navigation subsystem in a lab relying on mock-ups for the target and backgrounds.

Deliverables: Report, Software
Application/Need Date: Fly-around/ rendezvous missions and precision landing. **Mission Classification:** alpha, beta, gamma, delta
THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity F13 “Robustification of feature extraction and tracking-based image processing through machine learning”.

Artificial Intelligence

GNC, AOCS & Pointing

GT1I-809EC – Data learning model predictive control

Budget: 600 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 5

Objective:
To implement, test and port to space-hardware a Model Predictive Control (MPC) solution relying on data learning techniques
Description:
Recent prototypes of data-learning MPC systems have shown promise in use-cases characterised by complex or unknown dynamics, significant measurement noise, or challenging state constraints. In several space applications - such as those affected by flexible dynamics, sloshing, or atmospheric effects during satellite landing - these techniques can improve performance and robustness by compensating for hard-to-model plant behaviours, especially where constraints exist.

The activity foresees the development of a data-learning MPC solution such as Gaussian Process learning-based Model Predictive Control (GP-MPC), or a selected reinforcement learning method, its benchmark and test against a classical control approach, and the demonstration of its implementation in real-time on space hardware.

The proposed methods shall be fully compatible with modern GNC design practices that rely on automatic code generation. To validate flexibility and practical benefits, the developed methods shall be demonstrated on representative mission scenario featuring complex, constrained, optimal trajectory planning and control problems under uncertainty such as Mars heavy landers, and in-orbit servicing, assembly, and manufacturing missions requiring motion synchronisation.

The activity encompasses the following tasks:
- Review of state-of-the-art solutions and trade-off or justification of the retained solution from past prototyping.
- Selection of two space use-cases with the concurrence of the Agency.
- Implementation of the solution in a model.
- Benchmark and demonstration of the solution on the two use-cases against a classical controller with robust guarantees. This task shall use a high-fidelity dynamics environment capable of simulating the difficult dynamics of the selected use cases.
- Porting of the model to space-hardware and accelerator with possible hardware-software codesign to demonstrate real-time capabilities.

Deliverables: Report, Software

Application/Need Date: Mars heavy landers, in-orbit servicing, assembly, and manufacturing missions requiring motion synchronisation **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Artificial Intelligence

GNC, AOCS & Pointing

GT1I-810EC – Integrated large language models tools and simulation environments for faster AOCS design and prototyping

Budget: 700 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 5

Objective:
To develop a Large Language Models (LLM) based tool and simulation environment to assist rapid prototyping of AOCS architecture and design.

Description:
Traditionally, AOCS/GNC development follows a sequential and highly manual process: requirement engineering, use case analysis, worst-case scenario identification, design of system architectures, selection of sensors/actuators, and iterative design/prototyping of control algorithms. Tools such as MATLAB/Simulink, Python, and Model-Based Design frameworks are extensively used. However, the growing complexity and fast development requirements of modern space missions demands faster, more adaptive, and more intelligent design methodologies. Recent advances in Large Language Models (LLMs) and in various applications - from system engineering to structural design - offer a unique opportunity to augment the classical AOCS/GNC engineering process, providing capabilities like requirement refinement assistance, early-stage architecture suggestions, database-driven component selection / optimization, rapid prototyping of control algorithms, sensitivity analyses support, natural language interface with simulation environments. This activity will develop an LLM-powered toolset and simulation environment that supports and accelerates the AOCS design cycle, especially during early mission definition and prototyping phases.

This toolset and environment will:
- Assist Requirements Engineering: Help engineers refine and structure requirements and design-driving cases using LLM-based analysis.
- Support Worst-Case Analysis: Suggest and validate critical worst-case attitude/orbit scenarios.
- Propose Architectures: Generate first-cut AOCS architectures based on mission profile, constraints, and databases of sensors/actuators.
- Prototype Algorithms: Help quickly generate initial prototypes of attitude control and estimation algorithms in Matlab/Simulink, Python, or Julia.
- Conduct Sensitivity Analyses: Automate or semi-automate parametric sweeps to identify design sensitivities.
- Interact via Natural Language: Allow engineers to ask for simulation setups, parameter adjustments, and result interpretations in human language.
- Enhance Simulation Capabilities: Allow faster "what-if" exploration at system-level, including easier integration of new control strategies or sensor fusion techniques.
- Extend current environments (MATLAB/Simulink, Python-based frameworks, Julia-based optimization libraries).

The activity consists of the following tasks:
- Review of available LLMs and relevant coding / simulation interfaces.
- Environment design and identification of data resources.
- Prototype development.
- Two use case selection.
- Validation and demonstration.

Deliverables: Report, Software

Application/Need Date: To improve and accelerate the AOCS design cycle. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity F18 “Large Language Models for AOCS/GNC engineering”.

Artificial Intelligence

GNC, AOCS & Pointing

GT1I-811EC – Verification and validation of AOCS/GNC using AI on-board

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 5

Objective: To develop a verification and validation framework and dedicated processes for AOCS/GNC systems using AI techniques.
Description: Despite interest in the Neural Networks (NN) performance exhibited, neural networks does not feature reliability, robustness, efficiency and explainability at the level requested for real time application in AOCS and GNC systems, for adoption in mission-critical space applications. Broadly speaking, the use of AI techniques, NN or else, as part of on-board AOCS and GNC systems, would ask for a reformulation of the state-of-the-art Verification process, based on analytical formulation, frequency domain analyses, physical interpretations and deterministic behaviour. Hence, promising results in the use of AI for some functionalities/missions raise the need for development of verification and validation (V&V) tools to verify AOCS/GNC properties such as pointing/knowledge performance, robustness to unforeseen conditions. For example, Neural network verification refers to the problem of verifying whether the output of a neural network satisfies certain properties for a bounded set of input perturbations. A potential solution to those issues is the use of convex optimization for Neural networks, applying smooth approximations of the nonlinear activation functions to make the problems tractable without loss of conservatism. System robustness can be assessed by embedding adversarial noise and uncertainties ensuring the output map remains bounded. In the case of neural nets, major challenge is to develop and provide a series of efficient numerical verification and validation algorithms to resolve the large-scale type of decision problems imposed by the nature of neural nets. The nature of these algorithms results from a trade-off between computational speed versus accuracy. The activity will develop core building blocks to embed AI techniques in the AOCS/GNC systems development cycle, to enable their reliable, verifiable and efficient use for ground and more importantly on-board use. This will include the following blocks: AI-based algorithms or modelling requirements specification (incl. metrics, statistical confidence level, functional and validation requirements, etc.), hybridization of AI-based and standard control/estimation definition in terms of design and tuning, data products development to feed AOCS/GNC Learning process, Verification and Validation process for hybrid AOCS/GNC systems (to replace/complement stability analysis, robustness analysis, performance verification), availability of health status of AI-based algorithms. These building blocks are developed on one or several ad hoc use cases to be defined with the Contractor, securing the development of the building blocks on real cases. The aim is to develop standard framework and process, to support an end-to-end use of AI/ML algorithms and products in an AOCS/GNC system development cycle.

This activity encompasses the following tasks:
- Definition of AOCS/GNC systems to be considered as use cases for the activity.
- Trade-off of the most appropriate AI/ML techniques for AOCS/GNC systems (functional systems, AOCS/GNC equipment, AOCS/GNC validation facilities), including hybridisation with standard AOCS/GNC techniques.
- Development of AOCS/GNC systems requirements for identified cases (using both prior tasks to define representative set).
- Preliminary tuning of AOCS/GNC systems for identified cases, including failure monitoring, covering the selected AI/ML techniques design, tuning and possible training.
- Definition of the full verification and validation process and development of verification and validation framework for the defined AOCS/GNC systems.

This activity is proposed to be implemented in Bundle 5 – AI for verification and validation, together with the activity GT1I-813ED - Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform.
Deliverables: Report, Software
Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta
THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity F08 "AI-based GNC/AOCS systems validation and verification evolution".

Artificial Intelligence

GNC, AOCS & Pointing

GT1I-812EC – Enhanced on-board autonomy and agility using AI-based Guidance/Control optimization

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 5

Objective:
To develop, implement, test and port on processing hardware AI-based guidance and control suitable to AOCS applications requiring a high degree of on-board autonomy and/or agility.
Description:
Upcoming missions will require a higher degree of autonomy and agility of the AOCS sub-system, which challenges the capabilities of the traditional guidance and robust control approaches. New mission operational scenario requesting enhanced onboard, real-time autonomous decision (incl. guidance) and realisation (incl. control) are stemming from missions with real-time optimisation under constraints (e.g. optimal slews, optimal orbit manoeuvre), or with tight reactivity needs (e.g. agile missions, collision avoidance), or with adaptation need to coupled and complex dynamics in orbit (flexible appendages, sloshing). Recent advances in artificial intelligence architectures applied to AOCS/GNC have shown great promise in such contexts, enabling new levels of autonomy and performance when hybridised with deterministic techniques. Solutions based on neurocontrollers, imitation learning, reinforcement-learning and learning-based MPC for instance can offer significant complement and enhancement w.r.t. traditional approaches. Their assessment and adoption must now be accelerated via exposure to high-fidelity simulation environments and implementation on space processing hardware.

This activity aims to implement, test and port to space hardware guidance and control solutions using AI techniques (combined to non-AI techniques), suitable to embedded AOCS applications requiring a high degree of on-board autonomy and agility. The selected algorithms shall be benchmarked against state-of-the-art approaches, and their suitability shall be demonstrated by a real-time implementation on a space-graded computer. The emphasis shall be put on “frugal AI” solutions, under stringent computing resource constraints, to avoid inflating on-board computing hardware requirements. At least one of the following applications will be covered:
- High-agility attitude manoeuvre planning and execution.
- Optimal, autonomous and adaptable orbital manoeuvres with complex objectives (e.g. multi-objective optimisation, collision avoidance, formation control).
- On board target scheduling.
- Guidance in highly constrained attitude/orbit environment.
- Fault tolerant guidance/control.

This activity encompasses the following tasks:
- Review of state-of-the-art solutions, with a focus on frugal solutions compatible with embedded applications.
- Trade-off, selection and justification of the retained solutions based on fast prototyping.
- Selection and justification of two space use-cases, justified by current or future mission needs.
- Implementation of retained solutions.
- Benchmark and sensitivity analysis on high-fidelity benchmarks fully representative in terms of dynamics and sensor/actuator complexity, uncertainty distribution (typ. CDR-level benchmarks). The performance and stability of the developed algorithms shall be compared against state-of-the-art robust controllers.
- Real-time implementation on a space-graded AOCS computer to assess feasibility.
Deliverables: Report, Software

Application/Need Date: AOCS/GNC future missions in Earth Observation and Science. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity F05 “Near minimum-time autonomous AOCS with guidance and control neural networks”.

Artificial Intelligence

Avionics & EEE

GT1I-813ED – Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform

Budget: 700 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 1

Objective:
To develop a new concept for the Integration, Verification, Validation and Qualification phase to introduce ML & AI strategies for improving and speeding-up the process of validation and testing activities for data handling system in avionics platform test campaign.
Description:
In the fields of platform standardization needs, this initiative focuses on the creation and validation of a cohesive suite of integrated software tools aimed at automating and optimizing the verification and testing of the avionics hardware interfaces and data handling systems (e.g. OBC, Mass Memory, Reconfiguration Module, communication links, etc.). These tools will encompass the complete testing lifecycle, addressing test sequence generation, efficient processing and reporting of results, the compliance verification, and the identification and potential resolution of anomalies. Then, the integrated toolset will incorporate support for verifying coverage of the initial avionics requirements, and possibly TMTC database. By significantly reducing the time and cost associated with the test campaign while enhancing the precision and speed of problem detection, this effort directly addresses a crucial market demand for more efficient and dependable avionic subsystem testing solutions. By automating and optimizing key aspects of the test campaign – from test generation to problem identification and coverage checks – this initiative will address a critical need for more efficient and reliable testing solutions, ultimately reducing risks and costs.

The activity encompasses the following tasks:
-Developing a set of automated software tools to:
  - Generate test sequences on the base of input requirements, data handling design description, test plan, etc.
  - Process and manage test archives, producing comprehensive test results reports.
  - Identify issues with tests and/or effective anomalies and suggest potential resolutions to fix and close the test.
  - Implement support functionalities for verifying coverage of avionics hardware interfaces and data handling requirements as part of the test results analysis.
  - Evaluating the feasibility of an additional tool for emulating hardware units to aid in anomaly investigation, if needed.
- Integrating these individual software tools into a unified and cohesive suite that provides comprehensive support for the avionic and data handling test campaign.
- Rigorous testing and validation of the integrated toolset to ensure it meets defined performance targets.

This activity is proposed to be implemented in Bundle 5 – AI for verification and validation, together with the activity GT1I-811EC - Verification and validation of AOCS/GNC using AI on-board.

Deliverables: Report, Software

Application/Need Date: All missions. TRL6 by 2030 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Avionics Embedded Systems.

Artificial Intelligence

Avionics & EEE

GT1I-814ED – AI/ML framework for on-board inference

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 2

Objective:
To develop a flexible software library for efficient AI/machine learning inference and neuromorphic capabilities on space-grade hardware.
Description:
AI is being rapidly adopted in the space sector, creating a growing need for tools and software libraries optimized for space-grade hardware. This activity aims to develop a European space-related vendor-independent software library to support the execution of AI models on representative space platforms as primary objective traditional machine learning (ML) and as a secondary objective spiking neural networks (neuromorphic). Currently, no European inference software library exists that has been specifically tailored for space-graded hardware and can act as a reference solution reducing the complexity associated with the deployment of machine learning on-board. This activity will also support European strategic autonomy by developing a vendor-independent AI inference framework. Current solutions are largely proprietary and non-European, creating a dependency that this initiative seeks to eliminate. The proposed software library will prioritize flexibility and scalability, enabling the integration of increasingly complex models without requiring changes to the core architecture. It will also focus on
- adaptability across diverse hardware platforms,
- allowing its use in different types of devices with little effort required for its adaptation, and
- address the constraints of the space software, such as limited memory, specific memory management, deterministic execution.

To meet these objectives, the activity shall explore, among others, advanced compilation and optimization technologies, including LLVM compilers, Multi-Level Intermediate Representation (MLIR), and Apache TVM. These tools have shown strong potential in efficiently deploying machine learning (ML) models across heterogeneous computing environments. Furthermore, MLIR is emerging as a powerful technology not only to support the execution of classical ML solutions but also neuromorphic ones. The approach for requiring an open-source solution follows the existing strategy of other open-source flight software building blocks like operating systems (RTEMS and ROS), mathematical libraries (LibmCS) and multicore processing libraries (OpenMP). For machine learning most projects are using open-source solutions to support on-board inference but not yet tailored for space-graded hardware. Therefore, targeting an open-source inference library is considered to have a bigger impact on its adoption.

This activity encompasses the following tasks:
- Identification of relevant space-grade hardware and neuromorphic solutions.
- Selection of the most suited use cases covering spacecraft and robotic applications.
- Trade-off of machine learning inference frameworks.
- Adaptation of framework for selected hardware and compliance to flight software environments (e.g. RTEMS and ROS) and standards.
- Development of test framework based on identified use cases and validation on representative hardware including performance assessment to evaluate the framework's efficiency.
- Dissemination of the results of the activity.

Deliverables: Software, Reports

Application/Need Date: All missions. TRL 7 by 2030 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Enabling Artificial Intelligence for Space System Applications.

Artificial Intelligence

RF Payloads & Technology

GT1I-815EF – AI reconstructed radiation patterns from amplitude measurements

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 7

Objective:
To reconstruct the radiation phase pattern of an antenna, based on the type, the frequency range, the dimensions of the antenna and the database of years of past measurements using an AI algorithm.
Description:
Characterization of Antennas in anechoic chambers is based on the measurement of the amplitude and phase patterns. The phase is much more sensitive to the environment and needs a reference signal to have viable results. On the contrary, the amplitude pattern can be measured easily without any reference. Several systems to measure antennas are based on amplitude measurements only: drones, films, etc. due to the complexity to obtain a stable and robust reference for the phase measurements. The information contained in the phase measurements is very important and needs to be measured to perform the nearfield-farfield transformation, calculate the phase centre and the group delay. Several research has been conducted to retrieve the phase from the amplitude measurements without being successful so far. The Antenna Laboratory at ESTEC has measured different antennas for 40 years for a large frequency range using different measurement methods. Based on this database and the measurement of the amplitude pattern of an antenna under test, an AI algorithm will be developed to reconstruct the phase of the antenna under test and calculate the derived important parameters. The AI algorithm will be a fully connected neuronal network with supervised learning.

This activity is of interest for the space industry as the measurements of the solely amplitude patterns of antennas fast (results in seconds), accurate (not impacted easily by the drift) and easy to carry out (no reference signal needed). This will be the perfect tool to measure and validate antennas for satellite constellations. The developed AI algorithm could complete these measurements giving the estimate of the phase pattern and allowing the whole characterization of the antenna performances.

This activity encompasses the following tasks:
- Review of the measurement database at ESTEC identifying the type, the dimensions, the level of confidence which can be attributed to each measurement, etc.
- Development of an AI algorithm based on the database estimating the phase pattern of a very well characterized antenna on its field of view on a particular frequency range.
- Validation of the algorithm comparing measurements of phase in anechoic chamber and the results given by the algorithm.
- Extension of the algorithm to other frequency ranges, antenna types, etc.
- Recommendation for further developments and Roadmap.

Deliverables: Report, Software

Application/Need Date: Antenna Testing for all missions. TRL 5/6 by 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Array Antennas and Periodic Structures.

Artificial Intelligence

RF Payloads & Technology

GT1I-816EF – Testbed demonstration of payload processor applications using AI capable RF SoC Technology

Budget: 800 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
Develop a testbed to demonstrate AI-capable Radio Frequency System-on-Chip (RF SoC) technology for onboard payload processing. Evaluate interconnects, RF data interfaces, scalability and benchmarking.
Description:
Advances in SoC technology have enabled the integration of ultra-high-speed data converters with digital processors, allowing for efficient implementation through high integration. This efficiency is crucial for developing payload processors to support RF missions. Recent developments extend these capabilities by incorporating advanced accelerator units to execute AI algorithms on the same RF SoC device. Coupled with ultra-high-speed data converters, direct RF-domain interfacing/sampling with these AI-enabled devices is becoming possible.

In this activity a testbed will be developed using AI-capable RF SoC processor technology. The goal is to demonstrate improved efficiency in payload processing functions using this technology. The testbed will be benchmarked using AI models and algorithms designed to deliver real-time RF data insights (such as AI-enhanced beamforming, signal classification, target detection, sensor data optimization, interference detection and mitigation, edge computing, anomaly detection) based on user-defined requirements. AI algorithm performance will be validated using existing RF datasets. Most suitable fault mitigation methodologies shall be selected, and an assessment of thermal performance, power management and radiation tolerance shall be performed to assess suitability of such devices for a range of missions.

This activity encompasses the following tasks:
- Definition of application and requirements review.
- Breadboard preliminary design.
- Demonstration of solutions meeting key requirements.
- Breadboard detailed design and analysis.
- Digital, RF and thermal performance analysis.
- Definition of test and verification plan.
- Breadboard manufacturing and test.
- Benchmarking for the selected use case.
- Test against verification plan.
- Results analysis and evaluation.

Deliverables: Breadboard, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 3.5 “Revolutionizing Satellite Data Processing for Instant Insights”.

Artificial Intelligence

RF Payloads & Technology

GT1I-817EF – Sensor with AI based processing and quality improvement

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 5 - TD 1

Objective:
To develop an on-board processing algorithm for sensor quality improvement of high-level products using AI and demonstrate on breadboard of a representative sensor architecture these techniques.
Description:
Many commercial missions are moving toward onboard processing, in some cases even generating high-level products on-board the satellite. High-level products allow further processing of the sensor data and extract application-relevant information, at the same time making the entire system more sensitive to data quality degradation. Optical sensors can be affected by issues such as Modulation Transfer Function (MTF) degradation, defocusing, and ghost stray light. RF sensors can be degraded by jamming, signal compression, clutter, or interference from nearby sources, which can reduce the effective resolution and utility of the data.

Some of these issues can only be detected after launch or require long calibrations, making it crucial to have a solution in place on-board to generate and process higher-level products. The activity aims to develop an AI software/hardware solution capable of making immediate and autonomous adjustments to maintain sensor quality facilitates real-time, onboard processing. This ensures that the system maintains high performance, even when manual intervention is not feasible. For instance, sensors can leverage AI-based solutions to mitigate motion errors and environmental disturbances, while object detection algorithms can produce high-level situational awareness data. This approach optimizes system performance, enhancing overall functionality, and extending both availability and reliability. As part of this initiative, an onboard quality enhancement will be developed, focusing on representative hardware. This development aims to improve onboard processing capabilities, ensuring real-time adjustments to maintain optimal image quality across diverse environments.

This activity encompasses the following tasks:
- Identify target sensor types (e.g., optical, radio, radar) and typical degradation modes.
- Define mission scenarios for onboard generation of high-level products.
- Define functional architecture for high level product generation and correction on-board.
- Design the interaction between modules (e.g., degradation detection, correction, high-level analysis).
- Select, develop and train AI models for detection and correction of sensor degradation
- Implement techniques suitable for real-time onboard execution in a representative breadboard
- Test performance of critical functions under simulated mission scenarios and typical degradation conditions.
- Validate the algorithm implementation in hardware.

Deliverables: Breadboard, Report, Software

Application/Need Date: TRL 6 by 2027 for future EO (ERS, Phisat, national and commercial missions) and exploration missions with onboard processing capabilities **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.10 “AI sensors integration: Enhancing Sensor Capabilities with built-in AI”.

Artificial Intelligence

Structures, Mechanisms & Materials

GT1I-818MS – Stable diffusion algorithms and geometrical deep learning for space systems

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 24

Objective:
To develop denoising diffusion models, based on Stable Diffusion and Geometric Deep Learning (GDL), to improve analysis of complex geometric and spatial data in spacecraft and infrastructure development.
Description:
In space engineering, accurate sensor data and high-fidelity simulations are essential for designing and analysing spacecraft components such as thrusters, satellites platforms and launchers. However, sensor measurements often contain noise that obscures critical geometric deformation patterns needed for fatigue analysis. Traditional modelling approaches struggle to balance physical accuracy, geometric fidelity, and computational efficiency, especially in nonlinear regimes relevant to fatigue and hypervelocity impacts. Denoising diffusion models, particularly those based on Stable Diffusion and Geometric Deep Learning (GDL) stand out as a class of AI/ML methods which can create structure generatively and not based on simple parametric model parameters alone. This distinction already has revolutionized the field of medical drug design and will be essential for all types of material, structure and mechanisms designs.

Recent advancements in denoising diffusion models are transforming how we handle noisy sensor data and generate realistic 3D geometry. For example, these models clean up spacecraft vibration sensor data, making it more useful for fatigue analysis. They also enhance FEM and CFD mesh predictions in challenging conditions like turbulence and shock waves. When it comes to creating new 3D meshes, diffusion models can generate detailed, realistic structures based on text prompts, even if labelled aerospace data is scarce. By fine-tuning these models with geometric rules and compressing data into efficient formats, we achieve accurate, physics-compliant results quickly, supporting real-time design and analysis. Adversarial training ensures the synthetic meshes remain physically plausible, preserving essential geometric properties.

This activity encompasses the following tasks:
- Develop denoising diffusion models to enhance aerospace sensor data by removing noise while preserving critical geometric deformation patterns.
- Integrate probabilistic denoising and iterative refinement techniques to optimize FEM/CFD mesh predictions and air foil aerodynamic performance under complex flow conditions.
- Implement cross-modal conditioning to translate textual design requirements into corresponding geometric modifications and generate high-fidelity 3D meshes using Stable Diffusion.
- Embed physics constraints, such as Navier-Stokes equations via Physics-Informed Neural Networks, to ensure simulation outputs comply with fundamental fluid dynamics.
- Optimize computational efficiency by leveraging latent space representations and adversarial training to produce physically accurate, real-time synthetic meshes.

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.4 “Smart Manufacturing, Assembly, Integration and Testing”

Artificial Intelligence

Future Engineering

GT1I-819SF – AI accelerate ontology generation

Budget: 600 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 26

Objective:
To accelerate the development of the Space Systems Ontology leveraging Agentic Workflows, Natural Language Processing and Large Language Models.
Description:
The Space Systems Ontology (SSO) is an initiative launched in 2019. It allows interoperability among model-based and digital infrastructures and is therefore key to enabling the exchange of information between the various stakeholders involved in the development of a space system. It does so by formally modelling the semantics in a global data model expressed at conceptual level and is divided into various Universes of Discourse (UoD) (e.g., MBSE, requirements management, mission and operation, etc. but also for example digital procurement and quality assurance). The current developments of the Space Systems Ontology (SSO) are based on the existing information, tools and respective standards and are relying on discussions and agreements among various stakeholders and subsequent conceptual modelling.

The aim of this activity is to kick-start the definition of a new Universe of Discourse (UoD) by generating an initial draft ontology with Artificial Intelligence (AI). By leveraging agentic workflows, Natural Language Processing (NLP) and Large Language Models (LLMs), we can accelerate the identification of relevant concepts, their relationships and constraints within technical documents. Agentic workflows are automated processes in which AI agents autonomously plan, decide, and act to complete multi-step tasks with minimal human input, often coordinating with tools, systems, or other agents.

Hence this more automated process based on AI can significantly speed up the definition of new UoDs, ultimately leading to a quicker establishment of the Space System Ontology (SSO). This accelerated process is crucial for several engineering fields in the space domain.

This activity encompasses the following tasks:
- Conduct a state-of-the-art (SOTA) analysis of existing agentic workflows, and AI-based approaches to generate ontologies.
- Select the Universe of Discourses (UoDs) to be modelled as well as their respective available corpus of documents.
- To develop the architecture of the agentic workflow, testing and validating each agents in charge of subtasks such as data processing, entity recognition and relation extraction.
- To generate a draft UoD, including concepts of interest and their relations, maintaining semantic information extracted from documents.
- To evaluate the draft UoD produced.
- To quantify the impact of using NLP, LLMs and agentic workflows for the generation of ontologies.
- To define recommendations for future or complementary development.

Deliverables: Software, Report

Application/Need Date: All missions. TRL 5 by 2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Related to the Roadmap Model Based For System Engineering.

Artificial Intelligence

Future Engineering

GT1I-820SF – Spatiotemporal knowledge graph for insights from mission data

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 7 - TD 26

Objective:
To develop a robust and advanced data integration and analytics framework based on spatiotemporal Knowledge Graphs and Large Language Models.
Description:
A Spatiotemporal Knowledge Graph (STKG) is a sophisticated data structure that represents entities, their relationships, and how they change over both space and time. It integrates spatial (location-based) and temporal (time-based) dimensions into a graph format. STKGs are powerful tools for managing, reasoning over and extracting insights from complex and dynamic data. Many space applications rely on data evolving across both spatial and temporal dimensions, for instance, real-time event detection, disaster monitoring, constellation management, space situational awareness (SSA), space traffic management (STM), or cross-matching of astronomical events.

This activity aims to develop an innovative STKG pipeline specifically tailored to the space domain. Its goal is to unlock a competitive advantage by enabling deeper insights from space mission data, enhancing data interoperability and data fusion, and accelerating scientific discovery. The development of this pipeline should partly rely on Natural Language Processing (NLP) and Large Language Models (LLMs) method to process large amounts of data, complete the Knowledge Graph (KG), infer new relations and facilitate querying.

Creating this robust pipeline involves several key steps: (i) aggregating data from heterogeneous sources, (ii) mapping and validating the information into a STKG, (iii) building tools to interact with the STKG and extract new insights. This pipeline could for instance be applied to climate science, allowing for the tracking of changes in climate patterns and detecting emerging trends, or to science, linking observations of the same celestial object across different epochs and instruments. The main challenges would involve the harmonisation of data providing from different sources, various frames of references and semantics, handling spatiotemporal uncertainties, reasoning over uncertainty and uncomplete data, as well as challenges related to scalability of a larger KG.

This activity encompasses the following tasks:
- Conduct a state-of-the-art analysis to analyse the current applications of STKGs in the space field and other relevant fields to understand the latest advancements and trends.
- Select the appropriate tools and methods for building the STKG pipeline.
- Determine specific use cases for the validation phase, and to identify their respective relevant spatiotemporal data sources.
- Establish the ontology or schema for the STKG.
- Develop a use case agnostic framework to build a STKG relying on LLMs to enhance the integration and analytics capabilities.
- Evaluate the STKG pipeline across several use cases with an emphasis on the explainability of the pipeline's processes and outcomes.
- Benchmark the developed pipeline against existing methods or tools, to evaluate its effectiveness and identify potential areas for improvement.
- Provide recommendations for future or complementary developments.

Deliverables: Software
Application/Need Date: all missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 3.1 “Enabling scientific discoveries with AI”.

Artificial Intelligence

Future Engineering

GT1I-821SF – Generative design for early design phases

Budget: 900 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 26

Objective:
To use generative design and AI-based language tools to automate and accelerate early space mission design, enabling engineers to efficiently explore and evaluate a wider range of innovative solutions.
Description:
The early design phases of space missions and feasibility assessment are critical, as they set the foundation for the mission's success. During these phases, engineers must evaluate numerous design options and perform various trade-offs. Traditional design methods often limit the exploration of potential solutions due to time constraints and the complexity of the design space and systems. Generative design is a computational design approach leveraging algorithms, either Machine Learning-based or numerical, to automatically generate a wide range of optimized solutions based on defined goals, constraints, and parameters.

This activity will combine Generative Design and AI-based language processing approaches to expand the solution space of designs by partly automating the generation and evaluation of design options, thus supporting the experts in the early trade-offs and design iterations. This activity aims to accelerate the design process but also uncover innovative design solutions that would have been overlooked otherwise.

Expected input to the tool would include key performance objectives and constraints extracted from mission requirements, mission statement, study objective and mission drivers. Publicly available data may be used in the frame of this activity; however, the software architecture should enable the ingestion of additional and/or proprietary datasets. Expected outputs may include multiple optimized solutions and parameters that can support the system engineer in the consolidation of the mission architecture, mass and delta V budgets, Concepts of Operations (CONOPS), product trees, trade-offs (e.g., cost, mass, power, schedule, etc.). These outputs offer sufficient explainability for Users to trace how they were derived. The Generative Design part will focus on generating number-based outputs such as masses or delta-Vs, while the AI-language processing part will handle the interaction with the systems engineers and generate some text-based outputs such as risk tables or trade-off summaries.

The activity encompasses the following tasks:
- State-of-the-art analysis on generative design methods, AI-based language processing and their applications to space mission design and other relevant fields.
- Collect data sources for training and testing.
- Develop and test a mission agnostic and flexible pipeline to generate design options.
- Implement a frontend targeted at end-users.
- Investigate the generation of MBSE compatible outputs (e.g., ECSS-E-TM-10-25, Capella, SysML v2).
- Quantify the impact of applying Generative Design to the early design phases of spacecraft.
- Define recommendations for future or complementary development.

Deliverables: Software, Reports

Application/Need Date: All missions. TRL 7 by 2030 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 5.4 “Generative AI for data generation”.

Artificial Intelligence

Future Engineering

GT1I-822SF – On-the-fly data access for AI foundation models

Budget: 500 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 6 - TD 26

Objective:
To enable cost-effective large-scale training of AI Foundation Models on ESA and non-ESA data, by streaming data within widely used AI frameworks.
Description:
Access to a vast uniform archive of training data, easily processable by AI frameworks, is still one of the major blockers for the wide application of AI training for Earth Observation, Science and other space activities. Especially in the field of AI foundation models, large ESA datasets for AI training are built by physically replicating the data from the main data repository. With current technology solutions, such datasets can be defined in a virtual way, by simply storing pointers and metadata that allow to stage and instantiate the dataset when a model is being trained, directly via the AI frameworks used for the training, without the need to copy and duplicate such data in the training environment.

The objective of this activity is to create an on-the-fly data access library, to be integrated within widely used AI frameworks such as PyTorch and Tensorflow, allowing streaming access to the vast catalogue of data archives, from ESA Science, EO, Navigation and Telemetry. The activity will enable the effective use of ESA (e.g. ESA SpaceHPC) and non-ESA HPC (High-Performing Computing centre) resources for AI model training, by connecting it to the ESA Data Archives on-the-fly, without the need to move the data into the HPC.

The library is foreseen to be general, support ESA Data archives, starting with EO and then Navigation and Telemetry, and work in conjunction with existing streaming tools for non-ESA datasets to train very large AI models using ESA and non-ESA data. The first showcase for this tool will be supporting the creation of AI foundation models over Copernicus Sentinel-2 data, which can then be used to support general object detection over the entire archive of images.

The activity encompasses the following tasks:
- State-of-the-art analysis of ESA datasets, data models and their use, ESA SpaceHPC and High-Performing Computing Centres.
- To identify use cases: data, HPC(s) that will serve for the validation of the library developed during the activity.
- To define, design and develop the on-the-fly data access library for the selected data and HPC).
- To define, design and develop use cases to be deployed on the selected HPC(s).
- To integrate the on-the-fly data access library in the selected HPC(s) and deploy use cases on them.
- To validate the on-the-fly data access library by executing the use cases.
- To evaluate the advantages and drawbacks in the development, deployment and execution of applications.
- To define recommendations for future or complementary development.

Deliverables: Software, Reports.

Application/Need Date: All missions. TRL 6 by end 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Enabling Artificial Intelligence for Space System Applications.

Artificial Intelligence

Future Engineering

GT1I-823SF – AI library for holistic verification and validation of space systems

Budget: 1200 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 26

Objective:
To develop a library of AI models supporting the verification and validation of space systems and subsystems.
Description:
The verification and validation (V&V) of space systems and subsystems are critical activities that require a huge amount of effort and time to process and analyse all the data artefacts generated during the complete development process. Verification and validation activities can greatly benefit from the application of AI that has been proved particularly efficient in processing large volumes of data. Several solutions have already been successfully implemented on specific topics such as detecting anomalies is specific data sets. It is now time to use AI on more project data artefacts that are subject to verification and validation activities.
This activity aims to develop a library of AI models capable of analysing data and artefacts from a project to support the verification and validation process at system and subsystem levels. By adopting a holistic approach, the library will enhance the results of the verification and validation process. This activity aims at developing a first version of the library that will cover the analysis of the system and at least two subsystems.
The AI library shall be designed to be compatible with various test environments and data pipelines, ensuring a seamless adaptation and integration into existing tools and processes of different industries. The capabilities of the AI library shall be demonstrated on a representative project, through the integration in a toolbox or in an existing development environment. Concrete examples of models that could be part of the library:
- Verification of correct flow-down of requirements from system to subsystems and to design.
- Image Recognition and Tagging to identify and categorize images from tests.
- Intelligent Curve Analysis for insights.
- Anomaly Detection to detect irregularities and potential issues.
- Verification of compliance with respect to the list of applicable standards.
- Verification of correct closure of requirements by proposed design and tests (specification, procedures and results).
- Automatic generation of Validation Control Document.

Through these capabilities, the AI library will provide robust support for the verification and validation of space systems, ensuring accuracy and reliability in the V&V process.

This activity encompasses the following tasks:
- Review the state of the art of existing AI techniques and models able to support the verification and validation processes of space systems and related activities.
- Identify areas where the development of an AI model will be the most beneficial considering the expected reduction of effort, time and cost and the improvement of overall quality. Select at least three different areas for which to develop AI models to be included in the library considering availability of data. Identify the project used for the validation of the library.
- Design and development of the AI models to be included in the library. Creation of the library.
- Integration of the library into an existing development environment or design and development of a toolbox able to support the execution of the AI models contained in the library.
- Verification and validation of the library, including performance assessment to evaluate the AI models efficiency by executing models on the selected project.
- Creation of the data package for library distribution.
Deliverables: Software
Application/Need Date: All missions. TRL 7 by 2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.3 “System and subsystem verification, validation and qualification”.

Artificial Intelligence

Future Engineering

GT1I-824SF – AI based application for on-board planning and scheduling

Budget: 1200 k€ - **Duration**: 20 months - **Current / Targeted TRL:** 3 / 5 - TD 26

Objective:
To develop a flexible AI-based planning and scheduling application for on-board autonomy in space missions.
Description:
AI is increasingly being used for solving complex optimization problems, including planning and scheduling. This activity aims to develop a general-purpose planning and scheduling application that leverages AI techniques, such as reinforcement learning (RL), as alternatives or complements to classical approaches like linear programming, integer programming, and mixed-integer programming. The proposed application will be designed for space missions where communication with ground control is constrained and a high level of uncertainty exist, such as in planetary science operations, robotic exploration, or deep space missions. In these contexts, high levels of onboard autonomy are essential to ensure mission success and maximize mission return, especially in the face of communication delays, uncertainties, or unanticipated changes. The application shall be able to consider a variable number of different assets in the establishment of the planning. Typical assets are satellites being part of a constellation, different instruments on a spacecraft or different sensors and actuators of a robot. Therefore, the proposed solution shall provide enough flexibility to be adapted to different scenarios.

The activity will draw from recent progress in AI-based planning and scheduling applied to ground systems, as well as existing on-board planning and scheduling algorithms. The application will be designed to allow on-board deployment providing modular architecture that supports integration with future systems across a range of mission types.

This activity encompasses the following tasks:
- Review and analysis of classical and AI-based planning and scheduling techniques, focusing on on-board usage. Identification of relevant hardware, that could support the deployment of the algorithm including COTS and rad-hard/rad-tolerant hardware.
- Identify use cases where the AI-based application will be applied. At least, three different use cases will be proposed, to demonstrate the flexibility to adapt the framework to different situations. One for scientific mission, one for robotic exploration and one for earth observation.
- Design and development of the AI-based application, including the final selection of the onboard space hardware and the design specifications necessary for deploying the algorithm on it.
- Integration of the application into the different use cases, and deployment into the selected HW.
- Verification and validation of the application, including performance assessment to evaluate the application's efficiency, adaptability, and robustness compared with traditional planning and scheduling algorithms.
- Results dissemination.

Deliverables: Breadboard, Software, Reports

Application/Need Date: All missions. TRL 7 by 2030 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.7 “AI technologies for S/C on-board autonomy”.

Artificial Intelligence

Future Engineering

GT1I-825SF – Federate simulator for AI system verification and validation

Budget: 800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 26

Objective:
To develop a flexible, interoperable federated simulator, composed of a suite of software modules, for AI verification and validation, enhancing safety and robustness across various domains and applications.
Description:
Recent studies and use have shown that the nowadays usage of System-level simulators is insufficient for addressing the Verification and Validation of applications of AI in space applications.

This activity aims to design and prototype a modular federated simulator to better support the verification and validation (V&V) of AI technologies in space applications. As AI becomes increasingly used in space for both critical and non-critical applications, traditional simulation environments, which are typically monolithic and application-specific, struggle to handle the complexity and diversity of these emerging systems. The proposed activity addresses the current simulator limitations by adopting a modular approach with clearly defined interfaces, enabling the seamless integration of 1) dedicated verification and validation and 2) dedicated a modules). The core module will also coordinate and orchestrate the execution of all these federated modules. This approach enables the reuse of V&V techniques, and the seamlessly integration of new V&V module and application-specific modules, that can be developed by third parties, integrated and then executed in a federate manner, significantly reducing the time, resources, and complexity associated with the V&V of AI-enabled space systems.

This activity encompasses the following tasks:
- Comprehensive review and analysis of:
  - existing federated simulation architectures, and
  - AI verification and validation (V&V) techniques to be incorporated into the V&V modules. The techniques shall address different aspects of AI V&V, to be implemented into specifics V&V modules.
- Determine at least two distinct use cases to be implemented, including the data and AI solutions to be used as demonstrators for the V&V functionality of the federated simulator. These UC will be implemented into the application-specific modules.
- Design and develop the federated simulator, including:
  - Data management policy.
  - Secure and remote access policies to the simulator.
  - Design and development of the core module and their interfaces to allow the integration of new V&V and application-specific modules, and its coordinated execution.
  - Design and development of dedicated verification and validation modules (federated-plug-in), implementing specific techniques that can be integrated into the federate simulator, in accordance with the interface compatibility defined by the core module. The V&V modules shall be application-independent, enabling their reuse across different applications.
  - Design and development of dedicated application-specific modules (federated-plug-in), what will make use of the core module and the required V&V modules to perform the verification and validation of the specific UCs.
- Integrate the core module and the federated modules (plug-in modules previously developed) to perform the complete V&V of the selected UCs.
- Test and validate the complete federate simulator that integrates the core module and the federated modules.

Deliverables: Software

Application/Need Date: All missions. TRL 5 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 5.2 “Verification and Validation techniques for AI systems”.

Artificial Intelligence

Future Engineering

GT1I-826SF – Modular framework for enhanced perception and decision-making

Budget: 600 k€ - **Duration**: 16 months - **Current / Targeted TRL:** 3 / 5 - **TD** 26

Objective: To develop a software module based on visual agents to integrate and combine visual inputs processing and decision-making capabilities.

Description:

The use of artificial intelligence for computer vision tasks has been extensively explored and applied over the last decade, resulting in a myriad of industrial applications. However, these approaches have predominantly relied on traditional CNN-based systems, which are often rigid, task-specific, and lack decision-making capabilities. Recent advancements, such as multimodal large language models, visual transformers, and agentic workflows, have demonstrated the potential for more flexible and adaptable solutions, that can also combine both visual input processing and decision making. This activity aims to develop a software module that will:

- Define and implement a software layer (execution layer) to support the inference of visual agents and enable easy integration of new agents.
- Design and implement an agentic workflow to execute various computer vision tasks (i.e.: object localization, image segmentation) on top of the defined execution layer.
- Integrate decision-making capabilities into the agent implementation, leveraging the output of the vision tasks. For instance, planning and scheduling tasks or specific instruction to be executed, considering the output of the computer vison task processed by the agent.
- Identify and deploy the solution on space-representative hardware.

The software solution will integrate visual perception with decision-making capabilities that surpasses current functionalities provided by traditional convolutional neural network (CNN). Unlike traditional CNNs, the agentic software developed in this activity will utilize multimodal inputs that may include various types of sensors outputs, (e.g. optical, radar, time series, as textual instructions), enabling agents to perform diverse tasks. The software developed in this activity can impact multiple application domains, like Earth Observation missions. Additionally, the solution will enable onboard capabilities to be presented as functions or services that can be adapted based on evolving ground requirements. Therefore, at the system level (flight-ground), this solution will influence current interaction paradigms, allowing the ground segment to define new goals for a satellite dynamically. This is a key technology for the Space Domain Awareness (SDA) and for tracking activities on Earth, supporting disasters monitoring, civil protection, border control, and security applications.

The activity shall encompass the following tasks:

- Review and analyse traditional AI algorithms applied to computer vision tasks in space and identify new trends using multimodal transformers, visual transformers, and agentic workflows.
- Identify use cases where the visual agent framework will be applied. At least two different use cases shall be proposed to demonstrate the framework's flexibility to adapt to different situations and covering both the visual task and reasoning task to be implemented by agents.
- Design and develop the visual agent framework, including the final selection of onboard space hardware and the design specifications necessary for deploying the algorithm on it.
- Integrate the framework into the different use cases and deploy it onto the selected hardware.
- Verify and validate the framework, including performance assessment to evaluate the framework's efficiency, adaptability, and robustness compared with traditional algorithms.
- Disseminate the results of the activity.

Deliverables: Breadboard, Software

Application/Need Date: All missions. TRL 5 by 2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.7 “AI technologies for S/C on-board autonomy”.

Artificial Intelligence

Future Engineering

GT1I-827SF – AI enhanced digital twin simulators for operations and AIT

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 8

Objective:
To provide Digital Twin capabilities for dynamic spacecraft modelling in AIT and operations by integrating physics-based models, AI techniques, and mission data. The aim is to reduce manual calibration effort, improve accuracy throughout the spacecraft lifecycle, and enable faster, more reliable decision-making in testing and operations.
Description:
Spacecraft simulation environments are crucial for Assembly, Integration and Testing (AIT), as well as mission operations. Traditional simulators tend to be static and labour-intensive to update, making it hard to reflect the spacecraft's true, evolving state. As spacecraft become more advanced and mission timelines shorten, the need for high-fidelity, adaptable simulators grows. Digital Twins — combining physics-based models, AI surrogates, and real telemetry — present a promising solution. These advanced simulators provide a dynamic, accurate view of the system, improving early issue detection and decision-making throughout the spacecraft's lifecycle

This activity focuses on advancing Digital Twin-based simulation capabilities by integrating advanced calibration techniques to enhance the fidelity and dynamism of spacecraft simulators. Central to this work is the integration of heterogeneous sources, which include physics-based models, AI-driven surrogate models, telemetry, and test data, into a unified, continuously updated simulation framework. Each of these elements plays a vital role: physics-based models capture the fundamental laws governing the spacecraft's behaviour; AI-driven surrogate models provide fast, flexible approximations where full physics models are too computationally expensive; and telemetry and test data anchor the simulation in real-world measurements and operational realities. By fusing these diverse inputs, the Digital Twin evolves in step with the actual spacecraft, refining its estimates as new information becomes available. Through advanced state estimation methods, the simulator provides improved situational awareness for both ground operations and AIT processes. Incorporating real-time and historical data, the Digital Twin provides an accurate, dynamic model of the spacecraft, supporting decision-making and system evaluation under all conditions.

This activity encompasses the following tasks:
- Identify key AIT and operations scenarios that require enhanced simulation fidelity.
- Define requirements for dynamic calibration, data ingestion, and state awareness.
- Specify visualization needs (e.g., real-time status dashboards, anomaly indicators, state prediction timelines).
- Design the modular architecture to support hybrid models (physics-based + AI).
- Define data interfaces for ingesting telemetry, test data, and ground truth.
- Design visualization modules and integration points with user interfaces.
- Define the lifecycle management of the Digital Twin (model updates, validation hooks).
- Hybrid Models, Data Assimilation Engine combining physics-based and AI surrogate models.
- Visualization and Status Awareness Layer for Real-time spacecraft state estimation and predictions.
- Integration with mission environments.

Deliverables: Report, Software

Application/Need Date: TRL6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 2.3 "AI-Enhanced Modelling and Simulation (M&S) Capabilities".

Artificial Intelligence

Future Engineering

GT1I-828SF – Intelligent assistant for MBSE and simulation model generation, setting and evolution for mission preparation

Budget: 1800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - **TD** 8

Objective:
To develop an intelligent system to assist in the generation of system designs (with MBSE) and system simulators for space missions; and mission preparation, including data system and simulator configuration and procedures preparation.

Description:
Modern space missions face increasing complexity, making Model-Based Systems Engineering (MBSE) and advanced simulation environments crucial for effective management. However, building and maintaining these systems is still largely manual and time-consuming. With recent advances in generative AI—including large language models and symbolic reasoning—there is a major opportunity to automate these tasks. AI can help transform specifications into structured model elements, suggest simulation setups, map lifecycle data, and ensure consistency and alignment across tools and disciplines. When integrated with standardized modelling platforms such as the Simulation Modelling Platform (SMP), generative AI can dramatically reduce engineering effort and improve the flexibility and quality of the model-based process.
The project is structured in two main phases. Phase 1 will develop an AI-powered tool to assist systems engineers in creating, updating, and evolving system designs and simulators for space missions. This includes automated code generation, debugging facilitation, and consistent model updates using standardized interfaces, reducing manual workload and speeding up deployment. Phase 2 will deliver an intelligent system to support model configurations and mission preparation. This system will interact with previous mission data, map information across disciplines, and assist in preparing procedures and simulation models, acting as a "copilot" for engineers. The initiative will build on existing developments and use AI to automate model mapping and configuration generation, supported by a robust data foundation and toolchain.

This activity encompasses the following tasks:
Phase 1 (budget: 1000 k€, duration: 9 months):
- Identify relevant user needs and define system architecture (AI core, MBSE interface, simulator linkage).
- Implement intelligent assistant for MBSE and simulator model generation.
- Integrate SMP-based interfaces and AI suggestion engine.
- Enable tools to manage model evolution, propagate changes, and validate consistency across artifacts.

Phase 2 (budget: 800 k€, duration: 9 months):
- Develop assistant for mission preparation and configuration using generated models.
- Include procedures and data mapping capabilities.
- Interface with MBE Hub and OPOS data/services.
- Ensure semantic interoperability across tools and lifecycle phases.
- Apply tools in a representative mission case study (e.g., EO, telecom, or robotic mission).
- Demonstrate end-to-end workflow from design to mission preparation.
- Produce technical documentation and training material.

Deliverables: Report, Software

Application/Need Date: All missions. TRL 6 by 2028 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Model Based For System Engineering (2024) – Consistent with activity H08 "Intelligent assistant for mission preparation activities".

Artificial Intelligence

Future Engineering

GT1I-829SF – AI based sensor fusion

Budget: 900 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 26

Objective:
To implement a low-power AI-sensor fusion solution where raw data from multiple sensors is combined prior to AI-based prediction, generating both static and dynamic models of the environment.
Description:
The integration of AI-based sensor fusion has been successfully applied in the automotive industry, enhancing the reliability and accuracy of situational awareness. However, in the space domain, although some initial research has been conducted, current sensor fusion approaches heavily rely on dedicated co-processors, non-event-based sensors, and traditional AI algorithms, factors that limit their applicability. The current challenge is to develop a robust AI-sensor-fusion system capable of:
- Ingesting data from diverse sources, including both non–event-based sensors (e.g., radar, LiDAR, optical cameras, inertial sensors) and event-based sensors.
- Implementing a sensor fusion approach leveraging spiking neural networks (SNNs) or hybrid models combining SNNs and traditional neural networks (NNs).
- Deploying this solution on low-power neuromorphic hardware.

The added value of this activity lies in addressing a major limitation of current on-board sensor fusion systems, which depend on dedicated co-processors and conventional neural networks, leading to increased power consumption. In contrast, this activity proposes a unified hardware/software solution that leverages neuromorphic hardware and spiking neural networks, alongside event-based sensors, to deliver a more robust and energy-efficient alternative to systems relying solely on non-event-based sensors and traditional neural networks. The proposed solution aims to improve reliability across various data fusion domains, such as fusing optical and RF data for Earth Observation (EO) or enhancing situational awareness in Guidance, Navigation, and Control (GNC) applications, while remaining scalable from missions with generous power budgets down to small satellites with strict energy constraints.

The activity encompasses the following tasks:
- Review and analyse trends in AI-based sensor fusion and its potential applications in the space domain, with special emphasis on the benefits of incorporating event-based sensors. The review will also explore SNNs and hybrid (SNNs + traditional NNs) approaches to enhance robustness, reduce power consumption, and improve functional performance.
- Identify neuromorphic or alternative low-power hardware allowing the execution of even-based and SNNs solutions.
- Define at least three distinct use cases to demonstrate the benefits of the AI-based fusion approach (e.g., autonomous navigation, enhanced situational awareness, advanced remote sensing), including data requirements for training and validation.
- Design and develop the AI-sensor-fusion system, covering both the data fusion strategy and the implementation of SNN or hybrid algorithms for the identified use cases.
- Integrate the solution into each use case and deploy it on the selected hardware.
- Verify and validate the solution by assessing power consumption, hardware stack complexity, robustness, and overall performance against the defined objectives.

Deliverables: Breadboard, Software

Application/Need Date: All missions. TRL 5 by 2029 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Enabling Artificial Intelligence for Space System Applications (2025) – Consistent with activity 1.10 “AI sensors integration: Enhancing Sensor Capabilities with built-in AI”.

Artificial Intelligence

Future Engineering

GT1I-830SF – AI-powered assistant for SysML v2: enhancing MBSE

Budget: 600 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 4 / 7 - TD 8

Objective:
To enable automation and efficiency increase in MBSE processes by developing an integrated and interoperable environment leveraging SysMLv2 textual notation and Large Language Models.
Description:
Model-Based Systems Engineering (MBSE) is transforming how complex systems, like spacecraft, satellites, and ground control systems, are designed and managed. Instead of relying on static documents, MBSE uses digital models to represent system architecture, behaviour, and requirements.

The recent release of SysML v2, a major update to the Systems Modelling Language, introduces a textual notation and a standardized API. These features make it easier for software tools to interact with models programmatically and open the door to automation and intelligent assistance. At the same time, Artificial Intelligence (AI), especially Large Language Models (LLMs), has advanced rapidly. These models can understand and generate human-like text, making them ideal candidates for assisting engineers in navigating and manipulating complex digital models. Despite the benefits of MBSE, working with system models can still be time-consuming, repetitive, and difficult for non-experts. Engineers often face challenges such as:
- Repeating the same modelling tasks across projects.
- Difficulty in identifying reusable model components.
- Limited accessibility of model information for stakeholders unfamiliar with SysML.
- Inefficient ways to query and explore model data.

There is a clear need for a smart, AI-powered assistant that can help engineers interact with SysML v2 models more efficiently and intuitively—essentially, an AI co-pilot for MBSE.

This activity aims to develop and mature a digital assistant that leverages the SysML v2 API and AI technologies to enhance engineering productivity. The assistant will act as a bridge between human users and complex system models, offering support through:
- Automation of repetitive modelling tasks, such as creating standard components or updating model elements.
- Model generation and bootstrapping, where the assistant helps create initial models from templates or natural language descriptions.
- Knowledge base integration, enabling the assistant to identify and suggest reusable model constructs or libraries.
- Model synthesis for non-experts, by summarizing or translating model content into more accessible formats.
- Natural-language querying, allowing users to ask questions about the model and receive meaningful, context-aware answers.

The solution will build upon existing developments in related areas, such as the AI-powered Digital Assistant (AIDA), but will focus specifically on exploiting the capabilities of SysML v2 models.

This activity encompasses the following tasks:
- Analysis of user needs and use cases in terms of systems engineering with SysML v2.
- Assess the state of the art of (MB)SE assistants, SysML v2 technology and tools.
- Definition of the architecture for the copilot (tools used, interfaces, components to be developed, etc.).
- Development of components and integration of the copilot.
- Validation.

Deliverables: Report, Software
Application/Need Date: All missions. TRL 7 by 2027 **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Model Based for System Engineering.


Quantum
Technologies

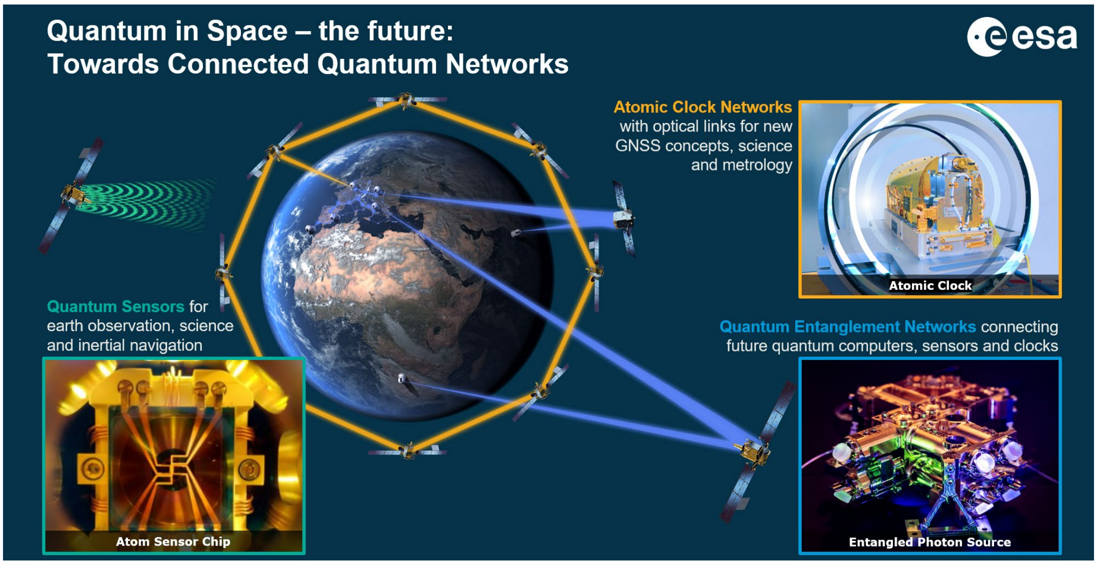
3.2.1. Introduction and selection criteria

Quantum Technologies activities of strategic importance for the future development of space system have been identified and form the basis of this compendium. The GSTP Quantum Industry Days organised this year facilitated and enabled the validation and consolidation of the Quantum Technologies proposed activities from ESA with Industry and Delegates.

Quantum technologies use quantum properties of physical systems (e.g. photons, atoms or more macroscopic objects) for practical purposes or scientific experiments. Examples of quantum phenomena are superposition, wave-particle duality, entanglement, and uncertainty. In the last century, these effects have been mostly studied in academic settings, but recent years have seen an increasing development towards real-world applications. To accelerate the development of industrial applications, over 30 BEUR global funding has been made available in the recent years Many governments have developed “quantum strategies” to strengthen development and deployment efforts especially in the areas of quantum computing, simulation, communication, and sensing & metrology.

Quantum technologies enable sensors with higher precision (e.g. gravity & acceleration, time & frequency, electromagnetic field), are leading to new type of devices (quantum computer, quantum cryptography) or enable new fundamental physics tests (exploring limits of quantum theory or quantum gravity). In many of these cases, space plays an important role.

In 2021, ESA listed quantum technologies as a strategically important topic as part of the ESA Agenda 2025, and it continues to be a priority within the ESA Strategy 2040. The ESA Quantum Technology Cross Cutting Initiative (QT-CCI) has been implemented to support industry, the scientific community and ESA programmes to strengthen the development and deployment of quantum technologies. The ESA QT-CCI Quantum white paper, published in 2022, discusses the landscape of quantum technologies for space, identifying areas of present and future applications and development goals. While the QT-CCI white paper reflects the overall ambitions of industry and the scientific community for all quantum technologies and applications, this compendium presents the specific implementations that shall be executed with the GSTP.



The figure above shows ESA’s vision for future quantum networks using atomic clocks, entanglement and quantum sensors. GSTP focuses on developments for clocks and sensor related quantum technologies while quantum communication and entanglement technologies are developed with ARTES ScyLight.

In the GSTP, ESA’s goals are to unlock new space capabilities with quantum clocks and sensing leveraging sensitivity, precision, stability and SWaP efficiency. Tailored instrument developments will de-risk future missions in the following technical areas:

High performance optical clocks. Developments on instrument level will be based on previous developments and enable optical clock ensembles on space platforms for metrology and scientific applications. Optical clocks will enable new tests of general relativity, the standard model of particle physics, search for time variation of fundamental constants and hunting for dark matter. Intercontinental clock comparisons using space-based time and frequency (T&F) links will help the monitoring of Earth's gravitational potential connected to phenomena like ocean circulation, ice melting, glacial adjustment, and land subsidence. Space based optical 'master' clocks can support time standards beyond the gravity variation limitations for ground-based metrology institutes.

Cold atom interferometer building blocks. Demonstrating critical requirements (performance, Size Weight and Power (SWaP)) of sub-systems will increase the TRL towards first applications for sensing or science applications. Matter-wave interferometers are utilizing the interaction of cold atoms with laser light to operate as inertial sensors or differential accelerometers with high long-term stability and well-known calibration factor.

Different quantum-based sensors for electro-magnetic fields or mass spectrometers will be developed to breadboard or engineering mode level. Rydberg atom-based quantum electric field sensors are capable of absolute measurement of blackbody brightness temperature. Quantum magnetometers have intrinsic vector-measurement capability and potentially uniquely low SWaP.

In addition, transversal technology developments will develop building blocks suitable for several types of instruments, especially to enable miniaturization using photonic technologies. This includes lasers, local oscillators, frequency combs, cryogenic detector technology and new tools to model quantum instruments in space environments. In the area of quantum computing, specific algorithms and tools (using terrestrial quantum computer hardware) will be developed to address computational use cases for space applications.

3.2.2. List of activities

Quantum Technologies

Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT1Q-800ED*	VCSELs for miniature quantum devices based on Rubidium vapour cells	1,600
	Total Avionics & EEE	**1,600**

*** Proposed Bundle 6 – Atomic clocks**
- GT1Q-800ED - VCSELs for miniature quantum devices based on Rubidium vapour cells
- GT1Q-820EF - Low noise ultra miniature local oscillator for chip scale quantum devices

Future Engineering

Programme Reference	Activity Title	Budget (k€)
GT1Q-801SF	Quantum computing algorithms development for space applications	750
	Total Future Engineering	**750**

Ground Station Engineering

Programme Reference	Activity Title	Budget (k€)
GT1Q-802GS	Hybrid ground clock-ensembles: toward the adoption of an operational ground optical clock	3,000
	Total Ground Station Engineering	**3,000**

Optics, Robotics & Life Sciences

Programme Reference	Activity Title	Budget (k€)
GT1Q-803MM	Laser sources for 480 nm (510nm) for Rb / Cs Rydberg sensors	1,000
GT1Q-804MM	Optical atomic clock for in-flight experiments	3,000
GT1Q-805MM	Quantum mass spectrometer using levitated nanoparticles	3,000
GT1Q-806MM	Optical vector magnetometer based on the Hanle effect	850
GT1Q-807MM	Miniaturization of nitrogen-vacancy center-based magnetometers	500
GT1Q-808MM	Reliable agile light-conditioning for quantum-sensors in space	500
GT1Q-809MM*	Continuous beam generation of ultra-cold atoms	2,000
GT1Q-810MM	Physics package for a hybrid cold atom interferometer and electrostatic accelerometer	2,000
GT1Q-811MM*	Robust high-flux source for ultra-cold atoms	2,000
GT1Q-812MM	Full-octave, optical waveguide integrated frequency comb with low SWaP	2,000

GT1Q-813MM	Nitrogen-vacancy vector magnetometer with low power consumption	500
GT1Q-814MM*	Agile laser system for cold atom interferometer technology with low SWaP	2,000
GT1Q-815MM*	Versatile control unit for cold atom interferometer technology	1,000
GT1Q-816MM*	Inertial reference platform for ultra precise cold atom interferometer inertial sensing	1,000
	Total Optics, Robotics & Life Sciences	**21,350**

*** Proposed Bundle 7 – Cold atom interferometer payload development**
- GT1Q-809MM - Continuous beam generation of ultra-cold atoms
- GT1Q-811MM - Robust high-flux source for ultra-cold atoms
- GT1Q-814MM - Agile laser system for cold atom interferometer technology with low SWaP
- GT1Q-815MM - Versatile control unit for cold atom interferometer technology
- GT1Q-816MM - Inertial reference platform for ultra precise cold atom interferometer inertial sensing

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1Q-817EP	Space environmental effects models and tools for quantum technologies	500
	Total Power Systems, EMC & Space Environment	**500**

RF Payloads & Technology

Programme Reference	Activity Title	Budget (k€)
GT1Q-818EF	Portable self-calibrated Rydberg electric field probe for antenna measurements	500
GT1Q-819EF	Compact high frequency local oscillator generation	2,000
GT1Q-820EF*	Low noise ultra miniature local oscillator for chip scale quantum devices	700
	Total RF Payloads & Technology	**3,200**

*** Proposed Bundle 6 – Atomic clocks**
- GT1Q-800ED - VCSELs for miniature quantum devices based on Rubidium vapour cells
- GT1Q-820EF - Low noise ultra miniature local oscillator for chip scale quantum devices

Thermal

Programme Reference	Activity Title	Budget (k€)
GT1Q-821MT	Cryocooler optimised for superconducting nanowire single-photon detectors	850
	Total Thermal	**850**

	Total Quantum Technologies	**31,250**

3.2.3. Description of activities

Quantum Technologies

Avionics & EEE

GT1Q-800ED – VCSELs for miniature quantum devices based on Rubidium vapour cells

Budget: 1600 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - TD 23

Objective:
To design and manufacture VCSELs suited for spaceborne quantum devices.

Description:
Vertical cavity surface emitting lasers (VCSELs) offer many advantages in terms of production, reliability and performance compared to other laser diode structures and are widely used in applications ranging from communications to spectroscopy. VCSELs have been one of the enabling elements for miniaturization of atomic sensors such as clocks and magnetometers. However, these applications have some specific performance requirements compared to VCSELs manufactured for terrestrial telecommunication applications. Most notably they need to be single mode (in frequency, spatial mode and polarization) and tunable to spectroscopic line of interest for the targeted atomic transition. In addition, given the very integrated design of the hosting devices, (proximity with micromachined hot vapor cell), it is required to have a higher than usual maximum operating temperature limit.

Despite a plethora of VCSELs being available on the commercial market, they are largely targeting terrestrial telecommunication applications (therefore different wavelengths, not suitable spatial/polarization modes) and not these types of quantum applications. In addition, the space environment and reliability challenges result in unavailability of European sources, and more generally nothing has been evaluated or qualified for space application.

This activity aims to develop 795nm single mode VCSELs suitable for integration in miniaturized devices based on warm Rubidium vapor cells, to be used in space.

This activity encompasses the following tasks:

- Definition of VCSEL performance requirements to target needs of Rb based miniaturised devices.
- Design and manufacture of VCSEL.
- Testing performances.
- Performing accelerated vacuum operation test.
- Complete an evaluation program of the component prior to qualification.

This activity is proposed to be implemented in Bundle 6 - Atomic clocks, together with the activity GT1Q-820EF - Low noise ultra miniature local oscillator for chip scale quantum devices.

Deliverables: Prototype, Report

Application/Need Date: Missions with atomic sensors such as clocks and magnetometers. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Future Engineering

GT1Q-801SF – Quantum computing algorithms development for space applications

Budget: 750 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 26

Objective:
To develop and assess quantum computing algorithms, methods and related tools for solving specific space related computational challenges with higher performance and reliability than the state of the art achieved with the classical technologies.
Description:
Quantum computing technologies are progressing rapidly, with early-stage commercial and academic platforms offering capabilities relevant to space applications. As one of the strategic sectors expected to benefit from quantum advantage, the space domain offers complex, high-impact challenges that can guide algorithm development and platform requirements. Lessons learned from these applications will be transferable to other sectors, enabling a broader quantum technology ecosystem. This activity aims to ensure European industry is quantum-ready and positioned to lead in the global race for space-enabled quantum technologies. It supports European strategic autonomy by fostering sovereign capabilities in quantum algorithm development, hybrid integration, and secure processing architectures tailored to ESA or nationalmission needs. The initiative also strengthens Europe’s industrial competitiveness by enabling use-case-driven innovation and demonstrating technical leadership in both terrestrial and space environments. This activity aims to encourage the usage of quantum computing potentials to address the challenges in space missions and technologies. It is expected to extensively use the precision and fast calculations to solve in the most innovative ways challenges related to, mission planning books, AI integration, cryptography, and onboard processing, optimisation algorithm for power batteries, among others. Technology focus areas can include but are not limited to quantum-enhanced simulation for propulsion, materials, thermal, power, satellite station keeping, radiation and orbital dynamics, optimization of tools for mission planning, routing, scheduling, debris avoidance and predictive maintenance, quantum Machine Learning (QML) for Earth Observation, anomaly detection, and FDIR, design of quantum-native pre-measurement processing flows, leveraging intrinsic quantum dynamics to condition mission-critical data within the quantum domain—prior to readout or classical interface, digital twin powered by quantum computing.

This activity encompasses the following tasks:
- Development and benchmarking of quantum algorithms tailored to space-relevant challenges considering gate-based, annealing or hybrid quantum computing methods
- Ground-based demonstrators for performance validation on quantum computers and/or classical high-performance computers
- Development of a software tool or product enabling the exploitation of the quantum algorithm for one or more specific space related use cases
- Analysis of the expected performance scaling with future generations of quantum computers

The tasks shall include clearly defined use cases and assess how quantum approaches—including hybrid methods and error mitigation techniques—perform under realistic computational loads and space environments, compared to classical baselines.

Deliverables: Software, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Big Data from Space (2023) – Consistent with activity B09 “Processing EO/Science products and Big Data using Quantum methods”.

Quantum Technologies

Ground Station Engineering

GT1Q-802GS – Hybrid ground clock-ensembles: toward the adoption of an operational ground optical clock

Budget: 3000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 7 - TD 12

Objective:
To develop and validate a pre-operational optical clock and deployment blueprint, enabling integration into critical infrastructure.

Description:
Optical clocks have demonstrated exceptional frequency stability and long-term accuracy, outperforming conventional microwave-based standards such as Active Hydrogen Masers (AHMs) and Cesium (Cs) clocks. Despite this, their use in operational ground-based infrastructures remains limited, primarily due to environmental sensitivities, integration complexity, and a lack of system-level validation.

The aim of this activity is to bridge the gap, between laboratory performance and field deployment and aim to transition optical clock technologies into pre-operational systems. A trade-off analysis of available optical clock technologies (e.g., optical lattice clocks, ion-based clocks, and cavity-stabilized lasers) is needed to select a suitable candidate for integration into hybrid timekeeping systems. The selected technology will then be developed into a pre-operational prototype capable of reliable performance under real-world conditions. Operational methods will be developed to ensure seamless integration with existing timing infrastructure based on AHMs and Cs clocks, including strategies for frequency steering and ensemble control. The optical clock will be validated in representative operational environments, enabling a realistic assessment of robustness, maintainability, and deployment readiness. A detailed deployment blueprint will be delivered to support commercial-scale rollout and standardization across high-precision timing infrastructures.

This activity encompasses the following tasks:
- Perform a trade-off analysis and select the most suitable optical clock technology for pre-operational deployment.
- Define interface and integration requirements for hybrid operation with Active Hydrogen Masers (AHMs) and Cesium (Cs) clocks.
- Develop operational procedures for frequency steering, performance monitoring, and redundancy handling.
- Manufacture ensemble compatible optical clock prototype.
- Validate the optical clock system under operational conditions at representative operational environments.
- Assess environmental, power, and infrastructure constraints relevant to deployment of the optical clock.
- Deliver a detailed blueprint for future operational deployment and commercial adoption.

Deliverables: Prototype, Report, Software

Application/Need Date: Missions using optical ground clocks. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-803MM – Laser sources for 480 nm (510nm) for Rb / Cs Rydberg sensors

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 17

Objective:
To develop an engineering model of Blue Laser source for next generation (Cold Atom) Rydberg Sensor.

Description:
Highly excited atoms, often known as Rydberg atoms, display extreme sensitivity to external electromagnetic (EM) fields, including the electric fields associated with black-body radiation (BBR). In addition to their use in quantum computing, Rydberg atoms can be configured for sensing applications. Such sensors make use of non-destructive read-out of Rydberg states, which is implemented optically, including fluorescence detection of low-lying Rydberg states. A range of different Rydberg sensing modalities have been developed; these include those developed for microwave standards and imaging in the Terahertz regime to sensitive digital communications techniques in radio and microwave bands. Next generation of Rydberg sensors, using Cold atom techniques, will allow better sensitivity and selectivity of the frequency range to be probed.

This activity encompasses the following tasks:
- Technology evaluation and state of the art for blue Laser system for Rb or Cs Rydberg sensors.
- Consolidation of technical requirements given application and preliminary design with down-selection of the best specie.
- Detailed Design of the laser system at 480 nm (Rb) and/or 510 nm (Cs).
- Manufacturing, integration and functional verification.
- Development plan to qualification, test campaign and outcomes.

Deliverables: Engineering/Qualification Model, technical reports

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-804MM – Optical atomic clock for in-flight experiments

Budget: 3000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 3 / 6 - TD 17

Objective:
To develop an optical atomic clock engineering Model (EM) with a precise, robust and technically mature system approach that can provide enhanced frequency, time and performance in the dynamic environment of space.
Description:
The optical systems can operate thanks to its higher carrier frequency (@ circa 400 THz) compared to the radio frequency (rf) system (operating @ circa 10 GHz only). In addition, the insensitivity to radio frequency interference (RFI) is a significant advantage of optical systems especially in the secure system applications.

The system performance of these optical devices had led to the need to develop a new set of verification techniques and associated systems to exploit the full potential of these optical systems. A redefinition of the SI second, based on an optical transition, is another major milestone expected to be implemented within a decade. Thus far, these systems are only operated in the well-controlled environments of thermally and vibrationally controlled laboratories. Typically, these systems are highly complex technical systems involving a substantial interplay between individual sub-system elements and requiring the operation by experienced scientists. Full system integration is viewed as a highly beneficial approach if the desired performance can be achieved with such a technological approach. Such system integration can lead to a level of simplification not possible with conventional breadboarding techniques. It will, therefore, have a direct impact on which clock type is proposed in this activity. Therefore, a detailed trade-off procedure (performance vs system simplification) must be undertaken enroute to selection of the clock type to be selected and developed to an EM.

To exploit these high-performance optical systems in a dynamic environment, an implementation of several system simplifications shall be addressed and solved. A high-performance frequency comparison infrastructure over short and long distance shall enable the proper dissemination and comparison of these ultra stable frequencies between optical clocks in development in Europe. An Engineering Model optical clock for flight experiments shall be developed leading to a high-performance flight clock in space,

This activity encompasses the following tasks:
- Up to 2-3 parallel technology shall be evaluated Lattice and single trapped ion systems.
- Atomic species to be selected from Sr, Rb, Sr+, Hg, Ca+, Al+ (quantum logic).
- Use of Integrated Photonic Systems to develop lasers and frequency and phase modulation device.
- Re-use of previous ESA-funded Atom/ion trapping/cooling sub-system approaches.
- Shared resources for Ultra-stable interrogation (sub Hz) laser system.
- Optical Frequency Comb using a planar photonic integrated approach.
- Implementation of an end-to-end fabrication modelling and design approach.

Deliverables: Engineering model, Reports

Application/Need Date: Science missions (e.g. Atomic Experiment for Dark Matter and Gravity Exploration (AEDGE), 2040 time frame). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-805MM – Quantum mass spectrometer using levitated nanoparticles

Budget: 3000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 17

Objective:
To design and manufacture a quantum mass spectrometer based on levitated nanoparticles EQM and test in relevant environment.

Description:
Levitodynamics Experiments for Neutral Density Sensing can collect real-time, in-situ data in space at submicrometric spatial and millisecond temporal scales. This sensor consists of levitated, neutral or charged, silica nanoparticles, whose levitodynamics in all three dimensions are tracked optically with quantum-limited precision. Quantum sensors are notoriously sensitive to their environments, typically requiring meticulous isolation. In this application, the quantum sensor exploits this high sensitivity of the trapped nanoparticle to the ambient environment to provide a novel measurement capability for in-situ particle density parameters. This allows to monitor the local chemical composition, with application of observations of the Earth's upper atmosphere, the interstellar medium composition or the gas composition of planets.

Contrary to standard mass spectrometer, this instrument is self-standing and does not need onboard accelerometer nor the knowledge of the satellite cross section for atomic density measurement. It can also provide fundamental physic test by probing the foundations of quantum physics at the interface with gravity.

This activity encompasses the following tasks:
- Technology review and consolidation of technical requirements for several use cases.
- Detailed design of a selected use case.
- Breadboarding and performance environment testing.

Deliverables: Breadboard, Reports

Application/Need Date: Earth Observation, Space Weather and Science missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-806MM – Optical vector magnetometer based on the Hanle effect

Budget: 850 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - TD 17

Objective:
To manufacture and test an optical vector magnetometer EQM based on the Hanle effect.

Description:
As fluxgate magnetometers show temperature and long-term drifts, they are usually accompanied by optical scalar magnetometers, which provide a reference point for calibration. Autonomous calibration of a vector magnetometer will reduce the SwaP and complexity of measuring magnetic field in space. A recent example is the Coupled Dark State Magnetometer (CDSM), part of the MAGSCA instrument aboard the JUICE mission, which measures the strength of the magnetic field by Coherent Population Trapping (CPT).

Utilising CPT in the so-called Hanle configuration to measure all vector components of the magnetic field optically allows a potential space instrument in the future, which can be combined with the existing CDSM. On its own, the 3D-Hanle magnetometer would complement the capabilities of the scalar CDSM by allowing near-zero magnetic field measurements in all vector components with negligible temperature and long-term offset drifts. Combining the two instruments in one would thus provide a sensor that can perform self-calibration during flight. This reduces the SWaP by having a single instrument and avoids science interruption needed for calibration process.

This activity encompasses the following tasks:
- Identify, develop and test of technology elements which are important for a compact and reliable instrument design.
- Design, development and test of a TRL 6 level sensor (qualification model) which can be verified under relevant environmental conditions (vibration, thermal-vacuum and radiation) for an outer planet mission.
- Design, development and test of a compact engineering model of the front-end electronics, including all relevant analog and digital control electronics based on EEE parts, which are functionally and electrically representative to flight components.
- AIT of the EQM optical vector magnetometer.

Deliverables: Engineering Qualification Model, Reports

Application/Need Date: All missions (e.g. L4 mission to Saturn/Enceladus, Space Situational Awareness).
Mission Classification: alpha, beta, gamma

THAG Roadmap: Photonics (2023) – Consistent with activity C05 “Optical Vector Magnetometer Based on The Hanle Effect”.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-807MM – Miniaturization of nitrogen-vacancy center-based magnetometers

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 17

Objective:
To develop a space proven NV-Centred Diamond magnetometer with reduced SWaP.

Description:
Magnetic field measurements from low Earth orbit are important for understanding Earth's magnetosphere, monitoring of space weather, and studying geophysical processes. These measurements provide insights into solar-terrestrial interactions and help mitigate risks to both orbital infrastructure and terrestrial systems. NV -centres magnetometers are recent developments that have been successfully implemented on the ISS. Next to their scientific interest, one of their applications lays in using the magnetic field for AOCS purposes in space. Using a magnetic map, the full attitude of the spacecraft can be followed. This has been demonstrated during the first Ariane launch where one of these systems could follow the first minutes of the trajectory of the rocket. There is now the necessity to increase the TRL level of such system to allow them to be utilized for navigation of spacecrafts or on other bodies (which have a magnetic field).

The development shall be focused on miniaturisation and reduction of resources (mostly power) and on increasing the resolution of the magnetometers to pT.

This activity encompasses the following tasks:
- Consolidation of technical requirements given application in miniature quantum devices and preliminary design.
- Detailed design of the magnetometer.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

Deliverables: Breadboard, Reports

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-808MM – Reliable agile light-conditioning for quantum-sensors in space

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 17

Objective:
Design and develop an engineering model for light-conditioning of Laser system for quantum inertial sensor to overcome the related challenges of frequency and power agility of such Laser system.

Description:
Atomic quantum sensors are showing, in the last decade, a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations. Whereas classical accelerometers suffer from high noise at low frequencies, cold atom interferometers (CAI) are highly accurate over the entire frequency range and do not need any external calibration. CAIs use the wave properties of atoms. For this purpose, atomic clouds are trapped and cooled in vacuum chambers using high precision lasers and magnetic fields. Manipulating the quantum states of the atoms using light pulses allows for creating interferometers with large spacetime areas being highly sensitive to inertial forces and to measure the desired properties of the environment or the atom properties themselves. Laser system needed are quite complex and need very stringent specifications with respect to frequency agility and power agility of the laser beams.

The main advantages of using double pass AOMs in CAI laser system are the following:
- Agile Frequency Switching / Tuning.
- Perfect Frequency Shifting.
- Agile Power Control.
- Large Power contrast.

This activity encompasses the following tasks:
- Technology assessment including the state-of-the-art mapping.
- Consolidation of technical requirements given application in low SWaP quantum devices and preliminary design.
- Detailed design of the fibered double pass AOM system.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

Deliverables: Engineering Model, Reports

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-809MM – Continuous beam generation of ultra-cold atoms

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 17

Objective:
To design and develop an engineering model for continuous generation of ultra-cold atoms beam to overcome the related challenges CAI intrinsic noise and aliasing

Description:
Atomic quantum sensors are showing in the last decade a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations. Whereas classical accelerometers suffer from high noise at low frequencies, cold atom interferometers (CAI) are highly accurate over the entire frequency range and do not need any external calibration. CAIs use the wave properties of atoms. For this purpose, atomic clouds are trapped and cooled in vacuum chambers using high precision lasers and magnetic fields. Manipulating the quantum states of the atoms using light pulses allows for creating interferometers with large spacetime areas being highly sensitive to inertial forces and to measure the desired properties of the environment or the atom properties themselves.

The ultra-cold atoms used for CAI are generated sequentially and therefore the inertial measurement is not continuous. This limitation compared to standard accelerometer leads to substantial noise due to aliasing. Having a continuous bean of ultracold atoms would circumvent this issue and leads to a unprecedent accurate and sensitive inertial instrument.

This activity encompasses the following tasks:
- Technology evaluation and state of the art. Consolidation of technical requirements given application in EOP and SCI and preliminary design.
- Detailed design of the continuous cold atom beam.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 7 – Cold atom interferometer payload development, together with the activities GT1Q-811MM - Robust high-flux source for ultra-cold atoms, GT1Q-814MM - Agile laser system for cold atom interferometer technology with low SWaP, GT1Q-815MM - Versatile control unit for cold atom interferometer technology and GT1Q-816MM - Inertial reference platform for ultra precise cold atom interferometer inertial sensing.

Deliverables: Engineering Model, Report

Application/Need Date: Earth observation and Science missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-810MM – Physics package for a hybrid cold atom interferometer and electrostatic accelerometer

Budget: 2000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 4 - TD 17

Objective:
To design and develop an overall instrument combining the benefits of a cold atom interferometers (CAI) accelerometer with an electrostatic one to validate the hybridisation concept in a campaign test.

Description:
The merging of two major technologies of inertial sensing, relying on electrostatic forces and on atom interferometer, show promising performances for future space missions dedicated to Earth observation (Gravimetry, EEI, Thermosphere density studies), Science mission and deep space navigation. Each of these two types of instruments have their own assets which are, for electrostatic sensors, their demonstrated short-term sensitivity and their maturity regarding space environment and for CAIs, amongst others, the absolute nature of the measurement and therefore no need for calibration processes. These two technologies are to a certain extend very complementary, where a hybrid sensor that combines both technologies performances could become a game changing for future space missions. The development of the sensor head merging the 2 technologies are of high priority.

This activity encompasses the following tasks:
- Technology evaluation of the different sub-systems of a hybrid CAI/Electrostatic accelerometer payload.
- Detailed design of the physics package (Including ultracold atom generation and atom chamber, CAI laser system, control unit, electrostatic accelerometer with its electronic unit,)
- Manufacturing, integration and functional verification.
- Test campaign (of the physics package), outcomes and development plan to qualification.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-811MM – Robust high-flux source for ultra-cold atoms

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - TD 17

Objective:
To design and develop an engineering model for a high flux of an ultra-cold atom source to increase maturity and robustness towards an engineering qualification model.

Description:
Atomic quantum sensors are showing in the last decade a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations. Whereas classical accelerometers suffer from high noise at low frequencies, cold atom interferometers (CAI) are highly accurate over the entire frequency range and do not need any external calibration. CAIs use the wave properties of atoms. For this purpose, atomic clouds are trapped and cooled in vacuum chambers using high precision lasers and magnetic fields. Manipulating the quantum states of the atoms using light pulses allows for creating interferometers with large spacetime areas being highly sensitive to inertial forces and to measure the desired properties of the environment or the atom properties themselves.

The sensor is ultimately limited by the quantum projection noise that scales with the square root of the number of ultracold atom. On top of that the ultra-cold atoms used for CAI are generated sequentially and therefore the inertial measurement is not continuous. This limitation compared to standard accelerometer leads to substantial noise due to aliasing. Having a high flux source will limit the aliasing on one hand and improve the sensitivity of the instrument to outperform classical accelerometer performance. On-ground hardware is existing and needs now to be tested and qualified for a space mission.

This activity encompasses the following tasks:
- Evaluation of the technology and assessment of the state-of-the-art.
- Consolidation of technical requirements given application in low SWaP quantum devices and preliminary design.
- Detailed Design of the high-flux source for ultra-cold atoms.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 7 – Cold atom interferometer payload development, together with the activities GT1Q-809MM - Continuous beam generation of ultra-cold atoms, GT1Q-814MM - Agile laser system for cold atom interferometer technology with low SWaP, GT1Q-815MM - Versatile control unit for cold atom interferometer technology and GT1Q-816MM - Inertial reference platform for ultra precise cold atom interferometer inertial sensing.

Deliverables: Engineering Qualification Model, Reports

Application/Need Date: Earth Observation and Science missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-812MM – Full-octave, optical waveguide integrated frequency comb with low SWaP engineering model

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - TD 17

Objective:
To design a low-power and reproducible self-referenced OFC compatible with compact optical frequency standard architectures.

Description:
Optical frequency combs (OFC) are being considered for many different space applications like optical atomic clocks, distance measurement, time and frequency transfer, optical instrument calibration, multiple wavelengths source and ultra-low phase noise microwave generation. Current technology involves a highly nonlinear optical fiber to broaden the spectrum and a nonlinear bulk crystal for frequency doubling which requires high power and free space optics. New technology based on photonics integrated circuit offers the prospect to massively reduce this power needs which would lead to frequency comb with better size, weight and power (SWaP) properties.

The activity will focus on developing a low-power and reproducible self-referenced OFC, to fulfil time and frequency applications, in particular miniature optical frequency standards.

This activity encompasses the following tasks:
- Literature review on OFC and consolidation of technical requirements for optical frequency standards with performances compatible with miniature optical clocks.
- Detailed Design of the self-referenced OFC.
- Manufacturing, integration and functional verification.
- Test campaign, including environmental testing evaluation and assessment on radiation sensitivity for usage in Space.

Deliverables: Engineering Model, Reports

Application/Need Date: All missions with optical clocks. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-813MM – Nitrogen-vacancy vector magnetometer with low power consumption

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 17

Objective:
To design and development of a Low power optimisation of NV-Centred Magnetometer for Space applications.

Description:
Magnetic field measurements from low Earth orbit are important for understanding Earth's magnetosphere, monitoring of space weather, and studying geophysical processes. These measurements provide insights into solar-terrestrial interactions and help mitigate risks to both orbital infrastructure and terrestrial systems. NV -centres magnetometers are recent developments that have been successfully implemented on the ISS. Further developments on NV centred instrument are required, combining different approaches: electrical excitation instead of laser, polarimetry instead of bias magnetic field and diamond chip technology for integrating primary optics and Microwave antenna.

This activity encompasses the following tasks:
- Technology review of the state of the art.
- Selection of use cases with enhanced techniques to reduce power consumption and preliminary design.
- Detailed Design of the NV magnetometer for the selected use case.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

Deliverables: Breadboard, Report

Application/Need Date: Earth Observation, Science and Space Situational Awareness missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-814MM – Agile laser system for cold atom interferometer technology with low SWaP

Budget: 2000 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 4 / 6 - TD 17

Objective:
To design and develop an engineering model for agile laser system for CAI with low SWaP to overcome the related challenges of mass and power consumption of CAI payloads

Description:
Atomic quantum sensors are showing in the last decade a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations. Whereas classical accelerometers suffer from high noise at low frequencies, cold atom interferometers (CAI) are highly accurate over the entire frequency range and do not need any external calibration. CAIs use the wave properties of atoms. For this purpose, atomic clouds are trapped and cooled in vacuum chambers using high precision lasers and magnetic fields. Manipulating the quantum states of the atoms using light pulses allows for creating interferometers with large spacetime areas being highly sensitive to inertial forces and to measure the desired properties of the environment or the atom properties themselves.

This activity encompasses the following tasks:
- Technology evaluation and state of the art. Consolidation of technical requirements given application in low SWaP quantum devices and preliminary design.
- Detailed Design of the laser system.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 7 – Cold atom interferometer payload development, together with the activities GT1Q-809MM - Continuous beam generation of ultra-cold atoms, GT1Q-811MM - Robust high-flux source for ultra-cold atoms, GT1Q-815MM - Versatile control unit for cold atom interferometer technology and GT1Q-816MM - Inertial reference platform for ultra precise cold atom interferometer inertial sensing.

Deliverables: Engineering Model, Reports

Application/Need Date: Earth Observation and Science missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-815MM – Versatile control unit for cold atom interferometer technology

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 5

Objective:
To design and develop an engineering model for a system control hardware and software of a space CAI instrument and validate it on a CAI experiment.

Description:
Atomic quantum sensors based on Cold Atom Interferometer (CAI) are showing in the last decade a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations. CAI requires a very complex and robust control of the various subsystem of the instrument to provide the ultimate performances of such inertial instrument. In particular the control unit needs to receive and send control signal to various units (Laser system, vacuum system, microwave generation). In a very fast and precise manner, which requires a high reliable HW electronics. Depending on the application this signal needs to be versatile to cope with various environment and science needs, which requires an efficient reprogrammable SW.

It is proposed to design and develop an EM of a control unit that will be able to perform a CAI with high tunability on the performance of the CAI instrument.

This activity encompasses the following tasks:
- Technology review of cold atom control unit hardware and software.
- Identify, develop and test of critical electronics which are important for a reliable and precise control of CAI sub-system.
- Develop and test SW reconfigurability for various CAI use cases.
- Detailed design of the control unit system.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 7 – Cold atom interferometer payload development, together with the activities GT1Q-809MM - Continuous beam generation of ultra-cold atoms, GT1Q-811MM - Robust high-flux source for ultra-cold atoms, GT1Q-814MM - Agile laser system for cold atom interferometer technology with low SWaP and GT1Q-816MM - Inertial reference platform for ultra precise cold atom interferometer inertial sensing.

Deliverables: Engineering Model, Reports

Application/Need Date: Earth Observation and Science missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Optics, Robotics & Life Sciences

GT1Q-816MM – Inertial reference platform for ultra precise cold atom interferometer inertial sensing

Budget: 1000 k€ - **Duration**: 16 months - **Current / Targeted TRL:** 3 / 5 - TD 5

Objective:
To design and develop a high performance AOCS combined with tip-tilt mirrors to overcome CAI limitation to rotation and vibrations.

Description:
Atomic quantum sensors based on Cold Atom Interferometer (CAI) are showing in the last decade a significant improvement on ground for accuracy and precision measurement of accelerations, gravity gradients or rotations.

One limitation of such instrument is the inertial environment. While vibrations will add noise, as to any inertial sensor instrument, rotation of the instrument leads to fundamental loss of contact of the interferometer and therefore loss of the measurement. Therefore, there is a specific need to control and measure any external inertial forces by means of an ultra-precise AOCS and rotation control of this inertial reference. The more sensitive the instrument is the more stringent the Inertial reference platform must be.

This activity encompasses the following tasks:
- Technology evaluation and state of the art. Consolidation of technical requirements given strict performance of the quantum devices and preliminary design.
- Detailed design of the Inertial reference platform.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 7 – Cold atom interferometer payload development, together with the activities GT1Q-809MM - Continuous beam generation of ultra-cold atoms, GT1Q-811MM - Robust high-flux source for ultra-cold atoms, GT1Q-814MM - Agile laser system for cold atom interferometer technology with low SWaP and GT1Q-815MM - Versatile control unit for cold atom interferometer technology.

Deliverables: Breadboard, Reports

Application/Need Date: Earth Observation and Science missions. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Power Systems, EMC & Space Environment

GT1Q-817EP – Space environmental effects models and tools for quantum technologies

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 7 - TD 4

Objective:
To analyse in detail the level of risk of space environmental effects (from radiation, plasma and microparticles) for the chosen quantum technologies in relevant orbital environments, develop new models and software for their analysis, and propose relevant testing campaigns and design strategies to mitigate these effects.

Description:
The various quantum technologies proposed under the GSTP Quantum Roadmap are sensitive and hence susceptible to intereference and other effects of the space environment they fly in (high-energy radiation, charging, microparticle strikes) that could drastically decrease their performances. Some of these effects may be similar to the effects observed on traditional technologies, while others can be novel due to the nature of the technologies employed and/or synergistic phenomena between the different environmental contributors.

This activity will analyse in detail the nature and level of risk of space environmental effects for the chosen quantum technologies in relevant orbital environments and mission categories, develop new models and simulation software for their analyses, and propose relevant testing campaigns and design strategies to mitigate such effects.

This activity encompasses the following tasks:
- Review and analysis of high interest quantum technology domains regarding their current and foreseen status in the context of space environmental effects. Such technologies include, but are not limited to, cold atom interferometers, high performance optical clocks, and photonic building blocks for quantum devices.
- Analyses of space environments (high-energy radiation, plasma/charging, microparticles) and their effects in relevant orbital environments. Possible effects include interference, noise, total radiation dose and Single Event Effects, displacement damage, charging and discharges, induced electric fields from internal charging, momentum changes, and possible novel and/or synergistic effects thus far not seen in classical technologies.
- Development of new space environmental models and software tools to analyse and make prediction of this effects on Quantum payloads in detail, with the aim of including the qualified models in the established analyses frameworks
- Establish recommendations for testing activities and design strategies to mitigate such effects.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

RF Payloads & Technology

GT1Q-818EF – Portable self-calibrated Rydberg electric field probe for antenna measurements

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 7

Objective:
To design and realize a first European portable self-calibrated SI-traceable Rydberg atom-based Electric field probe.

Description:
Accurate measurement of electric fields is crucial for the development and operation of RF communication systems, particularly in space missions. Conventional antennas need fine calibration for high-integrity reception. The calibration is performed by placing uncalibrated antennas in a known EM (electromagnetic) environment that is measured via a well-calibrated antenna which creates a dependency. Whereas Rydberg based receivers allow precise measurements of RF signals which is directly related to Planck's constant and hence can self-calibrate by relying on SI (international system of units)-traceable atomic standards

The activity aims to develop a portable, self-calibrated Rydberg atom based electric field probe.

The probe is based on the interaction of RF-fields with Rydberg atoms, where alkali atoms are excited optically to Rydberg states and the applied RF-field alters the resonant state of the atoms. For this probe, the Rydberg atoms are placed in a glass vapor cell. The vapor cell acts like an RF-to-optical transducer, converting an RF E-field to an optical frequency response. The probe utilizes the concept of Electromagnetic Induced Transition (EIT), where the RF transition in the four-level atomic system causes a split of the transition spectrum for the pump laser. This splitting is measured and is directly proportional to the applied RF field amplitude. Therefore, by measuring this splitting, we get a direct measurement of the RF E-field strength. This technique could allow traceable measurements over a large frequency band including 1-500GHz. The new approach has the following benefits: 1) it will allow direct SI units linked RF E-field measurements, 2) It is self-calibrating due to atomic resonances, 3) it will provide RF field measurements independent of current techniques, 4) the probe will not perturb the field during the measurement, since no metal is present in the probe.

This activity encompasses the following tasks:
- Assessment of different Rydberg atom-based methods of detection of RF fields.
- Requirement consolidation and performance assessment.
- Detailed concept design with analysis of critical functions.
- Development of the breadboard and verification in relevant environment.
- Development of a technology roadmap for an EQM.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Quantum Technologies

RF Payloads & Technology

GT1Q-819EF – Compact high frequency local oscillator generation

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 4 - TD 17

Objective:
To design, measure and characterize a highly integrated Local Oscillator based on a microwave photonics generation to scale up to sub-THz bands to address the quantum device applications such as LO (local oscillator) interrogation for atomic clocks.

Description:
The current state-of-art in terms of interrogation circuit used in ion trap atomic clocks are based on multiplication and synthesis stages, increasing the complexity and penalizing the phase noise performance as well associated considerable size and increased power consumption.

A highly integrated LO (Local Oscillator) based on photonic generation through a PIC (Photonic Integrated Circuit) for compact clock technologies can overcome the issue with existing accommodation constraining multiplication and synthesis chain, and improve overall integrability with benefit on the system Size, Weight and Power (SWaP),

LO shall be able to generate a signal from 40 to 200 GHz range, actual frequency to be defined within this range and given identified quantum application. Phase noise shall be comparable to the one achievable with state-of-the-art RF multiplication chain, but with compact design and lower power consumption. The LO shall be able to accept a low frequency signal (10 to 100 MHz) to inherit long term frequency stability. .

This activity encompasses the following tasks:
- Technology and quantum application review.
- Breadboard design.
- Manufacturing and testing.
- Conclusions and future roadmap.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

RF Payloads & Technology

GT1Q-820EF – Low noise ultra miniature local oscillator for chip scale quantum devices

Budget: 700 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 6

Objective:
To develop a low power consumption local oscillator in HF band (e.g. 10 MHz) with low phase noise to be used in miniature quantum devices. Power consumption shall be lower than 200 mW to preserve low-power properties of target quantum devices.

Description:
Ultra compact local oscillator with low power consumption is a key building block for current and future miniature quantum devices. For current devices (e.g. chip scale atomic clocks), it can significantly improve the output phase noise with benefit for system constraints, combining in one single device frequency stability and spectral purity of the output signal. For future quantum devices, the adoption of pulsed scheme for frequency standard performance improvements, requires the usage of low-phase noise local oscillators to avoid degradation due to aliasing effects (i.e. Dick noise).

Literature demonstrates that vacuum sealing design or other low-power solutions applied to Oven Controlled Cristal Oscillator (OCXO), can achieve outstanding phase noise with power consumption within the identified target. Other technologies (e.g. high-quality factor MEMS) may be also evaluated. Objective of this activity is the development of an engineering model (EM) able to fulfil low phase noise and power consumption requirements over standard OCXO operating temperature ranges. The EM shall be able to operate in vacuum conditions and selected technology shall be justified against LEO (Low Earth Orbit) radiation environment. Mechanical robustness against shock and vibration shall be also demonstrated by analysis.

This activity encompasses the following tasks:
- Technology evaluation and state of the art. Consolidation of technical requirements given application in miniature quantum devices and preliminary design.
- Detailed design of the local oscillator.
- Manufacturing, integration and functional verification.
- Test campaign, outcomes and development plan to qualification.

This activity is proposed to be implemented in Bundle 6 - Atomic clocks, together with the activity GT1Q-800ED - VCSELs for miniature quantum devices based on Rubidium vapour cells.

Deliverables: Engineering Model, Reports.

Application/Need Date: All missions embarking chip scale atomic clocks, e.g. future LEO-PNT missions.
Mission Classification: beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Frequency and Time Generation and Distribution.

Quantum Technologies

Thermal Division

GT1Q-821MT – Cryocooler optimised for superconducting nanowire single-photon detectors

Budget: 850 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - TD 21

Objective:
To miniaturise and optimise a 15K-20K cryocooler to enable a small platform implementing high temperature superconducting nanowire single-photon detectors.

Description:
High Temperature Superconducting Nanowire Single-Photon Detectors are an enabling technology for quantum applications on small satellite platforms. Current state-of-the-art requires at temperature below 4K for operation, but recent developments have shown that using high Tc (such as MgB2) superconductors for the nanowire material can enable a simpler cryocooling system.

The affordability of the technology, as well as the Size Weight and Power (SWaP), are blocking points for future adoption, and further development is needed in both areas. Specifically in this activity the SWaP shall be reduced by at least 50% when compared to a current spaceborne 4K Cooling System. By moving towards 15K-20K, a cryocooler as simple as a two-stage regenerative type cooler can be implemented, and with the right optimisations, power and mass can be further minimised. Another possible optimization is the use of an IDCA (Integrated Detector Cooler Assembly) architecture. In this activity an engineering model will be designed built and tested.

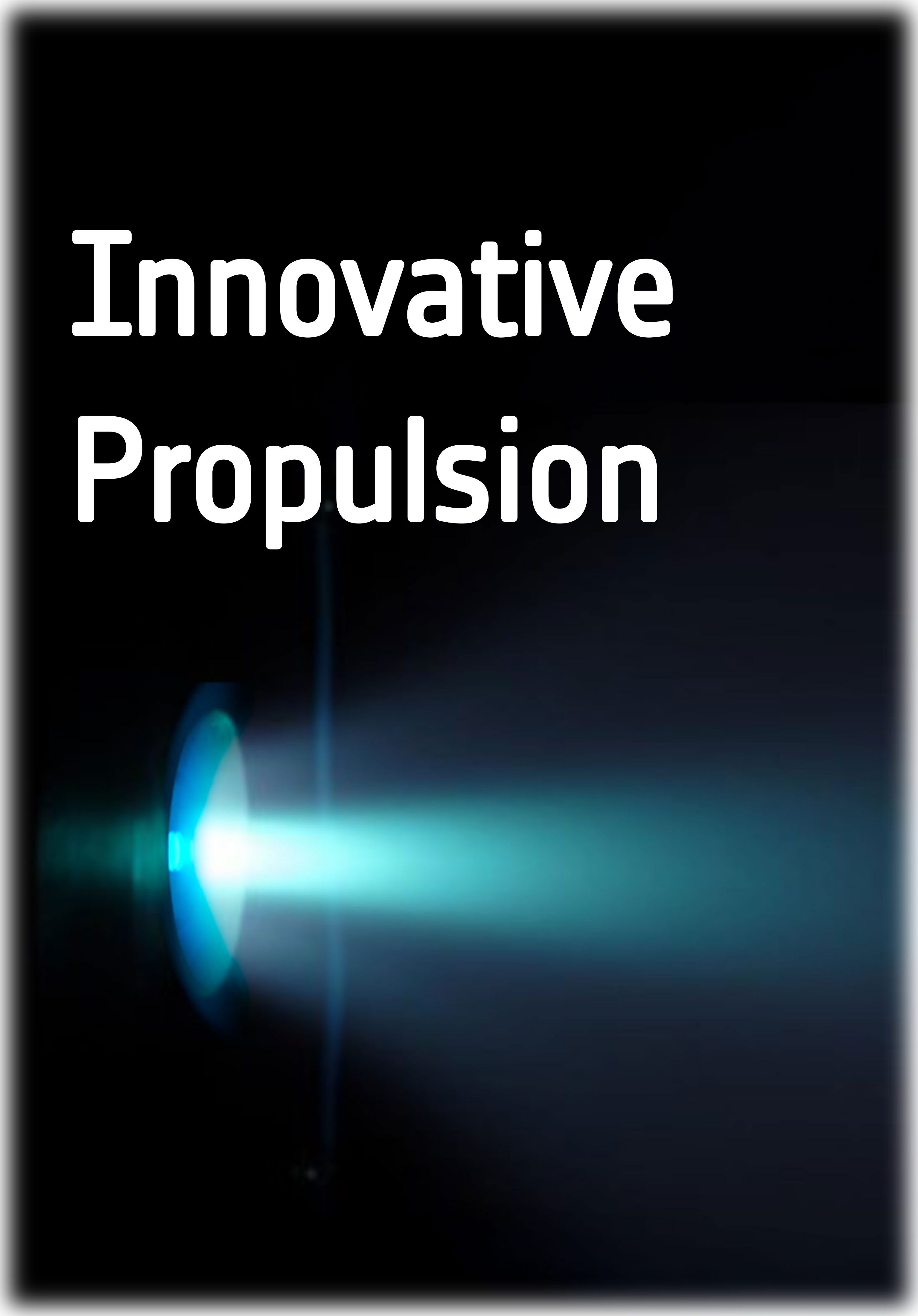
The activity encompasses the following tasks:
- Requirements consolidation.
- Technologies trade off review and baseline selection.
- Design and manufacturing of an engineering model.
- Validation testing of the engineering model.
- Way forward for industrialization.

Deliverables: Engineering Model, Report

Application/Need Date: Earth Observation and Science missions. TRL 6 by 2028. **Mission Classification:** alpha, beta, gamma

THAG Roadmap: Not relevant to any Harmonisation topic.

Innovative Propulsion



3.3.1. Introduction and selection criteria

As part of ESA's technology Vision and Strategy 2040 and within ESA's GSTP Element 1 Develop, Innovative Propulsion Cross-Cutting Initiative (IP-CCI) activities are at the forefront in enabling the next generation of European space missions. IP-CCI drives research and development in propulsion, fluid mechanics systems, space transportation, exploration, servicing, and (re-)entry vehicle technologies.

The IP-CCI theme is part of GSTP and is led by ESA's **Propulsion, Aerothermodynamics, and Flight Vehicles Engineering Division**, with the support of experts and mission leaders from multiple ESA directorates (TEC, CSC, HRE, STS, EOP, SCI, OPS, NAV, and CIC). The ambition of IP-CCI is twofold:

- **Advance Technology Readiness Levels (TRLs):** Accelerate the adoption of breakthrough technologies that deliver major gains in mass, performance, cost efficiency, and mission flexibility.
- **Enable system-level demonstration:** Prepare the ground for testing and validating innovative solutions on complete platforms, rather than in isolated subsystems.

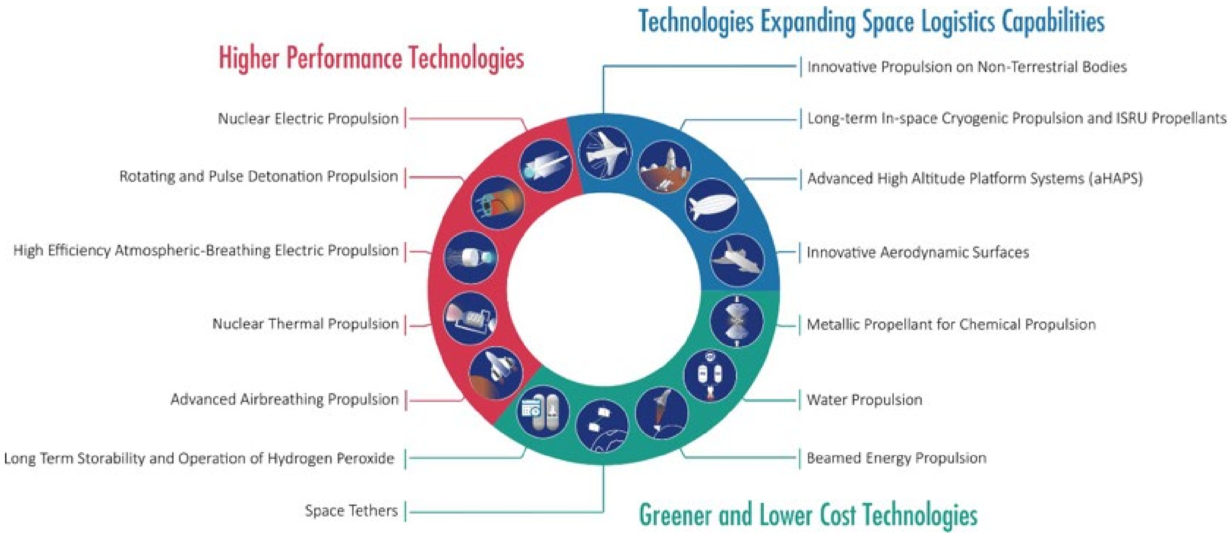
To achieve these objectives, IP-CCI leverages state-of-the-art simulation and test facilities, while advancing technologies across chemical propulsion, electric propulsion, and emerging concepts such as nuclear propulsion and hypersonic/air-breathing propulsion. Supporting domains include thermal management, materials and structures, mechanisms, guidance, navigation and control (GNC), and power generation and conversion.

Beyond these objectives, the IP-CCI also aims to:

- Identify and support the strategic interests of ESA Member States, industry, and academia.
- Stimulate the European propulsion ecosystem.
- Accelerate In-Orbit Validation and In-Orbit Demonstration (IOV/IO-D) missions while strengthening research, development, and testing capabilities.
- Provide greater visibility and coordination for innovative propulsion activities across Europe.

The IP-CCI is structured around three strategic technology groups:

- Higher-performance technologies.
- Greener and lower-cost technologies.
- Technologies expanding space logistics capabilities.



Within these groups, **15 priority technologies** have been identified. They are defined by their strong potential benefits, their maturity for engineering and medium-term demonstration, their ability to enable new missions and applications, their contribution to market creation, and their role in enhancing the competitiveness and reliability of European propulsion and flight vehicle products.

The main IP-CCI activity lines are the following:

- Developing propulsion (sub-)systems for emerging applications (e.g., non-toxic propulsion, in-space cryogenic propulsion, nuclear propulsion).
- Developing reusable propulsion technologies, components, and systems to enable orbital logistics, Earth and planetary re-entry & landing. (e.g. throttleability, fluid mechanics technologies like European Scalable Parachute Assembly System).
- Enhancing propulsion system reliability, competitiveness, and sustainability (e.g in orbit refilling technologies for spacecrafts, extended lifetime and reliable technologies for hydrogen peroxide in orbit, for cryogenic propulsion).
- Advancing thermal management, materials and structures, mechanisms, GNC, and power generation / conversion for innovative propulsion and fluid mechanics systems.
- Demonstrating and validating medium TRL solutions through advanced multi-physics simulation, testing, and diagnostic tools for propulsion and fluid mechanics.

Through the **GSTP Innovative Propulsion theme**, these activities both reinforce existing capabilities and drive the development of next-generation tools and systems, while targeting reduced recurring costs and increased reliability and sustainability of propulsion and fluid mechanics technologies. **The IP-CCI ensures that these activities are strategically prioritised, technologically ambitious, and positioned as key enablers for Europe’s future missions, spanning space transportation, orbital operations, and interplanetary exploration.**

3.3.2. List of activities

Innovative Propulsion

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1N-800EP	PPU for water propulsion electrolyser	2,500
	Total Power Systems, EMC & Space Environment	**2,500**

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

Programme Reference	Activity Title	Budget (k€)
GT1N-801MP	Reusability impact on fluidic components	1,000
GT1N-802MP	Reusable compact lightweight microchannel heat exchangers	1,000
GT1N-803MP	European scalable parachute assembly system	4,000
GT1N-804MP	European throttleable engine	4,000
GT1N-805MP	Hydrogen peroxide propulsion system - in orbit long term storability	2,000
GT1N-806MP	Hydrogen peroxide propulsion system 10N and 200N thrusters	3,000
GT1N-807MP	Spacecraft in-orbit chemical refiling technologies	5,000
GT1N-808MP	Electric propulsion thruster with metal or molecular propellants	1,000
GT1N-809MP	Cryogenic rotating detonation engine	2,000
	Total Propulsion, AeroThermodynamics and Flight Vehicles Engineering	**23,000**

Total Innovative Propulsion	**25,500**

3.3.3. Description of activities

Innovative Propulsion

Power Systems, EMC & Space Environment

GT1N-800EP – PPU for water propulsion electrolyser

Budget: 2500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - TD 19

Objective:
To develop and test an EM of the power processing unit (PPU) for operation of electrolysers in water electrolysis propulsion subsystems.

Description:
Based on current state of the art of cathode vapor feed PEM electrolysers, better performing concepts are to be developed with significant mass improvement.

Several electrolyzer types exist on the market, such as CVF, LAF and others. A benchmarking of those will identify the most promising technology candidate for water propulsion applications. The targeted power class is 50W.

Having identified the most suitable electrolyzer technology, a power processing unit shall be developed and tested to operate it.

The main tasks of the activity are:
- System level study and trade-off, definition of a system level architecture
- Requirements identification and identification of potential development gaps for the power processing unit
- Development of a power processing unit EM for the electrolyser, including a control and scheduling algorithms
- PPU EM testing and coupled test with an electrolyser in representative environment

Deliverables: Report, Engineering Model

Application/Need Date: Facilitating and enabling transition from toxic and carcinogenic hydrazine to green propulsion option of water propulsion. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Electric Propulsion Technologies (2023) – Consistent with activity C41 “Development of a water propulsion resistojet”.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-801MP – Reusability impact on fluidic components

Budget: 1000 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 3 / 5 - TD 18

Objective:
To develop models, test and correlate numerically the multi-phase phenomena in fluidic components relevant to reusable propulsion.

Description:
The reusability of flight vehicles requires operations and multiple re-starts at various conditions along the trajectory while being exposed to a wide thermo-mechanical environment covering both ground as flight in space vacuum and micro-gravity. This activity is in line with IP-CCI (topic Advanced Airbreathing Propulsion) as well as with the ESA Strategy and TEC-Vision 2040 (efficient high-speed vehicles for ground-to-orbit and deep-space missions.

The activity shall focus on the various heat transfer and flow phenomena encountered in the next generation of propulsive units, being liquid rocket or airbreathing engines, of reusable flight vehicles. Whether cryogenic or storable propellants are considered, the various components in the propulsion system from tank, over the feedlines, heat exchangers, injectors, combustion chambers up to the nozzles of the reusable engine are in scope of this activity. These include technologies for in-space conditioning of (non)-cryogenic systems related to re-ignitable and reusable engines, pre-chilling with phase change of tanks, feed and transfer lines, valves, thrust chambers, orifices.

To increase the physical understanding and practical implementation, the modelling and prediction capabilities for multi-phase flow phenomena in microgravity (e.g. boiling, cavitation, sloshing, heat transfer, wettability) need to be elaborated and validated. Based upon a gap analysis, dedicated experimental campaigns shall be defined under reduced gravity in function of their temporal exposure, i.e. drop towers, parabolic flight, space station. This shall include optical access and dedicated sensors to provide sufficient data for calibration and validation.

Likewise, multi-phase models (both implemented in engineering and high-fidelity tools) shall be elaborated and validated against these various flow phenomena of interest.

The activity encompasses the following tasks:
- Performance gap identification and definition of the multi-phase phenomena.
- Definition and execution of dedicated reduced gravity test campaigns to assess multi-phase flow phenomena.
- Elaboration, implementation and correlation of numerical modelling for multi-phase phenomena under reduced gravity.

Deliverables: Breadboard, Report, Software

Application/Need Date: Breakthrough performance technology in Advanced Airbreathing Propulsion **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Technologies for Fluid Mechanics.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-802MP – Reusable compact lightweight microchannel heat exchangers

Budget: 1,000 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 3 / 5 - TD 18

Objective:
To develop a micro-channel heat exchanger breadboard and demonstrate by functional testing the capability of the micro-channel heat exchanger to start up and reach the design stationary performance, and the capability to withstand the thermo-mechanical loads on the matrix.

Description:
The reusability of flight vehicles requires operations and multiple re-starts at various conditions along the trajectory while being exposed to a wide thermo-mechanical environment covering both ground as flight in space vacuum and micro-gravity. Achieving high power density for flight-weight high-performance microchannel heat exchangers (MCHXs) is an enabling technology of compact cost-effective cryogenic propulsion systems with prospective application to innovative airbreathing propulsion systems as well as liquid rocket engines. The vaporization of the coolant fluid possesses an additional challenge for the MCHX during the start-up transients, when the flow capacity and flow uniformity across the matrix may be reduced in the presence of two-phase flow.

Leveraging on the outcomes from previous developments, testing with cryogenics shall be undertaken to demonstrate the objectives. The design of the test shall be supported by analysis to confirm that the breadboard can withstand the loads.

This activity encompasses the following tasks:
- Design of the breadboard of the micro-channel heat exchanger.
- Breadboard development.
- Design and development of the experimental tests.
- Testing of the breadboard.

Deliverables: Breadboard, Report, Software

Application/Need Date: Breakthrough performance technology in Advanced Airbreathing Propulsion **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Technologies for Fluid Mechanics (2025) – Consistent with activity E02 “Heat transfer enhancement in micro-channel heat exchangers “.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-803MP – European scalable parachute assembly system

Budget: 4,000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 4 / 6 - TD 18

Objective:
To develop an EQM of a scalable European Parachute Assembly System (PAS) for applications in Exploration and Space Transportation.

Description:
Parachutes for European space missions are largely ad-hoc developments, mainly due to mission-specific requirements, leading to the common trope that "there is no such thing as a COTS space parachute". Nevertheless, commercial European entities are known to have procured non-European parachute systems, based on heritage, to avoid new developments. With the rise in (commercial) space transportation and re-entry service suppliers, the availability of such a European solution would enable those suppliers to avoid having to purchase from non-European actors.

The activity aims at the development of a fully European scalable Parachute Assembly System (PAS) (i.e. main & drogue parachutes, mortars, auxiliaries, etc.) that could be used as (quasi-) COTS for different types of missions. This will ensure the availability of such a European system for upcoming missions. The outcome will be a Parachute Assembly System family with as maximum of overlap as possible, and thus minimising needed additional or custom developments in future missions.

The following application use cases are targeted to drive the development of the PAS family:
- Earth re-entry capsules, with landing masses of up to 15t (for cargo, and potentially crewed missions at a later stage).
- Reusable launcher stages.
- Planetary exploration (re-)entry vehicles.

The activity encompasses the following tasks:
- Definition of performance requirements and operating environments.
- Definition of scalable PAS architecture by maximising technical design/operational/manufacturing overlaps.
- System-level modelling and analysis for design verification of the Parachute Assembly System(s).
- Manufacture (and perform development tests where needed) all elements (deployment mechanism, main parachute, reefing, structural components…) of the scalable Parachute Assembly System(s).
- Perform verification test campaigns (on-ground and in-flight) of the developed Parachute Assembly System(s).

The development shall provide a fully European parachute system family, ready to be used for Exploration and Space Transportation missions as (quasi-)COTS. In order to have a qualified family of PAS, following the successful completion of the proposed activity, a follow-on it is foreseen with targeted budget of 3 M€.

Deliverables: Breadboard, Engineering/Qualification Model, Report
Application/Need Date: LEO Cargo Return Service; re-usable launchers; sample return mission. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-804MP – European throttleable engine

Budget: 4,000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 19

Objective:
To develop the critical propulsion equipment required for a fully European lander propulsion system, in particular the activity will be focused on the development of the throttleable engine which is fundamental to achieving precise and controlled descent and landing.

Description:
Given the reliance of non-European propulsion technologies for Entry, Descent, and Landing (EDL), this activity proposes the development of a complete throttleable propulsion system to enable precision landing.

Precision landing demands fine control of thrust during the Entry, Descent, and Landing (EDL) phase, which requires enabling technologies including throttleable engine supported by a pressure regulation system (this last is not part of the present activity).

This activity encompasses the following tasks:
- Throttleable engine design.
- Throttleable engine breadboard development and verification, including performance testing under ambient conditions.
- Throttleable engine valve design, development and verification, ensuring compatibility and responsiveness across the required thrust range.

To achieve the integrated technologies demonstrations of the fully European throttleable propulsion systems for landers, additional follow up phases will be required after successful implementation of this activity. Two phases are foreseen as follows: phase 1 for the throttleable engine, 11,000 k€; phase 2 Enabling equipment's (Isolation valves, pyro valves, filters and propellant tanks) 10,000 k€.

Deliverables: Breadboard, Report, Software

Application/Need Date: Lander propulsion system **Mission Classification:** alpha, beta

THAG Roadmap: Relevant to the Roadmap Chemical Propulsion – Components.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-805MP – Hydrogen peroxide propulsion system - in orbit long term storability

Budget: 2000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop in orbit long term storability solutions to allow the safe and continuing use of Hydrogen Peroxide as an efficient "green" low-toxicity propellant as an alternative to hydrazine-based propellants.

Description:
Hydrogen peroxide has shown promise as an alternative to hydrazine-based propellants, with lower toxicity, however there are several characteristics that limit its adoption, and prevent it being the "plug and play" direct replacement for the current propellants like:
- Limited or no long-term material compatibility with the current generation of components.
- Relative long duration stability of the propellant composition is only achieved by the addition of stabilizing chemicals. Stabilizer chemicals affecting the thruster catalyst materials, reducing the engine performance and / or lifetime depending on elapsed time and / or propellant consumption.

To solve these short falls for the adoption of this technology, these factors need to resolve with fundamental development of the propellant itself, with consideration for the application, e.g. stabilizer species and quantities, then iterate with selected tank materials and engine development (not part of this present activities). Once appropriate materials are selected, these need to be confirmed suitable for the long term storage, for both the strength and corrosion resistance of the tank materials, the decomposition / gas generation stability of typical spacecraft flight times, and where new tank materials are selected any new manufacturing joining and inspection methods, processes and techniques will then need to be selected or developed, and then proven suitable.

The in-orbit long term storability solution consists of new stabilizer cocktails for the propellant, a tank up to 50L capable to store hydrogen peroxide over the operational life of the spacecraft (up to 4years), ensuring long term material compatibility and storability considering the new propellant formulations (including the stabilizer cocktails).

This activity encompasses the following tasks:
- Requirements definition for the in-orbit long term storability solution.
- Stabilizers cocktails design and test.
- Material testing to ensure the compatibility considering the new propellant formulations.
- Tank design and test plan definition.
- Tank manufacturing.
- Testing in relevant environment including aging test.

These are generic technologies required for future missions in multiple application areas and identified in the IP-CCI White paper.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Chemical Propulsion - Components (2024) – Consistent with activity A08 “Hydrogen Peroxide Propulsion System Processes qualification”.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-806MP – Hydrogen peroxide propulsion system 10N and 200N thrusters

Budget: 3,000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop 10 N and 200 N pulse-mode bipropellant thrusters using long term storable hydrogen peroxide as a green alternative to hydrazine.

Description:
Hydrogen peroxide has shown promise as an alternative to hydrazine-based propellants, with lower toxicity. However, there are several characteristics that limit its adoption, and prevent it being the "plug and play" direct replacement for the current propellants like:
- Limited or no long-term material compatibility with the current generation of components.
- Relative long duration stability of the propellant composition is only achieved by the addition of stabilizing chemicals.
- Stabilizer chemicals affecting the thruster catalyst materials, reducing the engine performance and / or lifetime depending on elapsed time and / or propellant consumption.

To solve these short falls for the adoption of this technology, these factors need to be resolved with fundamental development of the propellant itself, with consideration for the application, e.g. stabilizer species and quantities, then iterate with selected tank materials and engine development. Once appropriate materials are selected, these need to be confirmed suitable for the long term storage, for both the strength and corrosion resistance of the tank materials, the decomposition / gas generation stability of typical spacecraft flight times, and where new tank materials are selected any new manufacturing joining and inspection methods, processes and techniques will then need to be selected or developed, and then proven suitable.

The following tasks are foreseen:
- Development of engines up to TRL5 (10N and 200N pulse-mode bipropellant)
- Ensure compatibility with the new hydrogen peroxide formulation suitable for typical earth orbit attitude control and orbit maintenance and disposal operation.
- Demonstrate by functional testing critical functions in steady-state and pulse mode in high altitude test facilities or as close as possible to flight conditions

It is foreseen a continuation of the activity up to the qualification of the two thrusters targeting a budget of 5M€. These are generic technologies identified in the IP-CCI White paper.

Deliverables: Breadboard, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to Roadmap Chemical Propulsion - Components.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-807MP – Spacecraft in-orbit chemical refuelling technologies

Budget: 5,000 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop and test breadboard components for spacecraft chemical propulsion in orbit refiling system.

Description:
Spacecrafts refiling capability is considered as a strategic tool for in-orbit life extension, manoeuvring and change of orbit.

This activity scope is to validate & verify the refiling process on satellites, specifically ones which are not designed to be refill and derive critical propulsion subsystem building blocks necessary to build in orbit spacecraft refill service.

Coverage includes mechanical, fluidic, electrical, control, and automation functions for MON-x, MMH propellants, and pressurant gas transfer, in particular the demonstration of the technologies such as propellant management devices capable of expelling gas bubbles from the fluid in the receiving tank during in-orbit refiling operations.

The following components shall be developed up to a breadboard level and tested during the activity:
- Gauging and measurement technologies for determining, with at least 1% accuracy, the propellant volume in a tank under zero-g conditions for in-orbit refilling.
- Transfer components for liquid or gaseous propellants.
- Venting/purge components for clearing lines in a connected spacecraft or refilling vehicle.
- Approach, connection, and disconnection components for refilling target spacecraft, making maximum use of state-of-the-art techniques to minimize droplet formation or residuals (e.g., as used in ISS/ATV, ESPRIT).
- Development plan of the integrated refilling system.

The activity is intended to be continued up to qualification, with a target budget of 12,000k€. The activity is in line with IP-CCI White Paper and ESA ISAM objectives.

Deliverables: Breadboard Models of considered components, Test report of Technologies breadboards and development plan of the integrated system

Application/Need Date: Spacecraft missions with need of life extension and/or increase number of manoeuvre/change orbit **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Chemical Propulsion - Components.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-808MP – Electric propulsion thruster with metal or molecular propellants

Budget: 1,000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop a low power (<1kW) Electric Propulsion Thruster operating with metal vapour propellants (e.g. zinc, magnesium, bismuth, etc) or with molecular propellants (e.g. adamantane).

Description:
Electric Propulsion (EP) is crucial for deploying and maintaining large constellations of navigation, communication and earth observation satellites in orbit. EP thrusters perform various tasks, including orbit insertion, orbit changes, orbit maintenance, repositioning, collision avoidance, and deorbiting at the end of their operational life.

For over four decades, the noble gas Xenon has been the propellant of choice for EP thrusters because it is inert, storable, relatively heavy and easy to ionise. Xenon-fed low-power Hall Effect Thruster (HET) sub-systems are used in Low Earth Orbit (LEO) constellations. However, Xenon's high extraction cost, low environmental performance, and need for high-pressure storage have necessitated the development of alternatives. Gaseous Krypton, Iodine, and Water are viable alternatives, but maturing ~1kW EP systems operating with these alternative propellants has been challenging, and flight-ready EP systems are not yet available.

There is a strong interest in developing EP systems that utilize alternative propellants to Xenon, especially for small platforms with tighter budget, mass, and volume constraints.

Condensable solid metal propellants produce a high enough vapor pressure when heated at low ambient pressure to drive a gaseous flow through either a feed system or directly into the discharge chamber, providing a propellant in gaseous form that can then be ionized. These alternative propellants theoretically offer several advantages over Xenon: they are solid at ambient temperature making them easier to store and handle, are abundantly available and have low cost and are lighter which means that higher Isp may be achieved at lower voltage. In addition, magnesium is found on the surface of Mars and the Moon in the form of MgO, from which magnesium can be extracted in-situ.

Other potential Xenon propellant alternatives being explored mainly on Hall Effect Thrusters are molecular propellants such as adamantane (C10H16), which has a mass similar to that of Xenon, lower first ionization potential, low production costs, and can be brought into gas phase with only moderate heating.

This activity encompasses the following tasks:
- Requirement Definition, Design Definition and Justification of the EP thruster with metal or molecular propellant.
- Breadboard manufacture.
- Functional Breadboard testing to preliminary verify the performance based on parameters such as thrust, specific impulse, and efficiency.
- Preliminary thruster endurance testing of minimum 500 hours.

Deliverables: Breadboard, Report

Application/Need Date: SmallSat constellations for Earth observation, navigation and Communication.
Mission Classification: alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Electric Propulsion Technologies.

Innovative Propulsion

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1N-809MP – Cryogenic rotating detonation engine

Budget: 2,000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop and test a breadboard of a Rotating Detonation Engine (RDE) with a thrust level greater than 1kN.

Description: In Line with IP-CCi White Paper given State of the Art, Rotating Detonation Engines (RDEs) constitute game changers for in-space stages and landers due to efficiency and compactness. RDEs have potential applications in upper stages once stability and lifetime demonstrations are achieved.

Competitors have shown progress in recent years from subscales demo to several Kilo-Newtons, including a first spaceflight demo. To bring European RDEs as a strong option for future missions, the main challenges are in injectors, controls, stability/lifetime and performance demonstrations.

The focus of the proposed activity will be on enhancing combustion stability, performance, cost optimisation and ensuring industrial scalability.

The presented activity aims at reaching TRL 4-5 for RDE (>1KN) integrated technology demonstration and it is including the following tasks:
- Identify suitable technologies for detonation waves including ignitors and combustion chamber
- Demonstrate repeatability of combustion
- Develop and design liquid-liquid injection for detonation engines
- Demonstrate successful detonation wave combustion with liquid-liquid injection
- Define and select instrumentation for adapting test bench to dedicated rotating detonation wave test bench
- Define an advanced test bench for rotating detonation engine development and testing
- Define the test facility requirements for altitude operations, allowing for extended engine testing in relevant environment

To achieve integrated technology demonstrations and mature the technology to TRL 7–8, a continuation is foreseen following the successful completion of the proposed activity:

Deliverables: Breadboard, Report

Application/Need Date: Space Transportation and Exploration missions. **Mission Classification:** alpha, beta

THAG Roadmap: Relevant to the Roadmap Technologies for Fluid Mechanics.


Sustainable
Space

3.4.1. Introduction and selection criteria

The selection of technology activities for the Sustainable Space theme of the GSTP Compendium 2026–2028 is driven by the ESA strong goals to understand and mitigate environmental impacts of space missions on Earth's environment across their entire life cycle and preserve Earth's orbital environment as a safe zone free of debris, while contributing to a sustainable and competitive European space industry.

As "*ESA is dedicated to spearheading a Zero Debris future by 2030 and to enable the establishment of a circular economy in space from 2040*"[1], the Sustainable Space theme of the GSTP Compendium 2026-2028 aligns with ESA's sustainable strategy (including the Zero Debris Policy, the Green Agenda, ESA Strategy 2040) *"to lead in sustainable space operations, providing a competitive edge to the European space industry and ensuring the long-term sustainability of space activities"*[2]. This theme is structured around three main branches:

Ecodesign: *implement sustainable practices in satellite design (eco-design), production, and operations to minimise environmental impact throughout their lifecycle*[2].

Zero Debris: ESA aims to "implement a Zero Debris approach for its missions; and to encourage partners and other actors to pursue similar paths"[3], by developing and integrating innovative solutions to significantly limit the generation of debris in Earth and Lunar orbits for all future missions, programmes and activities by 2030.

Circular Economy in Space, and more broadly ISAM (In-Space Servicing, Assembly, and Manufacturing),: encompasses a range of disruptive technologies and in-orbit capabilities aimed at extending the lifespan of space assets essential for the satellite maintenance and space-based industry "enabling the future growth markets of in-orbit servicing, in-orbit assembly, in-orbit manufacturing and in-orbit recycling"[2]

Through a strong coordination effort that began in 2023 combining internal and external processes (e.g. Technology Harmonisation, workshops with companies and academia), ESA successfully completed a comprehensive mapping of sustainability-related activities across its various Directorates. This effort also resulted in the development of strategic roadmaps, shaped by input from TEC, OPS, and other ESA Directorates. Key outcomes include the cross-cutting Ecodesign and Zero Debris Roadmaps for 2026–2028, along with the final report from the In-Space Servicing, Assembly, and Manufacturing (ISAM) Working Group. The needs and priorities, identified and agreed upon across the Agency, were subsequently shared with ESA experts to serve as a key reference for shaping their technology activity proposals in alignment with these strategic objectives.

In line with a holistic approach, the selection of technology activities for the Sustainable Space theme of the GSTP Compendium 2026–2028 was driven by clearly defined user, market, and mission needs. The implementation of the activities is envisioned through a project-oriented approach: a strong coordination of the system level solutions and of the various technology developments across the ESA Programmes. This project-oriented integrated effort ensures an efficient and seamless pathway for direct integration of new technology with the right maturity level at the right time within ESA missions.

Ecodesign: The development of a sustainable solutions for space industry addresses a pressing need to minimise the environmental impacts in a growing market and is a key aspect for the European space sector long term competitiveness. The criteria for selection was based on previous Life Cycle Assessment (LCA) outcomes, prioritizing generic technology activities with the highest expected environmental benefits. In line with ESA Green Agenda objectives, the main drive is to achieve a 20 - 40% reduction of the environmental impacts of space missions by 2030. Two concrete goals were identified for this GSTP Compendium:

Enhance robust evaluation of the environmental impacts of space missions: development of a ground-based resonance lidar measuring metals from burned up debris.

Develop sustainable technologies expected to be highly recurrent: technologies that address recurrent environmental hotspots and replace pollutant or toxic substances. Key areas include sustainable and generic electronics, sustainable structure manufacturing, sustainable propulsion systems.

[1] ESA Strategy 2040, Objective 1.2, page 28
[2] ESA Strategy 2040, Objective 1.2, page 29
[3] Ministerial Conference of 2022

Zero Debris: With the exponential increase in space traffic in the Earth orbits in the last years, a Zero Debris approach is no longer an option but a necessity if the space sector wants to keep operating safely, in particular, in LEO and delivering reliable services to humankind. In line with the ESA Zero Debris Policy (ESA/ADMIN/IPOL(2023)1, Nov. 2023), the main drive is to drastically reduce debris generation by 2030. Priority was given to critical technologies that are supporting the evolution of the spacecraft platforms and the generic Space Surveillance and Tracking (SST) and Space Traffic Coordination (STC) technologies for collision avoidance. Two goals were identified for this GSTP Compendium:

Develop affordable disposal solutions for small spacecraft: under the ESA Space Debris Mitigation (SDM) Requirements (ESSB-ST-U-007 Issue 1), a LEO satellite must be disposed within 5 years from end of mission with at least a 90% success probability rate. The activity selected here targets the development and ground qualification of innovative and disruptive electric propulsion systems compatible with 6U to 27U LEO CubeSats.

Improve satellite resilience to impacts: this goal led to the selection of 7 activities covering, amongst others, the development of advanced collision and structural models integrated into operational simulators, the qualification of technologies that enhance the resilience of all satellites to hyper velocity impacts (HVI), automation of STC, data quality and accuracy in the STC process supporting spacecraft operations and technologies to enhance tracking of cm scale debris.

Circular Economy in Space: ISAM capabilities are essential for the future of space exploration, satellite maintenance, and space-based industries. They enable new mission concepts, can unlock new markets, enhance performances and maximise the value of space assets. This branch covers several key technology developments identified in the ESA Technology Vision 2040 and in the ISAM working group: validation of design and verification tools for safe Close Proximity Operations (CPO), developing interfaces to enable different IOS (assembly, refurbishment, repair), development of robotic tooling for IOS, development of highly autonomous rendezvous capabilities, or development of in-orbit material and equipment storages and processing technologies. In the frame of this GSTP Compendium 2026 - 2028, priority was given to enable new in-orbit services while ensuring safer Close Proximity Operations (CPO) to lower the risk of collision. This priority led to the selection of activities aiming at developing Photonic Integrated Circuit (PIC) Frequency Modulated Continuous Wave Lidar for the detection and capture of uncooperative targets (debris), GNC algorithms involved in the navigation and control or a robotic arm and servicer spacecraft involved in IOS task, navigation algorithms and hardware for CPO using multispectral imaging and ranging device, or small CPO range finder for rendezvous and docking.

3.4.2. List of activities

Sustainable Space

Ecodesign - Enhance robust evaluation of the environmental impacts of space missions

Optics, Robotics & Life Sciences

Programme Reference	Activity Title	Budget (k€)
GT1C-800MM	Ground-based resonance lidar for monitoring atmospheric space debris tracers	1,000
	Total Optics, Robotics & Life Sciences	**1,000**

Ecodesign - Develop sustainable technologies expected to be highly recurrent

Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT1C-801ED	Optimisation and evaluation of supercapacitors cells for space applications	300
	Total Avionics & EEE	**300**

Propulsion, Aerothermodynamics and Flight Vehicles Engineering

Programme Reference	Activity Title	Budget (k€)
GT1C-802MP	Iodine-fed, sub-kilowatt hall effect thruster subsystem for LEO missions	2,000
	Total Propulsion, Aerothermodynamics and Flight Vehicles Engineering	**2,000**

Structures, Mechanisms & Materials

Programme Reference	Activity Title	Budget (k€)
GT1C-803MS	Advanced sustainable manufacturing of space-qualified components from recycled high-value feedstock	1,000
GT1C-804MS	Manufacturing of composite structures with recycling, reprocessing and waste treatment capabilities	500
	Total Structures, Mechanisms & Materials	**1,500**

Total Ecodesign	**4,800**

Zero Debris - Develop affordable disposal solutions for small spacecraft

Propulsion, Aerothermodynamics and Flight Vehicles Engineering

Programme Reference	Activity Title	Budget (k€)
GT1C-805MP	Disruptive electric propulsion sub-system for cubesat de-orbiting	1,500
	Total Propulsion, Aerothermodynamics and Flight Vehicles Engineering	**1,500**

Zero Debris - Improve satellite resilience to impacts

Structures, Mechanisms & Materials

Programme Reference	Activity Title	Budget (k€)
GT1C-806MS	Advanced simulation for resilient collision modelling to enhance health monitoring and prognostics	2,000
GT1C-807MS	Innovative materials and structural concepts for a resilient satellite to debris collisions	2,500
	Total Structures, Mechanisms & Materials	**4,500**

Space Segment Systems Engineering

Programme Reference	Activity Title	Budget (k€)
GT1C-808SY	Autonomous collision avoidance system	1,200
	Total Space Segment Systems Engineering	**1,200**

Space Debris

Programme Reference	Activity Title	Budget (k€)
GT1C-809SD	Manoeuvre analytics software for space traffic management and coordination	500
GT1C-810SD	Data fusion for collision avoidance and space traffic coordination	500
	Total Space Debris	**1,000**

Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT1C-811ED	On-board event-based debris detection and collision avoidance for LEO satellites	1,000
	Total Avionics & EEE	**1,000**

RF Payloads & Technology

Programme Reference	Activity Title	Budget (k€)
GT1C-812EF	W-band gyro traveling wave tube for centimetre-level space debris detection	750
	Total RF Payloads & Technology	**750**

Total Zero Debris	**9,950**

Circular Economy in Space / ISAM - Ensure safer Close Proximity Operations (CPO) to lower the risk of collision

Optics, Robotics & Life Sciences

Programme Reference	Activity Title	Budget (k€)
GT1C-813MM	Frequency modulation continuous wave LiDARs based on photonic integrated circuits for debris identification	1,000
	Total Optics, Robotics & Life Sciences	**1,000**

GNC, AOCS & Pointing

Programme Reference	Activity Title	Budget (k€)
GT1C-814EC*	Close-proximity range finder for rendezvous and docking	800
GT1C-815EC*	Dual navigation and control of a servicer spacecraft and its robotic arm	1,800
GT1C-816EC*	Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations	1,000
	Total GNC, AOCS & Pointing	**3,600**

*** Proposed Bundle 8 – Enabling Proximity operations and rendezvous:**
- GT1C-814EC - Close-proximity range finder for rendezvous and docking
- GT1C-815EC - Dual navigation and control of a servicer spacecraft and its robotic arm
- GT1C-816EC - Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations

Total IOS / Circular Economy in Space	**4,600**

Total Sustainable Space	**19,350**

3.4.3. Description of activities

Sustainable Space

Ecodesign - Enhance robust evaluation of the environmental impacts of space missions

Optics, Robotics & Life Sciences

GT1C-800MM – Ground-based resonance lidar for monitoring atmospheric space debris tracers

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 17

Objective:
To develop a day-and-night operating resonance lidar within Fraunhofer lines for monitoring and in-situ measurement of reentered space debris tracers.

Description:
With rapidly increasing space activities and unprecedent number of objects in orbit, the clean-space initiative is important to ensured continued and sustainable access to space. Removing non-operational spacecrafts by controlled reentering in the atmosphere and the consequently burning up in high altitudes is the most efficient and only realistic way of limiting the number of objects. However, satellite deorbiting may have polluting effects, particularly in the middle atmosphere. Studies have linked various metals found in the atmosphere to the remnants of space debris. While recent research suggests that the atmospheric impact of burned-up satellites could be significant, it remains largely unquantified. Currently, there is very limited data on the environmental effects, such as changes in the atmospheric chemistry, associated with satellite reentry, and collecting in-situ measurements at the relevant altitudes remains a major challenge.

Using resonance lidar to measure metals from burned-up space debris is currently the only method that offers the necessary sensitivity, continuity, and efficiency. This technique, which detects metal layers such as potassium, sodium, and iron in the Mesosphere and Lower Thermosphere (MLT), is well established. However, the laser and lidar systems required for these measurements are not yet advanced enough for fully autonomous operation and still demand significant personnel effort.

This activity will develop a resonance lidar, using mature Alexandrite lasers and narrow-band filtering. Alexandrite offers the advantage to be able to tune the emitted wavelength into the absorption line of different metals.

This activity encompasses the following tasks:
- Review of requirements, e.g. re-entry materials (spacecraft common materials composition) and required wavelength for the identification of those components (laser absorption bands).
- Preliminary design of the resonant lidar setup for analyzing the suspended metals. The design shall include basic end-to-end optical calculation/performance of the instrument for the selected wavelengths and the required metals.
- Selection of commercial-off-the-shelf (COTS) critical function technologies (e.g. laser source and detector), and justification or de-risk of COTS components at realistic operational environments.
- Manufacture a lidar breadboard.
- Testing of analysis of relevant metals in a realistic scenario. e.g. atmospheric representative gas chamber with suspended metals for the study to demonstrate wavelength absorption at those bands.

Deliverables: Breadboard, Reports

Application/Need Date: TRL6 by 2028 for all LEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Big Data from Space.

Sustainable Space

Ecodesign - Develop sustainable technologies expected to be highly recurrent

Avionics & EEE

GT1C-801ED – Optimisation and evaluation of supercapacitors cells for space applications

Budget: 300 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 23

Objective:
To optimise supercapacitor cells manufacturing, increase their energy density and capacitance values and perform an evaluation for space applications.

Description:
Supercapacitors (SCs) are seen as promising high-power energy storage devices that can bridge the gap between traditional capacitors and batteries. They offer much higher energy density than typical capacitors, and much higher power density than batteries, making them ideal for applications that require quick energy delivery. In the space sector, SCs are considered a strong alternative to tantalum capacitors and batteries, especially for demanding, high-power applications. A key advantage of SCs is their material composition: they do not use lithium, tantalum, or other rare or conflict metals. This reduces environmental and supply chain risks. They also rely on liquid-based electrolytes, which eliminates the explosion risk associated with certain battery chemistries, making them safer for use in space systems. In high-power systems, the use of SCs can significantly reduce overall mass. Compared to batteries, they can deliver more powerful energy pulses, though they store less total energy. Additionally, their long operational life, capable of withstanding hundreds of thousands of charge and discharge cycles with minimal degradation, makes them particularly well suited for long-duration missions. Their fast charge and discharge capabilities are ideal for high-power needs.

This activity aims to optimise and industrialise supercapacitor cells with improved packaging and performance. The reliability of these cells will be verified through rigorous testing, including thermal vacuum tests. The activity also focuses on reducing environmental impact, including using safer, more sustainable materials, cutting emissions, and planning for better end-of-life disposal. A full Life Cycle Assessment (LCA) will be conducted to evaluate the environmental footprint of the new supercapacitor cells in comparison with conventional solutions like lithium-ion batteries or tantalum capacitors.

This activity encompasses the following tasks:
- Assessment of the state of the art of supercapacitors, including the use of sustainable materials. A technology baseline for the design and materials of the supercapacitors shall be stablished.
- Design, development, and manufacturing of the supercapacitor cells in a repeatable manner that enables industrialization. Characterisation testing (capacitance, leakage current, self-discharge) shall be performed.
- Performance of mechanical, thermal (including thermal vacuum) and endurance testing.
- Analysis of the system-level impact of integrating supercapacitor cells into a bank of supercapacitors (BOSC) and performance of a comprehensive Life Cycle Assessment (LCA) of the component.

Deliverables: Engineering Model, Other, Qualification Model, Reports

Application/Need Date: All missions, with a focus on high-power mechanism support, power supply for robotics, high-power radar systems with low duty cycles. TRL 7 by end of 2030. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Electrochemical Energy Storage.

Sustainable Space

Ecodesign - Develop sustainable technologies expected to be highly recurrent

Propulsion, Aerothermodynamics and Flight Vehicles Engineering

GT1C-802MP – Iodine-fed, sub-kilowatt hall effect thruster subsystem for LEO missions

Budget: 2000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 4 / 7 - **TD** 19

Objective:
To develop a sub-kilowatt Hall Effect Thruster (HET) sub-system operating with iodine as propellant for LEO missions.
Description:
Electric Propulsion (EP) is essential for deploying large constellations of Communication, Navigation, and Earth Observation satellites. These thrusters perform tasks such as orbit insertion, orbit changes, orbit maintenance, repositioning, collision avoidance, and deorbiting at the end of their operational life.

Traditionally, xenon-fed HET EP sub-systems have been deployed on LEO constellations. However, xenon's high extraction cost, low environmental performance, and need for high-pressure storage necessitate developing alternatives. Iodine has recently gained attention as a cost-effective and environmentally friendly alternative to xenon for use as a propellant. It offers several advantages compared to xenon. For instance, iodine and xenon have similar atomic masses and first ionization energies, which means that thrusters using iodine can achieve nearly the same performance in terms of thrust density, efficiency, and specific impulse. Moreover, iodine is solid at ambient temperature, making it easier to store and handle, and it is abundantly available for the intended applications.

Low power iodine-fed gridded ion engines have already been used on CubeSats and MicroSats in LEO. However, LEO missions baseline sub-kilowatt HETs because they offer a higher performance (thrust and thrust-to-power ratio). Although there are ongoing developments of low-power HETs using xenon and krypton, significant progress is still needed to make HETs compatible with iodine as propellant.

This activity shall focus on the development and qualification of an iodine-fed, sub-kilowatt HET sub-system with all its main components: thruster, power processing unit (PPU), storage and fluidic management system (FMS). This HET subsystem shall be designed following an ecodesign approach to reduce as much as possible its environmental impacts and benchmarked against xenon or krypton fed equivalent HET.

This activity is organized in two phases and encompasses the following tasks:
Phase 1 (budget: 800 k€, duration: 24 months):
- Design the propulsion subsystem composed of an iodine-fed sub-kilowatt HET thruster, a power processing unit (PPU), a storage and fluidic management system (SFMS).
- Manufacture and test development models (DMs) of the components to preliminary verify the performance and validate the components and sub-system design. This shall include a vacuum coupling subsystem test and a thruster endurance test of minimum 300 hours to demonstrate the subsystem performances.

Phase 2 (budget: 1200 k€, duration: 12 months):
- Manufacture a qualification model of the propulsion subsystem and perform a complete qualification including life test.
Deliverables: Breadboard, Qualification Model, Reports

Application/Need Date: TRL8 in 2029 for All LEO CubeSats and MicroSats missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Ecodesign - Develop sustainable technologies expected to be highly recurrent

Structures, Mechanisms & Materials

GT1C-803MS – Advanced sustainable manufacturing of space-qualified components from recycled high-value feedstock

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 24

Objective:
To identify, demonstrate, and implement recycling of high-value scrap into quality feedstock for additive manufacturing of components for space applications.

Description:
As the demand for spacecraft, satellites and launch systems grows, so too does the consumption of critical raw materials - many of which are finite, expensive and environmentally burdensome to extract. Recycling and reusing materials offer a strategic solution, reducing dependency on virgin resources, lowering carbon emissions and enhancing supply chain resilience. By integrating recycled materials into components, the space sector can not only reduce its environmental footprint but also pioneer circular economy practices that will be essential for long-term sustainability - both on Earth and in future off-world manufacturing ecosystems.

Furthermore, materials like titanium, aluminium and nickel alloys are in high demand in the space sector, yet these materials are primarily imported from non-European sources. Developing local recycling solutions for high-value, critical materials is a vital step towards achieving European material self-sufficiency, especially in the current geopolitical and economic context. Potentially significant sources of scrap include machining waste (e.g. swarf, off-cuts) and unused powder from additive manufacturing. Scrapped parts due to manufacturing defects are expected to be a secondary, less frequent, source but may become more relevant in the context of out-of-earth manufacturing. The space sector should also consider sourcing waste/scrap materials from other industries (e.g. aviation, automotive).

The activity will start by identifying hot spots of high-value material scrap across European manufacturing (space and non-space). A hot spot in this context would be defined as materials with high volume of scrap, high material value/cost, raw material being sourced outside Europe, and high demand by the space industry. Selected scrap will then be processed into quality raw materials, tested, and used to produce and evaluate a demonstrator part to be defined based on a business case (e.g. bracket, fuel tank).

This activity encompasses the following tasks:
- Identify sources of scrap material, material types relevant for space, and their possible use; down-selection of the most interesting cases (including economic, ecological and technical factors).
- Develop processing route(s) for high-value material waste identified in the first task to transform into high-quality raw materials for use as space hardware. This will most likely require dedicated alloy development to factor in what properties and quality can be realistically extracted from recycled materials.
- Develop the advanced manufacturing process(es) for the use of "up-cycled" materials, considering the influence of impurities, potential development of detrimental phases, etc.
- Test the produced material on a sample level (material characteristics, mechanical properties, etc.).
- Design and manufacture a relevant space demonstrator part for subsequent testing to check compliance with relevant functional requirements.
- Provide guidance on how to design components produced from recycled materials, seeking to identify potential trade-offs between quality requirements, safety factors, cost and leadtime.

Deliverables: Breadboard, Report

Application/Need Date: TRL 6 in 2028 enabling a viable solution for use of recyclable materials for a multitude of space missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Ecodesign - Develop sustainable technologies expected to be highly recurrent

Structures, Mechanisms & Materials

GT1C-804MS – Manufacturing of composite structures with recycling, reprocessing and waste treatment capabilities

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 24

Objective:
To implement manufacturing of space grade composite structures with recycling and reprocessing capabilities using vitrimer based resin and/or biobased resin.

Description:
Conventional thermoset composites are non-recyclable and non-reprocessable, leading to significant waste during manufacturing. Existing recycling methods are energy-intensive, toxic, and degrade material properties. Recent advances in vitrimer and biobased resins offer a breakthrough, enabling full carbon fibre recovery and resin depolymerisation under mild conditions. This opens the path to sustainable, high-performance composite reuse.

This activity aims to demonstrate that, beyond offering 3R capabilities (repair, reuse, recycle) these composite structures also fully meet the requirements for space and launcher applications. This shift in approach is fully applicable to the production of structural sandwich panels with honeycomb and all monolithic parts like cylinder, struts, cleats, brackets and any other Carbon Fiber Reinforced Polymer (CFRP) made object.

Finally, this activity could enable a circular approach for future Moon or Mars missions, where unused components can be reused, reprocessed, or repurposed into new functional parts. Additionally, the demisability of such materials is expected to improve, reducing the risk of space debris.

This activity encompasses the following tasks:
- Define high level mechanical and thermal requirements for space applications.
- Perform materials and processes trade-off that enable manufacturing, recycling, reprocessing and waste management of composites taking into account requirements for space applications.
- Produce a test plan to assess at sample level the manufacturing, recycling, reprocessing and waste management capabilities in addition to mechanical and thermal performance based on the trade-off (at least 2 solutions shall be implemented).
- Propose at least 2 designs based on space applications capable to demonstrate the advantage of this approach.
- Manufacture and test 2 different breadboards (reduce scale).
- Update design to produce and test full scale model.
- Perform an environmental assessment through preliminary Life Cycle Assessment (LCA) to prove the environmental savings.

Deliverables: Breadboards, Reports

Application/Need Date: TRL8 in 2030 for all missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Zero Debris - Develop affordable disposal solutions for small spacecraft

Propulsion, Aerothermodynamics and Flight Vehicles Engineering

GT1C-805MP – Disruptive electric propulsion sub-system for cubesat de-orbiting

Budget: 1500 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 3 / 7 - **TD** 19

Objective:
To develop a disruptive and reliable electric propulsion (EP) de-orbiting subsystem, compatible with 6U-16U+ CubeSats operating in LEO.

Description:
Market analysis forecasts the launch of over 2000 CubeSats this decade. To ensure sustainability in LEO, it is paramount to foresee the availability of reliable propulsion subsystem enabling also end-of-life disposal, especially of those CubeSats operating at high altitudes, where natural decay is insufficient to guarantee re-entry within the time recommended by the ESA Space Debris Mitigation Policy (5 years, with a success rate between 90% and 95% at the end of the mission).

Existing propulsion technologies for CubeSats (chemical or electric) have limited proven performance, which may prevent the full exploitation of future CubeSats at higher LEO altitudes and for multiyear lifetimes.

This activity targets the development and ground qualification of a disruptive electric propulsion subsystem compatible with 6U to 16U+ CubeSats operating in LEO. This subsystem shall be capable of delivering augmented performance compared to existing products or current subsystems. Specifically, it should achieve a delta-V greater than 100 m/s, maintain a form factor of less than 2U, exhibit a thrust-to-power ratio exceeding 0.5 mN/W, deliver a specific impulse (ISP) greater than 80 seconds, and demonstrate a reliability greater than 0.95. In addition, this EP subsystem shall include all necessary propulsion equipment, including main thruster, propellant, propellant storage and management system, power processing units with firmware/software, pipework, harness.

This activity is organized in two phases and encompasses the following tasks:
Phase 1 (800 k€, 24 months):
- Specify and design the complete Electric Propulsion subsystem.
- Manufacture and test development models (DMs) of the subsystem main elements to preliminary verify the performance and validate equipment and subsystem design. This shall include a vacuum coupling subsystem test and a thruster endurance test of minimum 300 hours to demonstrate the subsystem performances.

Phase 2 (700 k€, 12 months):
- Manufacture a qualification model of the EP subsystem and perform a complete qualification including life test.

Deliverables: Breadboard, Qualification Model, Reports

Application/Need Date: TRL8 in 2030 for all LEO 6U-16U+ CubeSats and MicroSats. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Electric Propulsion Technologies.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Structures, Mechanisms & Materials

GT1C-806MS – Advanced simulation for resilient collision modelling to enhance health monitoring and prognostics

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 20

Objective:
To develop advanced collision and structural models integrated into operational simulators to enhance impact detection, damage assessment, and satellite health monitoring.

Description:
The growing density of space objects, including debris and satellites, significantly increases the risk of collisions in orbit. Traditional collision models often lack the resolution and adaptability to predict and assess the impact of near-real-time collision events on spacecraft health. At the same time, satellite prognostics and health management (PHM) systems depend heavily on accurate scenario modelling to assess and anticipate degradation and failure modes. Integrating enhanced collision modelling with PHM tools can offer more robust, resilient simulation environments that allow spacecraft operators to test, validate, and calibrate satellite systems against realistic, high-risk environmental scenarios. Also, by embedding advanced and dynamic collision modelling into spacecraft simulation environments, this activity provides operators and designers with a realistic simulation environment for stress-testing resilience strategies, improving anomaly detection, and extending satellite operational life. It will include updated debris fields, event-triggered degradation modelling, and probabilistic damage assessment. The outcome will be an asset in supporting decision-making for mission planning, satellite autonomy, and reactive manoeuvres. The simulation environment will be modular and extensible, enabling integration with existing mission planning and PHM tools.

This activity is organized in two phases and encompasses the following tasks:
Phase 1 (budget: 1100k€, duration: 14 months):
- Definition of a representative structure, high-velocity impacts (HVI) and micro-meteoroids and orbital debris (MMOD) simulations, definition of structural health monitoring (SHM) system and assessment of requirements for the usage in testing and operations.
- Development of a simulation model of a digital twin representative structure able to analyze HVI and MMOD damages.
- Ground HVI testing of physical hardware (coupon/assembly level) to validate the HVI detection capability of the SHM, to assess real impacts/damage and potential failure, correlation with simulation. This task shall leverage the developed simulation model to optimize the needed physical testing.

Phase 2 (budget: 900k€, duration: 10 months):
- Integration of the developed model into a simulation platform (e.g., operational simulators) and ensure the compatibility with existing tools, PHM algorithms and monitoring dashboards.
- Validation of the structural module in the operational simulator, together with other operational environment (telemetry/housekeeping data) to demonstrate the capability of decision making after HVI.
- Deploy the framework and prototypes to end-users.
- Assess the performance and impact of the solutions.

Deliverables: Breadboard, Software, Report,

Application/Need Date: Collision assessment training for operators. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Structures, Mechanisms & Materials

GT1C-807MS – Innovative materials and structural concepts for a resilient satellite to debris collisions

Budget: 2500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 11

Objective:
To develop and qualify material(s) and structural concept(s) ensuring the satellite robustness to debris impact and minimize debris generations.

Description:
Space debris poses a significant risk to operational satellites, with hypervelocity impacts (HVIs) being a major concern. The velocity, size, mass, and frequency of such impacts vary across different orbital environments, and understanding these parameters is critical for mitigating the risk to satellite systems and developing resilient spacecraft that can withstand these impacts while maintaining their functionality in a sustainable way.

Current shielding technologies tend to add significant mass and volume, making them less viable for certain missions. Therefore, identifying more efficient, lightweight, and innovative technologies for shielding is necessary to enable shields that minimize debris generation whilst promoting long-term sustainability in space operations.

This activity will build upon the most promising technologies identified and tested in a the TDE activity Innovative resilient satellite space system design solutions for Zero Debris and further develop lightweight shielding technologies.

This activity is organized in two phases and encompasses the following tasks:

Phase 1 (budget: 1000k€, duration 10 months):
- Review the HVI technologies identified in the previous activity at conceptual stage and then trade off the most promising candidates considering the state-of-the-art material technologies available in the market.
- Perform screening using simulation and hypervelocity testing of samples at appropriate facilities to validate performance of identified material technologies, particularly the requirements to survive the HVI requirements while being as lightweight as possible.
- Perform another trade study of the candidate materials technologies accounting for the actual demonstrated performance against the different requirements in particular HVI resistance and potential applications that enable the HVI shield concepts identified from previous tasks.

Phase 2 (budget: 1500k€, duration 14 months):
- Develop different detail designs of HVI shields using the most promising material technologies down selected from the previous phase, accounting for combinations of passive structures and/or deployable mechanisms as needed to provide an optimised design that meets system requirements (e.g. Whipple shield bumper for front/rear wall, deployable structures for Whipple shield, integrated sensors).
- Design review to identify the most promising HVI shield design(s) that satisfy system requirements, particularly mass, volume and power constraints. At least one design shall be selected to build the demonstrator breadboard.
- Manufacture at least two versions of the demonstrator breadboard for each selected shield design and test them under hypervelocity impact conditions to assess their performance.

Deliverables: Breadboard, Reports

Application/Need Date: TRL8 in 2032 for all missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Avionics Embedded Systems.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Space Segment Systems Engineering

GT1C-808SY – Autonomous collision avoidance system

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - **TD** 11

Objective:
To develop an on-board autonomous decision system for risk assessment and manoeuvre selection leveraging on-board Global Navigation Satellite System (GNSS) data and Fault Detection Isolation and Recovery (FDIR) adaptation for constellation-level collision mitigation.

Description:
Collision avoidance for Low Earth Orbit (LEO) satellites currently largely depends on ground-based computation and decision-making, causing delays. Multiple communication cycles are needed between ground and spacecraft, often resulting in decisions based on outdated data. This increases the risk of suboptimal manoeuvres and leaves little time to react if a manoeuvre fails, raising the probability of collision. At the same time the number of spacecrafts grows, and collision calculations become increasingly intensive. In this context, there is a strong need to enhance the efficiency and accuracy of collision detection and conjunction estimation for large spacecraft fleets.

The proposed approach to be developed within this activity consists in autonomous on-board decision-making and pre-calculations, e.g. Graphics Processing Units (GPU)-based, to identify high-probability collision pairs early, providing a preemptive list of potential collision candidates, allows computational resources to focus where they are most needed. These selected cases can be analyzed using more sophisticated and resource-demanding models, including on-board GNSS data, but applied only to a significantly reduced subset of conjunctions and satellites in a constellation scenario.

This activity shall develop and demonstrate a system comprised of a ground component and an onboard component. The ground component aggregates conjunction and orbital data from multiple sources, processing it to identify high-risk events and potential mitigation manoeuvres. Pre-approved manoeuvres, along with their priority rankings and validity ranges, and critical Conjunction Data Messages (CDMs) are then uploaded to the spacecraft for further processing by the onboard component. The onboard component processes GNSS data in near real-time to refine conjunction risk assessments. If the risk remains critical at the decision point, the highest priority pre-loaded manoeuvre that still fits the criteria is executed. Additionally, on-board monitoring of Collision Avoidance Manoeuvre (CAM) execution allows satellites to verify manoeuvre success in real-time. If an anomaly is detected, the satellite can autonomously trigger a backup manoeuvre, eliminating delays caused by waiting for ground control confirmation.

This activity encompasses the following tasks:
- Define the architecture for a conjunction estimation and CAM computation system for large numbers of spacecraft (higher than 50000).
- Define and integrate a hardware test-bed.
- Implement a collision detection algorithm on GPU.
- Implement test-bed of the architecture for up to 10000 objects.
- Test the ground component which aggregates conjunction and orbital data, identifies high-risk events, and pre-validates manoeuvre options with associated priority and validity constraints ready for uplink.
- Test the space component which processes GNSS data in near real-time to contribute to on-board risk assessment and autonomous manoeuvre definition.

Deliverables: Engineering model, Software, Reports

Application/Need Date: TRL 8 in 2030, for all LEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Space Debris

GT1C-809SD – Manoeuvre analytics software for space traffic management and coordination

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - **TD** 11

Objective:
To build methods and software for manoeuvre detection and pattern of life reconstruction for supporting collision avoidance and space traffic management.

Description:
Detailed knowledge of the manoeuvre capabilities and manoeuvre patterns of space objects in the orbital environment is essential for effective collision avoidance and space traffic management. This information is important both at operator-level, for informing decision making in case of conjunctions with active neighbours, and more globally, in facilitating coordination and safeguarding the space environment. Moreover, monitoring global satellite behaviour, such as end-of-life disposal, is key to promoting sustainable space practices and ensuring compliance with debris mitigation requirements and guidelines, tracking progress towards Zero Debris.

This activity will develop methods and software to extract "pattern of life" information from surveillance data, including detailed manoeuvre analytics, to maintain a catalogue of manoeuvre information for all tracked objects. By integrating space surveillance data sources, the software will support automated, real-time monitoring, requiring minimal analyst intervention. This serves to enhance collision risk management, to predict whether an active satellite may manoeuvre into the path of another, as well as space traffic coordination between space operators in the case of high-risk close conjunctions (to judge manoeuvre responsibility). From a wider perspective, it will also enable tracking of global adherence to sustainable space practices to promote responsible behaviour in orbit and shape future policy, for example, in support of a "rules of the road". The tool shall also facilitate the reconstruction of manoeuvre histories based on available historical surveillance data, to be used for analysing global space debris mitigation compliance statistics and in the definition of a space traffic management rules of the road.

This activity encompasses the following tasks:
- Develop methods and software tools to extract pattern of life information on space objects including detailed manoeuvre analytics both during operational phase (not limited to classification of station keeping or collision avoidance manoeuvre, date of last manoeuvre, manoeuvre frequency, station keeping envelope, classification of chemical propulsion, electrical propulsion, or differential drag) and during the disposal phase (end of life forensics).
- Develop a flexible information fusion approach for aggregating difference sources of surveillance data for this task.
- Ensure a high degree of automation in the tool, minimising analyst intervention to support regular deployment for maintaining a catalogue of current manoeuvre activity information of all tracked objects.

Deliverables: Software, Reports.

Application/Need Date: TRL8 in 2030, for all missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Space Debris

GT1C-810SD – Data fusion for collision avoidance and space traffic coordination

Budget: 500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 11

Objective:
To develop an information fusion software integrating heterogeneous space surveillance data into a consistent catalogue for collision avoidance and space traffic coordination.

Description:
Timely and efficient planning, coordination and execution of collision avoidance manoeuvres requires accurate and consistent knowledge on the orbits. The Zero Debris targets on orbit accuracies are only achievable when increasing the data volume and mixing different data sources.

Surveillance systems may produce conflicting information due to different model assumptions, which cannot be merged with classical data fusion approaches. This is an adverse condition for space traffic coordination, especially when multiple parties rely on different data sources. The harmonisation of the data sources, estimation of biases between systems, identification of deterioration of specific sources, characterisation of combined uncertainties are mandatory requirements to achieve the accuracy levels. Typically, data fusion is performed on different product levels, e.g. raw measurements or derived products.

This activity encompasses the following tasks:
- Develop a flexible information fusion approach and software to ingest orbital and observation data from various sources to generate a consistent non-conflicting estimate.
- Alternative visualisation or recommendation systems must be derived based on multiple products to support operator decisions.
- Automated data governance, performance monitoring and degradation identification using machine learning approaches. This requires traceable adaptive calibration / weighting of solutions. The software shall allow inclusion of trusted and untrusted data sources, allowing ingesting data from currently not sources new sources: in-orbit data, new data types from non-traditional sensors.

Deliverables: Software, Reports

Application/Need Date: TRL7 in 2028, for all missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any harmonisation topic.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

Avionics & EEE

GT1C-811ED – On-board event-based debris detection and collision avoidance for LEO satellites

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 1

Objective:
To develop an on-board system for autonomous, real-time debris detection, tracking, and avoidance using a multi-sensory payload.

Description:
The growing presence of space debris in Low Earth Orbit (LEO) threatens satellite operations and space sustainability. Small, fast-moving debris is difficult to detect yet capable of inflicting significant damage. Current avoidance strategies heavily depend on ground-based tracking and human-in-the-loop decisions, which can be slow and constrained by communication delays. This proposed activity builds upon a precursory ESA activity NEU4SST – Neuromorphic Processing for Space Surveillance and Tracking, which produced valuable insights and a limited dataset. This dataset however was generated with a ground-based sensing focus and lacks realism, metadata richness, and orbital context required for space-borne applications.

The aim of this activity is to demonstrate autonomous collision avoidance capabilities through real-time space debris detection and avoidance in LEO using event-based data fusion. To achieve this, this activity will develop a realistic, high-fidelity simulated dataset that mimics the operational environment of a spacecraft equipped with an event-based camera and navigation sensors (GNSS, IMU). This includes dynamic scenes with sub-10 cm debris, representative lighting conditions, orbital motion, and synchronized multi-sensor outputs. This dataset will be used to validate real-time detection, tracking, and collision avoidance algorithms based on Spiking Neural Networks (SNNs). Edge-based processing algorithms will be demonstrated running on representative space-grade or space-relevant embedded hardware platforms to evaluate their performance and feasibility under hardware constraints. In addition to the simulation-based prototype, the activity will develop a testbed prototype that integrates a physical event-based camera with a processing unit capable of executing the developed algorithms. This setup will allow real-time, ground-based demonstration of key capabilities, such as detection of fast-moving objects, real-time data fusion, and autonomous decision-making logic, in a controlled lab environment.

This activity encompasses the following tasks:
- Design and implement an on-board autonomy architecture using an event-based optical camera to enable decentralized decision-making for collision avoidance across spacecraft.
- Develop edge-based processing algorithms for real-time debris detection and tracking, leveraging simulated and semi-simulated datasets from the on-board sensory suite to estimate debris size, assess conjunction risk, trigger alerts and to optimize autonomous manoeuvres.
- Implement on-board autonomous conjunction risk assessment and collision avoidance manoeuvre planning, including demonstration of the system on low-power space-grade/space-relevant hardware to evaluate performance and reliability in constrained environments.
- Simulate and validate high-fidelity LEO scenarios in controlled lab environments, including variable debris densities, orbital regimes, and dynamic interaction scenarios, to benchmark system performance under realistic operational conditions.

Deliverables: Prototype, Reports

Application/Need Date: TRL7 in 2030, for all LEO missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Sustainable Space

Zero Debris - Improve satellite resilience to impacts

RF Payloads & Technology

GT1C-812EF – W-band gyro traveling wave tube for centimetre-level space debris detection

Budget: 750 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 4 - **TD** 11

Objective:
To design, manufacture and test a W-band gyro-TWT slow-wave structure as a key RF amplifier component for high-resolution space debris tracking radar.

Description:
With the advent of mega-constellations, the number of objects in space is ever growing, especially in low-earth orbit (LEO). This continuous population of LEO orbits with satellites also inevitably results in a higher amount of small to large sized uncontrolled objects in orbit. For space safety, these objects, their size and their trajectories need to be carefully monitored to avoid any risks of collisions of these objects with operating satellites. ESA has set itself an ambition target with its Space Debris Mitigation policy, as described in ESSB-ST-U-007, which supports the need for improving detection and tracking capabilities of debris compared to current capabilities. To continuously monitor space debris in LEO, in day and night and under all weather conditions, a ground-based radar is the only option. At present, the European Space Surveillance and Tracking (SST) system can detect up to 20 % of objects with a size of down to 7 cm in LEO with the plan to increase the detection rate and sensitivity to detect even smaller objects.

To accurately monitor small objects of sizes down to 3 cm in LEO with a radar, a high-power and high-frequency RF generation unit is required. This RF generation can be realised using a W-band gyro-TWT (96 GHz), with 8 GHz of bandwidth and of >5 kW pulsed power at 10% duty cycle. Such devices have been demonstrated for research purposes and can be further developed to be used in ground-based radar systems.

A gyro-TWT is composed of multiple building blocks, namely an electron gun, a slow-wave structure and a collector. The most critical of these building blocks is the slow-wave structure, itself composed of an interaction region with input and output couplers/mode converters.

This activity encompasses the following tasks:
- Review of electron beam characteristics based on electron gun performance and definition of detailed gyro-TWT slow-wave structure requirements.
- Designing a slow-wave structure using CAD software with a focus on RF performance as well as thermo-mechanical aspects.
- Analyse the manufacturing techniques and materials required to realise such a structure with the right tolerances.
- Manufacture the gyro-TWT slow-wave structure.
- Measure the complete gyro-TWT slow-wave structure to confirm the design.
- Review the outcome of the measurements and compare with the original simulation results.

Deliverables: Breadboard, Reports

Application/Need Date: TRL6 in 2032 for an entire gyro-TWT. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Circular Economy in Space / ISAM - Ensure safer Close Proximity Operations (CPO) to lower the risk of collision

Optics, Robotics & Life Sciences

GT1C-813MM – Frequency modulation continuous wave LiDARs based on photonic integrated circuits for debris identification

Budget: 1000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 17

Objective:
To develop an Engineering Model (EM) Photonic Integrated Circuit (PIC) Frequency Modulated Continous Wave Lidar (FMCW) for the detection and capture of uncooperative targets (debris).

Description:
One of the most difficult tasks for the removal of space debris, is the identification and autonomously capture the objects without jeopardizing the integrity of the main spacecraft. Lidars offer the possibility to identify objects shape and distance with a single analysis and without the use of complex algorithms. Moreover, FMCW lidars provide the possibility to obtain object velocity and rotation with a single shot.

The technology is currently being used in autonomous driving thanks to the benefits of being operational at different illumination conditions and by providing also a 3D point cloud and velocity of the detected objects to distances up to 500 m. Similar systems could be implemented for autonomous detection and removal of debris.

Photonic Integrated Circuits (PIC) lidars offer the advantage of low SWaP (<1 kg and 1 U) while promising higher device accuracy. PIC lidar devices with a range between 0-500 m could be used for a wide range of applications as Clean Space, RV&D or in-orbit servicing. The main tasks of the activity will be to identify current PIC Lidar building blocks (e.g. photonic integrated emitting laser, antenna arrays, detection scheme) used in on-ground applications, to later tailor them to requirements and environment needs for the different space applications.

This activity will encompass the following main tasks:
- Identification of Clean Space requirements for detecting, tracking and capturing objects and of current building blocks used in the automotive industry.
- Trade-off different photonics building blocks used in the automotive industry tailored to requirements and space compatibility.
- Justify or test compatibility to space environments of the critical functions of the integrated photonic building blocks (e.g. thermal gradients or radiation resistance).
- Manufacture an engineering model of a system capable to fulfil clean space requirements (e.g. range, albedo, size, speed).
- Testing of an engineering model in relevant environment (e.g. validate detection and speed measurement of different size and albedo objects at distances between 0 to >300 m).

Deliverables: Engineering Model, Software, Reports

Application/Need Date: TRL8 in 2030 for all Active Debris Removal missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Sustainable Space

Circular Economy in Space / ISAM - Ensure safer Close Proximity Operations (CPO) to lower the risk of collision

GNC, AOCS & Pointing

GT1C-814EC – Close-proximity range finder for rendezvous and docking

Budget: 800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 7 - **TD** 5

Objective:
To adapt, manufacture and qualify a small form factor range finder able to provide distance measurements for docking operations.

Description:
With the growing number of in-orbit servicing and close-proximity operations, there is currently a missing sensor in the European portfolio: one capable of providing precise distance measurements near the docking area. This sensor is essential for accurately positioning a servicing spacecraft or its docking mechanism right up to the point of contact. Introducing such a sensor would significantly reduce the risk of collisions and debris generation during these operations. It would do so by offering an alternative source of positioning data to complement visual-based navigation systems.

As the sensor operational time would be very reduced, the proposed approach in this activity is to adapt an existing terrestrial sensor by characterising it for space conditions, specifically for radiation exposure and thermal vacuum environments. To ensure reliability, redundancy would be implemented using two or three units, especially given that these sensors are expected to be very compact.

The proposed sensor should operate over a range of a few meters down to direct contact, offering millimetric accuracy to ensure precise positioning during proximity operations. To meet the constraints of space missions, the sensor must have a mass of less than 300 grams and consume no more than 2 watts of power. It should also be characterised for resilience to Single Event Effects (SEE) to ensure reliability in the space radiation environment. This new sensor should be versatile and have multiple users e.g. not tailored to a single mission.

This activity encompasses the following tasks:
- Identification and characterisation of the existing terrestrial sensor, analysing current capabilities and assessing via gap analysis the suitability to space specificities.
- Implementation of the necessary modifications to the sensor, re-design if required, and conduction of design reviews.
- Manufacturing the final unit and conducting qualification tests to secure the survivability to launch environment (mechanical vibrations and shocks) and space environment (thermal cycling, vacuum, radiation environment and blinding).

This activity is proposed to be implemented in Bundle 8 – Enabling Proximity operations and rendezvous, together with the activities GT1C-815EC – Dual navigation and control of a servicer spacecraft and its robotic arm and GT1C-816EC – Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations.

Deliverables: Engineering/Qualification Model, Reports

Application/Need Date: All In Orbit Servicing missions (e.g. RISE). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Sustainable Space

Circular Economy in Space / ISAM - Ensure safer Close Proximity Operations (CPO) to lower the risk of collision

GNC, AOCS & Pointing

GT1C-815EC – Dual navigation and control of a servicer spacecraft and its robotic arm

Budget: 1800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 6 - **TD** 13

Objective:
To develop GNC algorithms involved in the navigation and control of a robotic arm and servicer spacecraft involved in in-orbit-servicing tasks.

Description:
As in-orbit servicing missions become more complex, the need for precise coordination between a servicer spacecraft and its onboard robotic arm becomes critical. Achieving reliable docking or capture requires advanced guidance, navigation, and control (GNC) strategies that manage the dynamic interaction between these two subsystems. The dynamical interaction between the servicer spacecraft platform control and the robotic arm operation needs to be managed to ensure the feasibility of docking / berthing / capture operations. Several architectures can be used to manage this interaction, from fully decoupled, with separate GNC systems for the servicer platform and robotic arm, to fully coupled ones, where a single GNC system controls both in coordination. These two architectures, and any in-between, represent a different combination of technical and programmatic attractiveness. This activity aims to assess the viability of these approaches and will target the parallel implementation and benchmark of the two paradigms to compare their performance and robustness:
- combined navigation and control: the robotic subsystem is a commandable appendage; a global control plant is synthesised as well as a unique state estimation filter.
- decoupled navigation and control: the robotic subsystem and servicer have separated navigation and control architectures and exchange either no or minimal information, e.g. feedforward of arm movements. This activity shall develop the GNC algorithms up to demonstration in a robotic facility using a target mock-up with representative free-flight and contact dynamics.
This activity is organized in two phases and encompasses the following tasks:

Phase 1 (budget: 1,000 k€, duration: 15 months):
- Detailed modelling and system dynamics analyses in order to define and consolidate AOCS/GNC requirements and a V&V plan.
- Design and development of AOCS/GNC approaches and algorithms that fulfil the identified requirements and needs, including detailed design and implementation of the proposed solutions and algorithms with special focus on performance, robustness, and optimization for embeddability in space-graded processors.
- V&V campaign of the proposed solutions and algorithms in model-in-the-loop and SW-in the-loop setups.
Phase 2 (budget: 800 k€, duration: 9 months):
- V&V campaign of the proposed solutions and algorithms in a processor-in-the-loop setup with emulation of hardware and focus on the assessment of real-time properties.
- Closed loop test campaign in a robotic facility with representative hardware-in-the-loop (e.g. EM of selected sensors or COTS sensors) and representative dynamics.

This activity is proposed to be implemented in Bundle 8 – Enabling Proximity operations and rendezvous, together with the activities GT1C-814EC - Close-proximity range finder for rendezvous and docking and GT1C-816EC – Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations.
Deliverables: Software, Reports

Application/Need Date: TRL 6 end 2028 for all In-orbit servicing (e.g. RISE) and InSPoC missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Sustainable Space

Circular Economy in Space / ISAM - Ensure safer Close Proximity Operations (CPO) to lower the risk of collision

GNC, AOCS & Pointing

GT1C-816EC – Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 5 - **TD** 5

Objective:
To develop navigation algorithms and hardware for rendezvous and proximity operations using multispectral sensors and a ranging device, and define hardware needs to achieve required navigation precision.

Description:
Missions with rendezvous and docking phases, such as In-Orbit Servicing, Assembly and Manufacturing, and In-Space Transportation, require navigation systems able to handle dynamic illumination conditions, radiation issues, and orbit-specific dynamics. If GNC systems rely on favourable illumination, operations can become complex and risky. Thus, developing navigation systems less sensitive to illumination constraints is crucial.

To tackle this issue, this activity will focus on the development of multispectral camera systems. For this, lessons learned from previous ESA activities will be considered, starting at the level of image-processing, data fusion with the ranging device, and navigation function, ultimately leading to the development of an elegant breadboard for a multispectral camera system. Hardware requirements will be driven by the needs of image-processing and navigation software.

This activity encompasses the following tasks:
- Specification of rendezvous scenarios that the navigation system will have to serve, including navigation requirements, derivation of required performance of the pose estimation, data fusion and image processing algorithms, derivation of hardware requirements for all hardware elements of the sensors and their outputs, identification of in-orbit tuneable parameters, derivation of the requirements on cameras to provide the required images.
- Preliminary design of the multispectral navigation camera and ranging system, including design of the full-fledged camera and ranging system.
- Development of the image-processing, data fusion and pose estimation, and navigation algorithms, including: trade-off and definition of working ranges for the vision-based navigation system, algorithm development, coding of the algorithms according to SAVOIR guidelines, implementation of parallelization in hardware of the relevant parts of the algorithms.
- Strategic commercial off-the-shelf (COTS) selection, including trade-off and selection of space-grade boards/FPGAs/OBCs exploring mature and cost-effective COTS.
- Assembly, integration and testing of the system, including functional and performance tests to demonstrate the ability to deploy the software developed onto processor boards that are space qualified for the relevant missions, perform profiling and verify satisfaction of footprint and speed, functional and performance testing of the whole unit in a test facility.

This activity is proposed to be implemented in Bundle 8 – Enabling Proximity operations and rendezvous, together with the activities GT1C-814EC - Close-proximity range finder for rendezvous and docking and GT1C-815EC – Dual navigation and control of a servicer spacecraft and its robotic arm.

Deliverables: Breadboard, Software, Reports

Application/Need Date: TRL 6 end 2027 for all In-Orbit Servicing, space transportation and refuelling missions (e.g. InSpoc-1/2, CAT-IOD). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Specific Areas

(Cyber)security

Serialisation / Standardised Sub-systems

Technologies for VLEO


(Cyber)security

4.1.1. Introduction and selection criteria

The current Compendium gathers the Technology Developments on Cyber Security for Space proposed by the European Space Agency for implementation in 2026-2028 timeframe. The activities were prepared and coordinated by an ESA cross-directorate Task Force and in collaboration with the ESA Cyber Security Coordination Board, to ensure coverage of different but transversal needs while avoiding duplications. The objective is these technology developments to constitute the cornerstone of the security protection of future space missions, for ESA, for its Member States and their Industry.

Space systems constitute part of our critical infrastructure; investment in space systems increases, but also interconnectivity with terrestrial systems and adoption of terrestrial technologies. More and more space applications are meant to directly reach end users, while payloads are shared between different users or stakeholders. In parallel, new technologies bring new opportunities but also require new security measures. EU recently published the draft of the new EU Space Act, where, among others, security specific capabilities are mandated which require some technology developments (like on board security monitoring). ESA proactively and accurately through the Task Force had identified these needs and included them in the current Compendium.

The activities of this Compendium constitute a continuation of an effort started already in 2019 and continued in 2022 with the corresponding Cyber Security Compendia and hence are not meant to start from scratch; on the contrary they will (also) use previous Basic Activities of a Lower TRL as a stepping stone to facilitate their development. They are envisaged to cover the most important pillars of the security protection of space systems. These include traditional areas, like protection of the communication links and ground segment cyber security, but also space segment cyber security and new areas like Space Situational Awareness and Cyber Threat Intelligence.

The activities proposed are part of the Cyber Security Building Blocks for Space Systems. Building block is a term used to express the notion of products (hardware or software) that constitute and can implement a security reference architecture in a modular way (i.e. adding or removing ad-hoc building blocks without having to re-design the whole architecture). This modular security reference architecture will be used as a starting point for all future space missions. Each mission, based on its needs and following a threat and security risk analysis process, will tailor the security architecture and identify the cyber security building blocks needed to protect it. While there is no one size that fits all in terms of SWaP (Size, Weight and Power), required assurance, etc., different instances of the same building block can be developed to cover the most important foreseeable mission needs. ESA vision is that instances of these building blocks to become available in the market as Commercial-Off-the-Shelf products that mission designers / integrators can buy and use, reducing the cost (since there will be no need for development as part of the mission) and eliminate potential delays. Moreover, this approach will accelerate commercialisation and industrialisation, while avoiding reinventing the wheel and reducing the risk of security pitfalls through the use of well-established and assessed security products.

Specifically, the activities proposed as part of this Cyber Security Compendium are the following:

Protection of Communication Links (RF) and User Data (Crypto):

Advanced Interference Detection and Protection Module, aiming at developing an interference detection and protection module which can be used as an on-board building block for next generation space Missions.

Application layer cryptographic capability for user data protection and segregation, aiming at implementing a “multi-user” satellite payload quantum resistant cryptographic unit for payloads for unicast, multicast and broadcast messages.

Ground Segment Security:

Ground Segment Confidential Computing for Space Safety Data Processing, to develop and validate confidential computing and privacy-preserving software for space safety and payload data processing applications.

Ground Segment Artificial Intelligence Security Monitoring for Space Operation, having the objective to develop and implement a high-confidence, reliable AI enabled security monitoring capability tailored for the space systems domain to improve on threat detection, incident response and system security monitoring capabilities for space missions.

Ground Segment BPSec Key Management Unit, to develop and validate a BPSec (Bundle Protocol Security) implementation, testbed and supporting key management capabilities in representative Solar System Internet scenarios.

Space Segment Cyber Security:

End-point security detection and protection, aiming at developing and validating an on-board end-point security system for satellite components and subsystems, enabling real-time detection and mitigation of cyber threats targeting software, firmware, and hardware interfaces.

Spacecraft digital forensic and incident handling capability, to develop a tamperproof isolated module that ensures the authenticity of on-board security-relevant information, enabling the reliable collection, storage, and downlinking of forensics evidence, and provide support for incident investigation and recovery. The activity will build on the outcome of a previous Cyber Security Basic Activity, the results of which will become available to interest industries.

Secure reconfigurable software defined satellites, aiming to implement, test and verify secure reconfigurable technologies needed to support software defined satellites and in-flight re-programmability capabilities that will enable avionic platforms to support many missions.

Space Situational Awareness and Threat Intelligence:

Optical Space Surveillance and Tracking unit for space assets security, to develop and implement an on-board optical Space Surveillance and Tracking (SST) unit that grabs and processes visual data to identify, track and characterize malicious objects targeting space assets.

Space Cyber Threat Intelligence module, that can be deployed as a dedicated spacecraft payload. The module shall be able to collect real-time Tactics, Techniques and Procedures (TTPs) of space cyberattacks and ensure a secure downlink for analysis on ground, to guide future cyber defence developments, inform operations for better defence of existing missions.

ESA believes that the Technology Developments on Cyber Security will help Industry, ESA and its Member States to further secure future space missions.

4.1.2. List of activities

Cybersecurity

System and Applications Engineering

Programme Reference	Activity Title	Budget (k€)
GT1Y-800GA	Ground segment BPSec key management unit	1,500
GT1Y-801GA	Ground segment confidential computing for space surveillance and payload data processing	1,200
GT1Y-802GA	Advanced techniques for space operations security monitoring	1,200
	Total System and Applications Engineering	**3,900**

End-to-End Systems

Programme Reference	Activity Title	Budget (k€)
GT1Y-803SE	Advanced interference detection and protection module	2,000
GT1Y-804SE	Application layer cryptographic capability for user data protection and segregation	3,000
GT1Y-805SE	Secure reconfigurable software defined satellites	2,000
GT1Y-806SE	Space cyber threat intelligence module	2,000
GT1Y-807SE	Optical space surveillance and tracking unit for space assets security	2,500
GT1Y-808SE	End-point security detection and protection	1,500
GT1Y-809SE	Spacecraft digital forensics and incident handling	2,000
	Total End-to-End Systems	**15,000**

Total Cybersecurity	**18,900**

4.1.3. Description of activities

Cybersecurity

System and Applications Engineering

GT1Y-800GA – Ground segment BPSec key management unit

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 8

Objective:
To develop and validate a BPSec implementation, testbed and supporting key management capabilities in representative solar system internet scenarios.

Description:
The Bundle Protocol (BP) is set to become a foundation for interplanetary networking and the solar system internet (SSI). The supporting BPSec protocol provides associated security services addressing bundle confidentiality and integrity. Reference BPSec implementations are still in early development stages and lack end-to-end validation including definition and testing of supporting security management processes (e.g. key management). The development of such architecture, BPSec key management systems on ground, and supporting operation concepts present several challenges, for example due to shared multi-agency networks with commercial or private relay nodes, disrupted line-of-sight and different service level agreements and policies, and the need to support post-quantum cryptography. Centralized and hierarchical key management or conventional public key infrastructure (PKI) solutions may not necessarily represent the best fit and other options such as decentralized, distributed or peer-to-peer solutions, merkle-tree and proof-of-inclusion concepts may offer solutions to be explored and developed to ensure future scalability.

Building on state-of-the-art developments and aligning with planned network management and higher-maturity on-board implementations, this activity shall develop and validate, within a representative testbed, a BPSec (key) management unit including supporting management concepts and procedures. The development will be subjected to security and stress testing through simulation and emulation of attacks and traffic in delay tolerant network (DTN) contexts. Fuzzing of BP decoders and various convergence layers will be included, as well as service and network route discovery mechanisms. Verification shall include new security key and certificate management concepts in DTN scenarios, deployed with representative network and data systems. The development shall be fully aligned with planned on-board module developments, ensuring an end-to-end, cohesive concept.

This activity encompasses the following tasks:
- Perform a critical analysis of state-of-the-art technologies related to BPSec testbeds, network management, on-board implementations, cryptography and protocol standardisation.
- Specify and design the BPSec scenario emulation environment and operational concept.
- Specify and design the ground segment BPSEC key management unit.
- Develop, deploy, verify and validate the BPSec key management unit within a fully representative environment, demonstrating the end-to-end key management functionalities and operational concepts as well as the security resilience of the solution.

Deliverables: Report, Software

Application/Need Date: TRL 8, by 2031 for future exploration and communication missions (e.g. Lunar Navigation and Communication Services, Gateway, Solar System Internet Pathfinder, Argonaut). **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Progressive adoption of Bundle Protocol based Networks.
Relevant to the Roadmap Functional Verification and Mission Operations System.

Cybersecurity

System and Applications Engineering

GT1Y-801GA – Ground segment confidential computing for space surveillance and payload data processing

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 9

Objective:
To develop and validate confidential computing and privacy-preserving software for space safety and payload data processing applications.

Description:
Confidential Computing (CC) and related technologies including homomorphic encryption, secure multi-party computation and zero knowledge proofs, facilitate secure computation across heterogeneous data sources without compromising individual dataset confidentiality and data owner control. As the number of data owners and data service processing service providers proliferates, and given needs for privacy-preserving, dataset-spanning analysis and processing capabilities, such technologies offer promising solutions to facilitate down-stream applications.

Priority use cases and scenarios which can benefit from such technologies have been identified in the domains of space surveillance and tracking (SST) and operational collision avoidance, and in payload data product processing for space safety and earth observation missions. In the case of SST, application scenarios include spacecraft conjunction calculations on confidential space assets orbital data. Given the increasing number of commercial and institutional SST data service providers, the overall number of assets in congested orbits, and the subset of these for which the operator requires anonymity/confidentiality, a capability for such confidentiality processing is becoming an increasing need. For space safety (for example space weather monitoring missions) and future high-security earth observation programme payload data, subsets of L2/L3 products are likely to have confidentiality constraints and may be provided by multiple data processing service providers. In order to perform computation, including distributed processing and application of federated machine learning on supersets of such data, CC solutions are required.

This activity encompasses the following tasks:
- Assess the state-of-the-art solutions in confidential computing, consolidate use cases with stakeholders and identify associated target source systems for data persistence and processing.
- Design integrated CC solutions targeting common systems to be tailored based on target source data interfaces and relevant processing algorithms and data formats for the identified use cases and scenarios.
- Develop and verify the integrated CC systems, integrating with the real or fully representative data source interfaces, and validating on representative data in terms of scope and scale.
- Perform a critical assessment of results achieved, lessons learned, and any necessary future development roadmap.

Deliverables: Report, Software

Application/Need Date: TRL 8 by 2030 for multiple Space Safety and Earth Observation missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Optical Communications for Space.

Cybersecurity

System and Applications Engineering

GT1Y-802GA – Advanced techniques for space operations security monitoring

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - **TD** 9

Objective:
To develop and implement a high-confidence, reliable AI enabled security monitoring capability tailored for the space systems domain in order to improve on threat detection, incident response and system security monitoring capabilities for space missions.

Description:
Artificial intelligence (AI) and subdomains such as machine learning (ML) can provide a promising solution to the processing of large amounts of security event data and the extraction of useful information in order to detect threats. Supported by appropriately selected and tailored models and algorithms, AI solutions have the potential to process and classify events faster than humans, detecting outliers which may be missed by traditional signature-based systems, and detect security-relevant novelties and patterns. Extended benefits are expected to be achieved in a cohesive security infrastructure that will fully integrate actionable threat intelligence and automate real-time advanced threat protection, thus enabling security teams to keep up with the rapidly evolving threat landscape.

Studies have been conducted to leverage these techniques and technologies to correctly tailor and improve security event detection in the space system domain, focused on mission operations and supporting data systems and including threat modelling, event heuristics, data labelling and algorithm training. Nevertheless, such tailored application remains a yet unproven capability for which technology de-risk is still necessary to improve on and realise operational deployments. Building on previous studies and experiences of onboarding space system security operation center (SOC) tenants, this development and implementation aims to achieve a high-confidence, reliable AI enabled security monitoring capability for the space systems domain, accounting for multiple tailored model and agents integration as well as specialised hardware resource needs. The development shall seek synergy with SOC evolution initiatives, delivering the space data system tailored models and algorithms for use within a wider SOC context.

This activity encompasses the following tasks:
- Perform a critical assessment and analysis of existing AI/ML techniques / technologies / algorithms applied to security monitoring, supporting threat models and ongoing and planned activities within the space data system and SOC context.
- Specify and design the AI monitoring solution accounting for necessary training data availability and emulation capabilities with respect to the target validation context.
- Develop and verify the AI-driven security monitoring system.
- Integrate and deploy the system in a representative SOC and data system context, simulating real data flow and validating incident and attack scenarios.
- Perform a critical assessment of results achieved, lessons learned, and any necessary future development roadmap.

Deliverables: Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Functional Verification and Mission Operations Systems.

Cybersecurity

End-to-End Systems

GT1Y-803SE – Advanced interference detection and protection module

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 6

Objective:
To develop interference detection and protection module which can be used as a building block to protect next generation space missions.

Description:
Intentional Radio Frequency interference (RFI) is a threat to space missions as it can disrupt the radio links between satellite and ground/users. RFI can be generated from ground or from other space systems, for example those in orbit lower than the targeted victims. Detection of RFI can be achieved from ground as well as from the space, the latter offering a prompter reaction and introducing/enabling mitigation or protection measures.

Specific Government or Military space system have funded, analyzed and developed tailored methods and techniques to counter RFI, however commercial or scientific missions typically do not implement any detection or protection mechanisms, yet they are sensitive to malicious actors and denial of service.

This activity will consider typical RF threats and investigate and develop suitable interference detection techniques that can be implemented on board. Furthermore, it will investigate and develop protection mechanisms that can be directly applied on the RF space links with minimal impact on the links themselves or without requiring a re-design of the interface. The main benefit is the development of “modules” that can be selected and added in a modular manner to standard spacecraft architecture for commercial or scientific missions.

The activity will develop a generic architecture with multiple RF inputs covering various RF bands to adapt to most space receivers. The building blocks will be designed to operate as transfer function between the antenna and the nominal equipment and will be introduced either in parallel or in series with the nominal RF chain to continuously monitor the link and to react and mitigate the RFI in case it is harmful. The equipment should also provide logs to support real-time or off-line, on-board or on-ground investigation about the interference source.

The activity is divided into two phases:
Phase 1 (budget: 600 k€, duration: 8 months):
- Define the use cases and the threat scenarios.
- Design of the RFI detection and mitigation modules, including investigating the integration challenges of such modules in existing architectures.
- Develop a proof-of-concept of the building blocks in hardware to demonstrate the feasibility and assess the performance of the proposed solutions.

Phase 2 (budget: 1400 k€, duration: 16 months):
- Develop a hardware prototype to demonstrate the effectiveness of the proposed techniques.
- Develop a testbed aimed at supporting the demonstration of the prototype capabilities including a reference transmitter, a reference receiver and interference generation module.
- Integrate the prototype and the testbed.
- Perform functional and performance testing of the prototype both under nominal and Interference conditions.

Deliverables: Breadboard, Prototype, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Cybersecurity

End-to-End Systems

GT1Y-804SE – Application layer cryptographic capability for user data protection and segregation

Budget: 3000 k€ - **Duration**: 36 months - **Current / Targeted TRL:** 3 / 5 - **TD** 1

Objective:
To develop and implement a “multi-user” satellite payload crypto unit.
Description:
Today, a satellite payload is usually operated by a single mission control center (MCC, or multiple redundancies thereof) via a secure (logical) link between MCC and the satellite payload. This secure link is established between a dedicated crypto unit in the MCC on ground and the payload crypto unit onboard. If the mission provides on-request services to multiple users, they make a service order request to the satellite via MCC and the requested product will be provided (once available) from the satellite through MCC to the user. In this setup, all requests and the requested products from all users pass through the MCC. Thus, no such system can provide user anonymity and privacy. One approach towards user privacy and anonymity is to allow for direct access to the payload by multiple (authorized) users and to de-couple protection of order requests from protection of product deliveries. In this scenario, each user has its own secure (logical) link to the payload crypto unit to issue order requests (and receive products). In addition, through secure multicast / broadcast a product can be delivered to a specific target user group. This activity shall focus on the cryptographic needs for the payload crypto unit when following the above-mentioned approach. The following use cases shall be addressed for n users (n in the order of 10 – 100):
- Direct, anonymous and private user access to the payload through the payload crypto unit: User privacy at MCC: except for the fact that a user has issued an order request, the details of the request remain private and user anonymity: at satellite payload, requests shall not be traceable to the user.
- Secure unicast, multicast and broadcast response provision: According to the selected response type, the payload crypto unit shall be able to protect the response for the selected target user group (the set of users that are in general able to request orders and to receive replies are identical). In case of multicast, the target user group of up to m=n users can be specified.
- System high to system high communication over inter satellite link (ISL): the payload crypto unit shall be able to exchange (user) data with the payload crypto unit on another satellite through ISL.

The activity is divided in phases and encompasses the following tasks:
Phase 1 – Proof of concept (budget: 1100 k€):
- Trade-off analysis and demonstration of candidate solutions of:
    - Suitable for intended purpose cryptographic schemes (symmetric vs. asymmetric key establishment, multicast / broadcast encryption schemes against key transport like encryption schemes, depending on choice of n and m, etc.)
    -Secure mechanisms to allow for dynamic group management (users may join and leave the group).
    - Establishment of sizing / performance requirements for multicast/broadcast encryption schemes and definition/specification of an appropriate protocol.
- Develop and define a quantum resistant cryptographic concept suitable for the above-described use cases with a desirable throughput of 50 Gbps for product delivery.
- Concept validation through prototype and relevant testbed implementation.

Phase 2 – Breadboard implementation (budget: 1900 k€):
- Definition of security targets
- Implementation and testing of an engineering model of the intended solution
- Validation through engineering breadboard implementation in relevant (computing) environment.
Deliverables: Breadboard, Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Functional Verification and Mission Operations Systems.

Cybersecurity

End-to-End Systems

GT1Y-805SE – Secure reconfigurable software defined satellites

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 2

Objective:
To design, develop, implement, test and verify secure reconfigurable technologies needed to support software defined satellites.

Description:
Europe's first commercial satellite capable of being completely reprogrammed while in space are already in commercial use. Such satellites offer unprecedented mission reconfiguration capacity and capabilities (including but not limited to beams reshaping and redirection). In addition, in-flight re-programmability capabilities that will enable avionic platforms to support many missions are also considered. The availability of system-on-chip, powerful micro-controllers and reprogrammable Field-Programmable Gate Array (FPGA) enables a high-level of flexibility while improving the performance of avionic systems.

However, all these software defined capabilities and flexibility require to be managed and implemented in a secure way, to ensure that cannot be exploited by potential attackers to compromise the confidentiality, integrity or availability of the provided services, of the software defined capabilities and of the platform itself. Security measures, including access controls, cryptographic based authentication, software authenticity protection using digital signatures or other suitable technology, secure booting based on root of trust, user / operator / management segregation, and other, must be implemented.

This activity is divided in phases and encompasses the following tasks:

Phase 1 (budget: 500 k€, duration: 8 months):
- Asses the state of the art of software defined satellites technologies and architectures.
- Security assessment of the risks of using these technologies and architectures, identification and assessment of security technologies that can be used to mitigate these risks; identification of suitable products (software / hardware) implementing these security technologies.
- Design a complete secure reconfigurable software defined satellite using the identified security technologies to address the security risks.
- Proof-of-concept development of selected functionalities, demonstration and testing.

Phase 2 (budget: 1500 k€, duration: 16 months):
- Develop the technologies that will be used for securing the configuration and authenticity of software defined satellites.
- Implement an end-to-end secure reconfigurable software defined satellite.
- Test, verify and demonstrate end-to-end secure reconfigurable software defined satellite.

Deliverables: Breadboard, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Functional Verification and Missions Operations Systems.

Cybersecurity

End-to-End Systems

GT1Y-806SE – Space cyber threat intelligence module

Budget: 2000 k€ - **Duration**: 27 months - **Current / Targeted TRL:** 3 / 5 - **TD** 1

Objective:
To develop a cyber threat intelligence module to be deployed as a dedicated spacecraft payload. The module shall be able to collect tactics, techniques and procedures (TTPs) of space cyberattacks and ensure a secure downlink for analysis on ground.

Description:
In cybersecurity, the understanding of attacker tactics, techniques and procedures (TTPs) has become a standard guiding the development of technology and processes for the defence against cyberattacks. For example, virtually all modern cybersecurity tools use the taxonomy provided by the MITRE ATT&CK framework, which maps out known TTPs. A solid understanding of attacker TTPs is therefore considered a basic requirement for effective cybersecurity. Honeypot systems have long played an important role in security research and the active defense of computer networks. ESA has invested into translating these proven cybersecurity techniques into the space domain by setting up SPACE-SHIELD as a space-focused cybersecurity framework and by conducting a feasibility study for satellite-based honeypot systems.

The next step to help learn more about real-life TTPs of cyberattacks on space infrastructure, a dedicated space cyber threat intelligence module, will provide the means of collecting this data. The focus of this activity is the definition and development of a module to be flown as an additional payload of future satellites, posing a target for attackers to interact with. A secure segregation from the satellite platform is the most critical requirement. Logs detailing the interactions must be securely stored and downlinked to a ground infrastructure. Logs must be analyzed to extract and classify TTPs and to obtain more concrete indicators of compromise (IoCs). This threat intelligence information is to be incorporated in (near) real-time for the reconfiguration of spacecraft and ground protection systems to protect against similar attacks. The module must be reconfigurable to emulate different spacecraft hard- and software corresponding to the relevant mission types (communication, earth observation, navigation, etc.). The module must be designed to be interoperable with existing space segment and ground segment infrastructure and cybersecurity tools.

The activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 550 k€, duration: 12 months):
- State of the art review, threat model.
- Requirements specification and initial development of an operational concept.
- Design of a space cyber threat intelligence module and corresponding end-to-end demonstrator, including a testbed suitable for functional and security verification.
- Definition of functional and security test scenarios.
- Proof of concept implementation.

Phase 2 (budget: 1450 k€, duration: 15 months):
- Refinement of module and testbed design.
- Development of the module and testbed, including required space and ground segment modules.
- Testing, verification and demonstration of the space cyber threat intelligence module.

Deliverables: Breadboard, Prototype, Report

Application/Need Date: Collecting real-life attack data (TTPs) of cyberattacks on space infrastructure. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Enabling Artificial Intelligence For Space Systems Applications.

Cybersecurity

End-to-End Systems

GT1Y-807SE – Optical space surveillance and tracking unit for space assets security

Budget: 2500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - **TD** 11

Objective:
To develop and implement an on-board optical Space Surveillance and Tracking (SST) unit that grabs and processes visual data to identify, track and characterize malicious objects targeting space assets.

Description:
In the last years, a large effort has been invested in the Space Situational Awareness (SSA) domain. SSA involves the comprehensive monitoring, tracking, and analysis of objects and activities in space, encompassing areas from Near-Earth Objects, space weather and space debris. Within SSA capabilities, SST segment focuses on detecting, predicting and warning about in orbit debris. From a security perspective SST can enable the detection, tracking, identification, and classification of potential threats, such as hostile debris, ensuring that mission owners can protect their critical infrastructure in space. While technology has been developed for unintentional space debris monitoring and collision avoidance, more development is needed to integrate security in SST including detecting, identifying, tracking and characterize malicious objects targeting space assets. Specifically, mechanisms that can identify objects targeting satellites, evaluate if such objects are malicious or not, gather related proof and predict malicious debris motion need to be developed and included in a product that can be integrated on-board of satellites.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 800 k€, duration: 8 months):
- Survey of the state of the art of:
    - optical sensors technology usable for space applications,
    - algorithms for object detection, tracking and characterisation.
- Identify and analyse the potential threats in terms of kinetic attacks that can impact space assets.
- Definition of system concept for SST focused on malicious object detection, tracking and characterisation.
- Identification of the interfaces needed for the optical SST unit to be integrated in satellite platforms.
- Based on the system concept and interface needs, derivation of the on-board optical SST unit requirements.
- Design of the optical SST unit.
- Proof-of-concept development and preliminary feasibility assessment of the functionalities.

Phase 2 (budget: 1700 k€, duration: 16 months):
- Development of an Engineering Model (EM) of the designed SST unit.
- Verification of the SST unit functionalities.
- Development of a testbed to demonstrate the system concept using the developed EM.
- Demonstration of functionalities using the EM and the testbed.

Deliverables: Breadboard, Engineering Model, Other, Report

Application/Need Date: Protection of space assets against kinetic based security threats. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Enabling Artificial Intelligence For Space Systems Applications.

Cybersecurity

End-to-End Systems

GT1Y-808SE – End-point security detection and protection

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 1

Objective:
To develop and validate an (on-board) end-point security system for satellite components and subsystems, enabling real-time detection and mitigation of cyber threats targeting software, firmware, and hardware interfaces.
Description:
Current space systems lack dedicated onboard end-point detection and protection tools, leaving critical software and hardware components exposed to vulnerabilities such as firmware tampering, malicious code injection, and unauthorized access. Existing security monitoring is often centralized or ground-based, introducing latency and reducing resilience against fast-evolving or targeted attacks.

This activity introduces a decentralized, real-time endpoint security layer for satellite systems, offering protection and response to cyber-attacks. It improves platform resilience and minimizes the attack surface, enabling compliance with emerging space cybersecurity standards. The proposed system will primarily employ rule-based logic for known threat signatures, firmware integrity checks, and real-time process monitoring. Optional use of AI/ML techniques may be explored to enhance behaviour-based anomaly detection, depending on mission and performance constraints.

This activity will be implemented in phases and encompasses the following tasks:
Phase 1 (budget: 700 k€, duration: 12 months):
- Threat model definition for on-board endpoints (e.g. OBC memory, avionics buses, communication interfaces, TM messages, component logs, etc.).
- Design of a lightweight, modular endpoint protection agent and corresponding management platform.
- Development of a test environment to simulate endpoint-level attacks (e.g. code injection, privilege escalation).
- Integration of anomaly detection, and optionally AI models, to classify events and trigger mitigation.
- Proof-of-concept flat-sat or hardware emulator-based testing to validate detection accuracy and system impact.

Phase 2 (budget: 800 k€, duration: 12 months):
- Full-scale development of the endpoint protection system with integrated detection and response modules.
- Optional enhancement with refined AI/ML anomaly detection based on Phase 1 evaluation.
- Deployment in a high-fidelity flat-sat or hardware-in-the-loop testbed.
- Validation through testing under multiple simulated attack scenarios.

Deliverables: Breadboard, Report, Software

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Cybersecurity

End-to-End Systems

GT1Y-809SE – Spacecraft digital forensics and incident handling

Budget: 2000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 1

Objective:
To develop a tamperproof isolated module that ensures the authenticity of on-board security-relevant information, enabling the reliable collection, storage, and downlinking of forensics evidence, and provides support for incident investigation and recovery.

Description:
In case of incidents, it is essential to have guarantee of the authenticity of all on-board security-relevant information. For supporting forensics investigations, it is necessary to ensure reliable collection of evidence, enabling to create a timeline of events and to enable traceability towards the source of anomalies or incident. However, a compromised onboard component could affect the integrity of the security evidence, before it is downlinked. The authenticity of the security evidence needs to be guaranteed, and their collection and downlinking needs to be ensured to be tamperproof, even in case of a compromise. Therefore, the authenticity of security evidence needs to be protected end-to-end, closest to the source generating the information.

This activity will develop a tamperproof isolated module for this purpose, which provides a dedicated protected downlink mechanism, and is isolated from the main platform, for obtaining critical information, which won't be tampered if the main platform is compromised. This isolated module, enables the reliable collection of specific forensics indicators of compromise, events, artifacts, to be stored in non-modifiable protected memory, until it is downlinked. The forensics module should provide functions which enable the targeted collection of both volatile and non-volatile data, obtained from various onboard components, supported with functionality including file, application, and memory integrity verification.

Based on such information, the module will provide capabilities to analyze the forensics evidence and to identify the source of the incident. The forensics analysis will further support the incident handling, by effectively recovering from the incident through an isolated trusted onboard component, by leveraging fault-detection, fault-isolation and recovery mechanisms and host integrity measures.

This activity is divided in phases and encompasses the following tasks:
Phase 1 (budget: 750 k€, duration: 12 months):
- Definition of baseline requirements for reliable forensics capabilities.
- Design of the onboard forensics module, and definition of interface with onboard components.
- Definition of functional and security test scenarios.
- Implementation of a proof of concept for a representative institutional satellite.

Phase 2 (budget: 1250 k€, duration: 12 months):
- Refinement of the forensics module design, enhancing the coverage and operability of the forensics capabilities, considering system-specific use cases.
- Development of the module and representative testbed, including required space (e.g. flatsat) and ground segment components.
- End-to-end validation of the forensics module on a representative hardware testbed environment, including functional and security testing to validate the forensics capabilities.

Deliverables: Breadboard, Report, Software

Application/Need Date: Institutional Missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Serialisation / Standardised Sub-Systems

4.2.1. Introduction and selection criteria

Today, many space application domains are being disrupted by the growing use of large constellations in Low Earth Orbit:

- The **Earth Observation (EO)**, Security and Resilience sectors benefit from higher revisit and faster data distribution from LEO. Governmental Agencies can manage supply chain in new ways and establish new industrial policies that facilitate competitiveness. ESA is also working with the EC, who is defining the EOGS (EO Governmental Services), to establish requirements for EO constellations in LEO orbits.
- **Navigation** applications are also concerned about resilience. Europe is currently leading in terms of performance with Galileo. Meanwhile, China is to launch BeiDou LEO satellites to supplement the existing MEO constellation within 2 years. ESA together with the EC is also developing GNSS demonstrators in LEO in preparation of a larger LEO-PNT (Position, Navigation, Timing) constellation which is under definition.
- At the forefront is the **telecom sector,** which benefits from low latency and for which **Starlink** has established a model of competitiveness. ESA and the EC are already developing the Infrastructure for Resilience, Interconnectivity and Security by Satellite (IRIS2) constellation in LEO.

In this context ESA has launched a dedicated initiative aimed at supporting European integrators and the European supply chain for mid-size satellites for LEO constellations: the initiative is referred to as M-IND (Mid-size satellites INDustrialisation) and encompasses 3 objectives, addressing Competitiveness, Industrialisation and Sovereignty, in line with ESA 2040 Strategy Objectives #3 and #4.

The M-IND WG has taken two steps to acquire information from industry after IPC TWAG delegates provided contacts from 30+ mid-size satellite integrators in ESA State Members:

- **A Survey** (opened 14/02/25 to 12/03/25): it was answered 29 times by **25 mid-size sat integrators** from 14 countries.
- **A Workshop (1 day, 20/03/25)** held at ESTEC and attended by ~30 mid-size sat integrators, Eurospace and several delegations. The workshop included 21 pitches by most of the integrators who answered the Survey, as well as 3-Panels around the aggregated statistics gathered in the Survey, as well as open questions and answers with the full audience.

The findings were consolidated in a report that included 5 recommendations with regards to the role to be played by the Agency (please also see **M-IND Initiative Report):**

1. **Industrialisation**: Define and systematise the adoption of Manufacturing Readiness Levels (MRL) to support the European Space Industry for the transition from highly customised low volume subsystems to serial production, including a pragmatic Product Policy compatible with this ambition.
2. Promote an **Industrial Policy** aim/ed at guaranteeing a European redundant supply for LEO platforms and subsystems, as enabler for future batch-procurement. Support the related **Commercialisation** effort.
3. **Reduce** and **Communalise Interfaces**, both for platform subsystems and payloads, to enable interchangeability and inter-operability.
4. **Sub-system adaptations** and **European-ization**: Foster a Competitive, Modular, European **Sovereign Supply Chain** Solutions with priority for 5 platform subsystems: Data Handling; AOCS; Power; Propulsion and Payload Data Downlink.
5. Address **Constellation Deployment & Launcher Services**, including launcher adaptors and dispensers, while ensuring compliance to ESA Space Debris Mitigation Policy.

The implementation of recommendation 3 and 4 is illustrated below and relies on the combination of the TDE (and GSTP) investment on identified technologies (European Supply Chain development) combined with the Application Directorates Programmes ‘system pull’ (platform requirements)

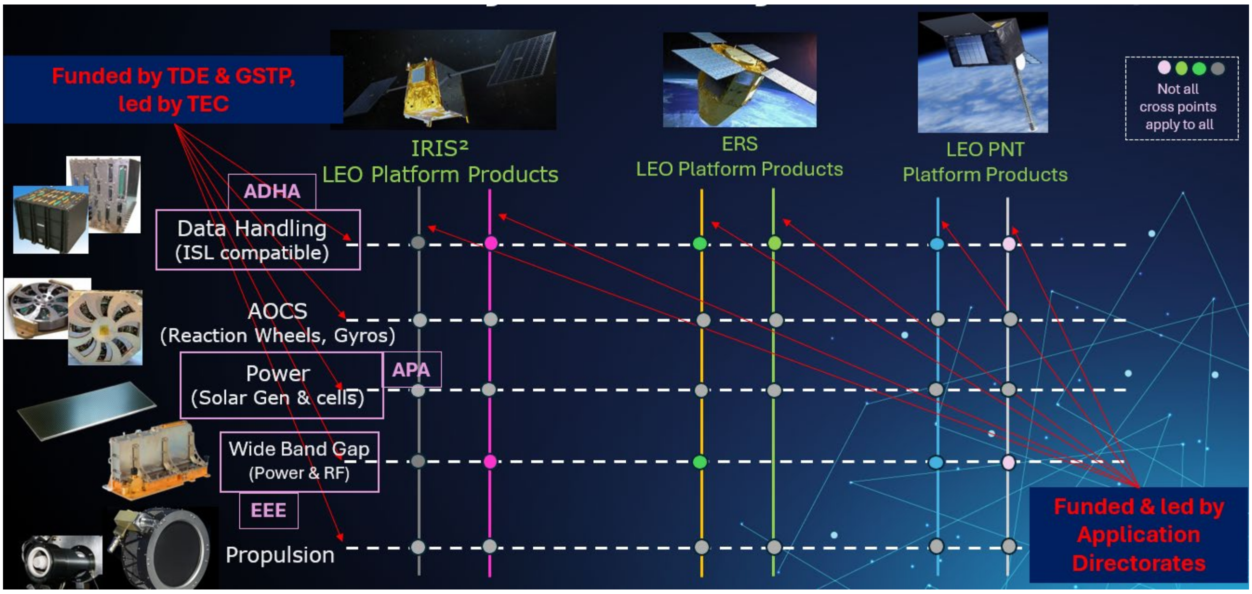


Figure-1: Priority subsystems development as recommended by European Integrators (see M-IND report)

As part of these compendia the main focus will be in maturing technologies aiming at increasing competitiveness and European sovereignty in the supply chain, with priority for enabling interchangeability and inter-operability of equipment and modules. Given the progress already made in this direction with Advanced Architecture in Data Handling (ADHA) and Power (APA) activities for large satellites, the focus will be to apply the ADHA-APA lessons learned and to adapt as necessary for mid-size satellites for LEO constellations.

This “Standardised Sub-Systems” section is based on information known at the time of the preparation of this compendium. However, the evolution of requirements for European sovereign (dual+ source) and resilient systems (e.g. to interconnect EO satellites with IRIS2 for near real-time data delivery) is progressing quickly, and therefore it is expected that new/updated technology roadmaps, also for other sub-systems than ADHA-APA, and priorities will be established together with ESA Programme Directorates in the coming months.

The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined -- as well as a standardized backplane connector and electrical interfaces.

The Advanced Power Architecture (APA) is a standard concept for the Power Conditioning and Distribution Unit (PCDU). APA follows the interoperable and interchangeable modularity approach applied for ADHA that started at least two years before APA. APA will be fully compatible with ADHA so that together they will form the kern of a satellite Platform. The advantage of having standard interfaces in APA and ADHA is that there are many functions that can be developed independently by different companies

4.2.2. List of activities

Serialisation / Standardised Sub-Systems

Advanced Data Handling Architecture (ADHA)

Avionics & EEE

Programme Reference	Activity Title	Budget (k€)
GT1M-800ED	ADHA on-board computer	1,500
GT1M-801ED	ADHA instrument controller unit	1,500
GT1M-802ED	ADHA mass-memory modules and unit integration	1,500
GT1M-803ED	ADHA EGSE and automated module testing products	800
GT1M-804ED	ADHA Ethernet I/O and router module	1,000
	Total ADHA - Avionics & EEE	**6,300**

RF Payloads & Technology

Programme Reference	Activity Title	Budget (k€)
GT1M-805EF	ADHA TTC transponder and payload transmitter module(s)	1,200
	Total ADHA - RF Payloads & Technology	**1,200**

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1M-806EP	ADHA high-performance power module	800
	Total ADHA - Power Systems, EMC & Space Environment	**800**

Total Advanced Data Handling Architecture (ADHA)	**8,300**

Advanced Power Architecture (APA)

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1M-807EP	APA integrated unit	400
GT1M-808EP	APA unit and module EGSE	500
GT1M-809EP	CAN backplane interface mezzanine board	250
GT1M-810EP	Active battery management system maturation	800
	Total APA - Power Systems, EMC & Space Environment	**1,950**

	Total Advanced Power Architecture (APA)	**1,950**

Others

GNC, AOCS & Pointing

Programme Reference	Activity Title	Budget (k€)
GT1M-811EC	Framework for automation of AOCS development verification and validation, based on automated code generation	600
	Total Others - GNC, AOCS & Pointing	**600**

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1M-812EP	Highly adaptable DC/DC converter for multipurpose applications	850
	Total Others - Power Systems, EMC & Space Environment	**850**

	Total Others	**1,450**

	Total Standardised Sub-Systems	**11,700**

4.2.3. Description of activities

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Avionics & EEE

GT1M-800ED – ADHA on-board computer

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 1

Objective:
To develop an ADHA OBC (on-board computer) with compatible modules usable standalone or with standard hardware in ADHA units for earth observation, exploration, science, and navigation missions.
Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined -- as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units.

This activity aims to develop an OBC unit, based on ADHA-compatible modules. The development will follow the standard requirements for the development of ADHA modules. The OBC design will be fully compliant with the ADHA generic specification for OBC modules. The development will include the development of the basic software (including boot-software). The developed ADHA OBC will support high-speed data reception and transfers, featuring a scalable architecture to meet the demands of future ESA missions in Earth Observation, Science, Exploration, and Navigation. Utilizing state-of-the-art processor technologies, this new OBC shall offer superior integration and processing performance compared to previous generations. These advancements will also enable the implementation of sophisticated functions, such as AI-based on-board Failure Detection, Isolation, and Recovery (FDIR) algorithms.

This activity encompasses the following tasks:
- Detailed module requirement analysis, based on the existing ADHA documents, including the ADHA generic OBC specification.
- ADHA OBC module (AOBCM) architectural design and key components trade-off.
- AOBCM engineering model design, including electrical, mechanical, thermal, FPGA design and software.
- AOBCM engineering model manufacturing and EGSE procurement.
- AOBCM engineering model validation campaign, including integration in an ADHA unit rack.
- Data-handling and storage demonstration, including performance benchmarks.

Following the successful implementation of the activity a follow-on activity is targeted to design, manufacture and test the EQM of the ADHA OBC (~2M€).

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: On-Board Computers, Data Handling Systems and Microelectronics (2021) – Consistent with Aim C “Design, develop, manufacture and qualify OBCDH modules”.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Avionics & EEE

GT1M-801ED – ADHA instrument controller unit

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 1

Objective:
To develop an ADHA-compliant Instrument Controller Unit (ICU) usable standalone or with standard ADHA hardware for Earth, exploration, science, and navigation missions.

Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined -- as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units. This activity aims to develop an ADHA-compatible ICU (Instrument Control Unit), with integrated Data Processing Unit(s) (DPUs). The modules target the overall control functions of an instrument with several secondary units (FEE units, mechanism control units, cryocooler unit, etc). The modules will at least include an ICU processor module, and instrument I/O module(s). The ICU processor module shall be capable to act as the system controller of an ADHA unit, running high-level instrument application software and providing standard high-level interfaces for connection to the spacecraft platform units, or other instruments and implementing all the necessary functions to control peripheral ADHA modules in the unit, including network routing capabilities. The I/O modules will cover lower-level interface functions such as additional low-speed digital interfaces (CAN, RS422/RS485, etc), and discrete interfaces (HPCs, thermistor readout, heater control, secondary power, etc) and others typically found in an ICU unit. The ICU will be based on state-of-the-art technologies, offering a higher level of integration and performance compared to previous generation products, and be able to perform e.g. advanced (AI-based) FDIR methods for the monitoring and control of a payload.

This activity encompasses the following tasks:
- Use case analysis of instrument control units for instruments on institutional missions.
- Modules requirements analysis (use case analysis, ADHA documents, ADHA ICU draft specification) - including functional split between ICU processor module and I/O modules.
- ICU architecture design and key components trade-off and selection.
- Modules engineering model design, including electrical, firmware, and base software.
- Module engineering model manufacturing
- Module engineering model functional and performance verification and validation.
- Validation testing of modules using an ADHA backplane, in an ADHA unit rack.
- Performance benchmarks and end-to-end data handling demonstration.

Deliverables: Engineering Model, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: On-Board Computers, Data Handling Systems and Microelectronics (2021) – Consistent with Aim C “Design, develop, manufacture and qualify OBCDH modules”.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Avionics & EEE

GT1M-802ED – ADHA mass-memory modules and unit integration

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 1

Objective:
To develop a complete ADHA mass-memory with compatible modules usable standalone or with standard hardware in ADHA units for Earth, exploration, science, and navigation missions.

Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined, as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units.

This activity aims to develop ADHA mass-memory (MM) modules, based on ADHA-compatible hardware modules. The development will follow the standard requirements for the development of ADHA modules. The mass-memory module design will be fully compliant with the ADHA generic specification for mass-memory modules. The development will include the development of the mass-memory software (including support for file-based operations, and CFDP Class-1 and Class-2). The developed ADHA mass memory module will support high-speed data reception and transfers, featuring a scalable architecture to meet the demands of future ESA missions in earth observation, science, exploration, and navigation. Utilizing state-of-the-art processing, memory storage and interconnect technologies, this new mass-memory shall offer superior integration, storage volume and processing performance compared to previous generations. These advancements will also enable the implementation of sophisticated functions such as artificial intelligence (AI) and machine learning (ML) algorithms for on-board data processing.

The activity encompasses the following tasks:
- Detailed module requirement analysis, based on the existing ADHA documents, including the ADHA generic mass-memory module specification.
- ADHA mass memory module(s) architectural design and key components trade-off.
- Engineering model design, including electrical, mechanical, thermal, FPGA design and software.
- Engineering model manufacturing and EGSE procurement.
- Engineering model validation campaign, including integration in an ADHA unit rack.
- Data-handling and storage demonstration, including performance benchmarks.

Following the successful implementation of the activity a follow-on activity is targeted to design, manufacture and test the EQM of the ADHA mass memory (~2M€).

Deliverables: Engineering Model, Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: On-Board Computers, Data Handling Systems and Microelectronics (2021) – Consistent with Aim C “Design, develop, manufacture and qualify OBCDH modules”.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Avionics & EEE

GT1M-803ED – ADHA EGSE and automated module testing products

Budget: 800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 1

Objective:
To develop standardized Electronic Ground Support Equipment (EGSE) and automated testing tools for ADHA modules and units to reduce non-recurrent engineering in data handling equipment development.

Description:
The activity focuses on the analysis of the ADHA EGSE needs and requirements for testing the ADHA architecture at unit and modules levels, across different configurations, including platform functions (on-board computer, RTU, mass memory, etc.), and payload functions (instrument control unit (ICU), payload data handling unit (PDHU), data processing unit (DPU), etc.). The main objective is to derive a standardized, modular EGSE architecture for ADHA, ensuring compatibility across different testing scenarios. The EGSE architecture will support testing of various module profiles, including system controllers and peripheral modules, and manage control signals and data links for both electrical functional and performance testing. Additionally, it will support the execution of test sequences and the archiving of test data, and the user interface must be available to execute and monitor any tests in real-time. The ADHA unit and modules performances must be validated at all, eventually introducing closed-loop tests in the EGSE development concept. This latter capability is essential for simulating realistic operational scenarios where ADHA is stimulated by and/or interacts with other subsystems in a fully integrated environment.

This activity encompasses the following tasks:
- Create the ADHA EGSE requirements specifications and the architecture design.
- Develop a comprehensive definition for automated ADHA module testing facilities and EGSE equipment.
- Define standard module testing facilities, referred to as "test racks," which are intended to stimulate and verify ADHA unit and modules, and ensure that the testing setup meets the starting operational requirements.
- Develop and manufacture a prototype ADHA EGSE and automated module tester.
- Produce a test rack prototype, which may be based on the ADHA-U1 backplane and mechanics. The prototype shall include modules designed to simulate the system controller and peripheral boards and enable functional testing, including testing of critical interfaces (e.g. CAN Bus, SpaceWire, etc.).

The test racks will provide essential capabilities, such as eventually hosting one or multiple ADHA boards, supplying power to the boards under testing, and offering electrical interfaces representative of the ADHA backplane at EICD level for integration testing. Producing an EGSE prototype will provide added value into the design and functionality of the Test Rack, facilitating the refinement of the testing infrastructure before full-scale implementation.
Deliverables: ADHA EGSE test-rack prototype, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: On-Board Computers, Data Handling Systems and Microelectronics (2021) – Consistent with activity E04 “EGSE to test an ADHA unit”.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Avionics & EEE

GT1M-804ED – ADHA Ethernet I/O and router module

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 1

Objective:
To develop an ADHA-compatible Ethernet I/O and router module that can be used integrated with other standard hardware modules in ADHA units for Earth, exploration, science, and navigation missions.

Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined -- as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units.

This activity aims to develop an ADHA-compatible Ethernet I/O and router module. The development will follow the standard requirements for the development of ADHA modules. The developed ADHA Ethernet I/O and router module will support high-speed data reception and transfers, featuring a scalable architecture to meet the demands of future ESA missions in Earth Observation, Science, Exploration, and Navigation. The module design will consider state-of-the-art microelectronics components, and Ethernet standardization, include high-rate Ethernet interface rates (10GbE and upwards) – and consider open standards for time-sensitive networks (TSN). The module design shall also include features to accurately monitor for packet-level errors and provide detailed housekeepnig telemetry (HKTM) on the status of the network.

This activity encompasses the following tasks:
- Detailed module requirement analysis, based on the existing ADHA documents.
- ADHA module architectural design and key components trade-off.
- Engineering model design, including electrical, mechanical, thermal, FPGA design and software.
- Engineering model manufacturing and EGSE procurement.
- Engineering model validation campaign, including integration in an ADHA unit rack.
- Data routeing demonstration, including performance benchmarks.

Deliverables: Engineering Model, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap On-Board Computers, Data Handling Systems and Microelectronics.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

RF Payloads & Technology

GT1M-805EF – ADHA TTC transponder and payload transmitter module(s)

Budget: 1200 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 6 - TD 6

Objective:
To develop ADHA compatible TTC transponder and payload transmitter module(s) for stand-alone use or integration with other standard hardware modules in ADHA units for Earth, exploration, science, and navigation missions.

Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined, as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units.

This activity aims to develop ADHA-compatible TM/TC transponder and payload transmitter module(s) hardware module. The development will follow the standard requirements for the development of ADHA modules. The modules will allow very high-speed data reception and transfers, as well having a scalable architecture, to allow to meet requirements for future earth observation, science and exploration missions. The modules will be implemented using state-of-the-art SDR (software-defined radio) methods, including support for e.g. variable data-rate/encoding schemes. The exact architecture and what parts of the RF chain (e.g. up-conversion) will be included in the module will be traded-off within the activity. The generic specifications for both the ADHA TM/TC transponder module and payload transmitter module (PDT) will be reviewed/derived, and a co-engineering with the ADHA system study consortia members should be organised to settle on the requirements and ensure that it fits in the end-to-end ADHA architecture.
In terms of hardware design, it shall be traded-off if the basic design of both the TM/TC Transponder Module and PDT Module can be the same, and the specific front-end/RF functions be implemented by exchanging the front-panel interfaces (e.g. utilising mezzanine boards) and different FPGA design -- or if the two functions require different hardware designs.

This activity encompasses the following tasks:
- Detailed module requirement analysis, including ADHA documents.
- SDR architectural design and key components trade-off.
- Engineering model modules design, including electrical, mechanical, thermal, FPGA design and software.
- Engineering model modules manufacturing.
- Engineering model modules RF performance and validation campaign, including demonstration and integration in an ADHA unit rack.
- Data-handling and storage demonstration, including performance benchmarks.

Deliverables: Engineering Model, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap TT&C Transponders and Payload Data Transmitters.

Standardised Sub-Systems & Platforms - Advanced Data Handling Architecture (ADHA)

Power Systems, EMC & Space Environment

GT1M-806EP – ADHA high-performance power module

Budget: 800 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 4 - TD 3

Objective:
To develop a high-performance ADHA power module to supply ADHA units containing modules with high current consumption, for Earth, exploration, science, and navigation missions.
Description:
The ADHA (Advanced Data Handling Architecture) is a collaboration between ESA, European satellite primes and data handling equipment suppliers. It includes the definition of an architecture for data handling unit based on standard module functions. A standard module procurement flow has been defined, as well as a standardized backplane connector and electrical interfaces. This allows independent development of interoperable modules that can be connected in ADHA units.

This activity aims to develop an ADHA-compatible high-performance power module. The development will follow the standard requirements for the development of ADHA modules. The developed ADHA power module will support a high number of ADHA payload modules with high consumption (at least 100W per module), to support a power supply of up to 1kW at unit level. The exact requirements will be refined within the activity, based on analysis of representative ADHA payload unit configurations.

The activity encompasses the following tasks:
- Detailed module requirement analysis, based on the existing ADHA documents.
- ADHA module architectural design and key components trade-off.
- Engineering model design, including electrical, mechanical, thermal, FPGA design and software.
- Engineering model manufacturing and EGSE procurement.
- Engineering model validation campaign, including integration in an ADHA unit rack.
- Data-handling and storage demonstration, including performance benchmarks.

Deliverables: Engineering Model, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap On-Board Computers, Data Handling Systems and Microelectronics.

Standardised Sub-Systems & Platforms - Advanced Power Architecture (APA)

Power Systems, EMC & Space Environment

GT1M-807EP – APA integrated unit

Budget: 400 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop the mechanical frame and the backplane for an APA unit and integrate the 5 functional modules to demonstrate the full APA unit performance.

Description:
The Advanced Power Architecture (APA) is a standard concept for the power conditioning and distribution unit (PCDU). APA follows the interoperable and interchangeable modularity approach applied for the project ADHA (Advanced Data Handling Architecture) that started in 2020. APA will be fully compatible with ADHA so that together they will form the kern of a satellite Platform. As ADHA, APA is composed of two elements: a standardized unit rack (element 1) that provides standard services to a set of power modules (element 2) that populates the rack to make an APA PCDU. The power unit can be built upon modules from different manufacturers, based on performances, cost, programmatic constraints.

This approach allows strong reduction of development time and recurrent costs, fast development and adoption of more performing technologies to name a few in great part enabled by the envisaged interoperable and interchangeable modules that compose the system. This activity aims at developing the mechanical frame of a unit and the electrical backplane to host the APA modules. Once the hosting frame is ready, the integrator must borrow the 5 functional modules of APA and integrate them in the unit to create

A fully functional APA unit. The 5 modules are SAR, control, battery, distribution and deployment. Once the unit is integrated, the unit must be tested to demonstrate its interchangeability and interoperability.

This activity encompasses the following tasks:
- Consolidation of APA requirements (generic and specific modules).
- Consolidation of the APA integration and test plan taking advantage of the information in the APA integrator package.
- Evaluation of the most promising technologies/components.
- Preliminary and detailed design definition.
- Manufacturing and testing the unit mechanical frame and backplane.
- Issue design documentation, unit technical specification and user manual.
- Unit test, including documentation and interoperability/interchangeability test.

Deliverables: Engineering Model, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Standardised Sub-Systems & Platforms - Advanced Power Architecture (APA)

Power Systems, EMC & Space Environment

GT1M-808EP – APA unit and module EGSE

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop test equipment (EGSE) for the APA units and modules.

Description:
The Advanced Power Architecture (APA) is a standard concept for the Power Conditioning and Distribution Unit (PCDU). APA follows the interoperable and interchangeable modularity approach applied for the project ADHA (Advanced Data Handling Architecture) that started in 2020. APA will be fully compatible with ADHA so that together they will form the kern of a satellite Platform. As ADHA, APA is composed of two elements: a standardized Unit rack (element 1) that provides standard services to a set of Power Modules (element 2) that populates the rack to make an APA PCDU. The power unit can be built upon modules from different manufacturers, based on performances, cost, programmatic constraints. This approach allows strong reduction of development time and recurrent costs, fast development and adoption of more performing technologies to name a few in great part enabled by the envisaged interoperable and interchangeable modules that compose the system. A key aspect to enhance industrialisation is to automatise the test processes. The advantage of having standard interfaces in APA and ADHA is that there are many functions that can be developed independently by different companies. Electronic Ground Support Equipment (EGSE) is a very good example. Thus, this activity focuses on developing EGSE that could perform automatic testing of an APA unit. Both the power and data interfaces are standard and hence, the EGSE can communicate with the unit, command it and test all functionality in a standard way. The EGSE will be used as well to test individual modules for the same reason explained above and, besides the test activities, will also support the assembly and integration activities of the APA unit.

The objective of the activity is to build on existing EGSE and develop the functionality that allows the automatic test process of the 5 functions of the APA unit: SAR, control, distribution, deployment and battery. The test data must be collected and stored in a way that easily allows building automatic reports.

This activity encompasses the following tasks:
- Consolidation of requirements.
- Definition of functions and automatic test sequences.
- Design of a EGSE breadboard.
- Manufacturing and test of a EGSE breadboard.
- Coordination with an APA unit integrator to perform test on an APA unit.

Deliverables: Breadboard, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Standardised Sub-Systems & Platforms - Advanced Power Architecture (APA)

Power Systems, EMC & Space Environment

GT1M-809EP – CAN backplane interface mezzanine board

Budget: 250 k€ - **Duration**: 12 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To build a mezzanine board to interface with a CAN bus on the units' backplane.

Description:
As part of the APA (Advanced Power Architecture) initiative, a standardised approach is being developed for internal communication between APA modules within an APA unit. This activity supports that effort by designing a compact, reusable TM/TC interface based on the CAN bus, known for its reliability and simplicity.

The mezzanine will serve as a common building block / solution for all APA modules, reducing integration effort and promoting consistency across designs. By consolidating all required components into a small footprint, it simplifies system architecture, supports modularity and eases procurement, manufacturing and recurrence.

This activity focuses on developing a compact mezzanine PCB that integrates five essential components required for CAN-based backplane communication (FPGA or microcontroller, ADC, voltage reference, CAN transceiver and XTAL/oscillator). The goal is to consolidate these elements into a small 30x30 mm footprint, supporting a range of external interfaces and operating on a 3.3V supply.

This modular solution aims to simplify internal communication across systems where CAN is the preferred protocol, offering a reusable and scalable interface.

This activity encompasses the following tasks:
- Consolidate the CAN protocol implementation.
- Architecture definition (evaluating if FPGA-based or microcontroller-based).
- Identify two suitable options for each one of the 5 required components, prioritising compact packaging and cost.
- Design a 30 mm x 30 mm PCB accommodating the 5 components and ensuring signal integrity and manufacturability, ready to be assembled into an APA module and to interface with CAN bus and the APA module.
- Implement the software or the VHDL code with the CAN protocol.
- Prototyping and testing, assemble and validate the board to confirm functionality and performance.

Deliverables: Breadboard, Report

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Standardised Sub-Systems & Platforms - Advanced Power Architecture (APA)

Power Systems, EMC & Space Environment

GT17-810EP – Active battery management system maturation

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 4 / 5 - TD 3

Objective:
To develop an active battery management system (BMS) based on advanced power architecture (APA) such that the BMS can be directly integrated in the power conditioning and distribution unit (PCDU) avoiding the needs for additional units related to the active BMS.

Description:
Conventional power systems in space missions have limited battery management functions, primarily focusing on voltage regulation within specified limits. Active battery management, widely used in terrestrial applications like automotive, includes active balancing of individual cell voltages and the provision of state of charge (SOC) and state of health (SOH) telemetry. These features are currently not available as telemetry and are of great value for operations, especially when mission extensions are considered. In a previous activity, a BMS was developed, demonstrating its basic functionality in a laboratory environment. This proposed activity aims to further develop the battery management system to enhance its performance, reliability, and efficiency in space missions, paving the way for more ambitious and longer-duration missions.

The active BMS will follow the APA requirements for direct integration in the PCDU, thus avoiding the needs for additional units related to the active BMS. The active BMS will be a separate module from the battery management module (BAT-M), which is taking care of regulating the main power bus. The active BMS to be developed in this activity, shall provide extra features like battery individual cell voltage balancing and provision of SOC and SOH telemetry. The active BMS shall comply with the APA standard.

The following tasks are foreseen for each module within this activity:
- Functional design.
- Electrical design and analysis.
- Thermal and mechanical design.
- Manufacturing.
- Functional and electrical testing.
- Support for module integration in an APA unit.
- Participation in APA related workshop and public reviews

Deliverables: Breadboard, Report, Software

Application/Need Date: All missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Electrochemical Energy Storage (2024) – Consistent with activity D06 “Active Battery Management System Maturation”.

Standardised Sub-Systems & Platforms - Others

GNC, AOCS & Pointing

GT1M-811EC – Framework for automation of AOCS development verification and validation, based on automated code generation

Budget: 600 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 6 - TD 5

Objective:
To develop the toolset and process for faster and sounder verification and validation of AOCS subsystems based on automated code generation, with tailoring per mission class.

Description:
Automated code generation from Simulink models has become a largely spread standard for AOCS applicative software development, replacing manual software development. Commercial (or internal corporate framework) tools are available to extend the automation of the AOCS applicative software production to other steps of the AOCS subsystem verification and validation (V&V), from the subsystem requirements definition to the compliance verification through automation of testing plan definition, realization and analyses. This extended process and framework will rely on model-based engineering tools (e.g. tools that link requirements to model implementation, analyses, correction, configuration and testing). As preliminarily performed through an ESA study (AIMONS), the automation of auto-coding guidelines check though the use of Mathworks SImulink Check contributes to the AOCS subsystem V&V acceleration.

It is noted that, though well spread in the higher ESA classes missions, this process would more importantly benefit to be developped for SmallSat AOCS development, with specific tailoring and corresponding adequacy of the process automation and supporting framework and methods. Given the timeline for SmallSat platform development, cost and schedule driven, this automation would bring more capacities for design adaptation and key verification aspects, by automating the more recurring and straightforward steps of the V&V. The objective is to progress from the state-of-the-art ecosystem in support of AOCS/GNC development and V&V, and to improve the toolset, methods and process for faster and sounder subsystem V&V.

This activity encompasses the following tasks:
- Conduct a gap analysis with respect to the process described in prior activities, state-of-the-art toolset and literature review (benchmark with aeronautic and automotive fields).
- Implement, with focus on automation, the process on an existing platform/mission, e.g. requirements at models' level, linked to model implementation and associated unit / closed loop, coverage (statement, decision, Modified Condition/Decision Coverage (MC/DC)) at Model-In-the-Loop (MIL) and Software-In-the-Loop (SIL) level, static code analysis, simulation and test scenario generation, simulation and test result analysis and corresponding compliance status.
- Implement the remaining custom checks to ensure proper coverage.
- Make a synthesis of the process, methods and means and automation level, possibly proposing further solutions on automation, with consultation of corporate and external stakeholders.
- Address continuous integration to enable more iterative / incremental development.
- Document the improved AOCS subsystem V&V process implemented and create knowledge management content (through generic examples and tutorials) to ease adoption of the process.

Deliverables: Report, Software

Application/Need Date: SmallSat missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity H07 "Framework toolset evolution in support of Automatic Code Generation & overall AOCS/GNC development ".

Standardised Sub-Systems & Platforms - Others

Power Systems, EMC & Space Environment

GT1M-812EP – Highly adaptable DC/DC converter for multipurpose applications

Budget: 850 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - **TD** 3

Objective:
To analyse DC/DC converter topologies with the highest possible versatility and to design and build an EM of the best topology and test it in highly variable conditions.

Description:
The power domain is still mostly based on ad-hoc developments that need modifications every time some specifications change. In most cases, trade-offs are made to maximise performance in very narrow operating conditions. However, it is possible to think about topologies that adapt well to changes and hence, built in adaptations can tune the converter to suit many different use cases.

The aim of this activity is to select DC/DC converter topologies from a perspective of adaptability to different operating conditions. Instead of maximising performance in a narrow band, the concept is to choose topologies with a good average performance in multiple conditions. There are multiple terrestrial applications with wide specifications and literature reports such concepts. Moreover, GaN switches allow using many topologies that were not used in the past and digital control based on FPGAs allow to control them and get a good performance adapting the control system to the situation. The activity target to develop a highly adaptable DC/DC converter, including all protections at the input and the output interfaces, plus internal protection mechanisms. The baseline would be to have a high input voltage range and a high output voltage range, paralleling capabilities and minimum size. Input voltage should adapt to all main power buses typical inputs and the output voltage should cope with the range 28 V - 50 V - 100 V. The power level per converter should be defined to maximise the capabilities of the semiconductors used, instead of defining them a priori. In any case, the minimum power level should be between 1 kW and 1.5 kW.

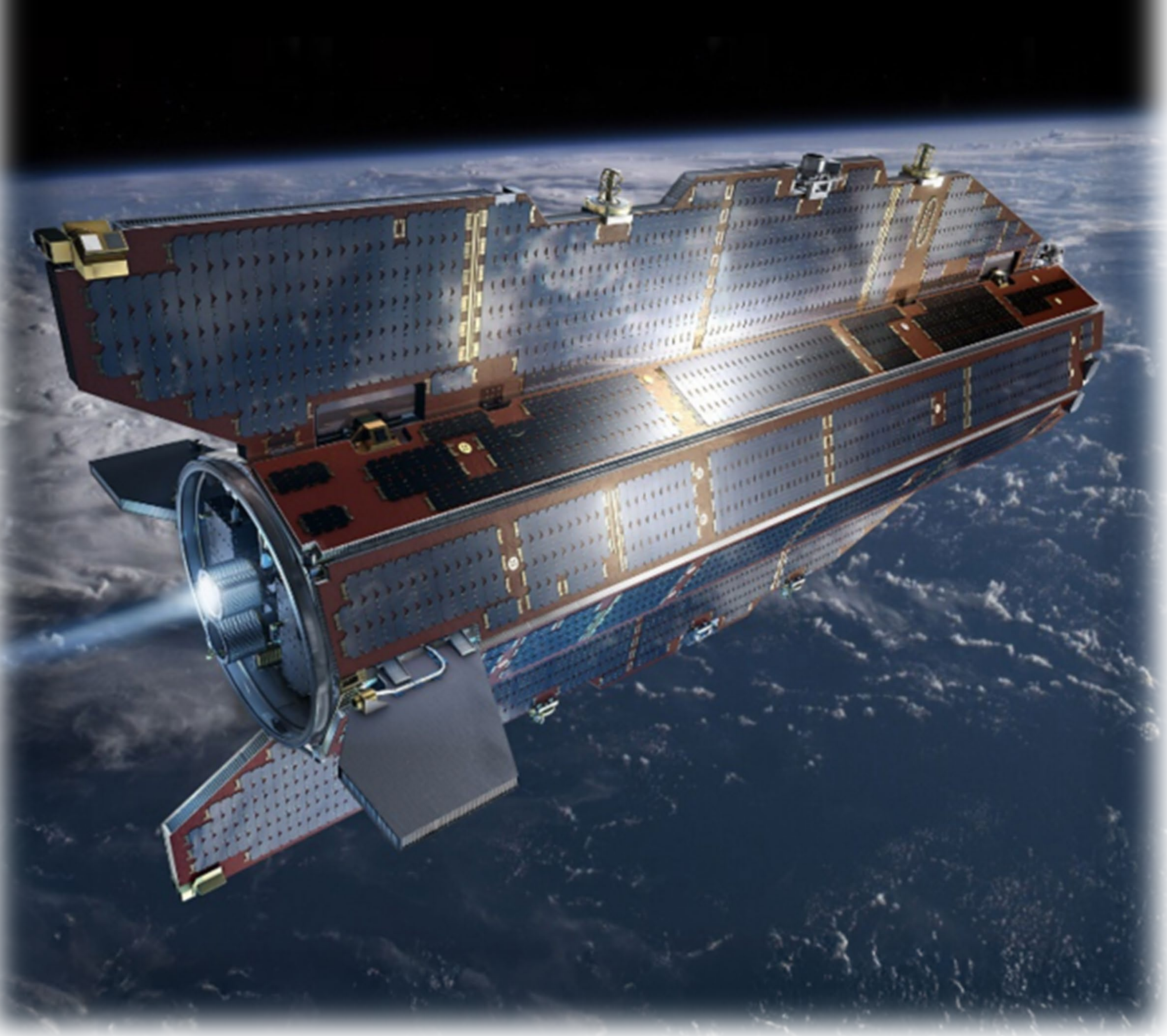
This activity encompasses the following tasks:
- Perform a trade-off to select highly adaptable topologies in terms of power level, input and output voltage range.
- Select the topology with the best average performance including protections.
- Build an engineering model of the topology.
- Design the control system based on an FPGA to maximise performance in the wide operating range.
- Test the engineering model in relevant conditions.

Deliverables: Engineering model, Report

Application/Need Date: All missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Power Management and Distribution.

Technologies for VLEO



4.3.1. Introduction and selection criteria

Very Low Earth Orbits (VLEOs) are of major importance for future missions for a huge variety of applications ranging from commercial telecommunication to atmospheric science. VLEO mission-enabling technologies form the basis of this Compendium.

Very Low Earth Orbits (approximately 120 km to 400 km altitude) offer major advantages for a wide range of future missions, mainly:

- increased resolution and/or reduced aperture size for optical payloads,
- improved link budgets and reduced latency for communication payloads,
- easier end of life compliance with new space debris regulations and ESA's Zero Debris Charter.

These advantages are of interest for a large variety of possible applications, from Earth observation, to telecommunications, to navigation, etc. Furthermore, at present, the exploitation of this altitude range is being of high interest for constellations, either standalone or in combination with higher orbits.

These missions are challenging in various aspects related to the residual atmosphere, i.e. ATomic OXygen (ATOX) as a dominant species and increased drag with large spatial and temporal variations, and the proximity to Earth's surface, i.e. higher relative speeds and shorter communication windows with single ground stations. The activities proposed as part of this VLEO Compendium are addressing major technological challenges and gaps. Specifically, the following areas/technologies are considered.

System Design and Simulation - The area addresses aspects related to environmental characterisation, modelling and system design optimisation, such as:

- In-situ sensor technology,
- Integrated aerodynamic/ aerothermodynamic modelling tools and test capabilities
- Satellite geometry and structural design optimised for minimal drag and aerodynamic efficiency
- High-efficiency power systems integrated to minimise surface drag & support aerodynamic design

Attitude and Orbit Control and Propulsion Technologies - This area includes technologies needed to enable precise station-keeping and long-duration operations in VLEO through drag-aware control and efficient propulsion systems:

- AOCS tailored to VLEO-specific disturbances (e.g. atmospheric drag, aerodynamic torques)
- Robust AOCS controller strategies to cope with disturbances uncertainties and their variations while ensuring Spacecraft safety with robust safe mode
- Drag compensation control laws for precise station keeping and continuous orbital adjustment
- Innovative compact and high-efficiency electric propulsion systems for sustained low-thrust drag compensation

VLEO compatible Materials & Structures - This area targets at the development of durable materials and structural solutions to resist VLEO environmental conditions:

- Materials with resistance to atomic oxygen exposure (e.g. polymers, composites, adhesives, coatings)
- Surface materials engineered for high aerodynamic performance and low drag
- Structural solutions ensuring long-term durability in the VLEO environment (thermal, mechanical, chemical)
- Atomic oxygen resistant photovoltaic assemblies including protection processes and materials

Payload Technology - This area addresses advanced compact, high-performance communication and sensing technologies tailored for the dynamic VLEO environment, such as:

- Electronic antenna steering (e.g. SAR, high relative velocity applications),
- VLEO Communication terminal technology,
- Ultra-stable compact imaging payload technologies (e.g. optical beam stabilisation, scanning, pointing), High-resolution detectors (SWIR TDI sensors, thermal IR sensors for high-temp operation).

4.3.2. List of activities

Very Low Earth Orbit (VLEO)

GNC, AOCS & Pointing

Programme Reference	Activity Title	Budget (k€)
GT1V-800EC	AOCS for VLEO with evolved reconfiguration, control recovery and actuators	1,000
	Total GNC, AOCS & Pointing	**1,000**

Power Systems, EMC & Space Environment

Programme Reference	Activity Title	Budget (k€)
GT1V-801EP	ATOX resistant development including protection processes and resistant materials	2,000
GT1V-802EP	In-situ environment monitor for VLEO platform	600
	Total Power Systems, EMC & Space Environment	**2,600**

Optics, Optoelectronics Robotics, Life Sciences

Programme Reference	Activity Title	Budget (k€)
GT1V-803MM	Compact and low drag dynamic targeting high resolution imager for VLEO	1,500
	Total Optics, Optoelectronics Robotics, Life Sciences	**1,500**

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

Programme Reference	Activity Title	Budget (k€)
GT1V-804MP	Enhanced modelling of complex gas-surface interaction in VLEO	800
GT1V-805MP	Atmospheric breathing electric propulsion system development	9,000
	Total Propulsion, AeroThermodynamics and Flight Vehicles Engineering	**9,800**

RF Payloads & Technology

Programme Reference	Activity Title	Budget (k€)
GT1V-806EF	Active antenna microwave payload for VLEO missions	1,500
	Total RF Payloads & Technology	**1,500**

Structures, Mechanisms & Materials

Programme Reference	Activity Title	Budget (k€)
GT1V-807MS	Modelling of VLEO atmosphere interactions with materials	650
GT1V-808MS	Drag-free materials for VLEO satellites	500
	Total Structures, Mechanisms & Materials	**1,150**

Total Very Low Earth Orbit (VLEO)	**17,550**

4.3.3. Description of activities

Very Low Earth Orbit (VLEO)

GNC, AOCS & Pointing

GT1V-800EC – AOCS for VLEO with evolved reconfiguration, control recovery and actuators

Budget: 1000 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 3 / 5 - TD 5

Objective:
To perform design trade-off, develop and test an AOCS for VLEO below 200 km with evolved reconfiguration, control recovery and actuators for guaranteed safety and robustness to solar activity, and enabled controlled reentry.
Description:
VLEO (usually below 400 km) may have significant advantages at mission and system level, however it has a major impact on the AOCS with higher complexity both on the hardware architecture (with sensors/actuators accommodation and sizing), the control algorithms (to cope with disturbances uncertainties and their variations while ensuring Spacecraft safety with robust safe mode), or both (for the compensation of increased and varying atmospheric drag and winds). Other aspects such as achievable pointing performance or de-orbiting concepts (as VLEO technologies enabling to reduce the last thrust and related propellant) to meet the new regulations while being robust to solar activity variations, pose additional technical challenges which need to be addressed. In order to solve these challenges, this activity will develop, breadboard and test an AOCS architecture targeting very demanding VLEO below 200km, with associated hardware accommodation and sizing (developing and testing use of Control surfaces/flaps in VLEO), on-board algorithms design, tuning, implementation and testing (adaptive guidance in closed loop for passive unloading while satisfying system needs with limited actuation, pseudo passive control for control recovery, and estimation of atmospheric density, drag and wind, including short term variations, with coupled autonomous reconfiguration), and finally assessment of the performances. This activity will use as inputs, existing models (high altitude wind models, solar activity) and their implementation (drag models with fine aerodynamics modelling in real-time simulation).

The activity encompasses the following tasks:
- Architecture trade-off and preliminary design (hardware accommodation and sizing, also including control surfaces/flaps actuator) based on control analysis (adaptive guidance and passive unloading considering power, thermal & other system needs), operational concepts definition including de-orbiting.
- AOCS & System reconfiguration detailed design and tuning considering solar activity and wind changes.
- AOCS Software development and its integration in a functional performance simulator (together with detailed models with dynamics, drag wind, all sensors and actuators including for flaps).
- Breadboard development and integration including SC shape mock-up, control surfaces and their actuators and AOCS Software integration on a representative processing board.
- Functional testing and detailed simulations and analysis of the performances, including pointing, momentum unloading and de-orbiting.
- Breadboard testing in wind tunnel for early demonstration of the guidance and control techniques and of the architecture design process.
- Results synthesis and identification of further required developments at hardware and software levels for feasibility& compliance, definition of recommendations for VLEO missions on the architecture design process (how to consider passive unloading, control surfaces and accommodation constraints for architecture definition and sizing), and assessment of the technology reuse for other missions with controlled reentry.

Deliverables: Prototype, Report, Software

Application/Need Date: All VLEO missions and indirectly all missions with controlled reeentry (as enabling to reduce the last thrust and related propellant) **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: AOCS and GNC Systems (2024) – Consistent with activity B13 “Integrated VLEO Aero-Control Surfaces for Communications and AOCS “.

Very Low Earth Orbit (VLEO)

Power Systems, EMC & Space Environment

GT1V-801EP – ATOX resistant PVA development including protection processes and resistant materials

Budget: 2000 k€ - **Duration**: 30 months - **Current / Targeted TRL:** 3 / 5 - TD 3

Objective:
To develop materials (e.g. interconnectors, coverglasses) and assembly processes enabling the ATOX protection of the European solar arrays and to qualify solar panel coupons against the ATOX environment and representative thermal cycles.

Description:
For VLEO applications the solar array needs to withstand the high exposure to the atomic oxygen (ATOX) environment expected in orbits of around 300 km altitude. In this environment, some of the critical elements for the performance of the photovoltaic assembly (PVA) require specific designs and/or protections.

Cell's interconnections require materials and design capable of surviving high ATOX erosion. Other elements which can be sensitive are protection diodes, busbars, coating of coverglasses, Kapton, cables, carbon fibre and its prepreg, paintings and coatings.

In addition to survive the ATOX environment, the technologies and materials proposed must demonstrate the survival to the fatigue behaviour caused by the large number of eclipses. In this respect, a full thermal cycling test campaign covering the most severe orbital configuration must be completed.

Therefore, this activity proposes the development of the PVA related technologies, including the substrate on which the PVA will be laid down, such that they can survive the ATOX environment encountered at 300 m altitude and the large number of thermal cycles expected in the worst-case orbit.

This activity encompasses the following tasks:
- Definition of specifications for an ATOX resistant PVA, including materials and manufacturing processes.
- Development of technology building blocks derived from the previously defined specifications.
- Manufacturing of a design verification test (DVT) coupon, including all the ATOX resistant PVA technologies and a suitable substrate for the ATOX environment.
- Execution of all related ATOX tests to demonstrate suitability of technologies and materials.
- Execution of the thermal cycling qualification test campaign.

Deliverables: Test coupons, Reports

Application/Need Date: VLEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Very Low Earth Orbit (VLEO)

Power Systems, EMC & Space Environment

GT1V-802EP – In-situ environment monitor for VLEO platform

Budget: 600 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 6 - TD 4

Objective:
To develop a low resources environment sensor (density, temperatures and atmospheric composition) compatible with operations in VLEO environments.

Description:
While the VLEO environment attracts more attention and interest for future space missions, the Earth lower thermosphere in altitude range 100 km to 300 km typically remains largely unexplored by spacecraft. Consequently, little in-situ environmental data have been gathered. Of particular importance, the temporal and spatial variability of the atmospheric density at very low altitude needs to be better constrained in order to improve atmospheric models used for mission design and reduce margins associated to attitude control and drag effects predictability in those regions of space.

This activity aims at filling this gap by providing a small resources/compact atmospheric environment sensor that shall be capable of providing as primary objectives in-situ environment data such as:
- total atmospheric density.
- atmospheric composition (with atomic oxygen as a driver).
- species temperature.

In addition, such sensor should be able to measure ionospheric components (species densities and temperatures) and optionally, ion drift. The technology can be based on either existing technologies to be modified to the VLEO needs and environmental constraints or on novel innovative sensor concepts.

This activity encompasses the following tasks:
- Critical review and consolidation of VLEO environments measurement requirements for neutral atmosphere and ionosphere.
- Definition of system requirements.
- Instrument design maturity and compliance assessment and design consolidation.
- Breadboarding (as needed).
- Sensor detailed design, including analyses (radiation assurance, thermal, EMC, structural/vibration, etc.).
- Fabrication of the environment sensor.
- Functional and environmental testing.
- Performance testing in dedicated chamber.

Deliverables: Engineering/Qualification Model, Report

Application/Need Date: VLEO missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Not relevant to any Harmonisation topic.

Very Low Earth Orbit (VLEO)

Optics, Optoelectronics Robotics, Life Sciences

GT1V-803MM – Compact and low drag dynamic targeting high resolution imager for VLEO

Budget: 1500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 16

Objective:
To develop a breadboard of an agile, multiple focal length, multispectral imaging system for space.

Description:
The overall payload objective is to address the needs for high resolution targeted observations from VLEO, with combining capabilities for detection of area of interest over large field of view (FOV) and observations at high resolution of automated selected areas. This combines the need for a large FOV multispectral imager, AI/intelligent on-board processing for detecting areas of interest and scanning/zooming function for imaging within the large FOV specific targeted areas at spatial high-resolution. The imagers are operating primarily in VIS/SWIR. The high-resolution imager provides sub-m resolution or lower capability. Such instrument is of particular interest in VLEO, as enabling in a compact form to provide both large area accessibility (thanks to the large FOV) while benefiting from the complementary high magnification and the low altitude to enhance spatial resolution. This is an element also benefiting for security and resilience objectives. Key challenges are in combining in a single instrument the imaging and detection over a large FOV, and the fast tracking and zooming function of a specific area within the imager FOV. The technical objective is to design the instrument consisting of the combined large field of view imager and the scanning/zooming high resolution imager. The development entails the system design, layout of the optical and mechanical design, breadboarding and testing of a prototype instrument. The activity is targeting a satellite flying at very low altitude with the need to design and optimise the payload and instrument shape to minimise drag, as well as providing scanning capability (cross and along track).

While this initial development will only address at design/concept level the on-board AI function, the longer-term development includes the development of the complete payload including the implementation of AI based control. As a first step, on board AI uses the large FOV system to select targets and command the high magnification camera with its scanner to image target areas within the large FOV. Even longer timescale development can include the investigation and implementation to include more spectral channels, other functionalities. Additionally, more complex agile operation modes can be explored and implemented, e.g. novel on-board AI techniques can be developed more complex acquisition schemes.

This activity encompasses the following tasks:
- Requirement consolidation.
- Design/analyse the system (including the automated AI detection concept), the optical and mechanical design and interface definition.
- Manufacturing of the core elements, implementing core functionalities.
- Integration of the prototype instrument, opto-mechanical and performance verification and testing.
- Evaluation and development plan.

Deliverables: Breadboard, Report

Application/Need Date: TRL6 by 2029 for VLEO missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap AOCS and GNC Systems.

Very Low Earth Orbit (VLEO)

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1V-804MP – Enhanced modelling of complex gas-surface interaction in VLEO

Budget: 800 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 18

Objective:
To enhance the gas-surface interaction predictive capabilities to enable high-efficiency intakes and aerodynamic concepts for airbreathing VLEO.

Description:
The correct modelling of gas-surface interactions is crucial for satellites operating in VLEO, where the residual atmosphere, primarily composed of atomic oxygen (ATOX) and nitrogen, significantly affects spacecraft (aero)dynamics. At these altitudes, the mean free path of atmospheric particles is comparable to the spacecraft dimensions, placing the vehicle in a transitional or free molecular flow regime. Inaccurate modelling of how gas particles interact with satellite surfaces could lead to errors in predicting aerodynamic drag, surface erosion, and thermal loads. These effects directly impact orbit stability, fuel consumption, and mission lifetime, making accurate gas-surface interaction modelling essential for reliable design and operation of VLEO spacecraft. Specifically for airbreathing VLEO spacecraft, where the efficiency and aerodynamic design of the intake is crucial to have a feasible aero-propulsive concept, accurate modelling becomes of critical importance.

Classical models of gas-surface interaction are commonly relying on Maxwellian velocity distributions, based on basic thermodynamical principles and assuming specular or diffusive reflections, which are independent on the actual surface molecular structure. They therefore don't always accurately predict the behaviour of real surfaces (which are for example affected by ATOX). More advanced models, such as "washboard models", could incorporate surface corrugations but some questions still arise on the effect of surface anisotropy, energy accommodation, etc.

The activity aims at enhancing the current state of the art (SOA) of gas-surface interaction models, supported by dedicated experimental campaigns, leading to the implementation and validation of such SOA models in Direct Simulation Monte Carlo (DSMC) codes. The use-case to keep in mind during the activity is model application to airbreathing VLEO internal and external aerodynamics.

The activity encompasses the following tasks:
- Review and gap analysis of the State-of-the-Art in gas-surface interaction modelling.
- Definition and execution of experimental test campaign(s) to support model improvement and validation.
- Improvement and implementation of enhanced models.
- Validation and verification of models for application use-case.

Deliverables: Breadboard, Report, Software

Application/Need Date: VLEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Technologies for fluid mechanics.

Very Low Earth Orbit (VLEO)

Propulsion, AeroThermodynamics and Flight Vehicles Engineering

GT1V-805MP – Atmospheric Breathing Electric Propulsion (ABEP) System Development

Budget: 9000 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 19

Objective:
To develop an Atmosphere Breathing Electric Propulsion (ABEP) system for extending mission lifetime in Very Low Earth Orbits (VLEO).
Description:
Operating in Very Low Earth Orbits (VLEOs) offers several advantages for various applications, including Earth observation, telecommunications, and navigation. Currently, VLEO exploitation is of high interest for constellations, either standalone or in combination with higher orbits. However, VLEO satellites face high residual atmospheric drag, leading to rapid orbit decay. This necessitates high-efficiency electric propulsion systems to maintain their orbits. Due to the high orbital maintenance needs and the requirement for compact systems, the lifetime of VLEO satellites is limited by the amount of propellant they can carry. To address this, the surrounding residual gas could be collected and used as propellant for “air breathing” electric propulsion systems, thereby increasing the mission lifetime. The maturation of ABEP systems to a TRL suitable for operation is planned in three phases:

1) ABEP system and subsystems’ definition and design, development models’ manufacturing and overall ABEP system’s development model’s assembly, integration and functional and performance verification,
2) ABEP system and subsystems’ qualification (environmental, lifetime, etc.),
3) ABEP system in-orbit demonstration and performances’ validation.
This proposal focuses on Phase 1, which aims to develop the subsystems of an ABEP system to TRL 5 and verify the overall ABEP system functionality and performance in an end-to-end test in a laboratory setup. The work logic for Phase 1 is based on an iterative engineering approach with two sub-phases:

Phase 1a - ABEP System Definition and Subsystems Design Specification (budget: 1.2 M€): to define the overall ABEP system design envelope, identify system-level design drivers and constraints, performance requirements, interfaces, and derive the design specifications for the ABEP subsystems (gas collector/intake and propellant management system, Thruster and Power Processing Unit) as detailed below:
- Aerothermodynamic and propulsive design of a generic satellite for VLEO applications, identification of potential operational scenarios, and derivation of the technical design and engineering specifications for the ABEP system elements.
- Derivation of minimum performance requirements for the ABEP system to ensure compatibility with orbital altitudes and spacecraft mass and aerodynamic characteristics.
- Modelling of collector/intake and prediction of collection efficiency and drag coefficients.
- Development of an overall ABEP system model to derive requirements for the thruster (e.g., specific impulse, thrust, power, thrust-to-power ratio, etc).
- Definition of detailed design specifications for the ABEP subsystems
- Definition of the specifications for test facilities and definition of the methods for on-ground testing and assessment of on-ground test limitations.

Phase 1b (budget: 7.8 M€): to design, manufacturing and testing of the ABEP subsystems (development models) and ABEP system’s development model’s assembly, integration and functional and performance verification:
- Design, manufacturing and test of a gas collector/intake and propellant management system (~2M€).
- Design, manufacturing and test of an ABEP Thruster (working with atmosphere gases mixture) (~2M€).
- Design, manufacturing and test of a Power Processing Unit for the ABEP thruster (~2 M€).
- Definition, preparation, and execution of the end-to-end ABEP system functional and performance test campaign, including the upgrade of test facilities as needed, to verify the aero-propulsive performance and environmental compatibility of the ABEP system (~1.8M€).

After completion of this activity, the maturation of ABEP system to a TRL suitable for operational utilization will require two additional follow-up phases:
- Phase 2: qualification (envisaged budget/duration: ~18 M€/24 months).
- Phase 3: in orbit demonstration (envisaged budget/duration: ~30 M€/18months).

Deliverables: Engineering Model, Report

Application/Need Date: VLEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap CubeSat Propulsion.

Very Low Earth Orbit (VLEO)

RF Payloads & Technology

GT1V-806EF – Active antenna microwave payload for VLEO missions

Budget: 1500 k€ - **Duration**: 18 months - **Current / Targeted TRL:** 4 / 5 - **TD** 7

Objective:
To design and development at (scaled) Engineering Model level of an active antenna microwave payload for VLEO missions demonstrating critical payload functionalities.

Description:
The potential of VLEO for space-borne microwave application missions has been widely acknowledged by space Industry and Agencies alike. For example, the reduced distance to the target area of interest allows a drastic reduction in the power required by payloads for Earth observation and telecommunications applications. However, to fully enable the prospects of VLEO's applications, the recent technological developments on platform side must be accompanied by a unified technology push on payload and instruments side. The stringent demands in terms of power, mass, size and thermal as well as plasma environment, enforced by the VLEO environment's properties, necessitate an adequate holistic payload design.

This activity will focus on the design, development and testing of a (scaled) engineering model of an active antenna microwave payload under the paradigm of VLEO scenario and an associated small platform. For this a dedicated mission scenario, utilizing the benefits of VLEO, shall be identified and a payload shall be designed. The payload shall be based on a reconfigurable active antenna solution with low profile (few cm), small surface area and low power consumption (few tens of watts), compatible with compact VLEO platforms. The actual dimensions of the payload shall be justified based on the considered application and platform. The operational frequency band shall be defined based on the proposed application in the range between X and Ka-band.

Attention shall be devoted to cost reduction implementing design to manufacture strategies, thereby reducing recurrent costs of the payload and enabling high volume production for a potential integration of the space segment into a constellation. This shall include the evaluation of existing ground/airborne-based technologies and COTS components for the instrument and its subsystems. A special focus shall be put on the usage of low power technology (e.g. SiGe) for the front-end, on integration of several RF functions on the same Integrated Circuit where possible, on the assessment of the optimum beamforming strategy, on highly dense integration techniques with multilayer PCB structures and on optimization of the losses.

Furthermore, advanced manufacturing techniques to enhance manufacturability and further optimized cost shall be investigated. Finally, the potential of standardized modular subsystems shall be evaluated, enabling plug and play modules that can be mass produced and exchanged at any point in the production. The impact on the payload design in terms of the minimization of mechanical interfaces as well as the usage of automated test benches and batch calibration shall also be evaluated.

RF as well as environmental aspects, such as thermo-mechanical behaviour and robustness to the harsh VLEO environment, shall be considered in the design and tested during the EM's experimental validation.

This activity encompasses the following tasks:
- Definition and justification of the selected application and consolidation of the requirements,
- Design and detailed analysis of the active antenna microwave payload,
- Manufacture of the scaled engineering model.
- RF and thermo-mechanical tests of the scaled engineering model.

Deliverables: Engineering model, Report

Application/Need Date: VLEO missions **Mission Classification: gamma,** delta

THAG Roadmap: Not relevant to any Harmonisation Roadmap.

Very Low Earth Orbit (VLEO)

Structures, Mechanisms & Materials

GT1V-807MS – Modelling of VLEO atmosphere interactions with materials

Budget: 650 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 24

Objective:
To develop a molecular dynamics (MD) modelling tool for determination of atomic oxygen materials erosion rates in VLEO atmosphere for early phase satellite design.

Description:
Flying a satellite in VLEO presents multiple advantages but is a technological challenge due to the density of the atmosphere at these altitudes. Of particular interest is the interaction between reactive chemical species present in VLEO, such as atomic oxygen, and spacecraft surfaces, that can erode and lose their useful functions once operated at such reactive environment. To study the interactions between surfaces and the atmosphere, a molecular dynamics simulation can be implemented, that allows for the simulation of erosion processes on the atomistic scale and, using the force field such as ReaxFF, can model the bond breaking and formation as chemical reactions occur. The MD simulations can also be used to simulate realistic atmospheric compositions, such as mixtures of oxygen and nitrogen, in a more straightforward way then experimental testing, providing a quick first de-risking assessment for spacecraft designers. The recent enhancements in AI technology will result in reducing computational cost, increasing accuracy and improving data analysis.

This activity encompasses the following tasks:
- Evaluation of available software solutions for molecular dynamics simulation.
- Requirements definition.
- Model creation for common spacecraft surfaces and materials (MLI, paints, coatings).
- Model optimization.
- Validation testing on various material samples.
- Design of user-friendly graphical interface for use by materials engineers and satellite designers.

Deliverables: Material samples, Software, Reports

Application/Need Date: VLEO missions **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap CubeSat Propulsion.

Very Low Earth Orbit (VLEO)

Structures, Mechanisms & Materials

GT1V-808MS – Drag-free materials for VLEO satellites

Budget: 500 k€ - **Duration**: 24 months - **Current / Targeted TRL:** 3 / 5 - TD 24

Objective:
To develop a coating or surface modifications with an effectively smooth gas-surface interaction potential to promote minimal accommodation and therefore minimal drag that can be applied on common spacecraft materials and the experimental validation of the technology in a space simulation facility.

Description:
In the low Earth orbit (LEO) and especially in the very low Earth orbit (VLEO), spacecraft operate in residual atmosphere, predominantly consisting of atomic oxygen (ATOX) and N2 which collide with ram surfaces at relative velocities of 8 km/s. The high densities of species cause sufficient drag to require an extra use of the propulsion system for a satellite to maintain orbit. Hence it is of interest to select materials with low drag characteristics and resistance to ATOX/N2 impingement to reduce the propellant's use and increase the mission lifetime. To achieve that, materials with low drag surface properties should be selected for the manufacturing of external surfaces. Broadly speaking if incident atoms or molecules scatter from a smooth surface in the forward direction while retaining most of their incident kinetic energy, then drag is significantly reduced. The key to assessing the drag potential of a surface is thus to characterize the scattering dynamics of O and N2 molecules that impinge on the surface with orbital velocities. This activity envisages the identification and characterization of materials, coatings and/or surface treatments to significantly reduce the drag acting on future VLEO spacecrafts.

This activity encompasses the following tasks:
- Definition of representative mission scenarios and the related environment and consolidation of requirements.
- Evaluation of available solutions on the market.
- Selection materials, coatings and/or surface treatment including related processes.
- Coupon manufacturing and testing focused on validation of the process for space applications.
- Definition and planning for representative testing
- Coupon testing in representative environment to demonstrate drag reduction properties and resistance to space environment.

Deliverables: Test coupons, Report

Application/Need Date: TRL 4 by 2028. VLEO missions. **Mission Classification:** alpha, beta, gamma, delta

THAG Roadmap: Relevant to the Roadmap Electric Propulsion Technologies.

5. Annex I – Proposed bundled activities

Bundle	Domain	Area	Progr. Ref.	Activity Title	Budget
Bundle 1 - Enabling in-orbit assembly of large antennas	Generic Technologies	GNC & AOCS & Pointing	GT17-809EC	Guidance, navigation and control system development for in-orbit assembly of very large antennas	800
		RF Payloads / Technology	GT17-821EF	In-orbit assembly and performance demonstration of a large antenna	1,900
		Structures, Mechanisms, Materials	GT17-848MS	Deployable support structure with feed horn and metrology functionality	900
Bundle 2 - Enabling New space - miniaturisation and serialisation	Generic Technologies	Optics, Optoelectronics Robotics, Life Sciences	GT17-838MM	Large duolithic mirrors for snap-together telescope assembly	1,000
			GT17-839MM	Low energy / high pulse repetition frequency based space lidars	1,000
			GT17-846MM	Integrated chip based high power c-band optical amplifier	1,500
Bundle 3 - Enabling high Infrared performances	Generic Technologies	Optics, Optoelectronics Robotics, Life Sciences	GT17-841MM	Infrared detector for next generation sounder instruments	1,500
			GT17-843MM	Coating development for lens-like samples in the infrared	500
Bundle 4 - LIDAR developments for enhanced vegetation monitoring	Generic Technologies	Optics, Optoelectronics Robotics, Life Sciences	GT17-840MM	Alexandrite based Lidar for the development of vegetation monitoring techniques	1,000
			GT17-844MM	Detector array for EO Lidar applications	1,000
Bundle 5 – AI for verification and validation	Artificial Intelligence	GNC, AOCS & Pointing	GT1I-811EC	Verification and validation of AOCS/GNC using AI on-board	1000
		Avionics & EEE	GT1I-813ED	Autonomous and standardized integration, verification, validation and qualification of data handling system in avionics platform	700

Bundle	Domain	Area	Progr. Ref.	Activity Title	Budget
Bundle 6 - Atomic clocks	Quantum Technologies	Avionics & EEE	GT1Q-800ED	VCSELs for miniature quantum devices based on Rubidium vapour cells	1,600
		RF Payloads / Technology	GT1Q-820EF	Low noise ultra miniature local oscillator for chip scale quantum devices	700
Bundle 7 - Cold atom interferometer payload development	Quantum Technologies	Optics, Robotics & Life Sciences	GT1Q-809MM	Continuous beam generation of ultra-cold atoms	2,000
			GT1Q-811MM	Robust high-flux source for ultra-cold atoms	2,000
			GT1Q-814MM	Agile laser system for cold atom interferometer technology with low SWaP	2,000
			GT1Q-815MM	Versatile control unit for cold atom interferometer technology	1,000
			GT1Q-816MM	Inertial reference platform for ultra precise cold atom interferometer inertial sensing	1,000
Bundle 8 – Enabling proximity operations and rendezvous	Sustainable Space	GNC, AOCS & Pointing	GT1C-814EC	Close-proximity range finder for rendezvous and docking	800
			GT1C-815EC	Dual navigation and control of a servicer spacecraft and its robotic arm	1,800
			GT1C-816EC	Multispectral imaging and ranging data fusion and navigation for rendezvous and proximity operations	1,000